Troubleshooting Manual

HYDRAULIC
MINING
SHOVEL

PC3000E-6

SERIAL NUMBER 06249 and up
SERIAL NUMBER 46225 and up

KOMATSU

COPYRIGHT

This material is property of Komatsu Mining Germany GmbH and is not to be reproduced, used, or disclosed except in accordance with the written authorization of Komatsu Mining Germany GmbH.
It is our policy to improve our products whenever it is possible and practical to do so. We reserve the right to make changes or add improvements at any time without incurring any obligation to install such changes on products sold previously.
Due to this continuous program of research and development, periodic revisions may be made to this publication. It is recommended that the customer contact their distributor for information on the latest revision.

INTRODUCTION 7

Introduction & general information for troubleshooting 8

Description of this troubleshooting manual 8
Precautions when carrying out any operation 11
Fundamental requirements for troubleshooting 12
Points to remember when troubleshooting 13
Sequence of events in troubleshooting 16
Checks before troubleshooting 17
Handling of electric equipment and hydraulic components 18
General working procedures 22
Information on the ETM Versions and the Electronic Pump Controller CR700 26
service level password 28

ELECTRICAL PARTS 31

Mounting locations of electrical parts, connector types and electrical standard values 32

Component locations 32
Location of electrical parts Capacitor / electric motor 34
Location of electrical parts on PTO, Suction Oil Reservoir and Central Control and Filter Panel 36
Location of electrical parts on Pilot Control Frame 38
Location of electrical parts on Main Hydraulic Oil Reservoir 40
Location of electrical parts on Switch Board X2 in Cab Base 42
Location of electrical parts of the hydraulically operated Access Ladder 44
Location of electrical parts of the automatic Lubrication Systems 46
Overview connector types and connector pin numbers 50
Standard value table for electrical components 53
PT100 temperature charts 61

TROUBLESHOOTING BY TROUBLE CODE 63

Message -1- Gear lubrication out of function (B17) 64
Message -2- Motor shutdown: Fire suppression system actuated 67
Message -3- Oil filter for control oil restricted (B22) 71
Message -4- Oil tank breather filter restricted (B24) 74
Message -5- Return oil filter restricted (B26) 77
Message -6- Leak oil filter restricted (B25) 80
Message -7- Hydraulic oil temperature too high (B15) 83
Message -8- (Not used) 89
Message -9- Main transformer temperature too high (1K4) 89
Message -10- Low voltage monitoring actuated 93
Message -11- (Not used) 97
Message -12- (Not used) 97
Message -13- (Not used) 97

Message -14- Motor-shutdown: Too low hydraulic oil level! (B4) 98
Message -15- Caution, slew gear house brake ON 102
Message -16- Caution, travel gear house brake ON 107
Message -17- No clearance for starting, wrong direction of motor rotation 112
Message -18- Central lubrication system fault 115
Message -19- Slew ring gear lubrication system fault 122
Message -20- Motor shutdown: Motor protection relay released (1K2) 130
Message -21- Central lubrication system, empty grease barrel 134
Message -22- Slew ring gear lubrication system, empty grease barrel 134
Message -23- Motor shutdown: Motor bearing temperature too high (1K14) 140
Message -24- Caution, Insulation monitoring actuated (1K10) 145
Message -25- No clearance for starting by motor protection relay (1K3) 148
Message -26- Attention! Overheating protection of cab heating has triggered 151
Message -27- (Not used) 155
Message -28- (Not used) 155
Message -29- (Not used) 155
Message -30- (Not used) 155
Message -31- (Not used) 155
Message -32- PTO-gear lubrication filter restricted (B27) 155
Message -33- Cooler fan drive filter restricted (B28) 158
Message -34- (Not used) 161
Message -35- (Not used) 161
Message -36- (Not used) 161
Message -37- (Not used) 161
Message -38- (Not used) 161
Message -39- Motor-shutdown: Shut-Off (gate) valve S31 closed 162
Message -40- Caution, switch-cabinet temperature too high (6B1, 6B2) 166
Message -41- Motor shutdown: Engine temperature too high (1K6) 170
Message -42- (Not used) 175
Message -43- Motor-shutdown: Emergency stop switch actuated 175
Message -44- Motor-shutdown: Safety switch actuated (1K58) 179
Message -45- Motor-shutdown: Capacitor failed (1K12) 183
Message -46- (Not used) 187
Message -47- (Not used) 187
Message -48- Too high PTO-gear oil temperature (B49) 187
Message -50- Caution, pull switch from ground man actuated 190
Message -51- Motor shutdown: pull switch from ground man actuated 191
Message -52- Load limiting clam defective 193
Message -53- Pressure switch of the central lubrication system actuated (B43) 196
Message -54- Pressure switch of the slew ring gear lubrication system actuated (B46) 196
Message -55- Refill hydraulic oil 197
Message -56- Pilot control cut out 201
Message -57- Pump regulation faulty 205
Message -58- Strainer oil cooler restricted (B165) 205
Message -59- Central lubrication system, refill grease barrel 207
Message -60- Slew ring gear lubrication system, refill grease barrel 207

TROUBLESHOOTING BY SYMPTOMS 213

Symptoms of the electric motor and related electrical system 214

SEM001 - Motor does not start 214
SEM002 - Motor does not stop 218
SEM003 - Excessive motor vibration 220
SEM004 - Abnormal noise from motor 221
SEM005 - Motor lacks output (low power) 222
SEM006 - Air to Air cooling system not sufficient 223
SEM007 - Motor shuts down during start 224
SEM008 - Sub Station main circuit breaker trips 226

Symptoms of the hydraulic system 228

SHY001 - All the work equipment shows lack of power or slow movement (especially the BOOM UP function) 228
SHY002 - No work equipment, travel or swing movement 230
SHY003 - Work equipment has too much hydraulic drift 231
SHY004 - Work equipment has big time lag 232
SHY005 - Machine deviates during travel movement 233
SHY006 - Machine does not swing 234
SHY007 - Swing acceleration is poor 235
SHY008 - Excessive overrun when stopping swing 237
SHY009 - Floating system without function 239
SHY010 - Abnormal noise coming from SRV or MRV 240
SHY011 - Track tensioning system without function 241
SHY012 - Ladder does not move, or slow movement 242
SHY013 - The hydraulic driven grease pump does not move 244
SHY014 - Abnormal track chain movement during digging 245
SHY015 - Swing parking brake without function 246
SHY016 - Travel parking Brake without function 247

Symptoms of the mechanical system 248

SME001 - Abnormal noise around the PTO / hydraulic pumps 248
SME002 - There is a big shock when stopping swing 250
SME003 - There is a big abnormal noise when moving/stopping swing 251
SME004 - Abnormal noise coming from a travel gearbox 253
SME005 - Abnormal noise coming from the work equipment (lack of grease) 255
SME006 - Abnormal idler movement 257
SME007 - Leak at the PTO shaft seal 258

Symptoms of additional systems 260

SAD001 - Air conditioner does not work 260
SAD002 - Air conditioner HV / LV cabinet does not work 262

INTRODUCTION

INTRODUCTION & GENERAL INFORMATION FOR TROUBLESHOOTING

DESCRIPTION OF THIS TROUBLESHOOTING MANUAL

In this manual you find troubleshooting information which should assist you when troubleshooting.
The content of this manual is divided in 5 parts, which are titled:

Table of contents

Introduction and general information for troubleshooting

Mounting locations, connector types, and electrical standard values

Troubleshooting by trouble code

Troubleshooting by symptoms

Part II contains Safety Notes and Precautions which should be followed during troubleshooting work. Furthermore fundamental requirements and points to remember when troubleshooting are given. Also you find recommendations concerning the sequence of working steps, checks which should be done before troubleshooting, how to handle electric and hydraulic equipment, and how to perform basic working procedures.

Part III provides you with information about mounting locations of components, different connector types as they are used in your machine, and standard values for electrical components as a quick reference.

In Part IV troubleshooting charts are listed in the numerical sequence of trouble codes as they are stored in the control system. If predetermined conditions occur, the ETM displays relevant trouble codes and messages.

In Part V troubleshooting charts are listed for various motor electrical, mechanical, and hydraulic problems. These charts are showing symptoms of any malfunction which the monitoring system can not detect under the particular conditions. Therefore these malfunctions are not indicated by stored trouble codes or messages via ETM display.

Nevertheless the troubleshooting charts in Part IV and Part V can not cover all eventual malfunctions which may possibly occur. In such cases a general inspection of the suspected system is necessary.

Furthermore, it is not necessary that you strictly follow the given troubleshooting charts in every case. It is important to find the cause for any trouble in the easiest way and in the shortest time. In most times the relevant electrical diagram will help you to be successful.

Safety Notes

Important safety notice

Proper service and repair are extremely important for safe machine operation. The service and repair techniques recommended by Komatsu and described in the shop manual are both effective and safe. Some of these techniques require the use of tools specially designed by Komatsu for the specific purpose.

⚠ WARNING

To prevent injury to workers, the symbol is used to mark safety precautions in this manual. The cautions accompanying these symbols should always be followed carefully. If any dangerous situation arises or may possibly arise, first consider safety, and take the necessary actions to deal with the situation.

General precautions

⚠ WARNING

Mistakes in operation are extremely dangerous. Read the Operation and Maintenance Manual carefully before operating the machine.

1. Before carrying out any maintenance or repairs, read all the safety plates affixed to the machine. For the locations of the safety plates and detailed explanation of precautions, see the Operation and Maintenance Manual.

2. Decide a place in the repair workshop to keep tools and removed parts. Always keep the tools and parts in their correct places. Always keep the work area clean and make sure that there is no dirt, water, or oil on the floor. Smoke only in the areas provided for smoking. Never smoke while working.

3. When carrying out any operation, always wear safety shoes and helmet. Do not wear loose work clothes, or clothes with buttons missing.

    - Always wear safety glasses when hitting parts with a hammer.
    - Always wear safety glasses when grinding parts with a grinder, etc.

4. When carrying out any operation with 2 or more workers, always agree on the operating procedure before starting. Always inform your fellow workers before starting any step of the operation. Before starting work, place a UNDER REPAIR warning sign in the operator's cab and set the Maintenance Safety Switch to -0- position. Secure this position with a padlock.

5. Only qualified workers must carry out work and operation which require license or qualification.

6. Keep all tools in good condition, learn the correct way to use them, and use the proper ones of them. Before starting work, thoroughly check the tools, machine, forklift, service car, etc.

7. If weld repairs are needed, always have a trained and experienced welder for carrying out the work. Always wear welding gloves, apron, shielding goggles, cap and other clothes suited for welding work. Observe the PROTECTIVE MEASURES BEFORE STARTING WELD REPAIRS given in the Operation and Maintenance Manual of your machine.

8. Before starting work, warm up your body thoroughly to start work under good condition.

Safety points

1	Good arrangement
2	Correct work clothes
3	Following work standard
4	Making and checking signs
5	Prohibition of operation and handling by unlicensed workers
6	Safety check before starting work
7	Wearing protective goggles (for cleaning or grinding work)
8	Wearing shielding goggles and protectors (for welding work)
9	Good physical condition and preparation
10	Precautions against work which you are not used to or you are used to too much

Preparations for work

1. Before adding oil or making any repairs, park the machine on hard and level ground. Isolate the machine according to local regulations.

2. Before starting work, make sure that the work equipment is lowered to the ground. Also make sure that the maintenance safety switch (S58) is switched OFF and locked with a padlock. In addition hang up warning signs.

3. When disassembling or assembling, support the respective unit with blocks, jacks, or stands as necessary before starting work.

4. Remove all mud and oil from the steps or other places used to get on and off the machine. Always use the handrails, ladders or steps when getting on or off the machine. Never jump on or off the machine. If it is impossible to use the handrails, ladders or steps, use a stand to provide safe footing.

Precautions during work

5. Before disconnecting or removing components of the oil, coolant, or air circuits, first release the pressure completely from the circuit. When removing the oil filler cap, a drain plug, or an oil pressure pickup plug, loosen it slowly to prevent the oil from spurting out.

6. The coolant and oil in the circuits are hot when the motor is stopped, so be careful not to get scalded. Wait for the oil and coolant to cool before carrying out any work on the oil or coolant circuits.

7. Before starting work, stop the motor. When working on or around a rotating part, in particular, stop the motor. When checking the machine without stopping the motor (measuring oil pressure, revolving speed, temperature, etc.), take extreme care not to get rolled or caught in rotating parts or moving parts.

8. Before starting work, remove the leads from the negative (–) terminal first.

9. When raising a heavy component (heavier than 25 kg), use a hoist or crane. Before starting work, check that the slings (wire ropes, chains, and hooks) are free from damage. Always use slings which have ample capacity and install them to proper places. Operate the hoist or crane slowly to prevent the component from hitting any other part. Do not work with any part still raised by the hoist or crane.

10. When removing a cover which is under internal pressure or under pressure from a spring, always leave 2 bolts in diagonal positions. Loosen those bolts gradually and alternately to release the pressure, and then remove the cover.

11. When removing components, be careful not to break or damage the electrical wiring. Damaged wiring may cause fire.

12. When removing piping, stop the fuel or oil from spilling out. If any fuel or oil drips onto the floor, wipe it up immediately. Fuel or oil on the floor can cause you to slip and can even start fires.

13. As a general rule, do not use gasoline to wash parts. Do not use it to clean electrical parts, in particular.

14. Be sure to assemble all parts again in their original places. Replace any damaged parts and parts which must not be reused with new parts. When installing hoses and wires, be sure that they will not be damaged by contact with other parts when the machine is operated.

15. When installing high pressure hoses, make sure that they are not twisted. Damaged tubes are dangerous, so be extremely careful when installing tubes for high pressure circuits. In addition, check that connecting parts are correctly installed.

16. When assembling or installing parts, always tighten them to the specified torques. When installing protective parts such as guards, or parts which vibrate violently or rotate at high speed, be particularly careful to check that they are installed correctly.

17. When aligning 2 holes, never insert your fingers or hand. Be careful not to get your fingers caught in a hole.

18. When measuring hydraulic pressure, check that the measuring tools are correctly assembled.

19. Take care when removing or installing the tracks. When removing the track, the track separates suddenly, so never let anyone stand at either end of the track.

20. If the motor is operated for a long time in a place which is not ventilated well, you may suffer from gas poisoning. Accordingly, open the windows and doors to ventilate well.

PRECAUTIONS WHEN CARRYING OUT ANY OPERATION

When carrying out removal or installation (disassembly or assembly) of units, be sure to follow the general precautions given below when carrying out the operation.

Precautions when carrying out removal work

- If the coolant contains antifreeze, dispose of it correctly.
- After disconnecting hoses or tubes, cover them or fit plugs to prevent dirt or dust from entering.
- When draining oil, prepare a container of adequate size to catch the oil.
- Confirm the match marks showing the installation position, and make match marks in the necessary places before removal to prevent any mistake when assembling.
- To prevent any excessive force from being applied to the wiring, always hold the connectors when disconnecting the connectors. Do not pull the wires.
- Fit wires and hoses with tags to show their installation position to prevent any mistake when installing.
- Check the number and thickness of the shims, and keep in a safe place.
- When raising components, be sure to use lifting equipment of ample strength.
- When using forcing screws to remove any components, tighten the forcing screws uniformly in turn.
- Before removing any unit, clean the surrounding area and fit a cover to prevent any dust or dirt from entering after removal.
- After disconnecting any piping during disassembly operations make sure to use the correct plugs and seals when connecting the piping again.

Precautions when carrying out installation work

- Tighten all bolts and nuts (sleeve nuts) to the specified torque.
- Install the hoses without twisting or interference and fix them with intermediate clamps, if there are any.
- Replace all gaskets, O-rings, cotter pins, and lock plates with new parts.
- Bend the cotter pins and lock plates securely.
- When coating with adhesive, clean the part and remove all oil and grease, then coat the threaded portion with 2 – 3 drops of adhesive.
- When coating with gasket sealant, clean the surface and remove all oil and grease, check that there is no dirt or damage, then coat uniformly with gasket sealant.
- Clean all parts, and correct any damage, dents, burrs, or rust.
- Coat rotating parts and sliding parts with motor oil.
- When press fitting parts, coat the surface with anti-friction compound (LM-P).

- After fitting snap rings, check that the snap ring is fitted securely in the ring groove.
- When connecting wiring connectors, clean the connector to remove all oil, dirt, or water, then connect securely.
- When using eyebolts, check that there is no deformation or deterioration, screw them in fully, and align the direction of the hook.
- When tightening split flanges, tighten uniformly in turn to prevent excessive tightening on one side.
- When operating the hydraulic cylinders for the first time after reassembling cylinders, pumps and other hydraulic equipment removed for repair, always bleed the air as follows:
    - Start the motor.
    - Operate the work equipment control lever to operate the hydraulic cylinder 4 - 5 times, stopping the cylinder 100 mm from the end of its stroke.
    - Next, operate the hydraulic cylinder 3 - 4 times to the end of its stroke.
- When using the machine for the first time after repair or long storage, follow the same procedure.

Precautions when completing the operation

- Refilling with coolant, oil and grease
- If the coolant has been drained, tighten the drain valve, and add coolant to the specified level. Run the motor to circulate the coolant through the system. Then check the coolant level again.
- If the piping or hydraulic equipment has been removed, always bleed the air from the system after reassembling the parts.
- For details, see “Bleeding air”.
- Add the specified grease to the work equipment parts.

FUNDAMENTAL REQUIREMENTS FOR TROUBLESHOOTING

Technical documentation

The personnel in charge of troubleshooting must have permanent and easy access to the following appropriate and up-to-date documents of the relevant hydraulic excavator:

- Electric Circuit Diagram
- Hydraulic Circuit Diagram
- Operation & Maintenance Manual
- Service Manual
- Operation & Maintenance Manual for the Siemens electric motor
- Inspection and Maintenance Manual of the equipped Fire Suppression System (optional equipment)
- Inspection and Maintenance Manual of the equipped Air Conditioning System

Without the above documents troubleshooting is ineffective and unreliable!

Tools

For proper troubleshooting and subsequent repairs it is essential to have access to complete facilities which include appropriate measuring instruments as well as necessary tools for the repair of a hydraulic excavator.

For testing electrical components the following tools are obligatory.

Test equipment	Part number
Electrical Test-Box for Deutsch Connectors	232 762 40
Clamp-on Ammeter	232 756 40

View onto the above mentioned tools

Clamp-on ammeter	Electrical Test-Box for Deutsch Connectors	
 Z 27353		 Z 27355

Personnel

Only authorized and trained personnel should perform troubleshooting and the resulting repairs.

Work on the electrical system and equipment of the machine must be carried out only by a skilled electrician or by instructed persons under the supervision and guidance of a skilled electrician and in accordance with electrical engineering rules and regulations.

Work on the hydraulic system must be carried out only by personnel with special knowledge and experience of the hydraulic equipment.

Work on the Fire Suppression System must be carried out only by certified persons.

POINTS TO REMEMBER WHEN TROUBLESHOOTING

1. Park the machine at a safe place on level and solid ground. Observe the instructions for parking the excavator in the Operation and Maintenance Manual of the machine.
2. When carrying out the operation with 2 or more workers, keep strictly to the agreed signals, and do not allow any unauthorized person to come near.
3. If the motor is hot wait for the motor to cool down before starting troubleshooting.
4. Be extremely careful not to touch any hot parts or to get caught in any rotating parts.
5. When disconnecting wiring, always disconnect the negative (–) terminal of the battery first.
6. When removing the plug or cap from a location which is under pressure from oil, water, or air, always release the internal pressure first. When installing measuring equipment, be sure to connect it properly.

The aim of troubleshooting is to pinpoint the basic cause of the failure, to carry out repairs swiftly, and to prevent reoccurrence of the failure. When carrying out troubleshooting, and important point is of course to understand the structure and function. However, a short cut to effective troubleshooting is to ask the operator various questions to form some idea of possible causes of the failure that would produce the reported symptoms.

When carrying out troubleshooting, do not hurry to disassemble the components.
If components are disassembled immediately any failure occurs:

- Parts that have no connection with the failure or other unnecessary parts will be disassembled.
- It will become impossible to find the cause of the failure.

It will also cause a waste of man hours, parts, or oil or grease, and at the same time, will also lose the confidence of the user or operator. For this reason, when carrying out troubleshooting, it is necessary to carry out thorough prior investigation and to carry out troubleshooting in accordance with the fixed procedure.

- Points to ask user or operator:
  - Have any other problems occurred apart from the problem that has been reported?
  - Was there anything strange about the machine before the failure occurred?
  - Did the failure occur suddenly, or were there problems with the machine condition before this?
  - Under what conditions did the failure occur?
  - Had any repairs been carried out before the failure?
    When were these repairs carried out?
  - Has the same kind of failure occurred before?

1. Check before troubleshooting

- Check the oil level
- Check for any external leakage of oil from the piping or hydraulic equipment
- Other maintenance items can be checked externally, so check any item that is considered to be necessary.

2. PM-CLINIC

- Check the basic adjustments according to the data given in the PM-Clinic Procedure Manual. Make sure the settings of the temperature switch units for hydraulic oil, PTO gear oil and coolant are within specified values.

3. Download of failure history

- Download the contents of Protocol and Statistics Memory. Refer to the Operation& Maintenance Manual for download procedure.
  - The recorded fault data may be helpful for finding the cause of the trouble condition.

4. Confirming failure

- Confirm the extent of the failure yourself, and judge whether to handle it as a real failure or as a problem with the method of operation etc.
  - When operating the machine to reenact the troubleshooting symptoms, do not carry out any investigation or measurement that may make the problem worse.

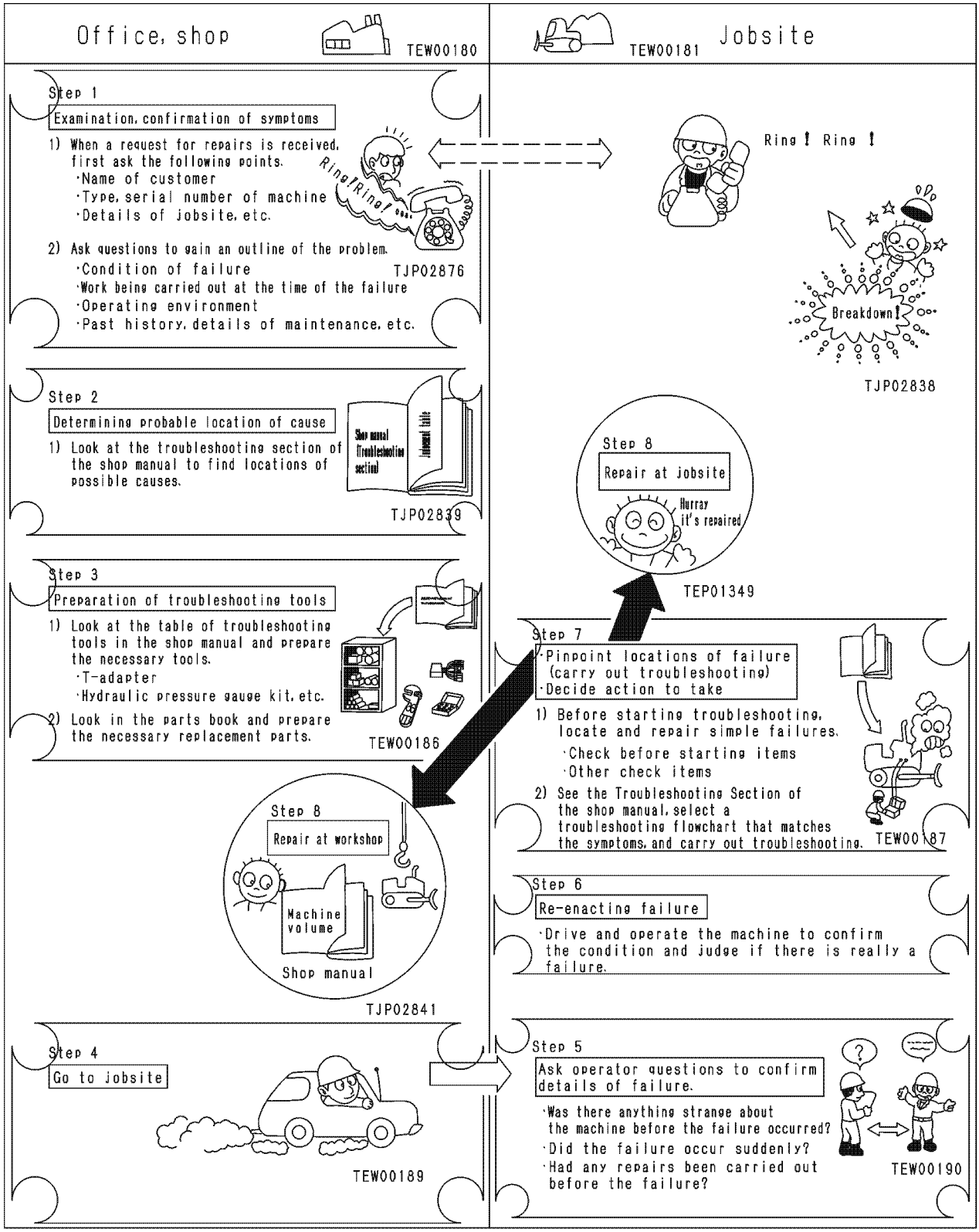
5. Troubleshooting

- Use the results of the investigation and inspection in items 2-5 to narrow down the causes of failure, then use the troubleshooting flowchart to locate the position of the failure exactly.
- The basic procedure for troubleshooting is as follows.
    - Start from the simple points.
    - Start from the most likely points.
    - Investigate other related parts or information.

6. Measures to remove the root cause of any failure

Even if the failure is repaired, if the root cause of the failure is not repaired, the same failure will occur again. To prevent this, always investigate why the problem occurred. Then, remove the root cause.

SEQUENCE OF EVENTS IN TROUBLESHOOTING



Z 26981

CHECKS BEFORE TROUBLESHOOTING

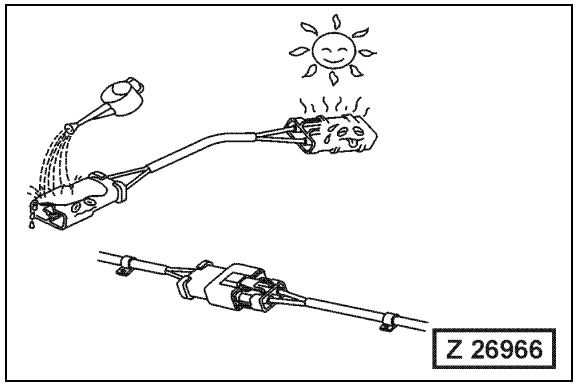
Item	Inspection	Criterion	Remedy
Lubricating oil	1. Check of level and type of fuel	-	Add fuel
	2. Check of fuel for foreign matter	-	Clean and drain
	3. Check of fuel filter	-	Replace
	4. Check of hydraulic oil level	-	Add oil
	5. Check of hydraulic oil strainer	-	Clean and drain
	6. Check of swing machinery oil level	-	Add oil
	7. Check of dust indicator for clogging	-	Clean or replace
	8. Check of hydraulic oil filter	-	Replace
Electrical equipment	1. Check of battery terminals and wiring for looseness and corrosion	-	Retighten or replace
	2. Check of alternator terminals and wiring for looseness and corrosion	-	Retighten or replace
	3. Check of starting motor terminals and wiring for looseness and corrosion	-	Retighten or replace
Hydraulic, mechanical equipment	1. Check for abnormal noise and smell	-	Repair
	2. Check for oil leakage	-	Repair
	3. Bleeding air	-	Bleed air
Electrical components	1. Check of battery voltage (with motor stopped)	22 - 24V	Replace
	2. Check level of battery electrolyte	-	Add or replace
	3. Check of wires for discoloration, burn, and removal of the isolation cover	-	Replace
	4. Check for released wire clamp and dropping wire	-	Repair
	5. Check of wires for wetness (in particular check connectors and terminals for wetness)	-	Disconnect the connectors and dry
	6. Check for broken or corroded fuses	-	Repair
	7. Check battery charging voltage (motor)	After operating for several minutes: 27.5 - 29.5V	Replace battery charger module
	8. Check battery relay for operating sound (when starting switch is turned ON or OFF)	-	Replace

HANDLING OF ELECTRIC EQUIPMENT AND HYDRAULIC COMPONENTS

Points to remember when handling electric equipment

Handling wiring harnesses and connectors

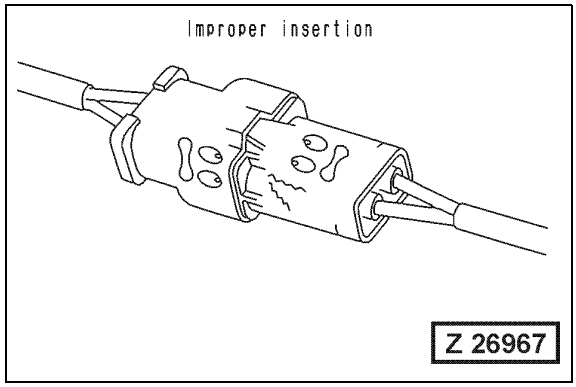
Wiring harnesses consist of wiring connecting one component to another component, connectors used for connecting and disconnecting one wire from another wire, and protectors or tubes used for protecting the wiring. Compared with other electrical components fitted in boxes or cases, wiring harnesses are more likely to be affected by the direct effects of rain, water, heat, or vibration. Furthermore, during inspection and repair operations, they are frequently removed and installed again, so they are likely to suffer deformation or damage. For this reason, it is necessary to be extremely careful when handling wiring harnesses.



Main failures occurring in wiring harness

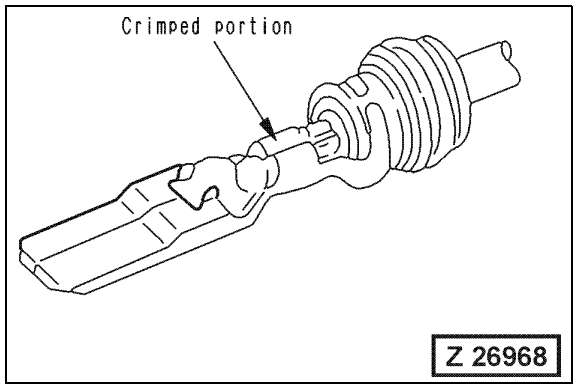
Defective contact of connectors (defective contact between male and female)

Problems with defective contact are likely to occur because the male connector is not properly inserted into the female connector, or because one or both of the connectors is deformed or the position is not correctly aligned, or because there is corrosion or oxidation of the contact surfaces.



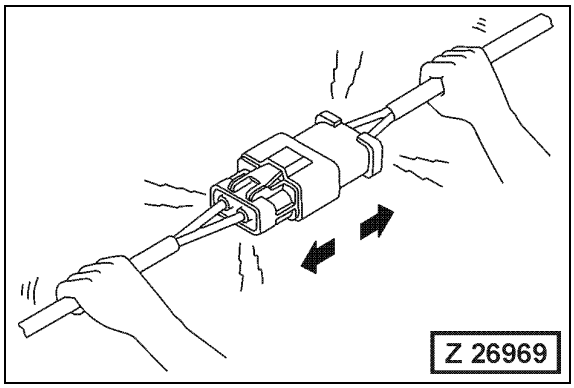
Defective crimping or soldering of connectors

The pins of the male and female connectors are in contact at the crimped terminal or soldered portion, but if there is excessive force brought to bear on the wiring, the plating at the joint will peel and cause improper connection or breakage.



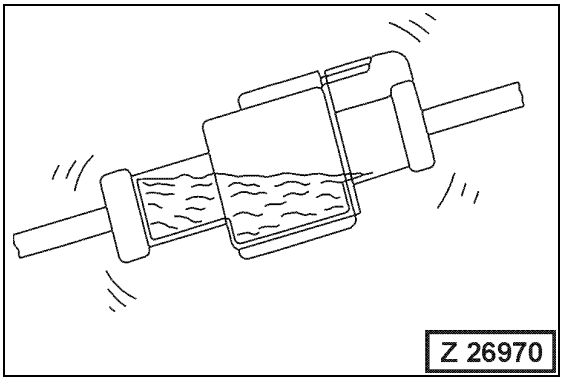
Disconnections in wiring

If the wiring is held and the connectors are pulled apart, or components are lifted with a crane with the wiring still connected, or a heavy object hits the wiring, the crimping of the connector may separate, or the soldering may be damaged, or the wiring may be broken.



High-pressure water entering connector

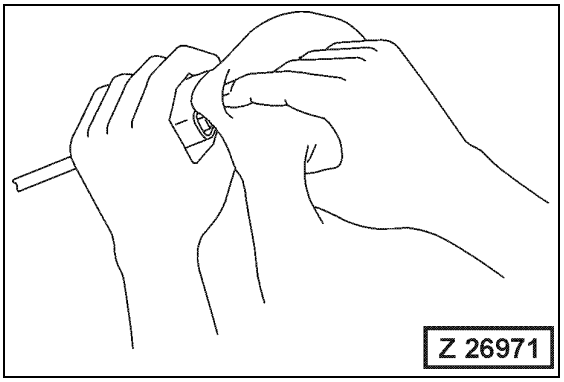
The connector is designed to make it difficult for water to enter (drip-proof structure), but if high-pressure water is sprayed directly on the connector, water may enter the connector, depending on the direction of the water jet. Accordingly, take care not to splash water over the connector. The connector is designed to prevent water from entering, but at the same time, if water does enter, it is difficult for it to be drained. Therefore, if water should get into the connector, the pins will be short-circuited by the water, so if any water gets in, immediately dry the connector or take other appropriate action before passing electricity through it.



Oil or dirt stuck to connector

If oil or grease are stuck to the connector and an oil film is formed on the mating surface between the male and female pins, the oil will not let the electricity pass, so there will be defective contact. If there is oil or grease stuck to the connector, wipe it off with a dry cloth or blow it dry with compressed air and spray it with a contact restorer.

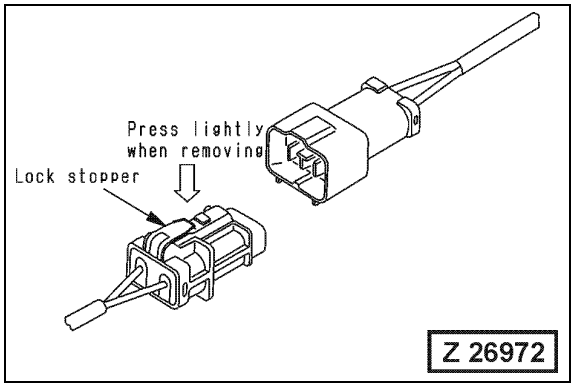
- When wiping the mating portion of the connector, be careful not to use excessive force or deform the pins.
- If there is oil or water in the compressed air, the contacts will become even dirtier, so remove the oil and water from the compressed air completely before cleaning with compressed air.



Disconnecting connectors

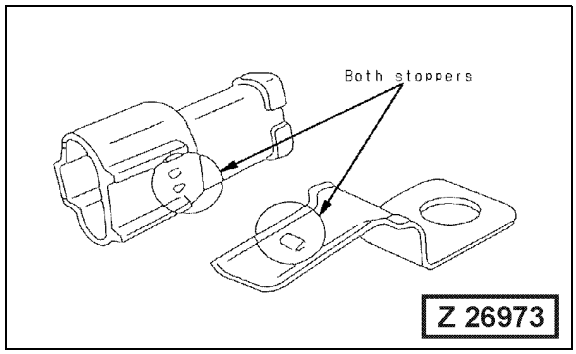
1. When disconnecting the connectors, hold the connectors. For connectors held by a screw, loosen the screw fully, then hold the male and female connectors in each hand and pull apart. For connectors which have a lock stopper, press down the stopper with your thumb and pull the connectors apart.

- Never pull with one hand.

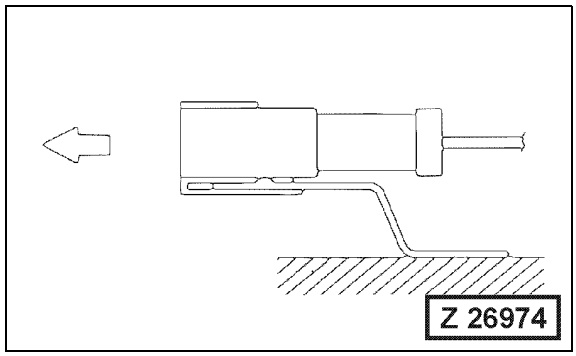


2. When removing from clips

- Both of the connector and clip have stoppers, which are engaged with each other when the connector is installed.

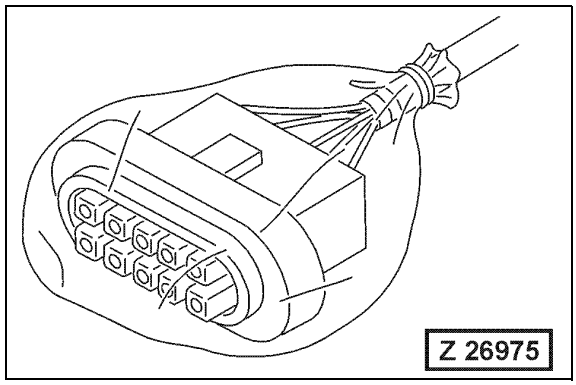


- When removing a connector from a clip, pull the connector in a parallel direction to the clip for removing stoppers.
  - If the connector is twisted up and down or to the left or right, the housing may break.



3. Action to take after removing connectors
   After removing any connector, cover it with a vinyl bag to prevent any dust, dirt, oil, or water from getting in the connector portion.

- If the machine is left disassembled for a long time, it is particularly easy for improper contact to occur, so always cover the connector.

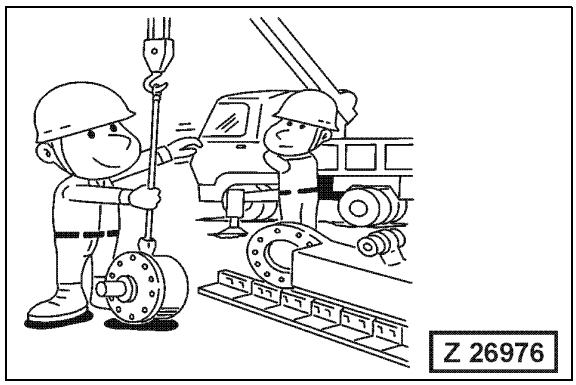


Points to remember when handling hydraulic equipment

With the increase in pressure and precision of hydraulic equipment, the most common cause of failure is dirt (foreign material) in the hydraulic circuit. When adding hydraulic oil, or when disassembling or assembling hydraulic equipment, it is necessary to be particularly careful.

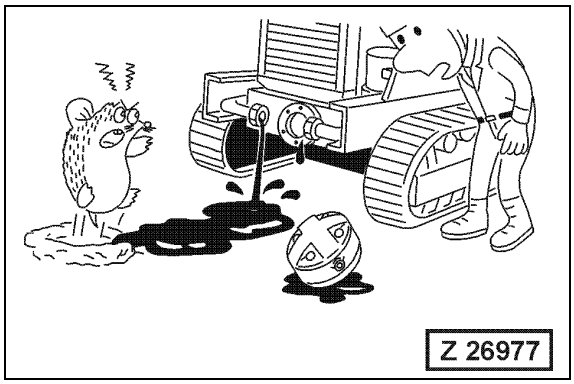
Be careful of the operating environment

Avoid adding hydraulic oil, replacing filters, or repairing the machine in rain or high winds, or places where there is a lot of dust.



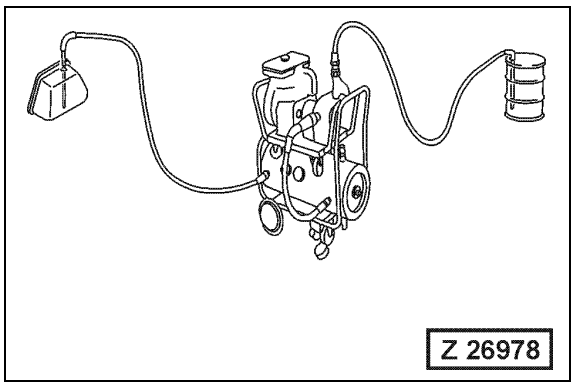
Disassembly and maintenance work in the field

If disassembly or maintenance work is carried out on hydraulic equipment in the field, there is danger of dust entering the equipment. It is also difficult to check the performance after repairs, so it is desirable to use unit exchange.
Disassembly and maintenance of hydraulic equipment should be carried out in a specially prepared dustproof workshop, and the performance should be checked with special test equipment.



Sealing openings

After any piping or equipment is removed, the openings should be sealed with caps, tapes, or vinyl bags to prevent any dirt or dust from entering. If the opening is left open or is blocked with a rag, there is danger of dirt entering or of the surrounding area being made dirty by leaking oil so never do this. Do not simply drain oil out onto the ground, but collect it and ask the customer to dispose of it, or take it back with you for disposal.



Do not let any dirt or dust get in during refilling operations

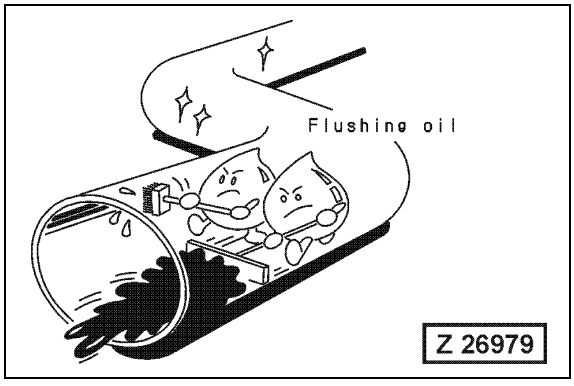
Be careful not to let any dirt or dust get in when refilling with hydraulic oil. Always keep the oil filler and the area around it clean, and also use clean pumps and oil containers. If an oil cleaning device is used, it is possible to filter out the dirt that has collected during storage, so this is an even more effective method.

Change hydraulic oil when the temperature is high

When hydraulic oil or other oil is warm, it flows easily. In addition, the sludge can also be drained out easily from the circuit together with the oil, so it is best to change the oil when it is still warm. When changing the oil, as much as possible of the old hydraulic oil must be drained out. (Drain the oil from the hydraulic tank; also drain the oil from the filter and from the drain plug in the circuit.) If any old oil is left, the contaminants and sludge in it will mix with the new oil and will shorten the life of the hydraulic oil.

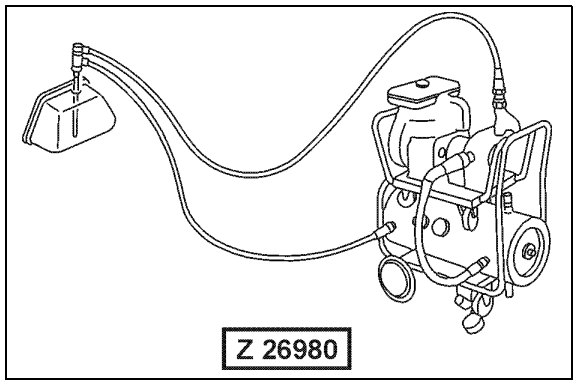
Flushing operations

After disassembling and assembling the equipment, or changing the oil, use flushing oil to remove the contaminants, sludge, and old oil from the hydraulic circuit. Normally, flushing is carried out twice: primary flushing is carried out with flushing oil, and secondary flushing is carried out with the specified hydraulic oil.



Cleaning operations

After repairing the hydraulic equipment (pump, control valve etc.) or when running the machine, carry out oil cleaning to remove the sludge or contaminants in the hydraulic oil circuit. The oil cleaning equipment is used to remove the ultra fine (about 3 μ) particles that the filter built in the hydraulic equipment cannot remove, so it is an extremely effective device.



GENERAL WORKING PROCEDURES

Air bleeding of various hydraulic parts

Air bleeding item / Contents of Work	Steps for air bleeding					
	1	2	3	4	5	6
	Hydraulic pump air bleeding	Engine start	Cylinder air bleeding	Swing motor air bleeding	Travel motor air bleeding	Operation start
• Change of hydraulic oil • Cleaning of strainer	○	○	○	○ (Note)	○ (Note)	○
• Replacement of return filter element		○				○
• Repair or replacement of hydraulic pump • Removing suction pipe	○	○	○			○
• Repair or replacement of control valve		○	○			○
• Replacement of cylinder • Removing cylinder piping		○	○			○
• Replacement of swing motor • Removing swing motor piping		○		○		○
• Replacement of travel motor swivel • Removing travel motor swivel piping		○			○	○

Note: Bleed air from the swing motor and travel motor only when oil in the casing is drained.

Z 27269

Air bleeding from the hydraulic pumps

The air bleeding procedures for the hydraulic pumps are described in the following manuals. Refer to these manuals as necessary.

- Operation & Maintenance Manual, section HYDRAULIC SYSTEM - CHANGE OIL, REPLACE SUCTION STRAINERS AND PULSATION DAMPER
- PARTS & SERVICE NEWS AH01513.

Air bleeding from hydraulic cylinders

1. Start the motor and keep it running for 5 minutes.
2. Running the motor raise and lower the respective cylinders 12 to 15 times.
   - Be careful not to apply the relief pressure, stopping the piston rod approx. 100 mm before its stroke end.
3. Apply the relief pressure smoothly by extending the piston rod to its stroke end and with the motor running.
4. For bleeding air from the arm cylinder and bucket cylinder, follow the same steps explained in items 2) and 3) above.
   - In case a cylinder is replaced with new one, it is advised to bleed air from the new one before mounting the work equipment. It is especially so with the boom cylinder, because its rod does not extend to the stroke end of LOWER side, after the work equipment is mounted.

Cylinder bypass test

The cylinder bypass test can be performed to check if a hydraulic cylinder has an internal leakage.

- A cylinder with an internal leakage also has a higher temperature (at working temperature of the machine) than a hydraulic cylinder which has no internal leakage.

⚠ WARNING

Always wear a safety harness when working at the attachment.

- First check hydraulic cylinder temperature using an infrared thermometer if an internal leakage is assumed. If the temperature is not higher than the temperature of the other cylinders, another reason e.g. leakage of a valve may be the reason for loosing pressure. If the temperature is higher than the temperature of the other cylinders, perform the following steps:

- Isolate the machine according to local regulations.
- Relieve the pressure in the hydraulic system, refer to the Operation & Maintenance Manual, section RELIEVE PRESSURE IN THE HYDRAULIC SYSTEM.

⚠ WARNING

Hot hydraulic oil!
Do not carry out the following work on a hot machine. Wait until the temperature of the hydraulic oil is below 50 °C. Disobedience may result in personal injury from heated oil.

⚠ WARNING

Danger of hydraulic pressure!
Blindness, serious injury, permanent disfigurement, or scaring may result. The hydraulic system may be pressurized. Make sure that the pressure is relieved before any hydraulic lines will be removed.

- Disconnect the hydraulic hose from the piston rod side of the respective cylinder.

    - When the hydraulic hose is disconnected, the oil inside the piping will flow out. Catch it in an oil pan. Cap the hydraulic hose with a blind plug to avoid contamination.

- Connect a hydraulic hose to the piston rod side of the cylinder.
  Put the open end of the hose into an oil pan.
- De-isolate the machine according to local regulations.

⚠ WARNING

Make sure that nobody is at close range of the hydraulic hose's open end when extracting the cylinder.

- At low idle, **slowly** extract the hydraulic cylinder.

    - The oil inside the cylinder will flow out of the hydraulic hose connected to the piston rod side of the cylinder. Catch it in the oil pan.

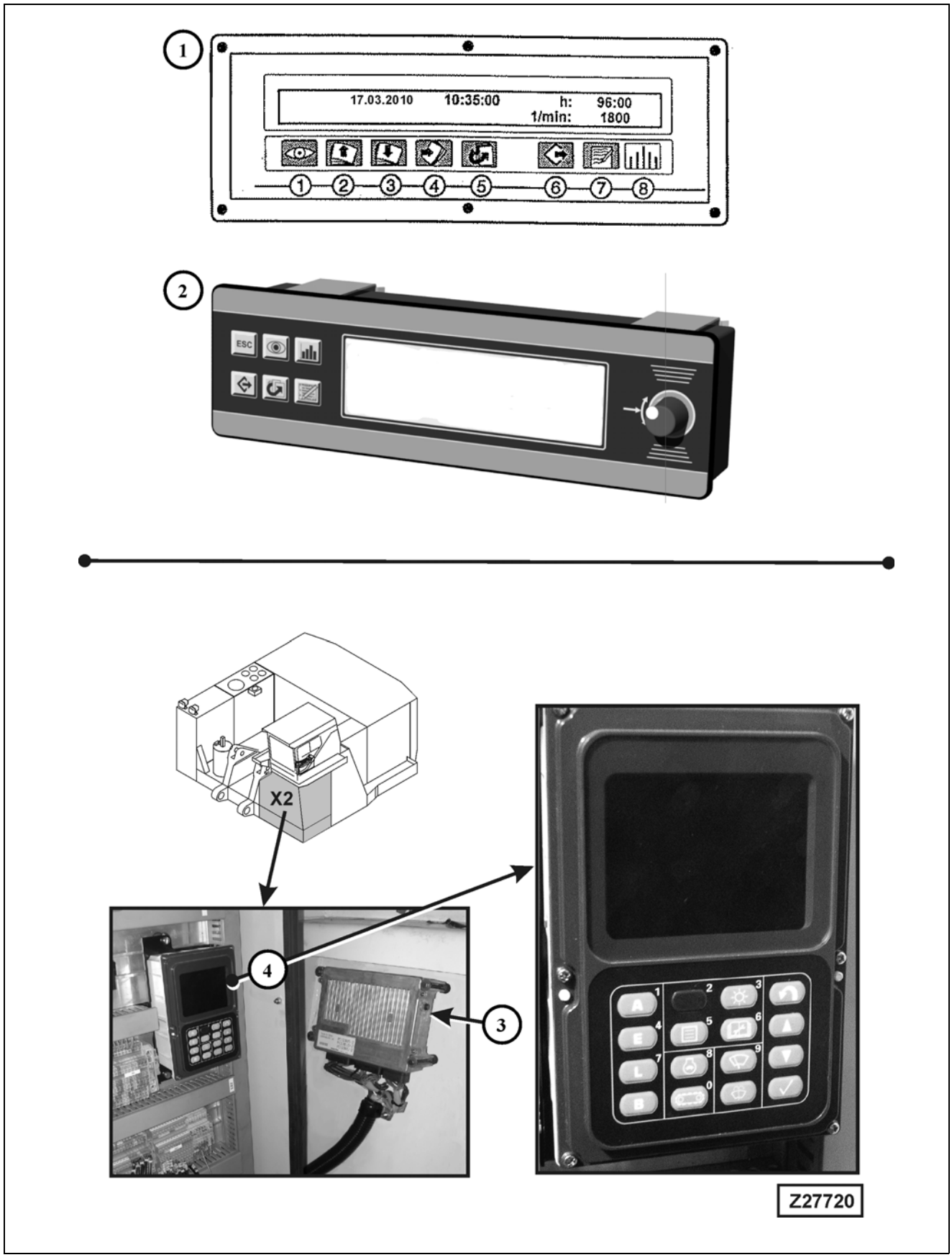
- When the piston of the cylinder reaches its mechanical stop, continue to extract the cylinder.

If the cylinder has an internal leakage, some oil will continuously flow out of the hydraulic hose connected to the piston rod side of the cylinder.
If the cylinder has no internal leakage, oil will stop flowing out of the hydraulic hose.

- After the cylinder bypass test is finished make sure that the machine is brought back to its normal condition.

Continue reading on the next page

INFORMATION ON THE ETM VERSIONS AND THE ELECTRONIC PUMP CONTROLLER CR700



Legend for illustration Z27720:

(1) Text display of ETM Version -A- used in older machines

(2) Text display of ETM Version -B-

- The main difference between ETM versions A and B is the design of the display and the mode of data transmission to the display. Version A utilizes input signal modules for data transfer, while Version B utilizes a CAN-Node.

  The sensors are the same for both versions. Refer to the Operation and Maintenance Manual of your machine for the implemented ETM version and operating instructions for the menu control.

(3) Electronic pump controller CR700

- For a general description and trouble shooting of the pump controller refer to the SERVICE MANUAL PC3000E-6.

(4) Multi monitor

- For a general description and adjustment of parameters at the multi monitor refer to the SERVICE MANUAL PC3000E-6.

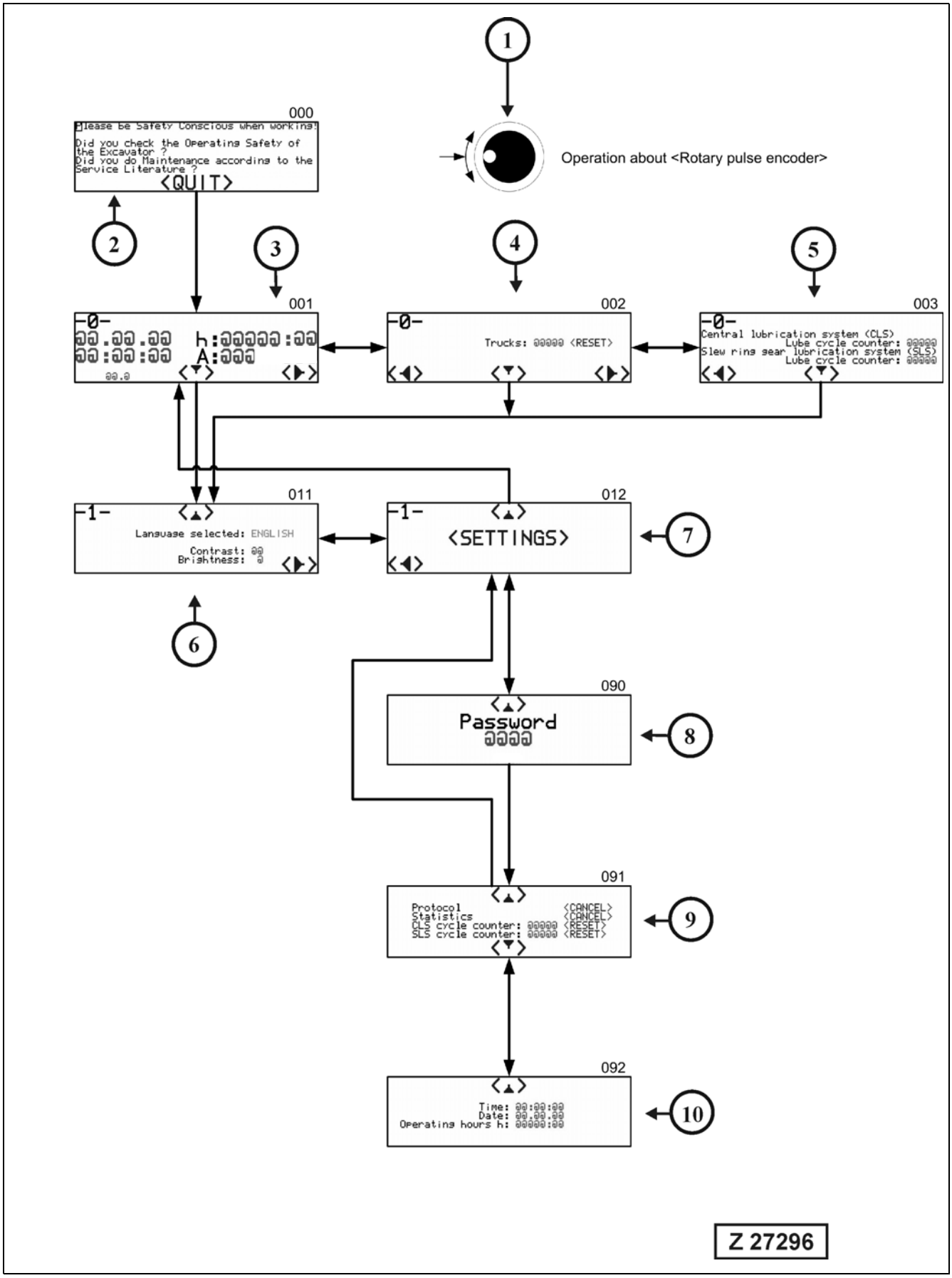
Variations of electrical diagrams

REMARK

During troubleshooting always make sure to use the electrical diagrams which are assigned to your machine. Due to variable requirements in many respects, the samples of wiring diagrams shown in this manual can not meet the actual state of each machine in every case. Always refer to the technical documentation delivered with the machine.

SERVICE LEVEL PASSWORD

Structure of the control menu



Legend for illustration Z27296:

(1)	Rotary Pulse Encoder (RPE) knob dial.
(2)	Screen providing basic information for general operation of machine.
(3)	Basic display showing date, time, operating hours and motor power.
(4)	Screen showing truck counter.
(5)	Screen showing lube cycle counters for Central lubrication System and Swing Circle Lubrication system.
(6)	Screen to adjust language, contrast and brightness.
(7)	Screen to enter the settings menu.
(8)	Screen to enter password for settings.
(9)	Screen allowing cancellation of Protocol and Statistic settings and the resetting of the CLS and SLS cycle counter.
(10)	Screen to alter time, date and operating hours.

On screen (9), the protocol and statistics memories can be cancelled and the Central Lubrication System and Swing Gear Lubrication System cycle counters can be reset.

Screen (10) allows the time, date and operating hours to be reset.

These settings are password protected and accessible for authorised service personnel only.

Enter the service password

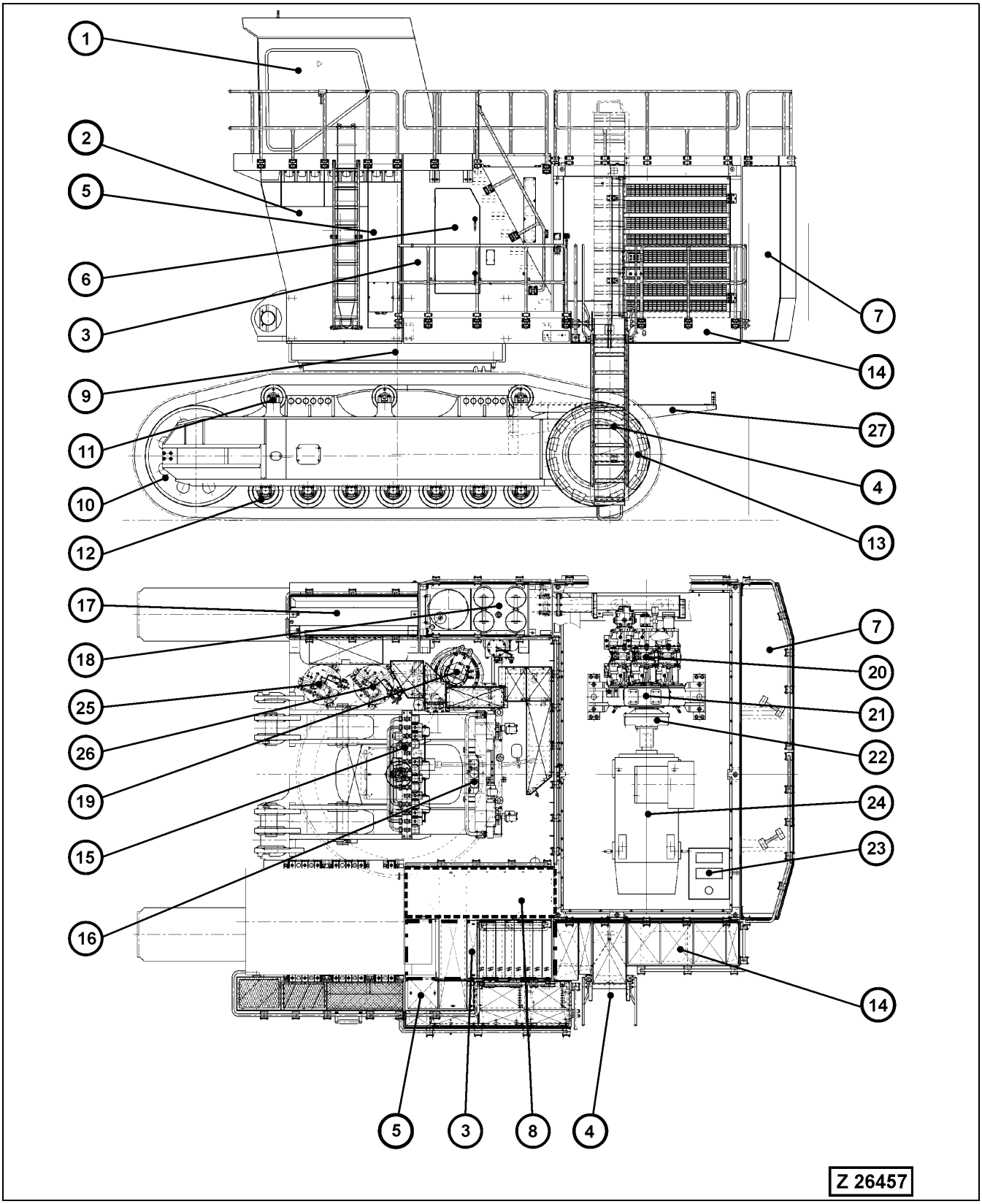
1. Navigate through the menu until the Settings (7) menu appears. Select **<SETTINGS>** and press the knob dial. The Password screen (8) appears.

2. To set the password, press the knob dial to select the first digit. Rotate the knob dial to the correct number and press the knob dial in to confirm. Repeat this procedure until the four digits are complete.
   Enter the service password: ***8125***
   Screens (9) and (10) can now be accessed.

ELECTRICAL PARTS

MOUNTING LOCATIONS OF ELECTRICAL PARTS, CONNECTOR TYPES AND ELECTRICAL STANDARD VALUES

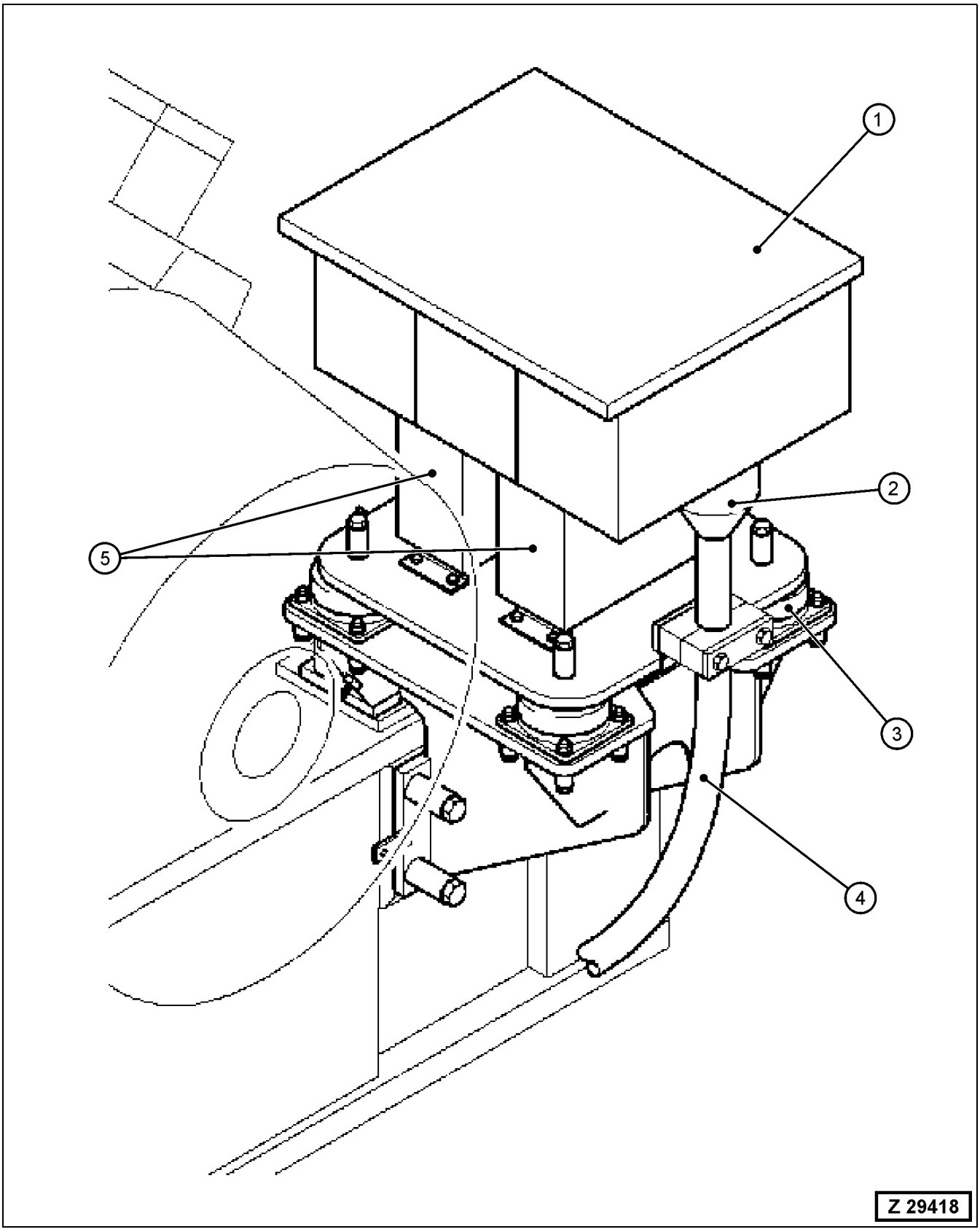
COMPONENT LOCATIONS



Legend for illustration Z 26457:

(1)	Operator’s cab
(2)	Cab base contains board net control panel (X2) and hydraulic pilot control components
(3)	High voltage switch cabinet (HV-cabinet)
(4)	Hydraulically operated access ladder
(5)	Gas bottle chamber for fire suppression system
(6)	High voltage cabinet access door
(7)	Counterweight
(8)	Medium voltage cabinet (motor control & monitoring)
(9)	Swing circle
(10)	Guide wheel (idler)
(11)	Carrier roller (top roller)
(12)	Track roller (bottom roller)
(13)	Sprocket and travel gear
(14)	Batteries inside of pedestal
(15)	Distribution manifold of attachment hydraulic
(16)	Main control valves
(17)	Hydraulic oil cooler with hydraulic driven fans
(18)	Main hydraulic oil tank
(19)	Swing gearbox with hydraulic motor and brakes
(20)	Hydraulic pumps
(21)	PTO (pump distributor gear)
(22)	Flexible coupling
(23)	Capacitor for power factor correction
(24)	Main electric motor
(25)	Central lubrication system
(26)	Swing ring gear lubrication system
(27)	Feeder cable support

LOCATION OF ELECTRICAL PARTS CAPACITOR / ELECTRIC MOTOR



Legend for illustration Z29418:

(1) Connecting box

(2) cable cland

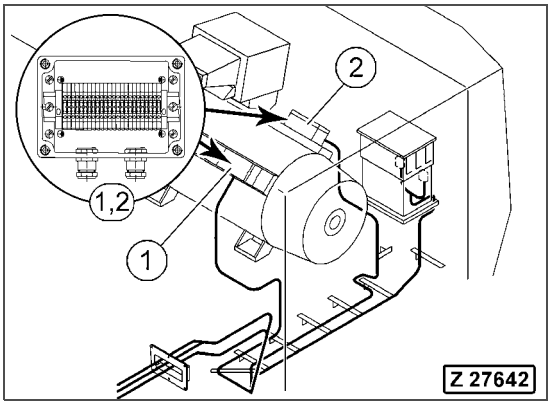
(3) Shock absorber

(4) Junction cable

(5) capacitor bank

Motor

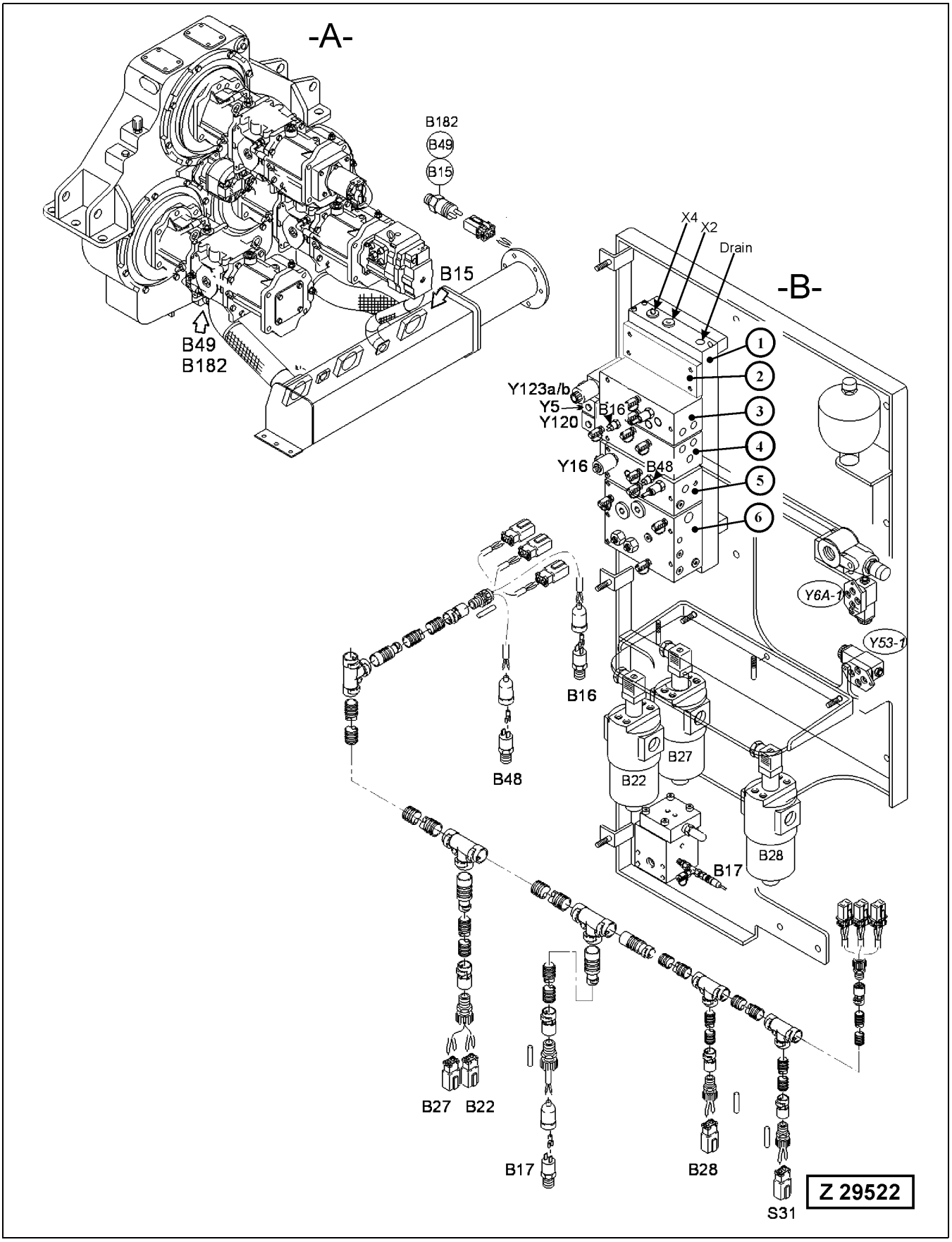
The wiring ends of the bearing temperature sensors are combined in a junction box (1) and the wiring ends of the winding temperature sensors are combined in a junction box (2).



REMARK

Refer to the Electrical Diagram of the machine for additional information.

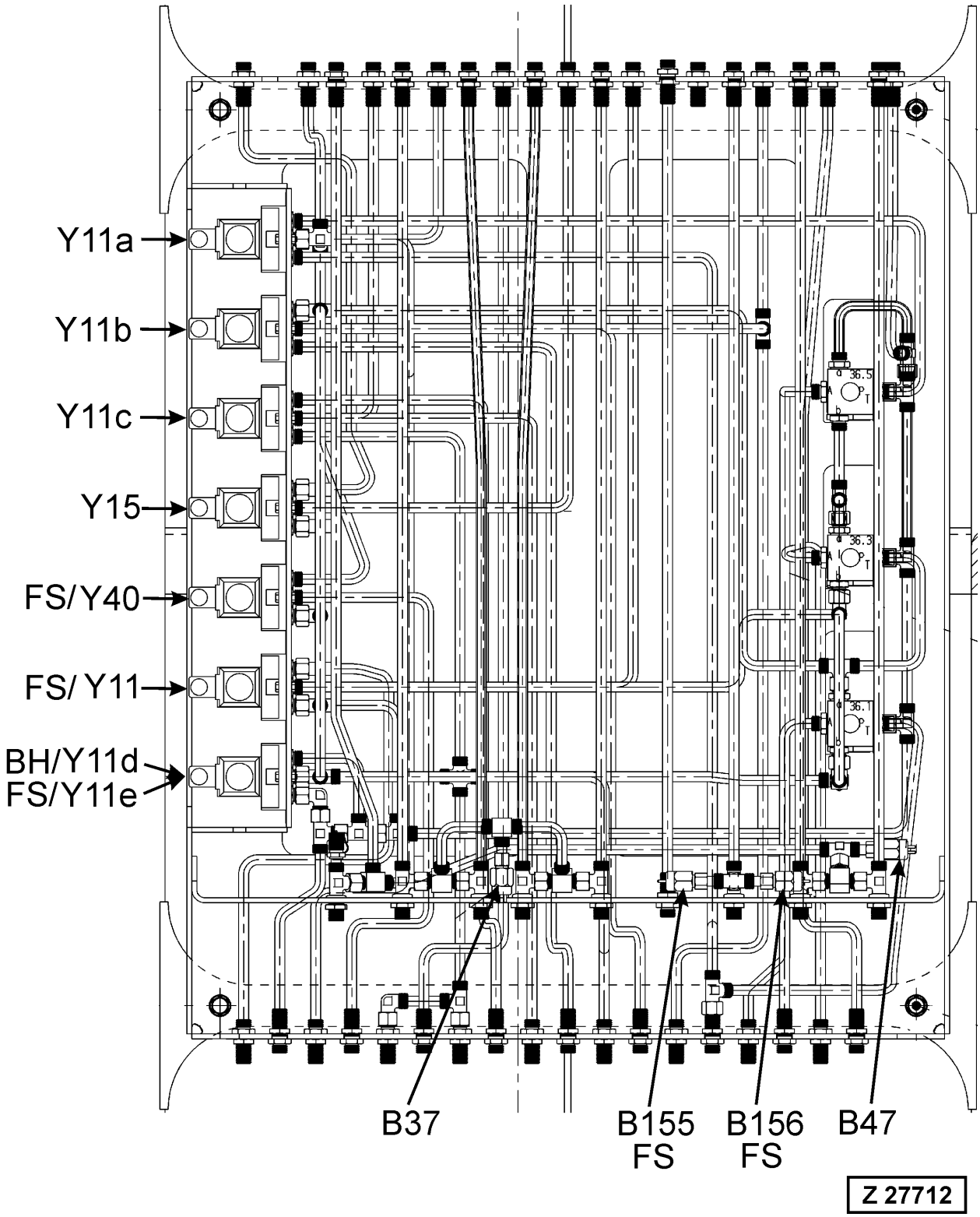
LOCATION OF ELECTRICAL PARTS ON PTO, SUCTION OIL RESERVOIR AND CENTRAL CONTROL AND FILTER PANEL



Legend for illustration Z29522:

-A-	**PTO and Suction oil Reservoir**
(B15)	Hydraulic oil temperature sensor (PT100). Signal to switch unit K39 >Warning message Hydraulic Oil Temperature too high< and bucket motion cut-off.
(B49)	PTO gear oil temperature sensor (PT100). Signal to switch unit K43 >Warning message PTO Gear Oil Temperature too high<
(B182)	PTO gear oil temperature sensor (PT100). Signal to switch unit K182 >Pressure relief of gear oil cooler<
-B-	**Central Control and Filter Panel**
(1)	Valve bank base plate
(2)	Blind plate
(3)	Control block ladder
(4)	Control block swing brake
(5)	Control block travel brake and track tensioning
(6)	Control oil block
(B16)	Pressure switch, swing parking brake ON. Switch point 24 bar.
(B17)	Pressure switch, PTO gear lubrication failure. Switch point 0.5 bar.
(B22)	Differential pressure switch, control oil filter restricted. Switch point 5.0 bar.
(B27)	Differential pressure switch, PTO lubrication oil filter restricted. Switch point 5.0 bar.
(B28)	Differential pressure switch, cooler fan drive oil filter restricted. Switch point 5.0 bar.
(B48)	Pressure switch, travel parking brake ON. Switch point 24 bar.
(S31)	Connector for proximity switch S31 on main shut off gate valve
(X2)	Control pressure, 35 bar
(X4)	Control pressure, 60 bar
(Y5)	Solenoid valve, swing parking brake control
(Y6A)	Solenoid valve, hydraulic oil cooler fan speed minimal (oil cool)
(Y16)	Solenoid valve, travel parking brake control
(Y53-1)	Solenoid valve, PTO oil cooler relief
(Y120)	Solenoid valve, swing service brake control
(Y123a/b)	Solenoid valve, ladder a) raise, b) lower.

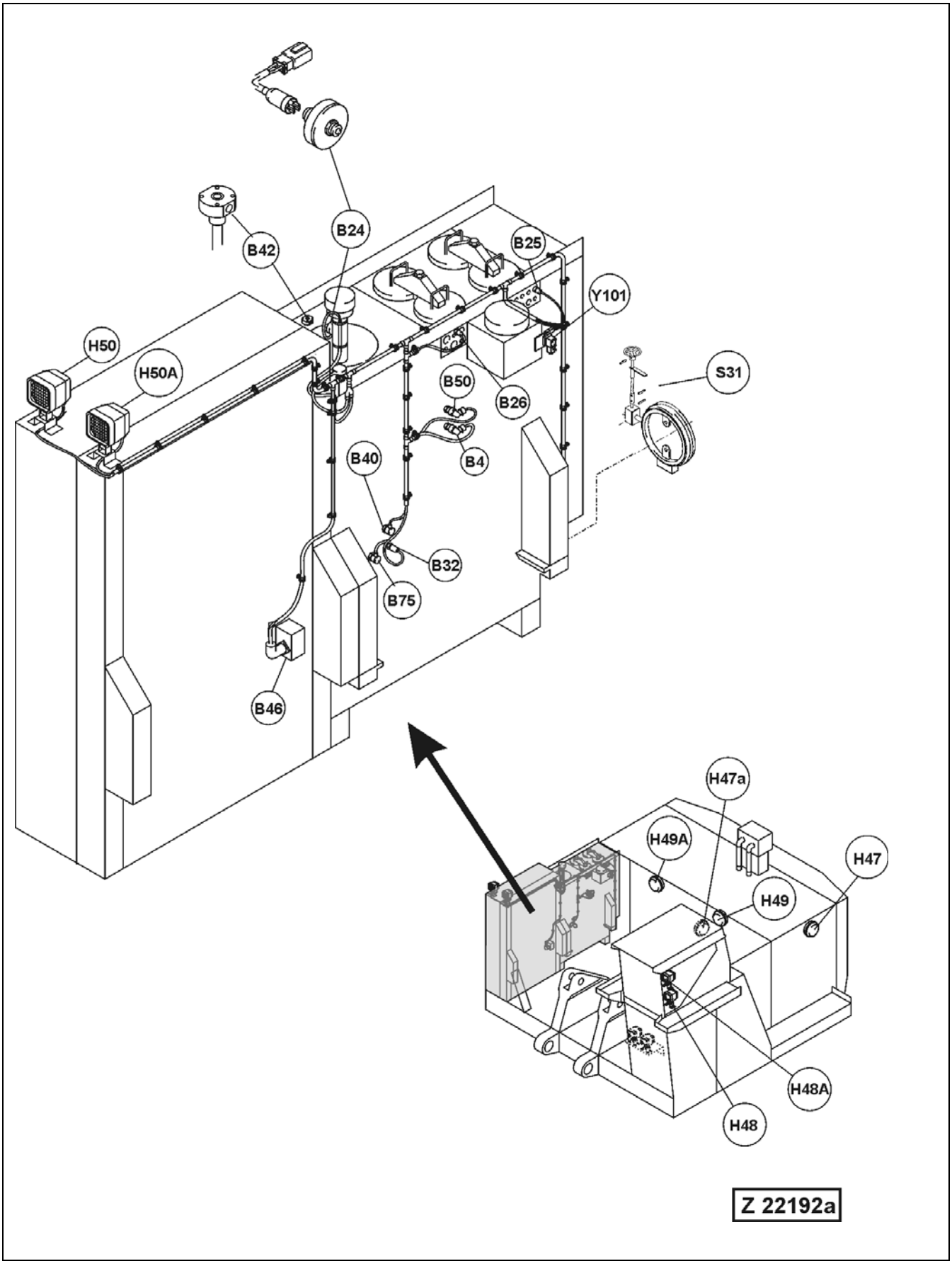
LOCATION OF ELECTRICAL PARTS ON PILOT CONTROL FRAME



Legend for illustration Z27712:

(B37)	Pressure switch, motion detection travelling. Switch point 7 bar
(B47)	Pressure switch, motion detection slewing. Switch point 7 bar
(B155)	Pressure switch, controls pressure cut-off of bucket clam closing (main pump 1 in Qmin position), switch point 8.5 bar.
(B156)	Pressure switch, monitors function of pressure cut-off of bucket clam closing (main pump 1 in Qmin position), switch point 22 bar. Malfunction generates message no. 52.
(Y11)	Solenoid valve, pilot control locking bucket clam (Face Shovel only).
(Y11a)	Solenoid valve, pilot control locking bucket
(Y11b)	Solenoid valve, pilot control locking stick
(Y11c)	Solenoid valve, pilot control locking boom
(Y11d)	Solenoid valve, pilot control release (Backhoe only).
(Y11e)	Solenoid valve, pilot control release (Face Shovel only).
(Y15)	Solenoid valve, pilot control locking swing service brake.
(Y40)	Solenoid valve, bucket motion cut-off. Hydraulic oil temperature too high or automatic lube system trouble.

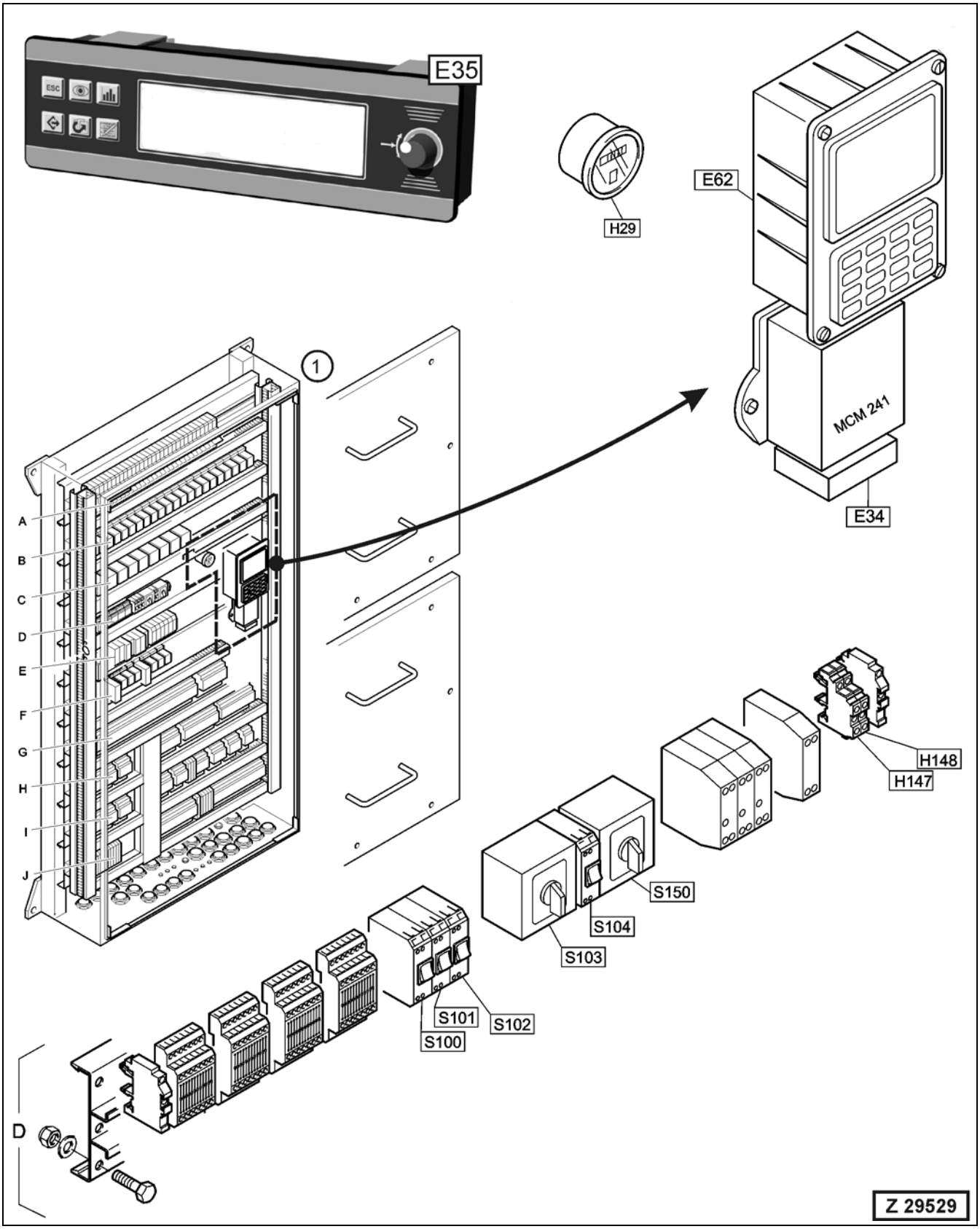
LOCATION OF ELECTRICAL PARTS ON MAIN HYDRAULIC OIL RESERVOIR



Legend for illustration Z 22192a:

(B4)	Hydraulic oil level switch, level too low
(B24)	Low pressure switch, hydraulic tank breather filter monitoring. Switch point 0.08 bar (negative pressure).
(B25)	Pressure switch, leakage oil filter monitoring. Switch point 0.5 bar.
(B26)	Pressure switch, return oil filter monitoring. Switch point 2.0 bar.
(B32)	Temperature transmitter (analogue) hydraulic oil temperature.
(B40)	Temperature sensor, hydraulic oil cooler fan speed minimal (oil cool)
(B42)	Oil level switch, refilling system maximum filling (lamp H52).
(B46)	End line pressure switch, swing ring lubrication system.
(B50)	Oil level switch, refill level.
(B75)	Temperature sensor, low hydraulic oil temperature. Causes pump derate to 1/2 Qmax, reduction of the return oil back pressure through Y101 and fade out of the auxiliary filter monitoring.
(B171)	Not shown. Temperature sensor, hydraulic oil cooler fan speed medium (oil temperature medium). Option for low temperature versions only.
(S31)	Proximity switch at the main gate valve monitors valve position. Closed gate valve causing start prevention or shutdown and generates a warning message.
(Y101)	Solenoid valve for reducing the return oil back pressure (B75 controlled).
(H47)	Light at the access ladder.
(H47a)	Light at the cabin entrance and steps.
(H48)	Working light cabin, old with front lights.
(H48A)	Working light cabin, old with front lights.
(H49)	Service light in front of the control blocks.
(H49A)	Service light in font of the control blocks.
(H50)	Working light, on top of the oil cooler frame.
(H50A)	Working light, on top of the oil cooler frame.

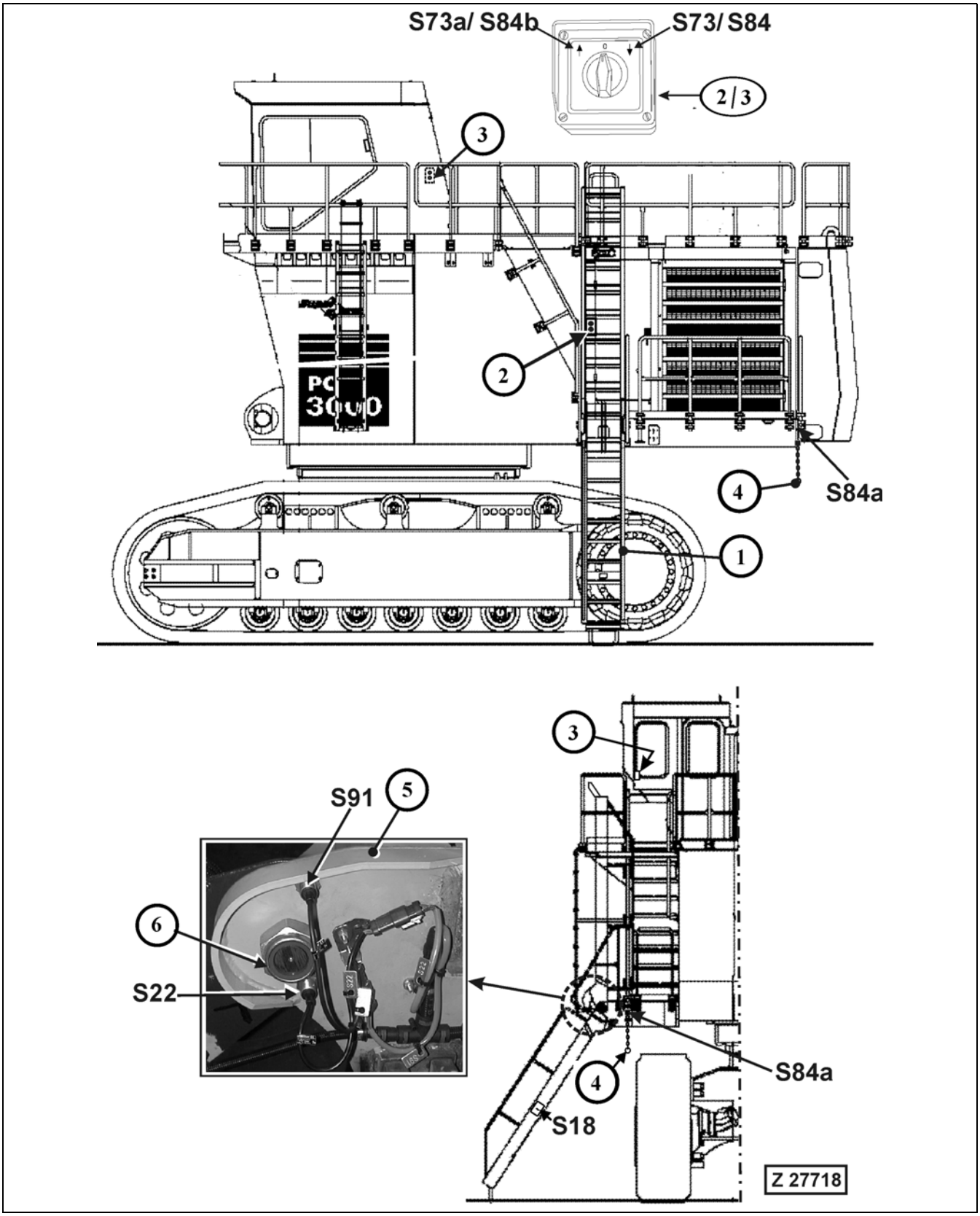
LOCATION OF ELECTRICAL PARTS ON SWITCH BOARD X2 IN CAB BASE



Legend for illustration Z29529:

(1)	X2 box inside the cab base
(E34)	Input signal module (CAN-Node) of the ETM
(E35)	Text display of ETM
(E62)	Multi monitor for diagnostic of the electronic pump control system CR700. For a general description and adjustment of parameters at the multi monitor refer to the SERVICE MANUAL PC3000E-6.
(H29)	Hour meter for travel operation. The hour meter indicates the total number of hours of travelling operation.
(H147 - H148)	Fault lamps for the electronic pump control system CR700. For a general description and trouble shooting of the pump control system, refer to the SERVICE MANUAL PC3000E-6.
(S100)	Service switch for the electronic pump control system CR700 (adjustment mode)
(S101)	Service switch for the electronic pump control system CR700 (selection down)
(S102)	Service switch for the electronic pump control system CR700 (selection up)
(S103)	Service switch for the electronic pump control system CR700 (test mode selection position 2)
(S104)	Service switch for the electronic pump control system CR700 (primary P-factor)
(S150)	Service switch for Q_{min} position of main hydraulic pumps. Position 1 normal pump regulation. Position 2 Q_{min} position.

LOCATION OF ELECTRICAL PARTS OF THE HYDRAULICALLY OPERATED ACCESS LADDER



Legend for illustration Z27718:

(1) Access ladder

(2) Ladder control switch (S73 and S73a) at the machinery house door

(3) Ladder control switch (S84 and S84b) at the operator's cab

(4) Pull chain for emergency lowering of the access ladder

⚠ WARNING

Use this chain only in emergency cases, when the Operator does not respond to other communication signals.

When the chain (4) is being pulled down with the motor running, the pilot control system is made inoperative by pull switch (S84a), preventing further movements of the Shovel.

(5) Ladder pivot bracket

(6) Lock nut for ladder pivot ball bearing

(S18) Light switch for access area lighting

(S22) Safety sensor located on ladder pivot bracket. Function of sensor (S22): Cut out of the pilot control system and actuation of the hydraulic swing brake with the ladder in lowered position.

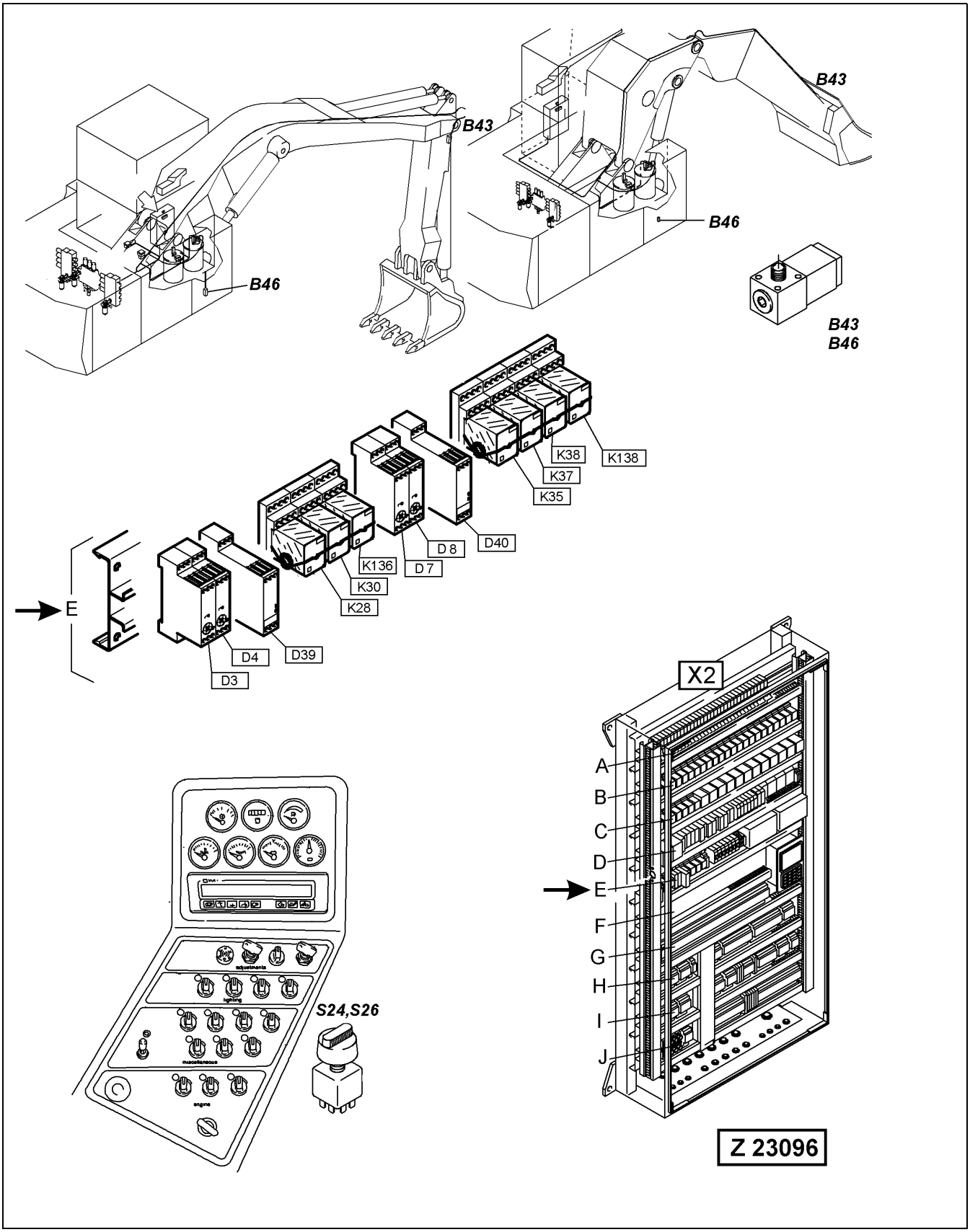
(S73a / S84b) Switch direction for lifting the ladder

(S73 / S84) Switch direction for lowering the ladder

(S84a) Pull switch for lowering the ladder in emergency cases.

(S91) Monitor and control sensor. Function of the sensor: This sensor monitors the ladder position and controls the moving speed of the ladder. In case the sensor (S22) fails to function properly, the sensor (S91) prevents unintended movement of the ladder.

LOCATION OF ELECTRICAL PARTS OF THE AUTOMATIC LUBRICATION SYSTEMS



General

The shovel is equipped with two automatic lubrication systems:

1. Central Lubrication System -CLS- for lubrication of the attachment bearings, the slewing connection bearing and the slew gear bearing.

2. Slew ring gear pinion Lubrication System -SLS- for lubrication of the slew ring teeth.

These systems work automatically as soon as the motor is running. The slew ring gear lubrication system is only activated during slewing movement of the superstructure. The ETM System monitors the function of both lubrication systems. Trouble conditions of the lubrication systems are indicated through fault messages on the ETM display.

Legend for illustration Z23096:

Electrical parts of the Central Lubrication System (CLS) located on the X2 switch board in the cab base

D3	Pause time relay, factory setting 60 minutes
D4	Monitoring time relay, factory setting 3 minutes
D39	Time relay for switching off the bucket motion if the lubrication system fails to function over a period longer than 4.2 hours
K28	Relay initiates a fault message if current supply to the lubrication system is interrupted
K30	Relay initiates a fault message if the lubrication cycle is not carried out in the adjusted time period
K136	Relay initiates a fault message if the grease barrel is empty

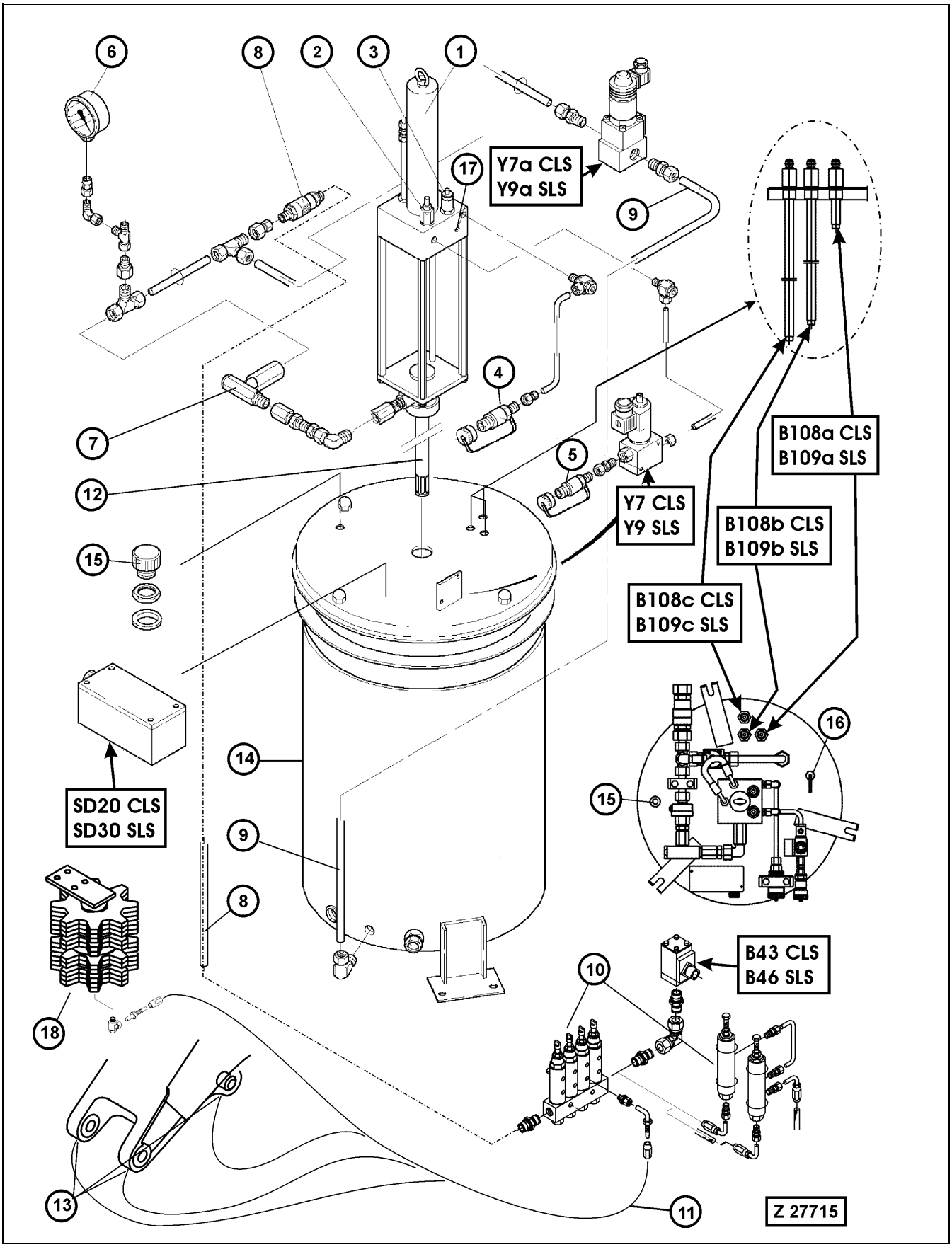
Electrical components of the Slew ring gear Lubrication System (SLS)

D7	Pause time relay, factory setting 9 minutes
D8	Monitoring time relay, factory setting 2 minutes
D40	Time relay for switching off the bucket motion if the lubrication system fails to function over a period longer than 4.2 hours
K35	Relay initiates a fault message if current supply to the lubrication system is interrupted
K37	Relay activates the slew ring gear lubrication system during slewing of the superstructure
K38	Relay initiates a fault message if the lubrication cycle is not carried out in the adjusted time period
K138	Relay initiates a fault message if the grease barrel is empty

B43	Line pressure switch of CLS interrupts lubrication cycle
B46	Line pressure switch of SLS interrupts lubrication cycle
S24	Rotary switch for manual actuation of the CLS
S26	Rotary switch for manual actuation of the SLS

REMARK

For location of electrical parts on the grease containers, refer to the next page.


6
8
2
3
1
17
Y7a CLS
Y9a SLS
9
4
7
5
12
Y7 CLS
Y9 SLS
B108a CLS
B109a SLS
B108b CLS
B109b SLS
B108c CLS
B109c SLS
15
16
14
SD20 CLS
SD30 SLS
9
8
18
10
B43 CLS
B46 SLS
13
11
Z 27715

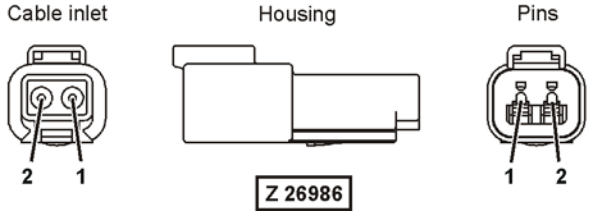
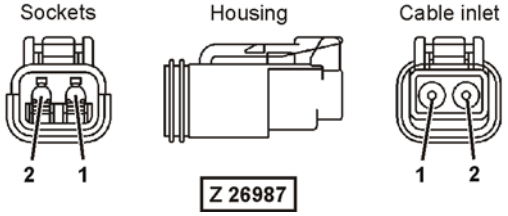
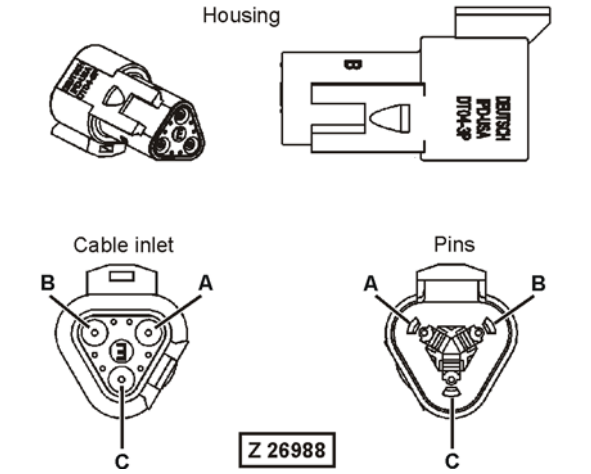
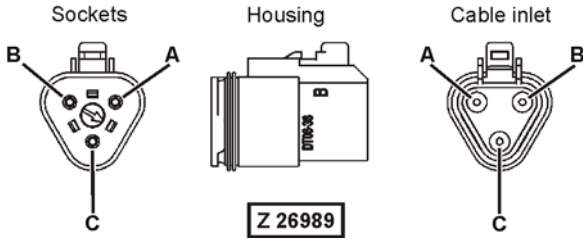
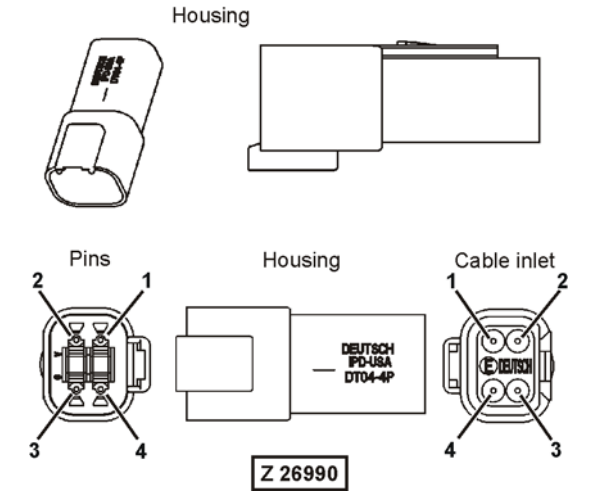
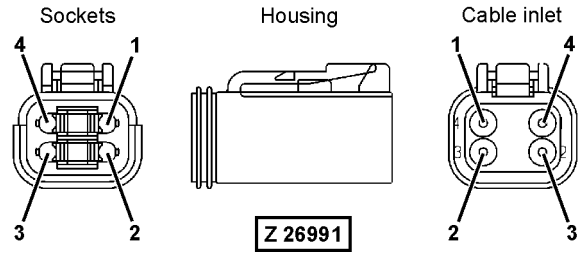
Legend for illustration Z27715:

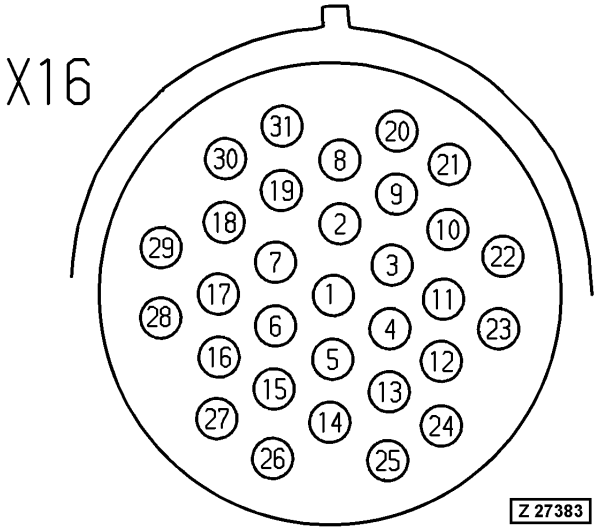
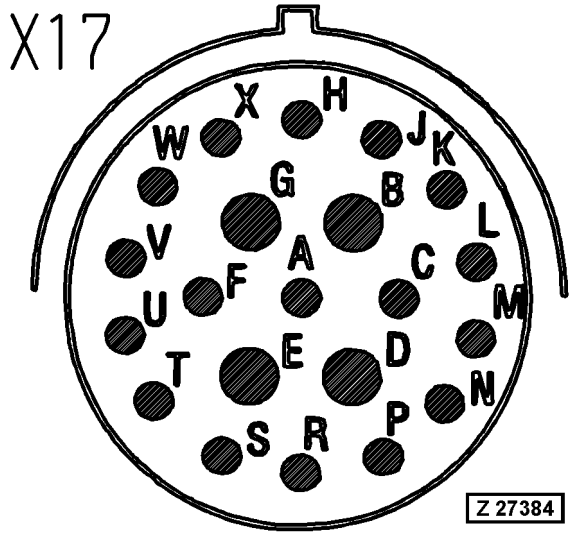
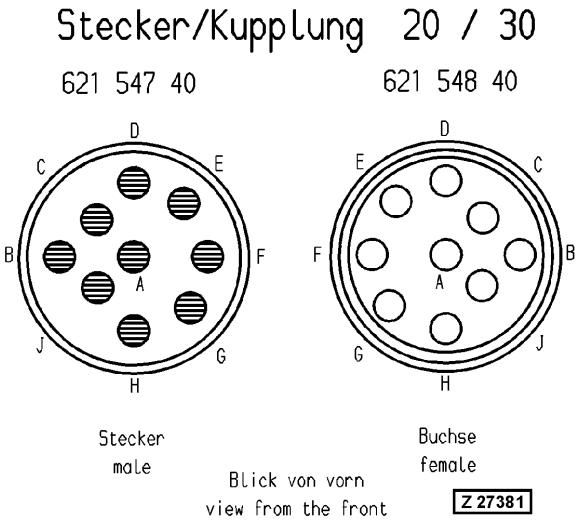
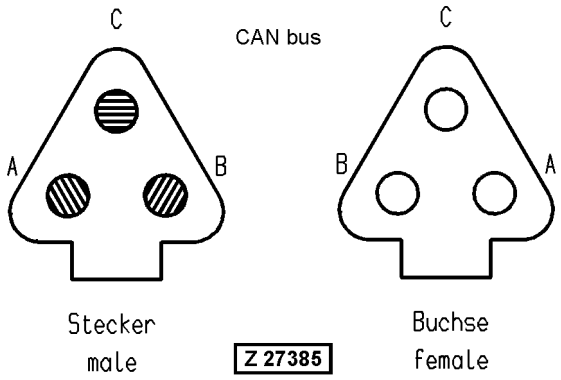
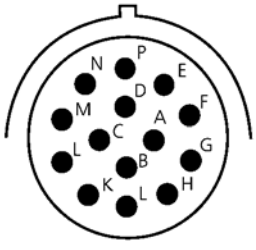
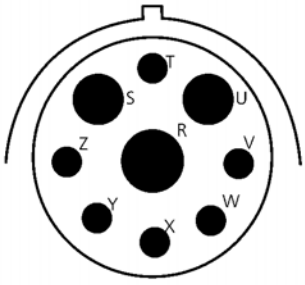
Electrical parts on the Grease Barrels of Central Lubrication System -CLS- and Slew ring gear Lubrication System -SLS-

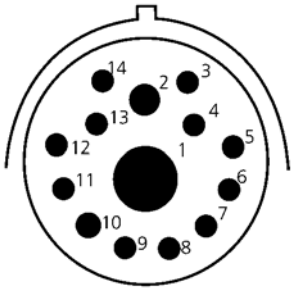
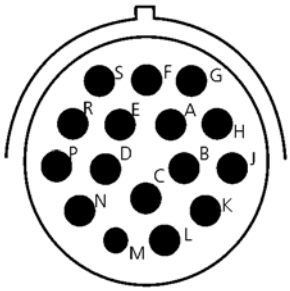
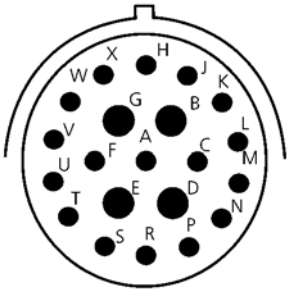
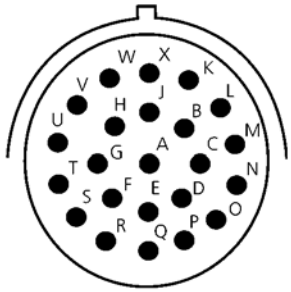
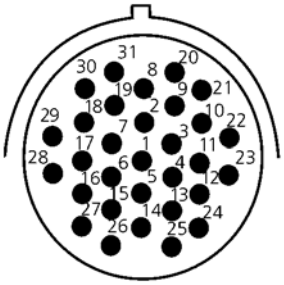
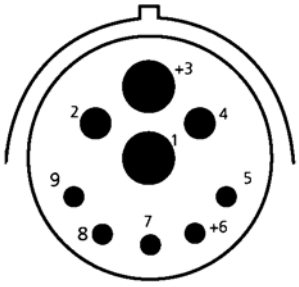
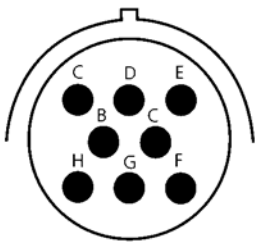
(B43 / B46)	End-line-pressure switch, shuts off lubrication cycle
(B108a / B109a)	Switch grease container FULL, activates indicator lamp on the service arm.
(B108b / B109b)	Switch grease level LOW, generates warning message -Refill grease container-.
(B108c / B109c)	Switch grease container EMPTY, generates warning message -Grease container empty- and activates time relay D39 of CLS and/or D40 of SLS. The time relays will initiate bucket motion cut off after a period of 4.2 operating hours, if the grease container has not been refilled within this time period.
(SD20 / SD30)	Electrical terminal box
(Y7 / Y9)	Pump control solenoid valve (pump on / off)
(Y7a / Y9a)	Grease pressure release solenoid valve
(1)	Hydraulically driven grease pump
(2)	Flow control valve
(3)	Pressure relief valve (45 bar)
(4)	Hydraulic oil return line
(5)	Hydraulic oil supply line (60 bar X4-pressure)
(6)	Grease pressure gauge
(7)	Grease in line filter
(8)	Grease supply line from pump to injectors
(9)	Grease release line
(10)	Grease injector
(11)	Grease feed line from injector to the bearings
(12)	Grease pump
(13)	Lubrication points (bearings) CLS only
(14)	Grease container
(15)	Breather filter for grease container
(16)	Grease level dipstick
(17)	Test port grease pump drive pressure
(18)	Lubrication pinion SLS only

OVERVIEW CONNECTOR TYPES AND CONNECTOR PIN NUMBERS

- The terms male and female refer to the connector pins, while the terms male housing and female housing refer to the mating portion of the connector housing parts.

Male (female housing)	Female (male housing)
2 pins DT04-2P (Deutsch)	2 sockets DT06-2S (Deutsch)
	
3 pins DT04-3P (Deutsch)	3 sockets DT06-3S (Deutsch)
	
4 pins DT04-6P (Deutsch)	4 sockets DT06-6S (Deutsch)
	

31-pole connector with aluminium housing	21-pole connector with plastic housing
	
Grease barrel connector	CAN bus connector
	
Deutsch connectors HD 34/36-18-14-S…	Deutsch connectors HD 34/36-24-9-S…
Picture shows socket rear insert. 	Picture shows socket rear insert. 

Deutsch connectors HD 34/36-24-14-S…	Deutsch connectors HD 34/36-24-16-S…
Picture shows socket rear insert.  Z 28195	Picture shows socket rear insert.  Z 28196
Deutsch connectors HD 34/36-24-21-S…	**Deutsch connectors HD 34/36-24-23-S…**
Picture shows socket rear insert.  Z 28197	Picture shows socket rear insert.  Z 28198
Deutsch connectors HD 34/36-24-31-S…	**Deutsch connectors HD 34/36-24-91-S…**
Picture shows socket rear insert.  Z 28199	Picture shows socket rear insert.  Z 28200
Deutsch connectors HD 34/36-18-8-S…	
Picture shows socket rear insert.  Z 28192	

STANDARD VALUE TABLE FOR ELECTRICAL COMPONENTS

Abbreviations

Diff. press.	Differential pressure
Mot	Motor
N.C.	Normally Closed
N.O.	Normally Open
1S1	Main key switch 1S1
SM	Service Manual
T-box	Electrical test box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Electric motor and related parts	Bearing temperature sensor *(PT100 type)*	OB32-1 OB 32-2	Measure resistance	If the condition is within the range shown in the table below, it is normal Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Winding temperature sensor *(PT100 type)*	B64	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Bearing temperature sensor *(PT100 type)*	OB3 OB4 OB5	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Capacitor pressure switch *[N.C. type]*	OS1 OS2	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (3): 22 - 24V Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
PTO & related parts	PTO lube oil pressure switch *[N.C. type]*	B17	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Insert T-box 3) 1S1ON 4) Open terminals 1 & 2 at T-box 5) Motor OFF
				Between (1) - (2): 22 - 24V	6) Motor running 7) When filter clogged differential pressure >0.5bar
	PTO lube oil filter clogging switch *[N.O. type]*	B27	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): min. 1MΩ	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	PTO lube oil temperature sensor	B49	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	PTO lube oil temperature sensor *[Cooler relief]*	B182	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	PTO preload pressure solenoid valve	Y53	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 23 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Ladder	Ladder limit switch *[N.O. type]*	Up: S22 Down: S91	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 - 24V Between (1) - (3): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box 6) Motor OFF
				Between (1) - (3): 22 - 24V	7) Motor running
	Ladder solenoid valve → lower	Y123b	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Ladder solenoid valve → fast	Y125	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 35 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Ladder	Ladder solenoid valve → raise	Y123a	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Hydraulic system	Swing parking brake solenoid valve	Y5	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Oil cooler fan drive solenoid valve	Y6a Y6b	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Travel parking brake solenoid valve	Y16	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 30 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Hydraulic system	Oil cooler preload pressure solenoid valve	Y101	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Hydraulic brake solenoid valve *(emergency)*	Y15 Y120	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Hydraulic system	FS Bucket cut off solenoid valve	Y40	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 20 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Pilot control cut off solenoid valve	Y11 Y11a Y11b Y11c Y11d Y11e	Measure resistance	If the condition is within the range shown in the table below, it is normal. *Between (1) - (2): 20 ± 1.0Ω*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Pump angle solenoid valve	Y158f Y158r Y159f Y159r Y160f Y160r	Measure resistance	If the condition is within the range shown in the table below, it is normal. *Between (1) - (2): 9.7 ± 0.5Ω*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Hydraulic system	Hydraulic oil level sensor *(level: low)* → with oil	B4	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between 1 - 3: 22 - 24V Between 1 - 2: 22 - 24V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
	→ without oil			Between 1 - 3: 22 - 24V Between 1 - 2: 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Hydraulic system	Hydraulic oil level sensor *(level: refill)* → with oil	B50	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between 1 - 3: 22 - 24V Between 1 - 2: 22 - 24V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
	→ without oil			If the condition is within the range shown in the table below, it is normal. Between 1 - 3: 22 - 24V Between 1 - 2: 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
	Oil reservoir breather filter clogging switch *[N.O. type]*	B24	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open disconnect-terminals 1 & 2 at the T-box 6) Motor running
				Between (1) - (2): 22 - 24V	7) Filters: clogged
	Leakage oil filter clogging switch *[N.O. type]*	B25	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open disconnect-terminals 1 & 2 at the T-box 6) Motor running
				Between (1) - (2): 22 - 24V	7) Filters: clogged
	Hydr. oil temperature sensor *(PT100 type)*	B15	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Hydraulic system	Hydr. oil tank shut off gate valve *(proximity switch)*	S31	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (3): 22 - 24V Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box 6) Main gate closed
				Between (1) - (2): 22 - 24V	7) Main gate open
	Floating solenoid valve	Y132b Stick Y132c Boom	Measure resistance	If the condition is within the range shown in the table below, it is normal. *Between (1) - (2): 20 ± 1.0Ω*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
Hydraulic system	Filter pump control clogging switch *[N.O. type]*	B22	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When filter clogged differential pressure >5bar
	Slew brake pressure switch *[N.C. type]*	B16	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) Switch point 24bar
	Travel brake pressure switch *[N.C. type]*	B48	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) Switch point 24bar

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Hydraulic system	Return filter clogging switch *[N.O. type]*	B26	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When filter clogged differential pressure >2bar
	Filter oil cooler fan drive clogging switch *[N.O. type]*	B28	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When filter clogged differential pressure >5bar
	Pressure switch motion detection travelling *[N.O. type]*	B37	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When travelling pressure >7 bar
	Pressure switch motion detection slewing *[N.O. type]*	B47	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When slewing pressure >7 bar
	Temperature sensor oil cooler fan speed minimal, oil cool. *(PT100 type)*	B40	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Hydraulic system	Temperature sensor too low oil temperature, oil cool. *(PT100 type)*	B75	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): *Refer to PT100 temperature charts*	1) 1S1OFF 2) Disconnect connector 3) Insert T-box
	Pressure switch pressure cut off clam closing *[N.O. type]*	B155	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) pressure cut off pressure >8.5 bar
	Pressure switch monitors B155 *[N.O. type]*	B156	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) fault message pressure >22 bar
	Pressure transducer *(analog)*	B158-a B158-b B158-c	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 5V Between (1) - (3): 0.4 - 0.5V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (3): 0.5 - 4.5V	6) Motor running
	Pressure switch oil cooler strainer plugged *[N.O. type]*	B165	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 0V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 22 - 24V	6) Motor running 7) When filter clogged differential pressure >5bar

System	Component name	Component code	Inspection method	Judgement table	Measurement conditions
Lubrication systems	End line switch grease pressure	B43 B46	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 - 24V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box
				Between (1) - (2): 0V	6) Motor running 7) Grease pump running 8) Grease pressure 180bar available
	Grease pump solenoid valve	Y7 Y9	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector
	Grease line solenoid valve *(Vent valve)*	Y7a Y9a	Measure resistance	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 ± 1.0Ω	1) 1S1OFF 2) Disconnect connector
	Grease barrel level switch	B108a B109a	Measure voltage	If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 - 24V	1) 1S1OFF 2) Disconnect connector 3) Insert T-box 4) 1S1ON 5) Open terminals 1, 2 & 3 at T-box 6) Grease barrel full
		B108b B109b		If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 - 24V	Grease level advance warning
		B108c B109c		If the condition is within the range shown in the table below, it is normal. Between (1) - (2): 22 - 24V	Grease Barrel empty

PT100 TEMPERATURE CHARTS

Basic resistance values (Ohm) for measuring the resistance of PT100 sensors in relation to the temperature (degree centigrade)

Resistance values according to DIN 43 76

° C	-0	-1	-2	-3	-4	-5	-6	-7	-8	-9
-50	80,31	79,91	79,51	79,11	78,72	78,32	77,92	77,52	77,13	76,73
-40	84,27	83,88	83,48	83,08	82,69	82,29	81,89	81,50	81,10	80,70
-30	88,22	87,83	87,43	87,04	86,64	86,25	85,85	85,46	85,06	84,67
-20	92,16	91,77	91,37	90,98	90,59	90,19	89,80	89,40	89,01	88,62
-10	96,09	95,69	95,30	94,91	94,52	94,12	93,73	93,34	92,95	92,55
0	100,00	99,61	99,22	98,83	98,44	98,04	97,65	97,26	96,87	96,48

° C	0	1	2	3	4	5	6	7	8	9
0	100,00	100,39	100,78	101,17	101,56	101,95	102,34	102,73	103,12	103,51
10	103,90	104,29	104,68	105,07	105,46	105,85	106,24	106,63	107,02	107,40
20	107,79	108,18	108,57	108,96	109,35	109,73	110,12	110,51	110,90	111,28
30	111,67	112,06	112,45	112,83	113,22	113,61	113,99	114,38	114,77	115,15
40	115,54	115,93	116,31	116,70	117,08	117,47	117,85	118,24	118,62	119,01
50	119,40	119,78	120,16	120,55	120,93	121,32	121,70	122,09	122,47	122,86
60	123,24	123,62	124,01,	124,39	124,77	125,16	125,54	125,92	126,31	126,69
70	127,07	127,45	127,84	128,22	128,60	128,98	129,37	129,75	130,13	130,51
80	130,89	131,27	131,66	132,04	132,42	132,80	133,18	133,56	133,94	134,32
90	134,70	135,08	135,46	135,84	136,22	136,60	136,98	137,36	137,47	138,12
100	138,50	138,88	139,26	139,64	140,02	140,39	140,77	141,15	141,53	141,91
110	142,29	142,66	143,04	143,42	143,80	144,17	144,55	144,93	145,31	145,68
120	146,06	146,44	146,81	147,19	147,57	147,94	148,32	148,70	149,07	149,45
130	149,82	150,20	150,57	150,95	151,33	151,70	152,08	152,45	152,83	153,20
140	153,58	153,95	154,32	154,70	155,07	155,45	155,82	156,19	156,57	156,94
150	157,31	157,69	158,06	158,43	158,81	159,18	159,55	159,93	160,30	160,67

Z 27270

TROUBLESHOOTING BY TROUBLE CODE

MESSAGE -1- GEAR LUBRICATION OUT OF FUNCTION (B17)

[1/3]

Trouble code **Message page 1**	Trouble	Gear lubrication out of function (B17)
Contents of trouble	• Pressure switch B17 (NC*) signals, that there is no pressure in the PTO gear lubrication system, even with the motor running. *NC = Normally Closed	
Related information	• A problem with the pressure sensor B17 or its wiring may also cause the trouble. Basic switch data (B17)	
Problem that appears on machine	• Risk, to operate the machine without PTO gear lubrication (overheating / bearing seizure).	

No pressure present	0 V	Switch closed
Pressure present	24 V	Switch open
Switch point	0.5 bar	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Defective pressure switch B17	1	Check Pressure switch B17 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck if the PTO gear lubrication pressure is in normal range.	
Incorrect or no supply voltage at the pressure switch B17	2	► Motor running.			
		Using the T-box, measure the supply voltage for B17 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B17. (disconnection or defective contact in any connector)	3	► Motor stopped and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B17 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 820Ω
			No	Disconnection, short circuit, or fault in the wiring between connector X61A and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	

Message -1- Gear lubrication out of function (B17)

[2/3]

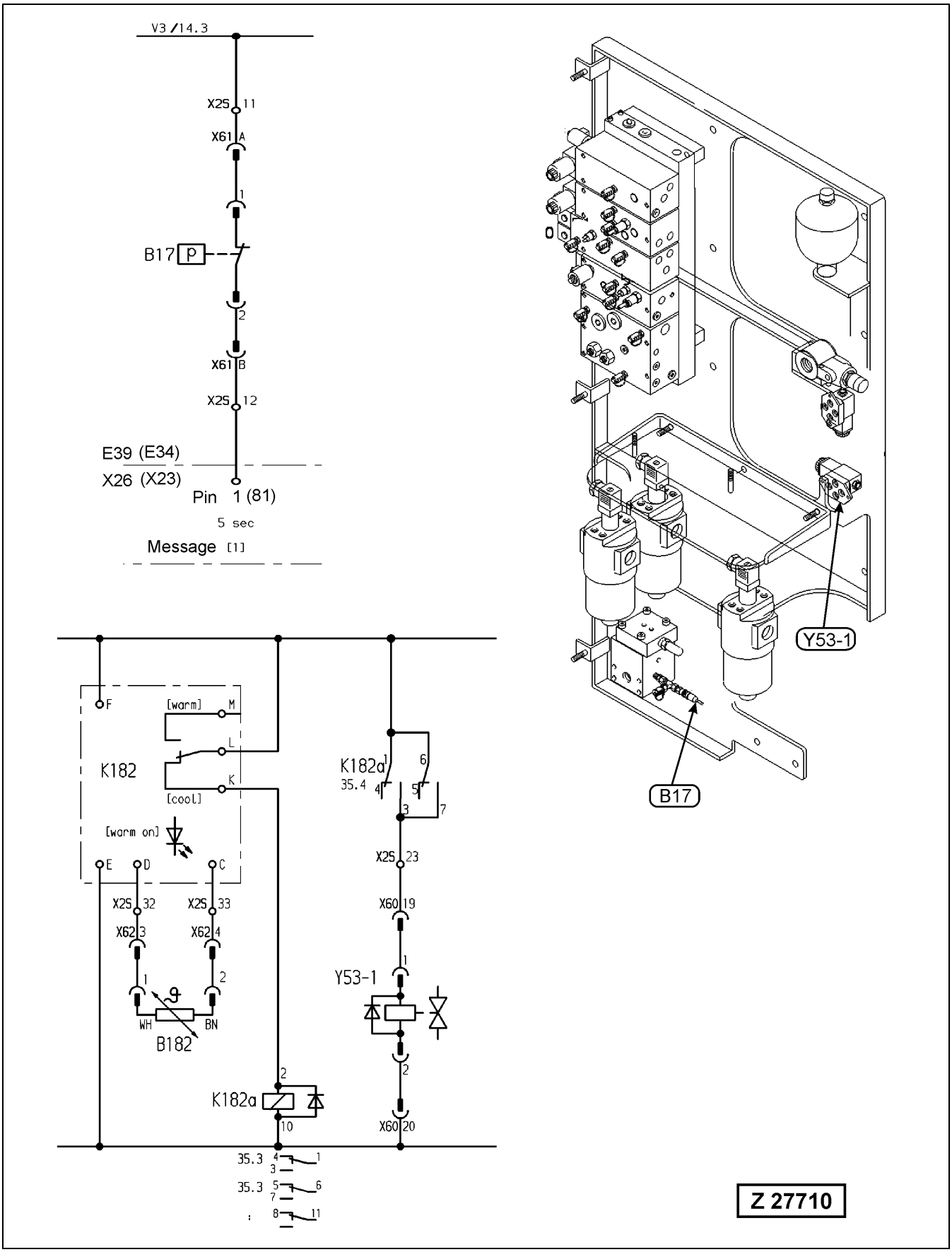
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the signal wiring between pressure switch B17 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	4	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 1. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 81. Is the signal present?	Yes	The signal wiring between pressure switch B17 and Input signal module is OK. Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B17 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Adjustment of PTO valve block incorrect	5	Refer to the Service Manual section 2 -PTO LUBRICATION AND COOLING- and follow the given adjustment procedures. Are the pressure adjustments as specified?	Yes	Go to next step.	
			No	Carry out adjustments as necessary. After that recheck if the PTO gear lubrication pressure is in normal range.	
Setting of temperature switch unit K182 for oil cooler relief incorrect.	6	Check the PTO gear oil settings (viscosity) according to the table below. Are the oil settings correct?	Yes	Recheck pressure switch B17 for continuity and rectify as necessary.	
			No	Adjust K182 to the correct switching point and verify that trouble message 1 -Gear lubrication out of function- is not set again.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

Adjust switch unit K182 to the respective oil viscosity grade as follows:

Oil viscosity grade according to ISO	HLP T32	CLP 150	SAE 80W-90
-K182- Adjust temperature limit value to switching point °C	+15°	+48°	+54°

Wiring Diagram related to trouble message -1-

[3/3]



MESSAGE -2- MOTOR SHUTDOWN: FIRE SUPPRESSION SYSTEM ACTUATED

[1/4]

<table>
<tr><td>Trouble code
Message page 2</td><td>Trouble</td><td>Motor shutdown: Fire suppression system actuated</td></tr>
<tr><td>Contents of trouble</td><td colspan="2">- The message is displayed when the fire suppression system (CO_2) for the LV and HV switch cabinet is activated.
- Once the system is triggered, it works in both switch cabinets (LV and HV) at the same time.
- The control module (9E2) is located in the operator’s cab.
- The gas cylinder is located in the LV switch cabinet.
- The same message is displayed if the optional fire warning system SC-N in the machinery house is triggered.</td></tr>
<tr><td>Related information</td><td colspan="2">Components of the fire suppression system:
<table>
<tr><td>Comp.</td><td>Function</td><td>Location</td><td>Remark</td></tr>
<tr><td>9B1</td><td>Temperature sensor</td><td>LV switch cabinet</td><td></td></tr>
<tr><td>9B2</td><td>Temperature sensor</td><td>HV switch cabinet</td><td></td></tr>
<tr><td>9E1</td><td>Control module</td><td>Cab</td><td>Optional fire warning system in the machinery house</td></tr>
<tr><td>9E2</td><td>Control module</td><td>Cab</td><td>Fire suppression system (CO_2) in the LV & HV switch cabinet</td></tr>
<tr><td>9K1</td><td>Actuator relay</td><td>-</td><td>For optional fire warning system in the machinery house</td></tr>
<tr><td>9K3</td><td>Actuator relay</td><td>-</td><td>For fire suppression system (CO_2) in the LV & HV switch cabinet</td></tr>
</table>
9E1 → Motor shutdown delay = 3 seconds
9E2 → Motor shutdown delay = 0 seconds

Discharge delay = 15 seconds</td></tr>
<tr><td>Problem that appears on machine</td><td colspan="2">Case A - Fire activated the fire suppression system.
Case B - False alarm caused the activation of the fire suppression system.
Case C - False alarm was cancelled by the operator.</td></tr>
</table>

Possible cause	Step	Test condition & Inspection	Procedure	Nominal value
Fire alarm was initiated and the fire suppression system was activated in order to extinguish a fire in the power house	A	Make sure that no danger or risk of injury exists before entering the machinery house. Identify the cause of the fire in the machinery house. **NOTE:** Also refer to the troubleshooting procedures which are described in the manual of the system manufacturer.	Remove the residues of the extinguishing agent and clean the machinery house as necessary. The detection and actuation system and the fire suppression systems must be checked, recharged, and approved by authorized service personnel immediately after operation. Rectify the cause for the fire.	

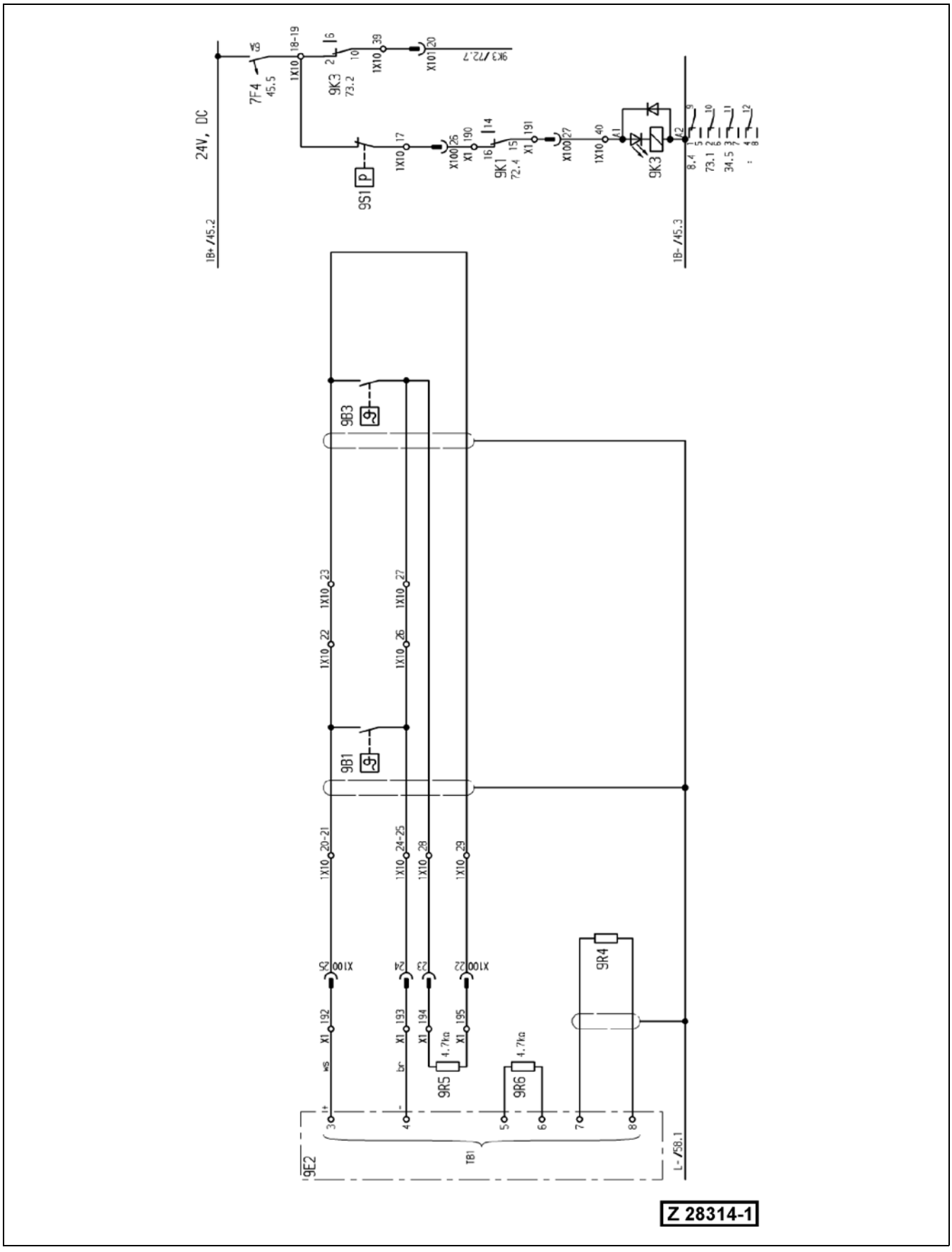
Message -2- Motor shutdown: Fire suppression system actuated

[2/4]

Possible cause	Step	Test condition & Inspection	Procedure	Nominal value
Fire alarm was initiated due to a false alarm and the fire suppression system was activated	B	Make sure that no danger or risk of injury exists before entering the machinery house. Identify the cause of the false alarm in the machinery house. **NOTE:** Also refer to the troubleshooting procedures which are described in the manual of the system manufacturer. Check the electric circuit of 9K1 for the optional fire warning system (SC-N). In case of any trouble repair or replace wiring/components as necessary. Check the electric circuit of 9K3 for the fire suppression system (CO_2). In case of any trouble repair or replace wiring/components as necessary.	Remove the residues of the extinguishing agent and clean the machinery house as necessary. The detection and actuation system and the fire suppression systems must be checked, recharged, and approved by authorized service personnel immediately after operation. Rectify the cause for the false fire alarm.	
False fire alarm was initiated and the operator cancelled the alarm by pushing the reset button at the control module of the fire suppression system. The fire suppression system was not activated.	C	Identify the cause of the false fire alarm. **NOTE:** Also refer to the troubleshooting procedures which are described in the manual of the system manufacturer.	The detection and actuation system and the fire suppression systems must be checked and approved by authorized service personnel only. Rectify the cause for the false fire alarm.	
Final check		- Bring the machine back into the original condition. - Start the machine and verify that the trouble message does not arise again.		

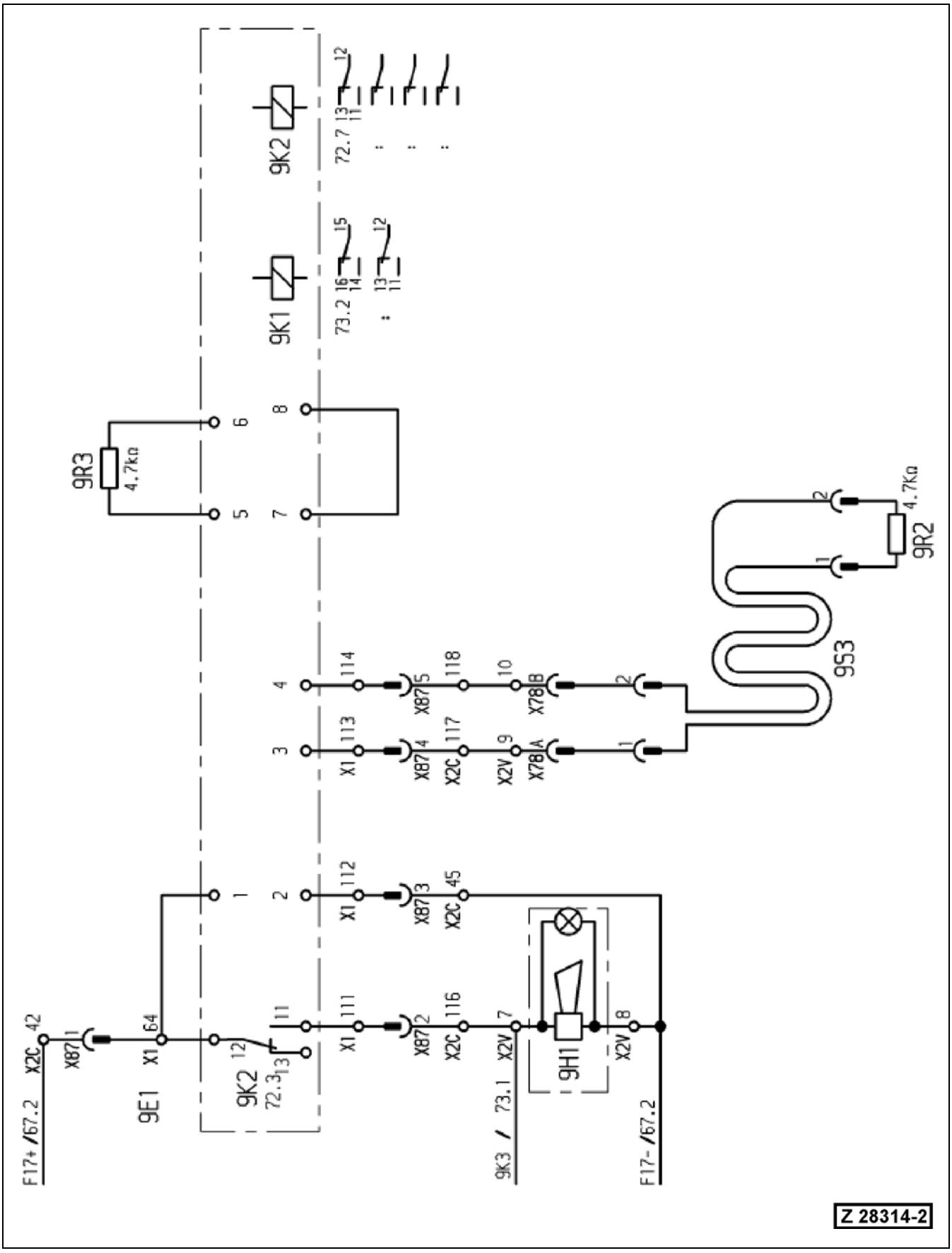
Wiring Diagram (CO2 Fire suppression system) related to trouble message -2-

[3/4]



Wiring Diagram (SC-N Fire warning system) related to trouble message -2-

[4/4]



MESSAGE -3- OIL FILTER FOR CONTROL OIL RESTRICTED (B22)

[1/3]

Trouble code **Message page 3**	Trouble	Oil filter for control oil restricted (B22)
Contents of trouble	• Pressure switch B22 (NO*) signals, that the oil filter for the control pressures (X2 and X4) is clogged. *NO = Normally Open	
Related information	• A problem with the pressure switch B22 or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. • The message 3 is not set until the hydraulic oil temperature reaches T2, see table below.	

Temperatur in °C temperature in °C	Viskosität nach ISO viscosity grade according to ISO				
	VG22	VG32	VG46	VG68	VG100
TypNr. / type no.	1	2	3	4	5
T1 (1000 cSt)	-20	-14	-6	0	5
T2 (100 cSt)	9	16	25	32	39
T3 (30 cSt)	32	41	50	58	67
T4 (10 cSt -5°C)	58	69	79	85	85

Z 27262

for example
Hydraulic oil:VG100
T2: 39°C

In this case the trouble message is not set until the hydraulic oil temperature reaches:
39°C

Problem that appears on machine	- Unfiltered control oil flows through the bypass of the filter.

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The oil filter for the X2 and X4 control pressures is clogged	1	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		Screw off the filter case and check for clogging. Is the filter element clogged?	Yes	Replace or clean the filter element. If the filter element was clogged check the hydraulic pump for control pressures (X2 and X4) for wear.	
			No	Go to next step.	
The pressure switch B22 is defective	2	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B22 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved? After that recheck the system for proper function.	Yes	Got to final step 7.	
			No	Go to next step.	
Defective pressure switch B22	3	Check Pressure switch B22 according to the Standard value table in this section. Is the test result OK?	Yes	Continue with step 4 of table [2/3].	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	

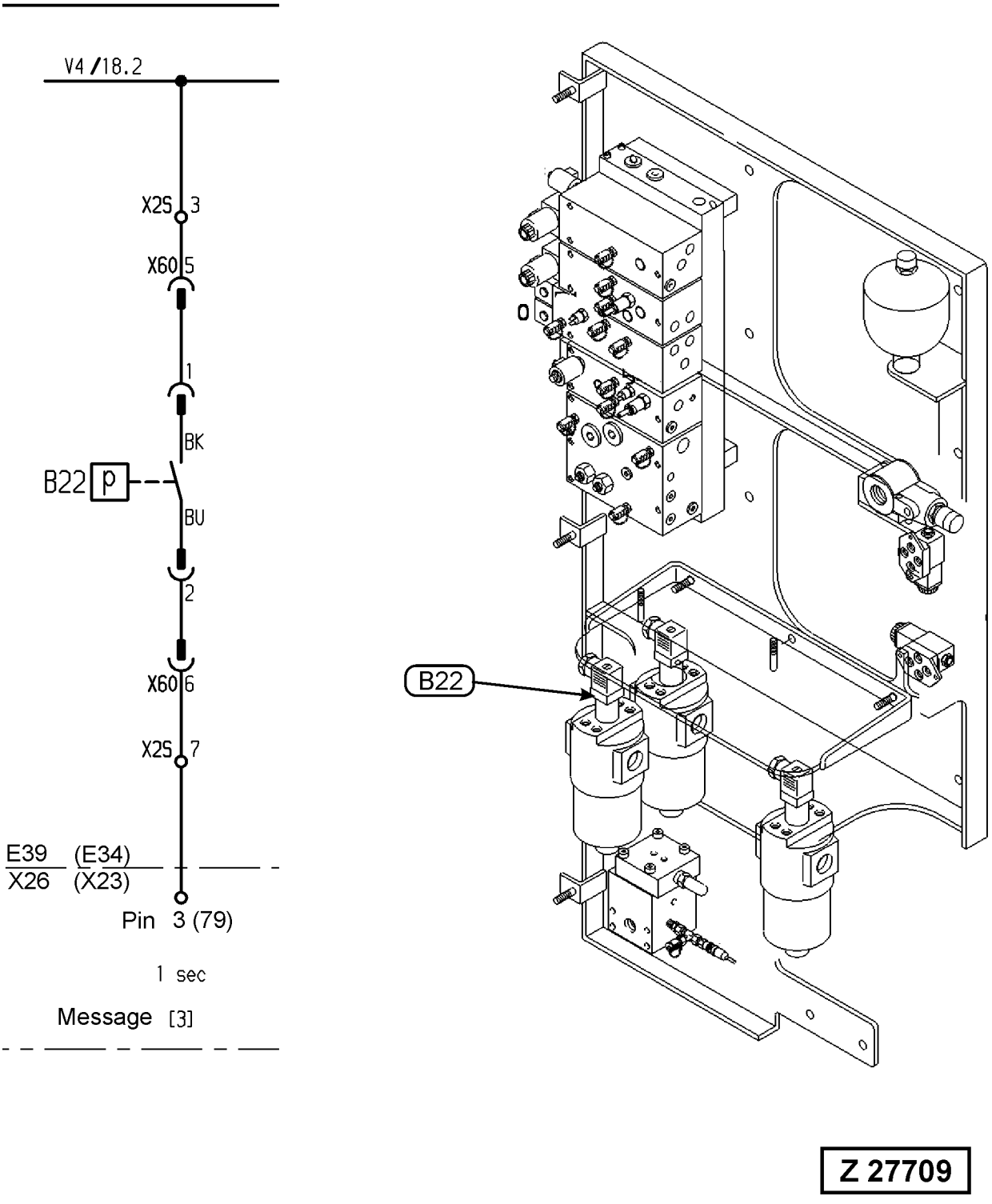
Message -3- Oil filter for control oil restricted (B22)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Incorrect or no supply voltage at the pressure switch B22	4	► Start motor.			
		Using the T-box, measure the supply voltage for B22 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 6.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B22. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B22 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 820Ω
			No	Disconnection, short circuit, or fault in the wiring between connector X60/5 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B22 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	6	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 3. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 79. Is the signal present?	Yes	The signal wiring between pressure switch B22 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B22 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -3-

[3/3]



MESSAGE -4- OIL TANK BREATHER FILTER RESTRICTED (B24)

[1/3]

Trouble code **Message page 4**	Trouble	Oil tank breather filter restricted (B24)
Contents of trouble	• The message is displayed if pressure switch B24 (NO*) detects a clogged breather filter (1), illust. Z27791. *NO = Normally Open	
Related information	• A problem with the pressure switch B24 or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. • Switch B24 watches the negative pressure of the breather filter and the ventilation of the hydraulic oil tank.	
Problem that appears on machine	• A clogged breather filter adversely affects the circulation of the hydraulic oil in the tank.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Hydraulic tank breather filter is clogged	1	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		Replace the breather filter element (C), illust. Z27791 as necessary. For details refer to the OPERATION & MAINTENANCE MANUAL, section HYDRAULIC SYSTEM - FILTER SERVICE. Remark: Use only filter elements with metal frame. If this trouble message appears frequently in-between the replacement intervals, make sure that: (a) the hydraulic oil tank is not overfilled when the machine is in its defined position, and (b) consider the dependency on the temperature range. Is the problem solved after replacement of the filter element?	Yes	Go to step 3.	
			No	Go to next step.	
The pressure switch B24 is defective	2	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B24 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved?	Yes	Got to final step 7.	
			No	Continue with step 3 of table [2/3].	

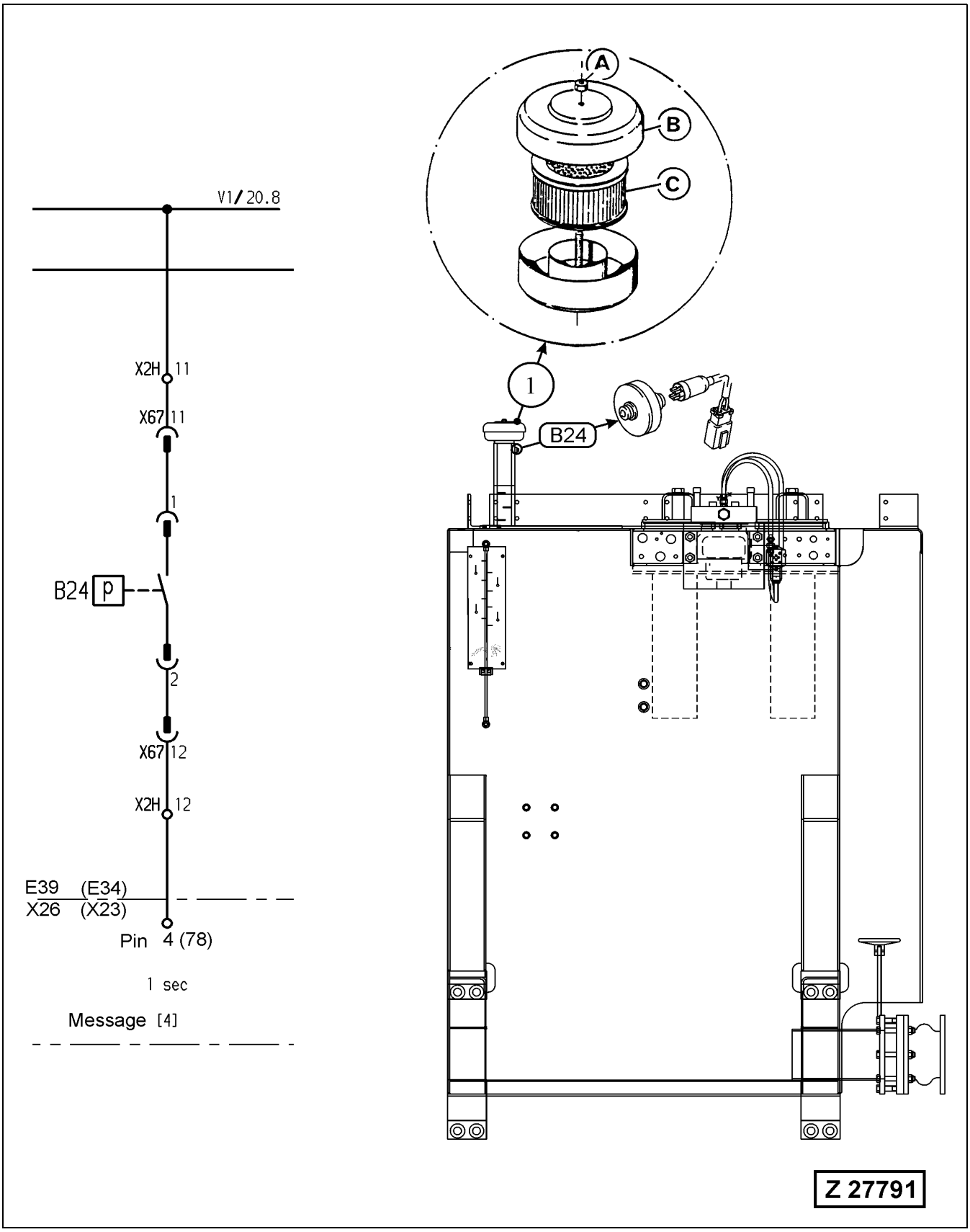
Message -4- Oil tank breather filter restricted (B24)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Defective pressure switch B24	3	Check Pressure switch B24 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	
Incorrect or no supply voltage at the pressure switch B24	4	► Start motor.			
		Using the T-box, measure the supply voltage for B24 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 6.	V_{min} 22V
					V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B24. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B24 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 820Ω
			No	Disconnection, short circuit, or fault in the wiring between connector X67/11 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B24 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	6	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 4. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 78. Is the signal present?	Yes	The signal wiring between pressure switch B24 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B24 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -4-

[3/3]



MESSAGE -5- RETURN OIL FILTER RESTRICTED (B26)

[1/3]

Trouble code **Message page 5**	Trouble	Return oil filter restricted (B26)
Contents of trouble	• Pressure switch B26 (NO*) signals, that the back pressure in the return oil filter exceeds 2 bar. *NO = Normally Open	
Related information	• A problem with the pressure switch B26 or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. • The message 5 is not set until the hydraulic oil temperature reaches T2, see table below.	
Problem that appears on machine	• Unfiltered return oil flows through the bypass of the filter into the hydraulic oil tank.	

Temperatur in °C temperature in °C	Viskosität nach ISO viscosity grade according to ISO				
	VG22	VG32	VG46	VG68	(VG100)
TypNr. / type no.	1	2	3	4	5
T1 (1000 cSt)	-20	-14	-6	0	5
(T2) (100 cSt)	9	16	25	32	(39)
T3 (30 cSt)	32	41	50	58	67
T4 (10 cSt -5°C)	58	69	79	85	85

Z 27262

for example
Hydraulic oil:VG100
T2: 39°C

In this case the trouble message is not set until the hydraulic oil temperature reaches:
39°C

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The return oil filter is clogged	1	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		Remove filter cover retainers on top of the hydraulic oil tank. For details refer to the OPERATION & MAINTENANCE MANUAL, section HYDRAULIC SYSTEM - FILTER SERVICE. Are one or more filters clogged?	Yes	Replace return oil filters as described in the OPERATION & MAINTENANCE MANUAL. Locate the cause of the contamination and rectify the problem.	
			No	Go to next step.	
The pressure switch B26 is defective	2	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B26 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved?	Yes	Got to final step 7.	
			No	Continue with step 3 of table [2/3].	

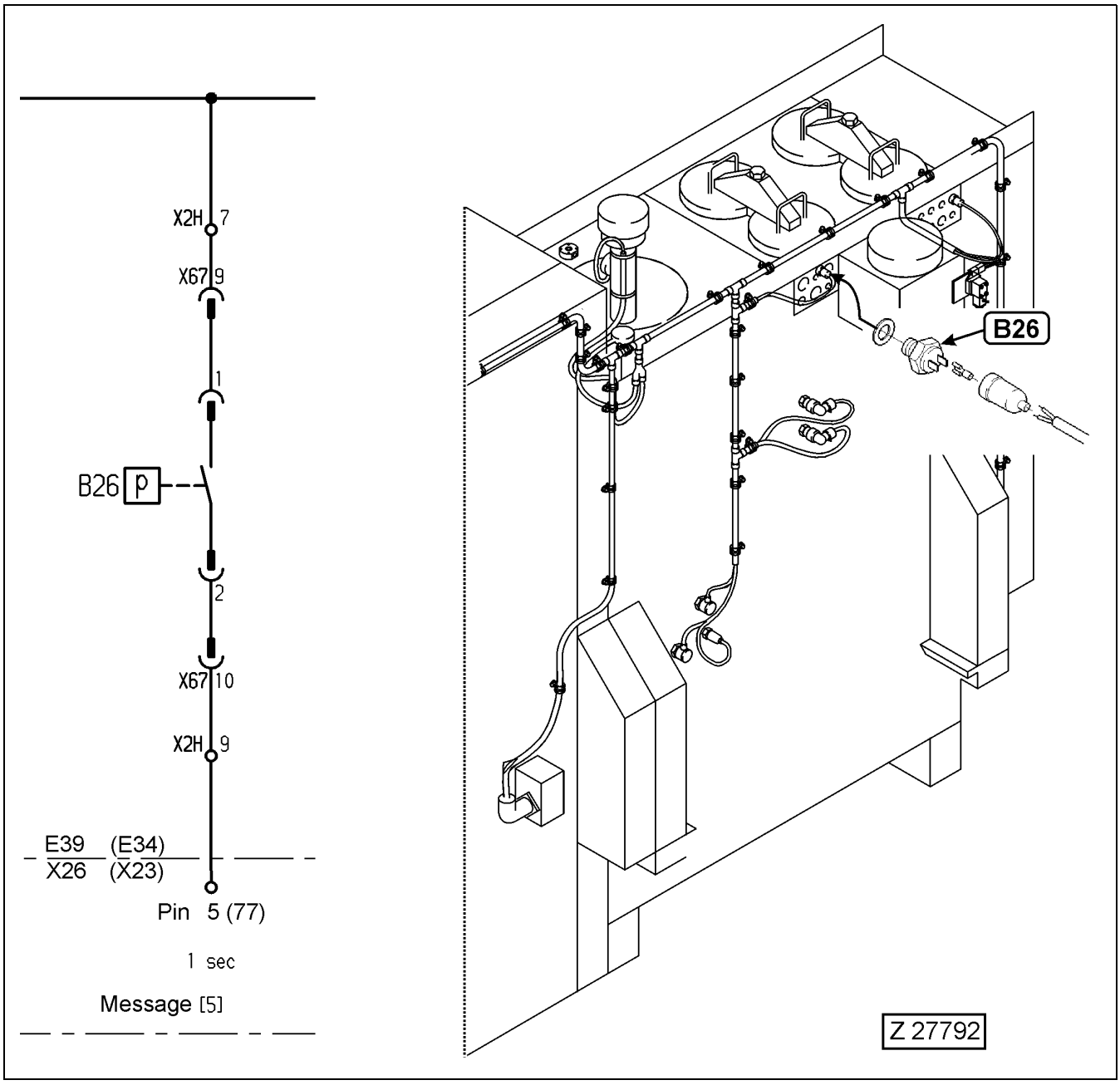
Message -5- Return oil filter restricted (B26)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Defective pressure switch B26	3	Check Pressure switch B26 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	
Incorrect or no supply voltage at the pressure switch B26	4	► Start motor.			
		Using the T-box, measure the supply voltage for B26 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 6.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B26. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B26 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring between connector X67/9 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B26 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	6	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 5. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 77. Is the signal present?	Yes	The signal wiring between pressure switch B26 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B26 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -5-

[3/3]



MESSAGE -6- LEAK OIL FILTER RESTRICTED (B25)

[1/3]

Trouble code **Message page 6**	Trouble	Leak oil filter restricted (B25)
Contents of trouble	• Pressure switch B25 (NO*) signals, that the back pressure in the leak oil filter exceeds 0.5 bar. *NO = Normally Open	
Related information	• A problem with the pressure switch B25 or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. • The message 6 is not set until the hydraulic oil temperature reaches T2, see table below.	

Temperatur in °C temperature in °C	Viskosität nach ISO viscosity grade according to ISO				
	VG22	VG32	VG46	VG68	VG100
TypNr. / type no.	1	2	3	4	5
T1 (1000 cSt)	-20	-14	-6	0	5
T2 (100 cSt)	9	16	25	32	39
T3 (30 cSt)	32	41	50	58	67
T4 (10 cSt -5°C)	58	69	79	85	85

Z 27262

for example
Hydraulic oil:VG100
T2: 39°C

In this case the trouble message is not set until the hydraulic oil temperature reaches:
39°C

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The leak oil filter is clogged	1	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		Remove filter cover retainer on top of the hydraulic oil tank. For details refer to the OPERATION & MAINTENANCE MANUAL, section HYDRAULIC SYSTEM - FILTER SERVICE. Is the leak oil filter clogged?	Yes	Replace leak oil filter as described in the OPERATION & MAINTENANCE MANUAL. Locate the cause of the contamination and rectify the problem.	
			No	Go to next step.	
The pressure switch B25 is defective	2	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B25 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved?	Yes	Go to final step 7.	
			No	Go to next step.	
	3	Check Pressure switch B25 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	
Incorrect or no supply voltage at the pressure switch B25	4	► Start motor.			
		Using the T-box, measure the supply voltage for B25 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 6.	V_{min} 22V V_{max} 24V
			No	Go to next step.	

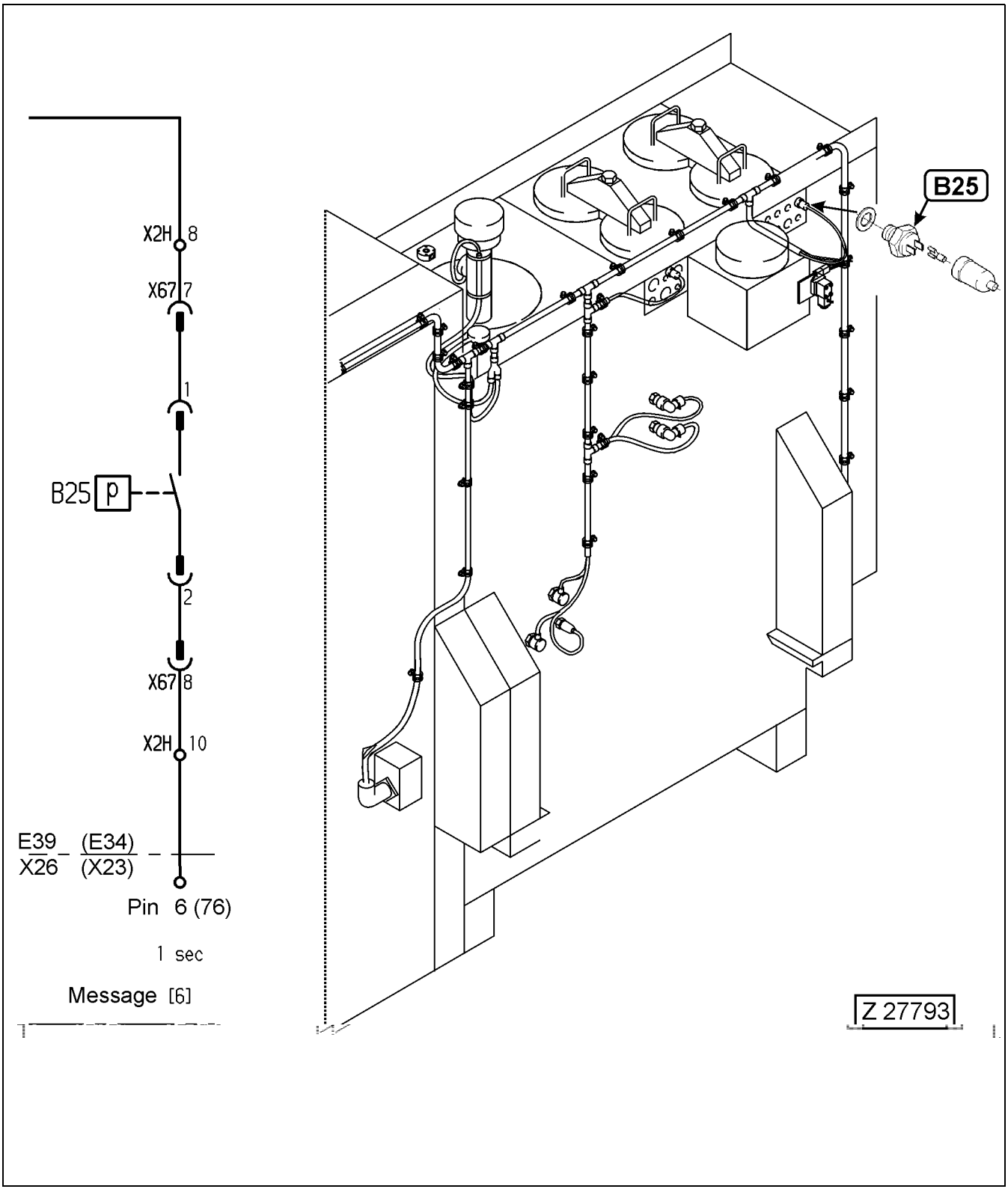
Message -6- Leak oil filter restricted (B25)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B25. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B25 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring between connector X67/7 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B25 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	6	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 6. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 76. Is the signal present?	Yes	The signal wiring between pressure switch B24 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B25 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -6-

[3/3]



MESSAGE -7- HYDRAULIC OIL TEMPERATURE TOO HIGH (B15)

[1/6]

Trouble code **Message page 7**	Trouble	Hydraulic oil temperature too high (B15)
Contents of trouble	• Overheated hydraulic oil is detected and the bucket function is disabled.	
Related information	• The hydraulic oil temperature sensor B15 detects an oil temperature above the maximum permissible operating temperature as shown in the temperature chart 1. • Compare temperature with gauge P3 on the dashboard. • A problem with the hydraulic oil temperature sensor B15 (PT100 type) or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. For more information see also the illustrations on page [4/6 - 6/6].	
Problem that appears on machine	• The hydraulic oil is overheated and the bucket movement is shut off.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The hydraulic oil is overheated causing a bucket motion shut-off	1	► Keep the motor running, with max. fan speed.			
		Recheck the hydraulic oil temperature at the temperature sensor B15, see illust. Z27794 using an infrared temperature gauge. For the maximum permissible operating temperature refer to the temperature chart 1 below. Does the measured hydraulic oil temperature indicate overheating as well?	Yes	Go to next step.	
			No	Check for other related faults starting with step 3 of table [2/3].	
Wrong setting of temperature switch point at switch unit K39 in relation to the hydraulic oil viscosity.	2	► Motor stopped and main key switch 1S1in OFF position.			
		Check the temperature setting in relation to the hydraulic oil viscosity used in the machine. Values are given in the temperature chart 1. Are the oil viscosity settings correct?	Yes	Continue with step 3 of table [2/3].	
			No	Adjust K39 to the correct switching point. Refer to the Operation and Maintenance Manual for setting instructions. Verify that trouble message 7 - Hydraulic oil temperature too high- is not set again.	

Temperature Chart 1
Adjust switch unit K39 to the respective oil viscosity grade as follows:

Oil viscosity grade according to ISO	VG 22	VG 32	VG 46	VG 68	VG 100	Shell Tellus Arctic (stabilized)
-K39- Adjust temperature limit value to switching point °C	+58°	+69°	+79°	+85°	+85°	+81°

Message -7- Hydraulic oil temperature too high (B15)

[2/6]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Depending on the state of construction a general problem with the hydraulic oil temperature sensor B15, the associated relays, or the belonging wiring may exist	3	Refer to the following PARTS & SERVICE NEWS (latest version) • AH07527 According to the state of construction of your machine, follow the procedures given in the PARTS & SERVICE NEWS. Is the problem rectified?	Yes	Go to final step 11.	
			No	Go to next step.	
Wrong setting of temperature switch point at switch unit K100, illust. Z27819 in relation to the hydraulic oil viscosity.	4	► Motor stopped and main key switch 1S1in OFF position.			
		Check the temperature setting in relation to the hydraulic oil viscosity used in the machine. Values are given in temperature chart 2. Are the oil viscosity settings correct?	Yes	Continue with step 5.	
			No	Adjust K100 to the correct switching point. Refer to the Operation and Maintenance Manual for setting instructions. Verify that trouble message 7 - Hydraulic oil temperature too high- is not set again.	
Defective temperature sensor B75 (PT100 type).	5	Check temperature sensor B75 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace temperature sensor. After that recheck that the trouble condition does no longer exist.	
Defective solenoid valve Y101 for back pressure valve control.	6	Check solenoid valve according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace solenoid valve Y101. After that recheck that the trouble condition does no longer exist.	
Wrong back pressure valve adjustment.	7	Check the pressure setting of the back pressure valve according to the instructions given in the Service Manual, section 4.3 BACK PRESSURE VALVE ADJUSTMENT. Is the pressure setting correct?	Yes	Go to next step.	
			No	Adjust back pressure valve to the correct pressure. Refer to the Service Manual for setting instructions. Verify that trouble message 7 -Hydraulic oil temperature too high- is not set again.	

Message -7- Hydraulic oil temperature too high (B15)

[3/6]

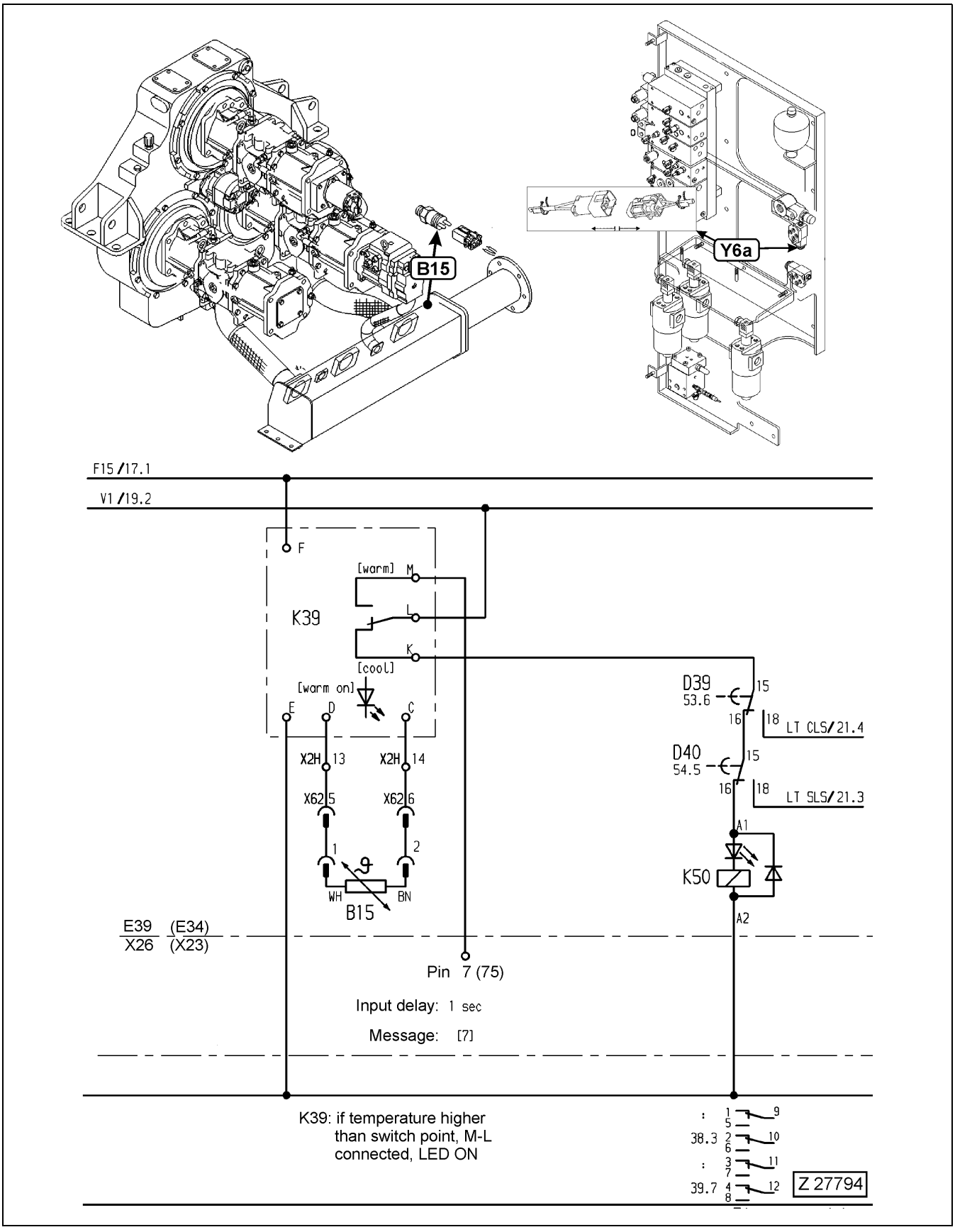
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The hydraulic oil cooler fans are not driven with maximum speed	8	► Keep the motor running.			
		Maximum fan speed is achieved if no voltage (0 V) is applied at solenoid valve Y6a, see illust. Z27794. Therefore disconnect the cable connector from solenoid valve Y6a. If the fan speed does not increase, check the hydraulic settings for the fan speed (refer to the Service Manual). Connect cable connector to the solenoid valve Y6a. Is the setting for max. fan speed correct?	Yes	Go to next step.	
			No	Refer to the Service Manual, section Hydraulic Oil Cooling, and perform the setting of the max. fan speed. If the setting is not as specified follow the adjustment and/or inspection procedures in the Service Manual as necessary.	RPM max. 1300 ± 50
Any clogged hydraulic oil cooler caused the overheating of the hydraulic oil	9	► Motor stopped - Main key switch 1S1OFF - Maintenance safety switch 1S58 OFF & locked.			
		⚠ This test step must only be performed with the motor at standstill and maintenance safety switch in OFF position and locked. Perform a visual inspection of the hydraulic oil cooler. Also open the swing out doors of the cooler frame and check for deposits between these door wings. Is intense pollution present?	Yes	Clean the hydraulic oil cooler with a steam cleaner as necessary. Continue to work and monitor the hydraulic oil temperature on the analog gauge.	
			No	Go to next step.	
Overheating due to any malfunction in the hydraulic system	10	► Run the motor with max. fan speed.			
		Check the hydraulic system by performing the PM-Clinic.		Continue to work and monitor the hydraulic oil temperature on the analog gauge.	
Final check	11	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Temperature Chart 2
Adjust switch unit K100 to the respective oil viscosity grade as follows:

Oil viscosity grade according to ISO	VG 22	VG 32	VG 46	VG 68	VG 100	Shell Tellus Arctic (stabilized)
-K100- Adjust temperature limit value to switching point °C	+9°	+16°	+25°	+32°	+39°	+5°

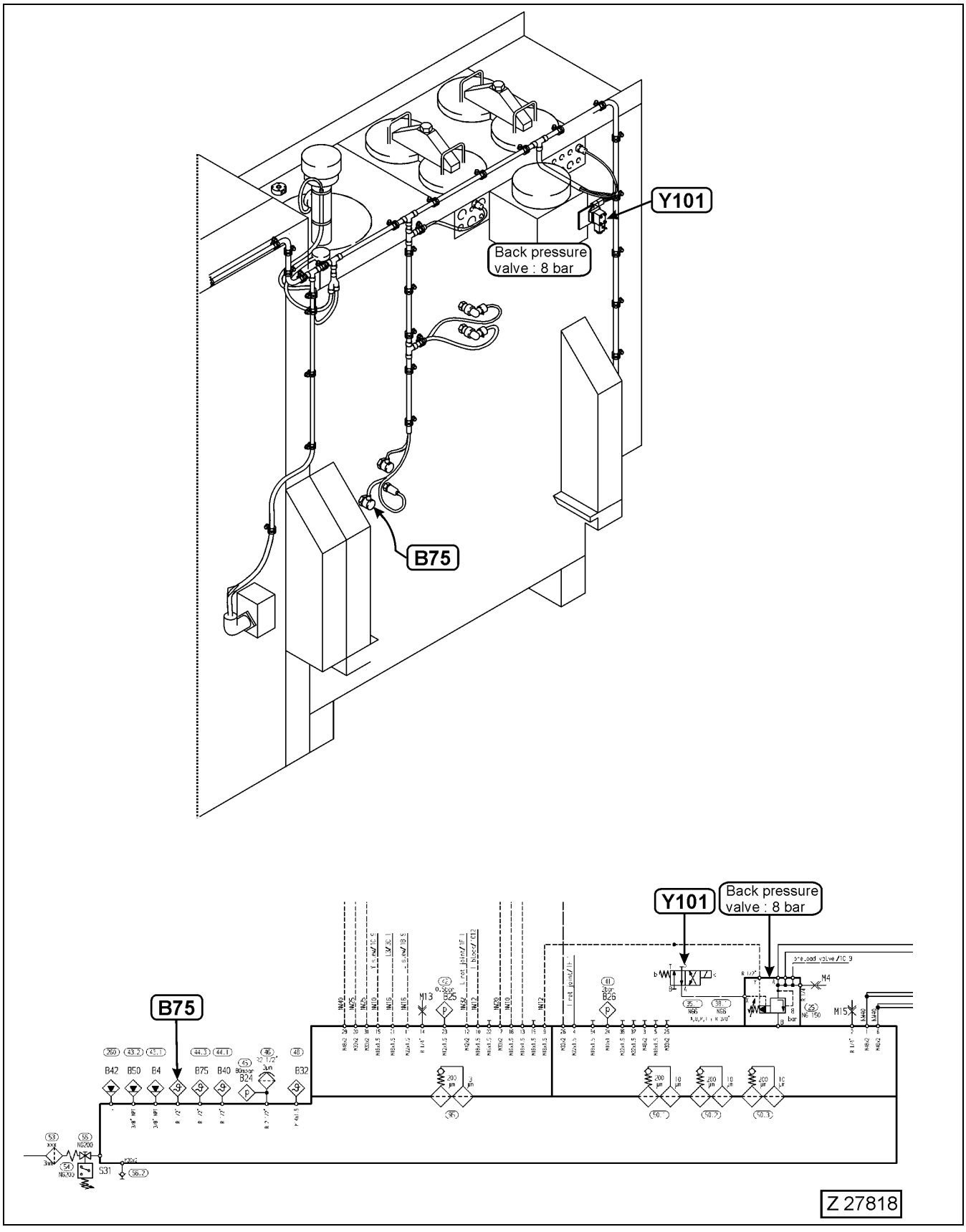
Illustrations related to trouble message -7-

[4/6]



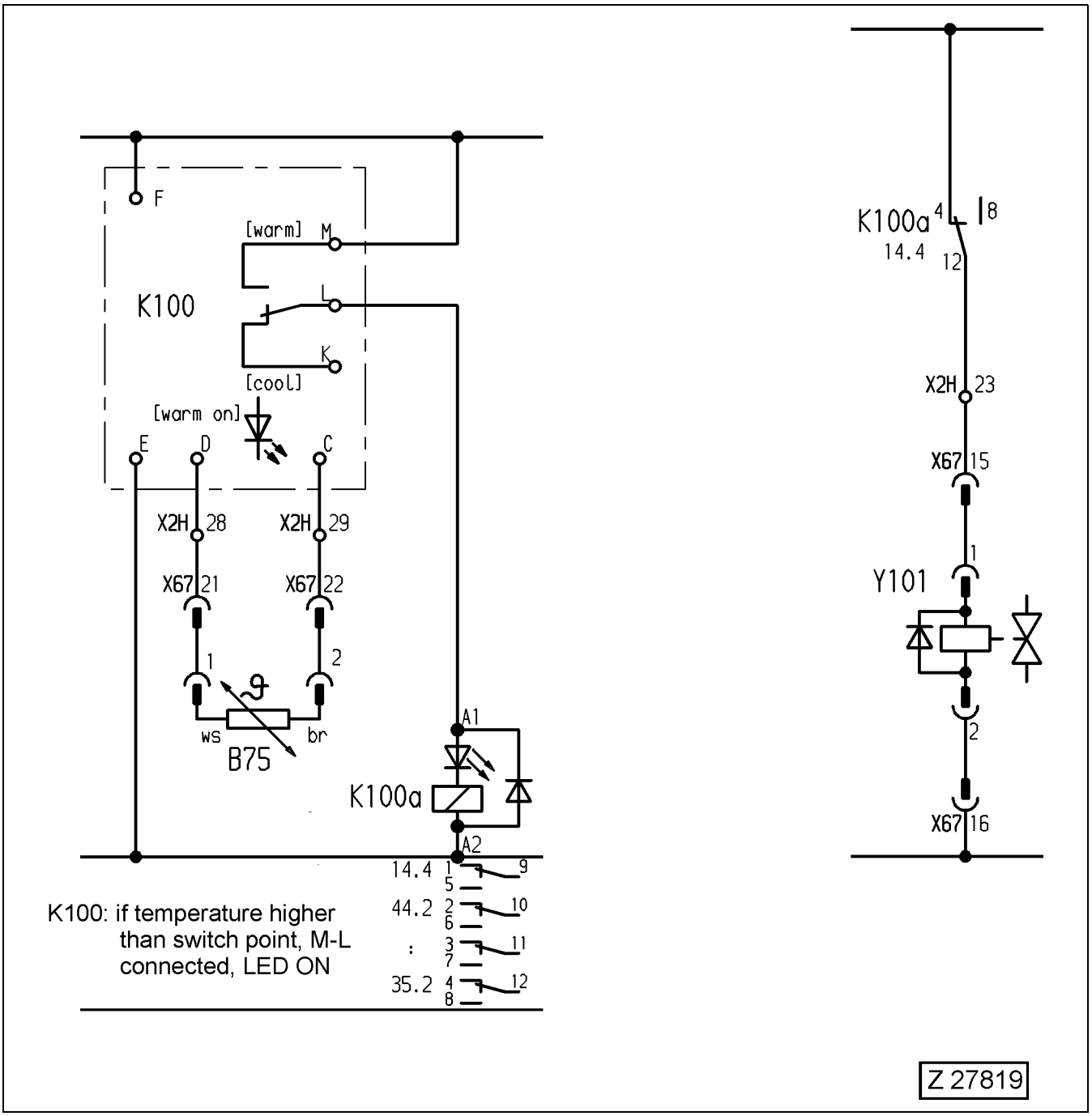
Illustrations related to trouble message -7-

[5/6]



Wiring Diagram related to trouble message -7-

[6/6]



MESSAGE -8- (NOT USED)

MESSAGE -9- MAIN TRANSFORMER TEMPERATURE TOO HIGH (1K4)

[1/4]

Trouble code **Message page 9**	Trouble	Main transformer temperature too high (1K4)
Contents of trouble		**HAZARDOUS VOLTAGE!** **High voltage is applied at the main transformer.** **There is danger of serious injury or death.** **Access to the main transformer is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** - Main transformer 0T6 is overheated (>120°C). - One or several temperature sensors (PT100 type) signal overheating (R >146.07 Ω).
Related information	In case of overheating warning light #1 at the LED monitor module 6V1 is ON. **REMARK** The LED monitor units 6V1 and 6V2 in the LV-cabinet indicate certain electric problems independent of ETM. For detailed information refer to the OPERATION & MAINTENANCE MANUAL.	
Problem that appears on machine	Motor shutdown due to overheated main transformer.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Main transformer overheated	1	► Main key switch 1S1 ON.			
		Measure voltage to ground at terminal X2Y 13 (leading to the pickup module E34, connector X23, pin 73). 24V → overheating & motor shutdown 0V → 1K4 open at contacts 31/32 & motor ON (no overheat) Do you measure 24V?	Yes	Go to next step.	22 - 24 V
			No	Motor is running and not overheated. Message is wrongly indicated. In this case relay 1K4 is stuck or its wiring is faulty. → Refer to the Wiring Diagram, check the wiring, and repair as necessary. → Or replace relay 1K4 if necessary.	
Switch module (1A1) for temperature sensors or its wiring is defective (location: LV board)	2	► Main key switch 1S1 ON.			
		Measure voltage (24V) between terminals A1 and A2 at module 1A1. Do you measure 24V?	Yes	Go to next step.	22 - 24 V
			No	→ Refer to the Wiring Diagram, check the wiring of 1A1, and repair as necessary. → Or replace module 1A1 if necessary.	

Message -9- Main transformer temperature too high (1k4)

[2/4]

<table>
<tr><th>Possible cause</th><th>Step</th><th colspan="2">Test condition & Inspection</th><th>Procedure</th><th>Nominal value</th></tr>
<tr><td rowspan="3">Temperature sensor (0B3, 0B4, 0B5) or its wiring is defective (location: HV cabinet)</td><td rowspan="3">3</td><td colspan="3">► Main key switch 1S1 ON.</td><td></td></tr>
<tr><td rowspan="2">HAZARDOUS VOLTAGE!
Access to the main transformer is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.

- Using an IR thermometer gun, measure the temperature in the area of the sensors.
- Disconnect the sensor cables at 0X5 terminals 1 &2.
- Measure the sensor output (W).
- Use the conversion table (W ~ °C) to determine the sensor temperature.
- Compare the temperature of the IR thermometer gun with sensor temperature.

Do the IR thermometer gun and the sensor output confirm overheating?</td><td>Yes</td><td>The sensor is OK.
Go to next step.</td><td></td></tr>
<tr><td>No</td><td>Refer to the Wiring Diagram, check the sensor wiring, and repair as necessary.

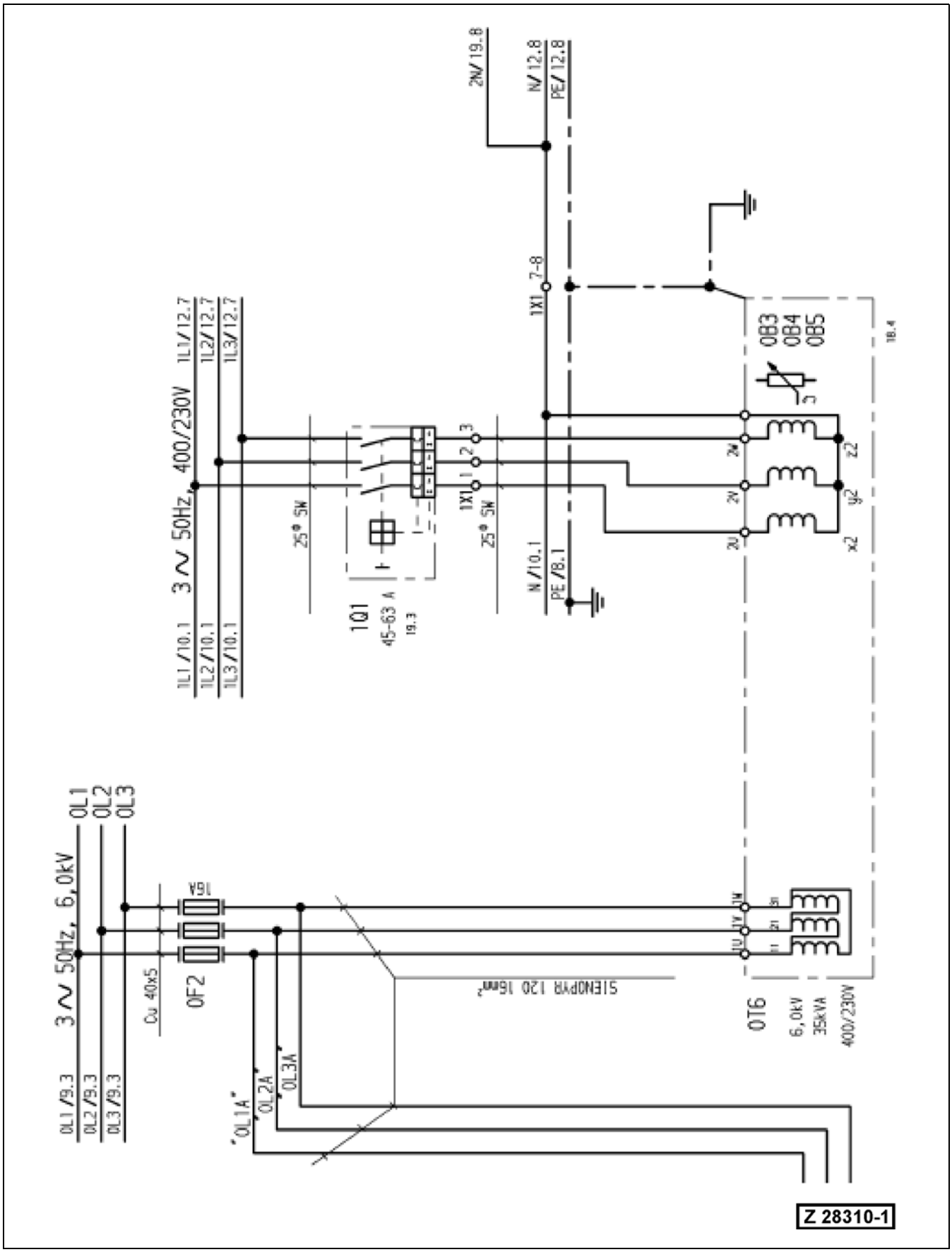
If wiring OK replace the temperature sensors.</td><td></td></tr>
<tr><td rowspan="2">Main transformer overheated due to overload</td><td rowspan="2">4</td><td colspan="3">► Main key switch 1S1 ON.</td><td></td></tr>
<tr><td colspan="3">HAZARDOUS VOLTAGE!
Access to the main transformer is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.

Reasons for overload are for example:
- Instable netline (voltage).
- Main transformer 0T6 defective.

The HV department needs to check netline and main transformer (0T6).
Repair or replacement needs to be done as necessary.</td><td></td></tr>
<tr><td>Final check</td><td>5</td><td colspan="3">Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.</td><td></td></tr>
</table>

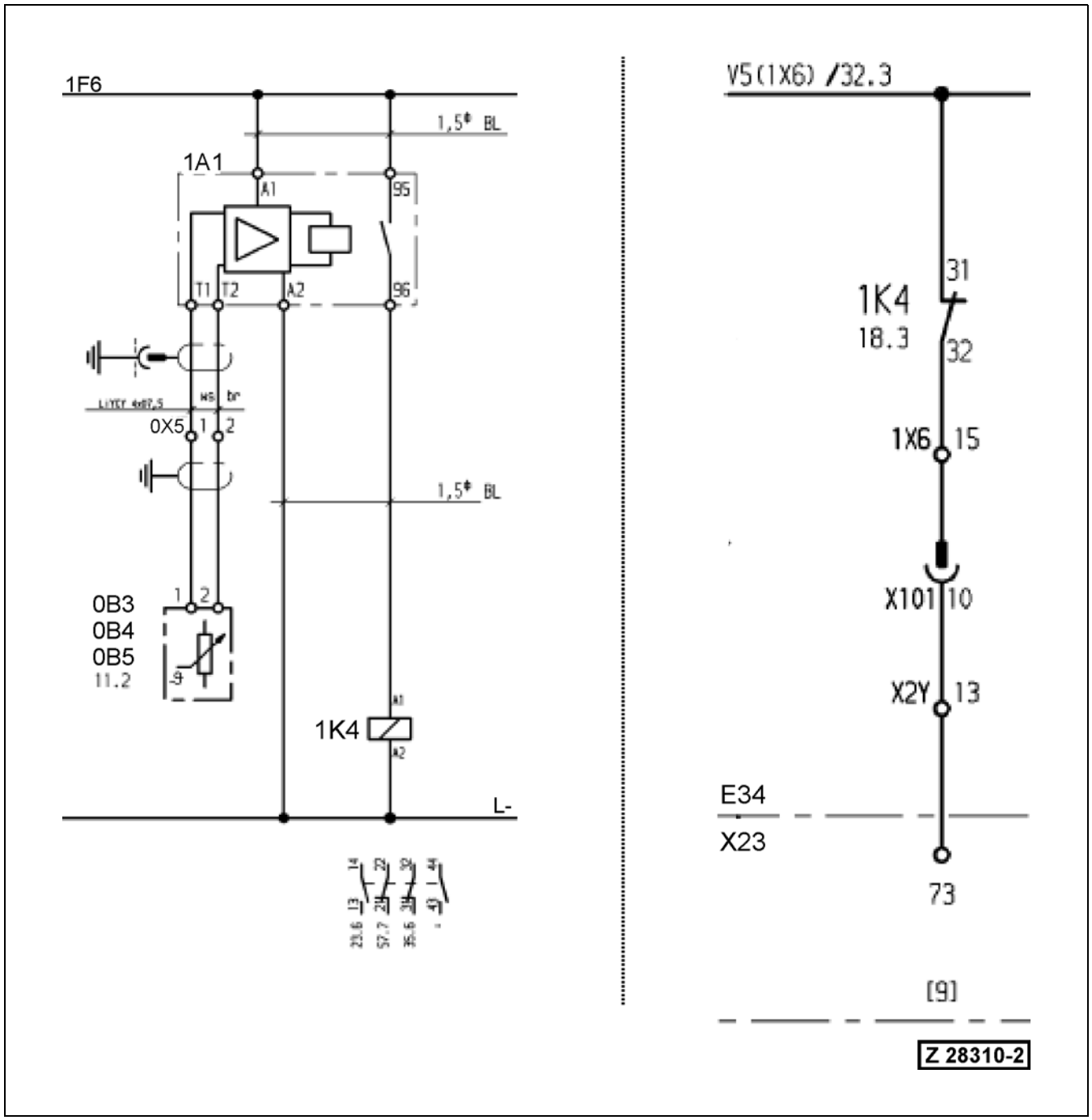
Wiring Diagram 1 related to trouble message -9-

[3/4]



Wiring Diagram 2 related to trouble message -9-

[4/4]



MESSAGE -10- LOW VOLTAGE MONITORING ACTUATED

[1/4]

Trouble code **Message page 10**	Trouble	Low voltage monitoring actuated
Contents of trouble	- The main power supply is monitored by module 1K13. - If the main supply voltage drops 15% below the specified voltage, the module triggers relay 1K27 and the motor is shutdown.	
Related information	Motor restart is possible, if the voltage drop is max. 10% of the target voltage.	
Problem that appears on machine	Motor shutdown.	

Settings on module 1K13 (start protection after voltage interruption) - 2 module types

Setting information: U = 0.324 hysteresis = 0.975 No jumper between terminals X3 and X4!
For the individual setting values check your Wiring Diagram.



Message -10- Low voltage monitoring actuated

[2/4]

Module 1K13 is preset ex factory. Use beside information if you need to double check the setting.

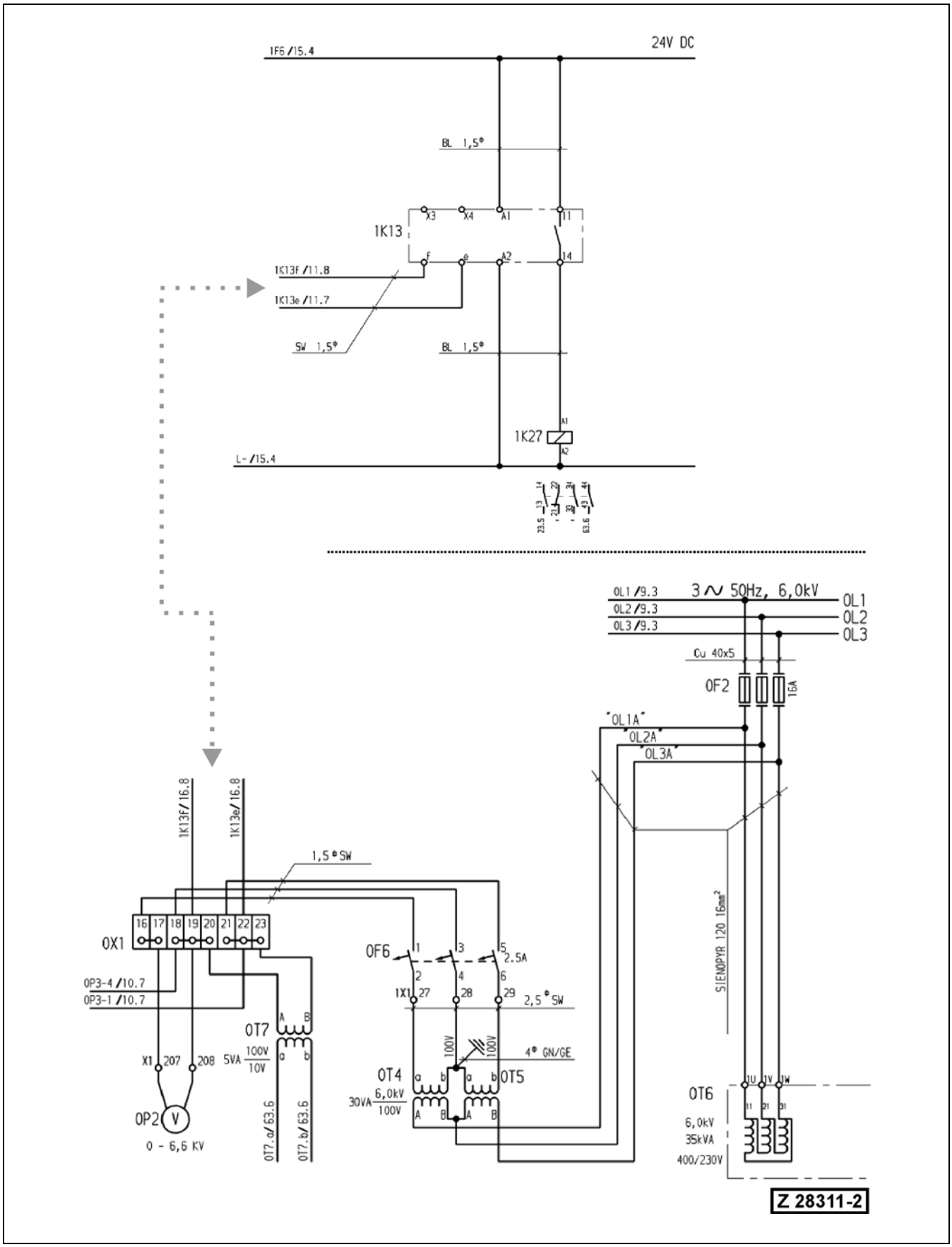
If re-setting is necessary and you are not sure about the procedure, contact your dealer or the KMG service.

LED	Status	Condition
Green	ON	Supply voltage present
Yellow or Red	ON	Activated output relay

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Low supply voltage detected via 1K13	1	► Main key switch 1S1 ON.			
		Check the voltmeter (0P2) in the cab. Example: Specified supply voltage = 6.0 kV 6.0 kV - 15% = 5.1 kV If the reading of 0P2 ≤ 5.1 kV, voltage dropped 15% or more. Does the voltmeter indicate undervoltage (15% or more)?	Yes	Contact the HV department and report the trouble.	Specified supply voltage
			No	a) Signal was triggered by netline variation. If specified supply voltage is present again, start motor. b) Go to next step.	
Module 1K13 or its wiring including relay 1K27 faulty (Motor shutdown was triggered even the netline voltage is OK)	2	► Main key switch 1S1 ON.			
		Check if relay 1K27 is energized (24V at A1 and A2) Are 24V present?	Yes	Replace the defective relay 1K27.	0 V
			No	Measure the voltage between relay pin 14 and GND (supply voltage of 1K13). If 24V are not present, repair or replace the wiring. If the cable is OK, go to next step.	
Module 1K13 or its wiring including relay 1K27 faulty (Basic setting is wrong)	3	► Main key switch 1S1 ON.			
		Take measurements at module 1K13: - Between terminals A1 and A2 = 24V (target) - Between terminals 11 and 14 = 24V (target) If 0V between terminals 11 and 14: - Check the wiring of 1K13 (target = 24V) - Check the wiring between terminals "e" & "f" (100 V): **HAZARDOUS VOLTAGE!** **Access to the main transformer is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** - Take the measurement at connector 0X1 between terminals 19 and 22. If the measurements are OK, resetting of module 1K13 is necessary. Refer to the above setting information and to your Wiring Diagram.			
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

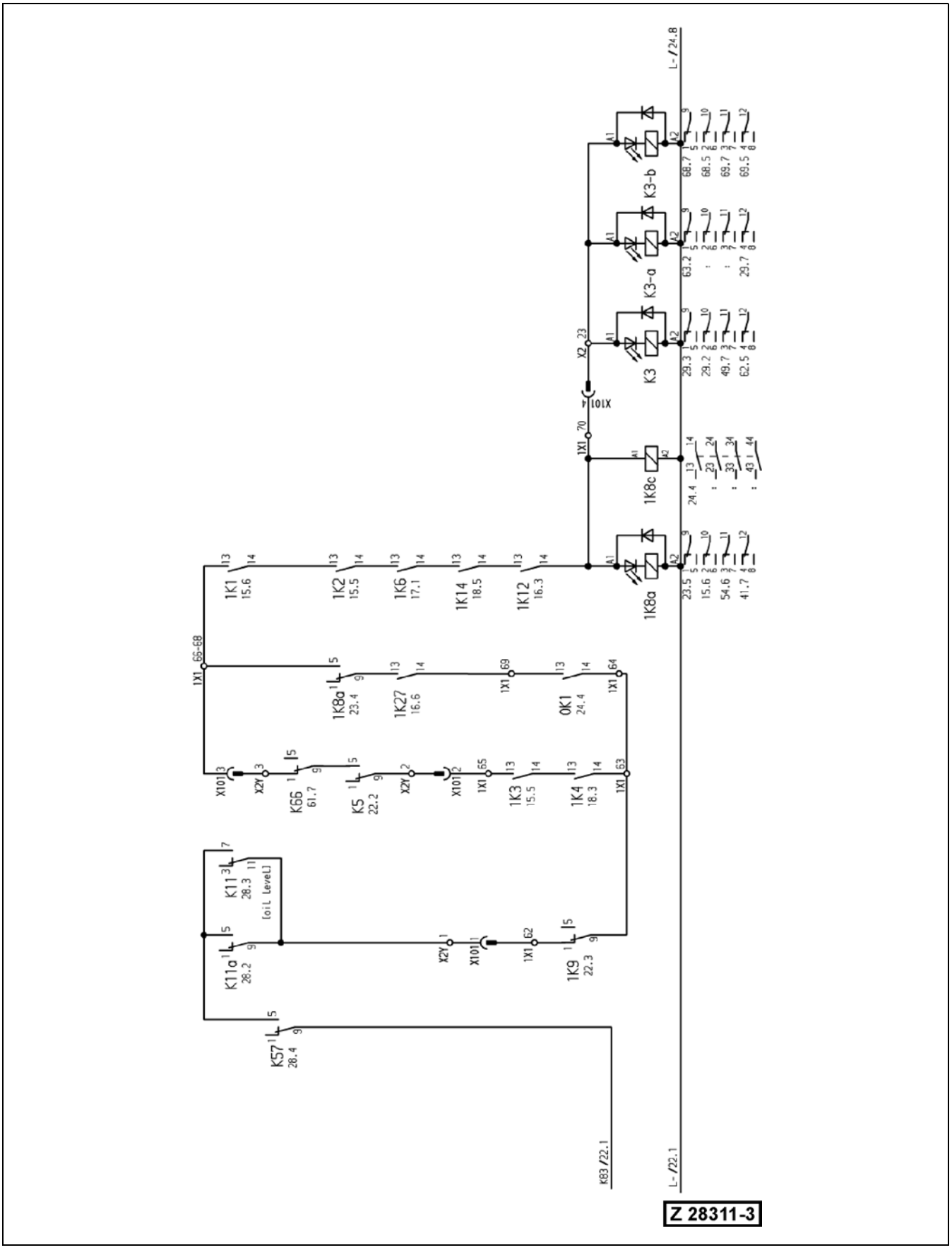
Wiring Diagram 1 related to trouble message -10-

[3/4]



Wiring Diagram 2 (1K27 in the motor start/stop and interlocking circuit) related to trouble message -10-

[4/4]



MESSAGE -11- (NOT USED)

MESSAGE -12- (NOT USED)

MESSAGE -13- (NOT USED)

MESSAGE -14- MOTOR-SHUTDOWN: TOO LOW HYDRAULIC OIL LEVEL! (B4)

[1/4]

Trouble code **Message page 14**	Trouble	Motor-shutdown: Too low hydraulic oil level! (B4)
Contents of trouble	• Sensor B4 indicates that the hydraulic oil level is below the minimum refill level. • This trouble condition should only occur, if the Operator has ignored the warning message 55 -Refill hydraulic oil-	
Related information	• If the sensor B4 does not sense oil, its signal voltage turns to 0 V. • The motor is stopped and buzzer H22 sounds continuously. • A problem with the sensor or connected wiring may also cause the trouble, see table [2/4] for testing procedure of the signal wiring and connections. • For more information see also the illustrations on page [3/4 and 4/4].	
Problem that appears on machine	• Automatic motor shutdown.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The hydraulic oil level is below the minimum level	1	► Check with stopped motor and the attachment in its defined position.			
		Verify if the hydraulic oil level in the main oil reservoir is above or below the oil level sensor B4, see illustration Z27799 on page 3 of this chart. Refer also to illustration Z28000 on page 4 of this troubleshooting chart, which shows the hydraulic oil tank label with the defined positions at the bottom of the label and the relevant oil levels on the left and right side of the label. Is the oil level below the sensor B4?	Yes	Refill specified hydraulic oil and verify that trouble message 14 (55) is not displayed again.	
			No	Go to next step.	
Incorrect or no supply voltage in the circuit of the oil level sensor B4	2	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Disconnect the cable connector from the oil level sensor B4, insert the T-box, and measure the supply voltage at the cable connector terminal A. Is supply voltage present?	Yes	Go to step 5.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
	3	Check if supply voltage is present at the power supply line F12. Is supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Disconnection or short circuit in any wire segment (including connectors) between sensor B4 and the main power supply from F1. Locate the problem and repair as necessary	

Message -14- Motor-shutdown: Too low hydraulic oil level! (B4)

[2/4]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the wiring segments between the oil level sensor B4 and the power supply line of circuit breaker F12 (disconnection or defective contact in any connector)	4	► Keep the main key switch 1S1in the OFF position.			
		Refer to the wiring diagram and check the wiring segments between oil level sensor B4 and the power supply line F12 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment (including connectors) between oil level sensor B4 and the line of the circuit breaker F12. Locate the problem and repair as necessary.	
Defective ground connection at the oil level sensor B4	5	► Keep the main key switch 1S1in the OFF position.			
		Disconnect the cable connector from the oil level sensor B4 and check for proper ground connection at the connector terminal B. Is the ground connection OK?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment (including connectors) between oil level sensor B4 and the ground line (-). Locate the problem and repair as necessary.	
Defective oil level sensor B4	6	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Using the T-box measure the signal voltage at the connector terminal C of the oil level sensor B4 when the sensor is dipped into oil (sensor closed), and when it is not touching any oil (sensor open). Do the measurements meet the specifications?	Yes	A sporadic malfunction may exist. Further inspection is required.	V_{Signal} open: 0V V_{Signal} closed: 22 - 24V
			No	Replace the oil level sensor B4 and verify that the repair was successful. Go to final step 7.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -14-

[3/4]

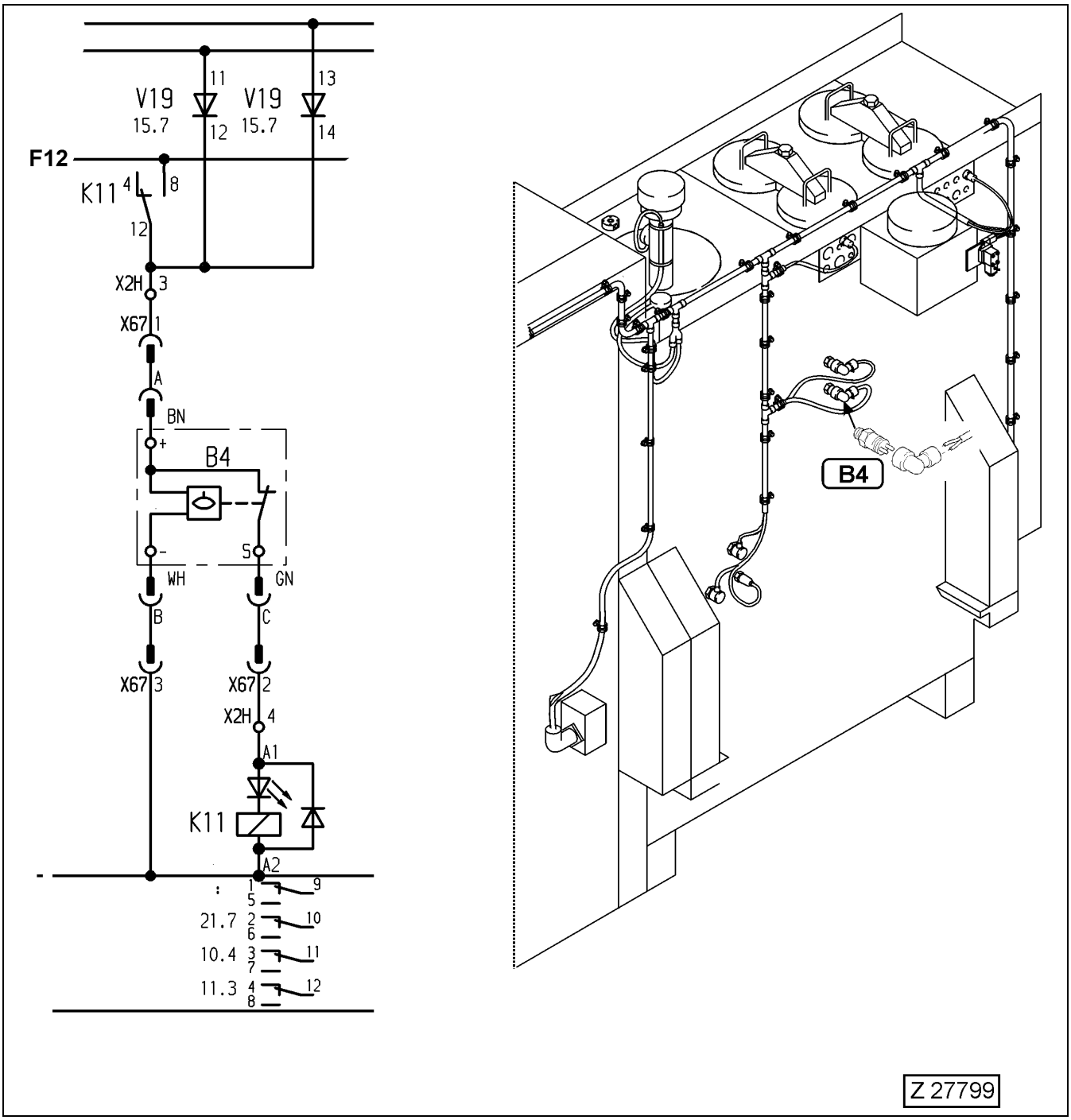
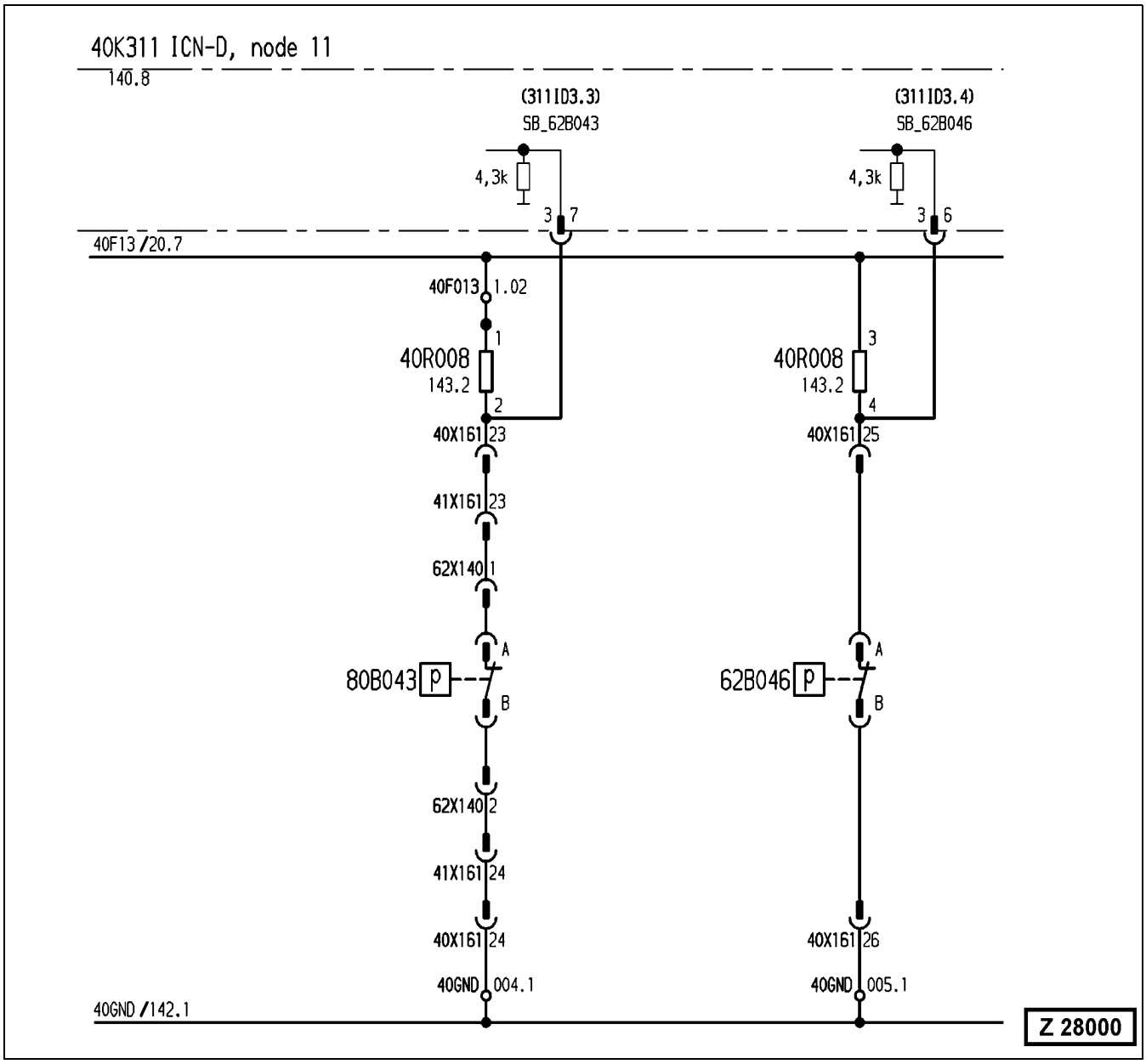


Illustration related to trouble message -14-

[4/4]



MESSAGE -15- CAUTION, SLEW GEAR HOUSE BRAKE ON

[1/5]

Trouble code **Message page 15**	Trouble	Caution, slew gear house brake ON
Contents of trouble	• Pressure switch B16 monitors the pressure of the swing gear house brake circuit. • If the swing lock switch S29 is switched to position 1 while the motor is running, the swing parking brake is locked.	
Related information	• The system operates as a so called spring-loaded parking brake. Hydraulic pressure is NOT present:Spring force locks the brake • Hydraulic pressure is present: The brake is released. • A problem with the pressure switch B16 or connected wiring may cause the trouble • Condition: The motor is running • The pressure switch B16 is closed or its wiring is defective. Basic sensor data (B16) • For functional test of the swing parking brake refer also to the Service Manual.	
Problem that appears on machine	• If the swing parking brake is locked, switch S29 in position 1, swing movement is impossible. • Swinging is not possible with motor running and swing lock switch S29 in brake release position 0.	

No pressure present	0 V	Switch closed
Pressure present	24 V	Switch open
Switch point	24 bar	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Switch S29 is set to position 1. The swing gear parking brake is locked.	1	► Keep the motor running.			
		Release the swing parking brake by switching S29 to position 0 (swing movement is possible).		In case the swing parking brake is activated (switch S29 in position 1), but swing movement is still possible, check the electric circuit, starting with step 2. If the problem is not solved, go to page 3, step 7 and follow the respective instructions.	
Incorrect or no supply voltage at the pressure switch B16	2	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Measure the supply voltage at connector X61C, terminal 1. Is the correct supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Go to next step.	

Message -15- Caution, slew gear house brake ON

[2/5]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B16 (disconnection or defective contact in any connector)	3	► Keep the main key switch 1S1in the OFF position.			
		Refer to the wiring diagram. Check the wiring between the pressure switch B16 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors) between connector X61C terminal 1 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B16 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	4	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 15. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 67. Is the signal present?	Yes	The signal wiring between pressure switch B24 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B16 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Defective ground connection at the pressure switch B16	5	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Disconnect the cable connector X61D and check for proper ground connection at the connector terminal 2. Is the ground connection OK?	Yes	Go to next step.	
			No	Locate the problem and repair as necessary.	
Defective pressure switch B16 (switch does not close)	6	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Disconnect connectors X61C and X61D. Measure the continuity of the pressure switch B16 between the connector pins 1 and 2. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Replace the pressure switch B16 and verify that the repair was successful. Go to final step 9.	

Message -15- Caution, slew gear house brake ON

[3/5]

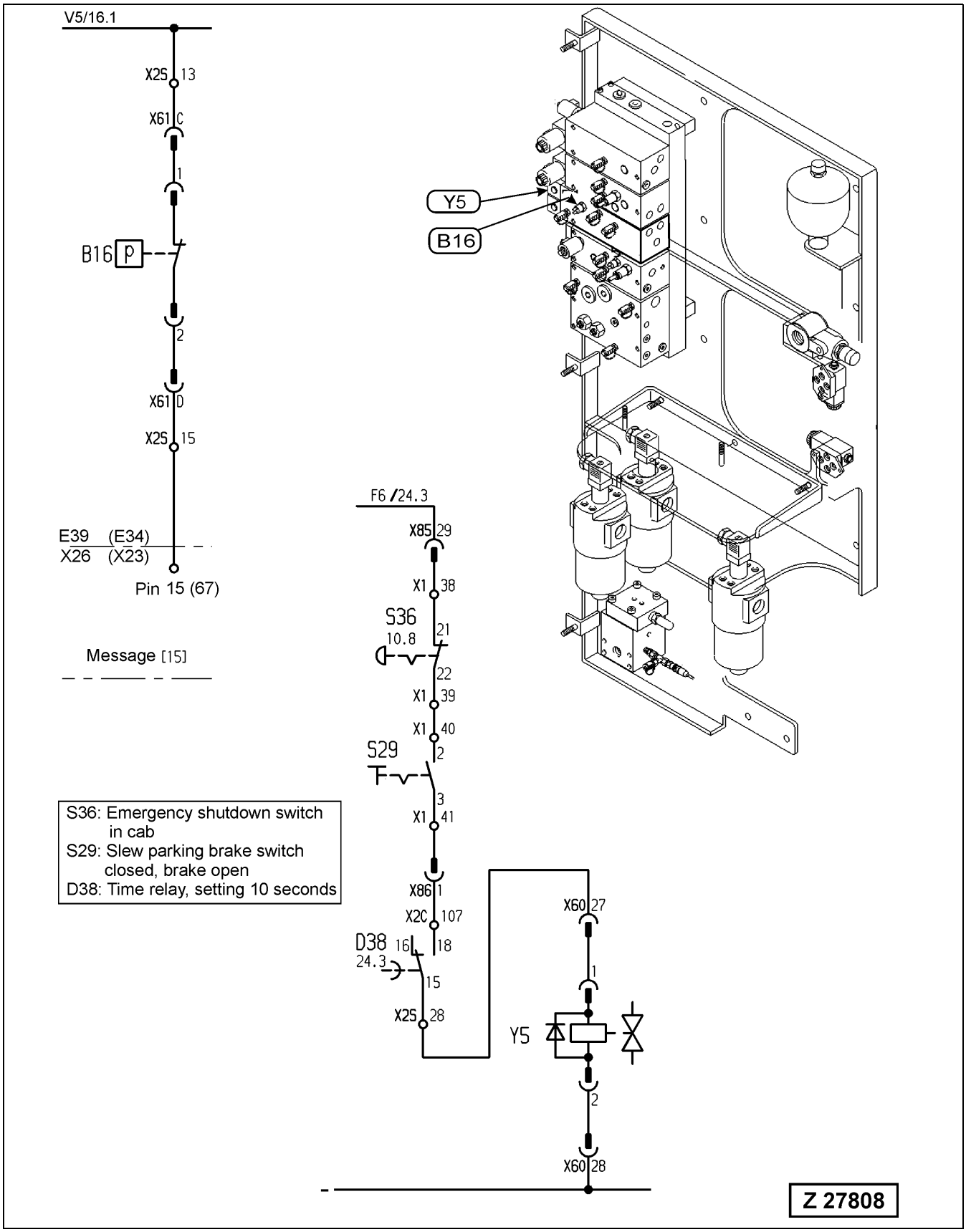
Mechanical or hydraulic problems may be the cause of the trouble

Contents of trouble	• The swing parking brake is inoperative (the superstructure moves) although the swing parking brake switch S29 is switched to position 1 (brake closed). • The X2 pressure circuit is already tested and OK (X2 pressure 35 bar at valve Y5). • The electrical circuit is already tested and OK.
Related information	• The swing parking brake is a multiple disc brake, mounted on the top of each swing gearbox. • The Siebenhaar swing gearbox has an external swing brake housing with a separate oil filling. • The L&S swing gearbox has an integrated swing brake system which is lubricated by the gearbox oil. • The swing parking brake works as a Spring Loaded System. That means the pilot pressure opens the swing parking brake. Without pilot pressure the springs close the swing parking brake.
Problem that appears on machine	• The swing parking brake is inoperative.
Possible causes	• The pilot pressure in the swing parking brake circuit is not released. • The brake discs are burnt and/or the piston seals are defective.

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The pilot pressure in the swing parking brake circuit is not released	7	► Main key switch 1S1in OFF position & maintenance safety switch 1S58 locked.			
		The solenoid valve Y5 is mechanically stuck. The pilot pressure in the swing parking brake line is not released to the hydraulic oil tank. Is the piston of the solenoid valve stuck?	Yes	Replace the solenoid valve Y5.	0 bar
			No	Go to next step.	
The brake discs are burnt and the piston seals are defective	8	► Main key switch 1S1in OFF position & maintenance safety switch 1S58 locked.			
		Perform the following steps: - Disconnect the pilot pressure supply hose for the swing parking brake from the swing gearbox. - Install a manual hydraulic pump with 60 bar pressure gauge in order to test the swing parking brake piston seals. - Use the manual pump and increase the pressure up to 35 bar. - The built-up pressure must be held for 15 minutes. Can you build-up and hold the pressure for 15 minutes?	Yes	Check for leaks or mixed-up connections in the pilot supply lines for the swing parking brake system.	35 bar for 15 minutes
			No	Repair or replace the swing parking brake. Refer to the Service Manual section 8.4 for additinal information. Check the swing parking brake for proper operation.	
Final check	9	Bring the machine back into the original condition. Start the machine and verify that the problem is solved.			

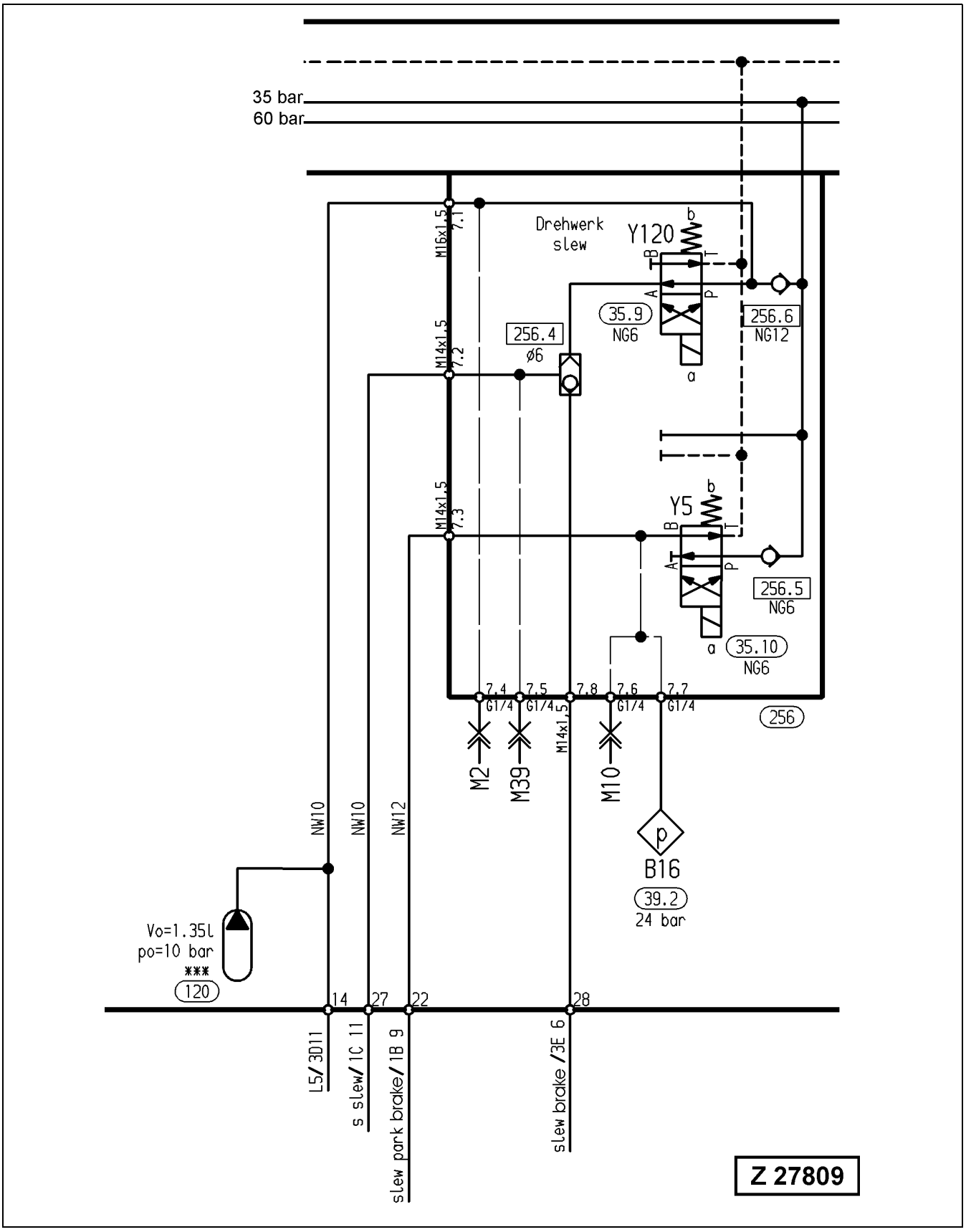
Illustrations related to trouble message -15-

[4/5]



Illustrations related to trouble message -15-

[5/5]



MESSAGE -16- CAUTION, TRAVEL GEAR HOUSE BRAKE ON

[1/5]

Trouble code **Message page 16**	Trouble	Caution, travel gear house brake ON
Contents of trouble	• Pressure switch B48 monitors the pressure of the travel gear house brakes. • The travel parking brakes are locked although the motor is running.	
Related information	• The system operates as a so called spring-loaded parking brake. Hydraulic pressure is NOT present:Spring force locks the brakes • Hydraulic pressure is present: The brakes are released. • A problem with the pressure switch B48 or connected wiring may cause the trouble • Condition: The motor is running • The pressure switch B48 is closed or its wiring is defective. Basic sensor data (B48) No pressure present / 0 V / Switch closed Pressure present / 24 V / Switch open Switch point / 24 bar / • For functional test of the travel parking brakes refer also to the Service Manual, section 5.3 TRAVEL PARKING BRAKE.	
Problem that appears on machine	• Traveling is not possible due to locked travel brakes.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Incorrect or no supply voltage at the pressure switch B48	2	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Measure the supply voltage at connector X61E, terminal 1. Is the correct supply voltage present?	Yes	Go to step 3.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B48 (disconnection or defective contact in any connector)	2	► Keep the main key switch 1S1in the OFF position.			
		Refer to the wiring diagram. Check the wiring between the pressure switch B48 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors) between connector X61E terminal 1 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	

Message -16- Caution, travel gear house brake ON

[2/5]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the signal wiring between pressure switch B48 and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-.	3	► Main key switch 1S1in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E39 (slave) pin 16. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 66. Is the signal present?	Yes	The signal wiring between pressure switch B24 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B25 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Defective ground connection at the pressure switch B48	4	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Disconnect the cable connector X61F and check for proper ground connection at the connector terminal 2. Is the ground connection OK?	Yes	Go to next step.	
			No	Locate the problem and repair as necessary.	
Defective pressure switch B48 (switch does not close)	5	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Disconnect connectors X61E and X61F. Measure the continuity of the pressure switch B48 between the connector pins 1 and 2. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Replace the pressure switch B48 and verify that the repair was successful. Go to final step 10.	
The travel parking brake does not release even the motor is running	6	► Keep the motor running.			
		Check if battery voltage (approx. 24 V) is present at terminal 8 of relay K3. Is battery voltage (approx. 24 V) present?	Yes	Go to next step.	V_{min} 22V V_{max} 24V
			No	Check if supply voltage is present at circuit breaker F21. Check the relay K3 and its wiring. Repair or replace as necessary.	

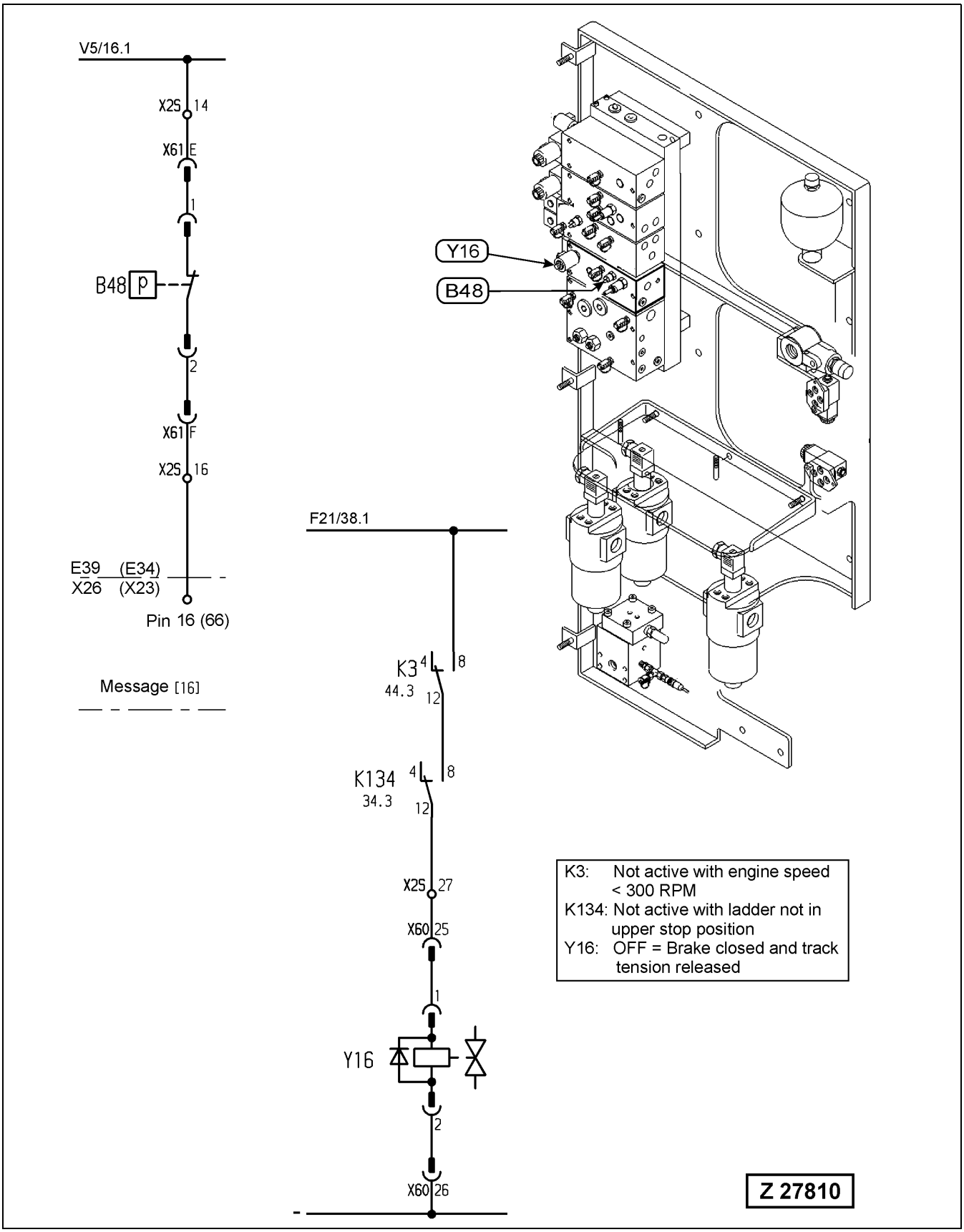
Message -16- Caution, travel gear house brake ON

[3/5]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The wiring of the solenoid valve Y16 is defective	7	► Keep the motor running.			
		Disconnect the connector and insert the T-box. Measure the supply voltage and check for proper ground connection. Are supply voltage and ground connection OK?	Yes	Go to next step.	V_{min} 22V V_{max} 24V
			No	Locate the problem and repair as necessary.	
The solenoid valve Y16 is defective	8	► Keep the main key switch 1S1in OFF position.			
		Using the T-box measure the resistance of Y16. Does the measurement reading meet the specification?	Yes	Go to next step.	R_{Y16} 30 ± 1Ω
			No	Replace Y16 (always replace the complete coil & valve unit) and recheck the travel parking brake for proper function.	
The Y16 piston is mechanically stuck	9	► Keep the main key switch 1S1in OFF position.			
		Replace Y16 (always replace the complete coil & valve unit) and recheck the travel parking brake for proper function.		Go to final step 10.	
Final check	10	Bring the machine back into the original condition. Start the machine and verify that the problem is solved.			

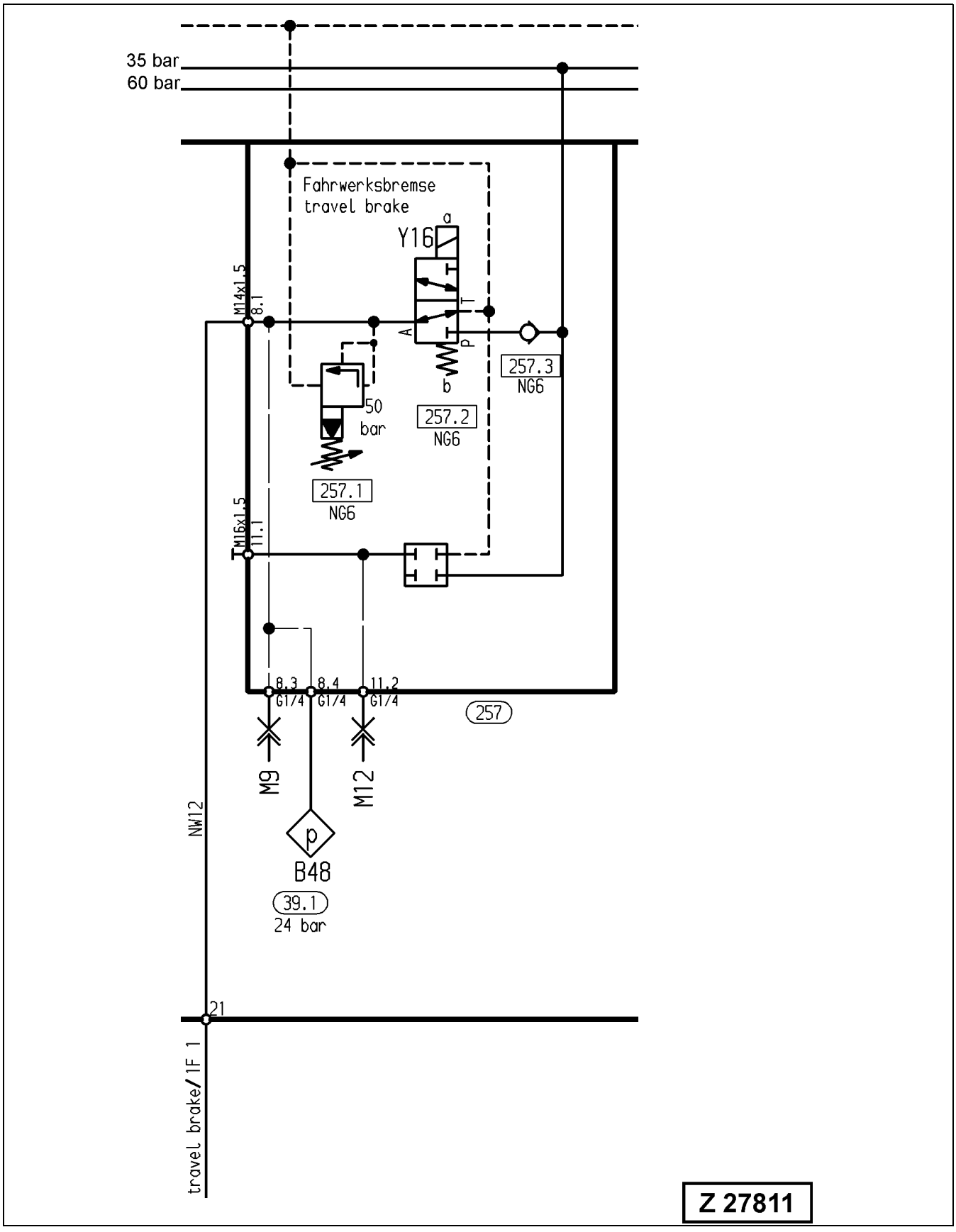
Illustrations related to trouble message -16-

[4/5]



Illustrations related to trouble message -16-

[5/5]



MESSAGE -17- NO CLEARANCE FOR STARTING, WRONG DIRECTION OF MOTOR ROTATION

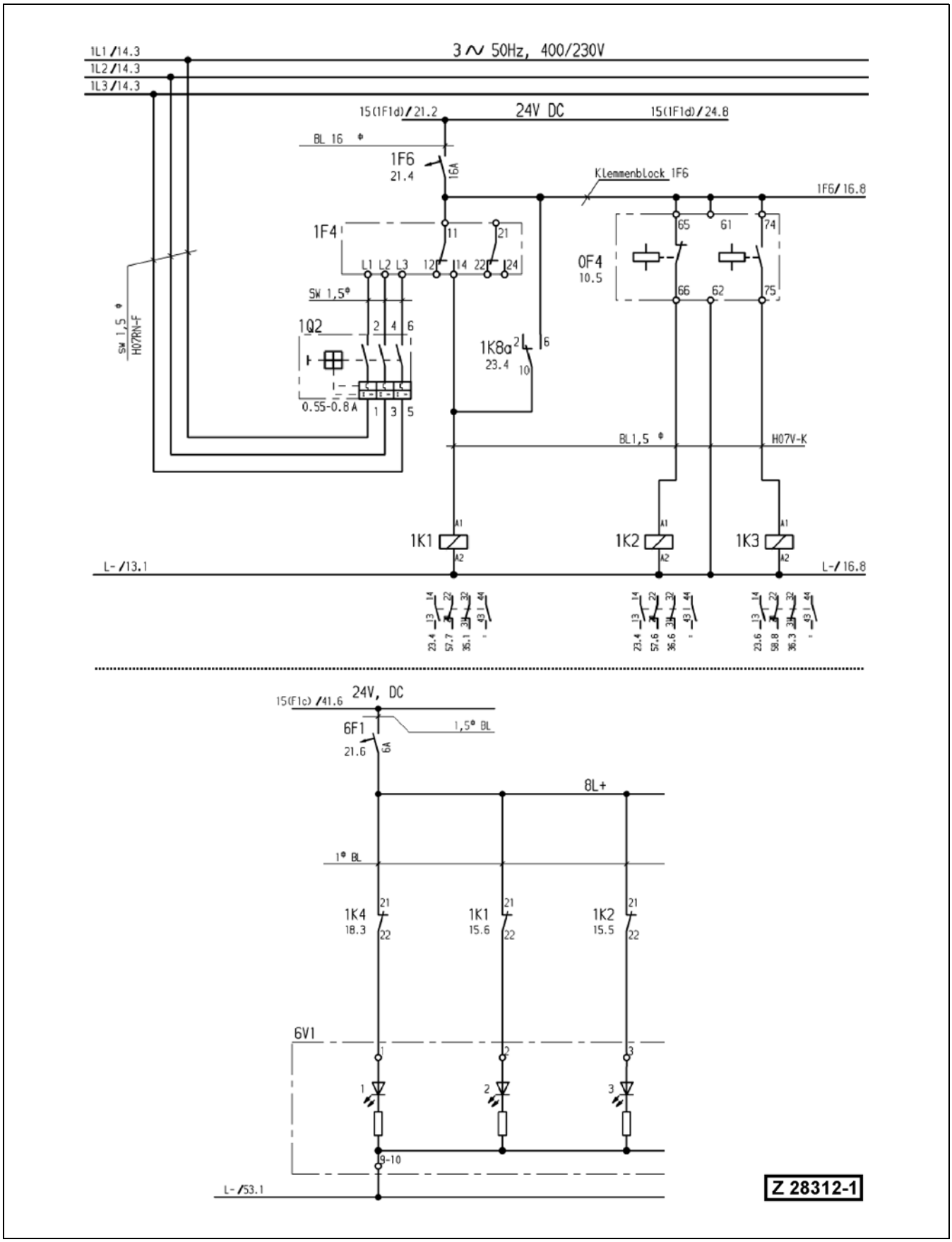
[1/3]

Trouble code **Message page 17**	Trouble	No clearance for starting, wrong direction of motor rotation
Contents of trouble	- The phase sequence monitoring module 1F4 detects wrong connection of L1, L2, L3. - If the connection sequence is wrong, the module triggers relay 1K1 and motor start is blocked.	
Related information	- In case of inverted phases the motor would rotate in wrong direction. - Accidental inversion of the 3 phases in the main supply cable may happen during normal machine relocation, when the cable is reconnected to a terminal box or to a substation. - In case of inverted phases warning light 2 at the LED monitor module 6V1 is ON.	
Problem that appears on machine	Motor start is blocked.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Phases are connected in wrong sequence	1	► Main key switch 1S1 ON.			
		 **Call for the HV department!** The phases L1, L2, L3 need to be checked and connected in correct sequence: - at the substation - or at the terminal box (mine side) Could you rectify the problem?	Yes	Go to final step 4.	
			No	Go to next step.	
Relay 1K1 or its wiring is faulty	2	► Main key switch 1S1 ON.			
		Measure the voltage at the relay between terminals A1 and A2. Are 24V present (relay energized)?	Yes	Go to next step.	
			No	Replace the relay or check and repair the wiring as necessary.	22 - 24 V
The signal from the phase sequence monitoring module 1F4 is missing	3	► Main key switch 1S1 ON.			
		Measure the output voltage at pin 14 (24V = target, 0V = error). Reading = 0V: - Module 1F4 and/or its wiring may be defective. Check the module and its wiring. Repair the wiring as necessary or replace module 1F4.  **Call for the HV department!** - The HV switch module 1Q2 and/or its wiring may be defective. Repair the wiring as necessary or replace module 1Q2.			
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

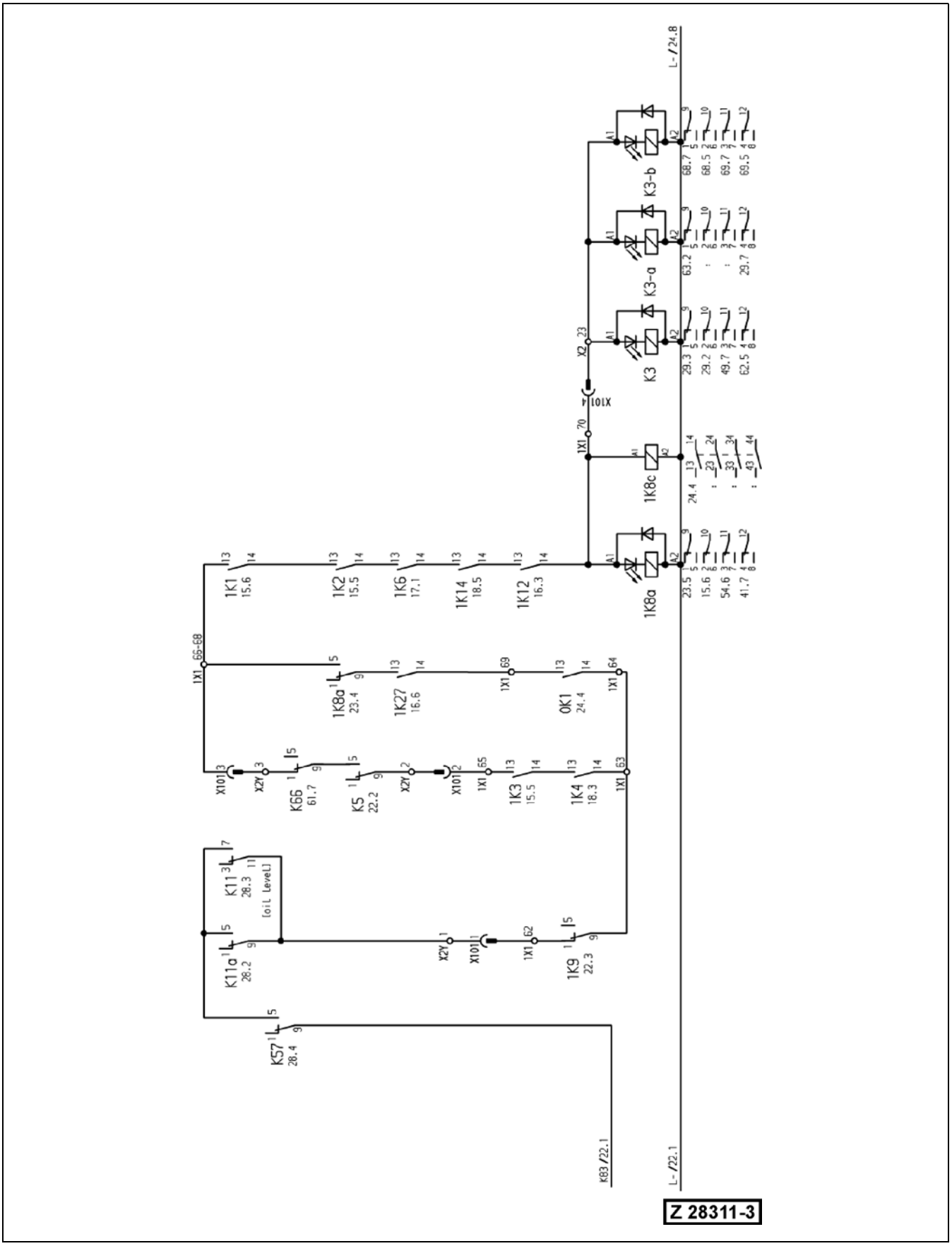
Wiring Diagram 1 related to trouble message -17-

[2/3]



Wiring Diagram 2 (1K1 in the motor start/stop and interlocking circuit) related to trouble message -17-

[3/3]



MESSAGE -18- CENTRAL LUBRICATION SYSTEM FAULT

[1/7]

Trouble code **Message page 18**	Trouble	Central lubrication system fault
Contents of trouble	• The lubrication cycle time is too long. • No power supply. Interrupted wiring connections.	
Related information	• The lubrication cycle is not carried out in the pre-set time frame. • Check to make sure that the grease level in the barrel is within the normal range, use the dipstick provided on the grease barrel cover. It is assumed, that the message 59 -REFILL- has not been displayed. • For further details concerning the lubrication system refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM. • The illustration Z27802 on page [6/7] of this chart shows the interacting parts of the control circuit. Use this illustration for testing procedure of the signal wiring and connections. • The illustrations on page [4/7 - 5/7] show the location of the related parts on the machine.	
Problem that appears on machine	• Risk of damages due to insufficient lubrication. • If the machine is operated further on, after fault message 18 is set, the bucket function will be disabled after 4.2 hours.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Automatic control circuit disturbed.	1	► Keep the motor running.			
		Manually activate the central lubrication system with switch S24 on the dashboard. Does the system work?	Yes	Go to next step.	
			No	Go to step 8.	
Malfunction of interacting relays in the wiring loop.	2	► motor OFF and main key switch 1S1OFF and then in ON position.			
		Check the relevant relays in the wiring loop for proper functioning. Refer to the illustration Z27802 on page [6/7] for typical configuration of the controlling relays. Use the applying electrical diagram and the Service Manual of your machine for the checking procedure. Are the relays in proper working order?	Yes	Go to next step.	
			No	Replace defective relays. Verify that trouble message 18 -Central lubrication system fault- is not set again. Go to final step 11.	
The electrical function of the grease pressure release valve Y7a is disturbed	3	Check the electrical function of the grease pressure release solenoid valve Y7a. Is the electrical function OK?	Yes	Go to next step.	
			No	Replace the grease pressure release valve Y7a.	
The grease may be soiled	4	Refill the grease barrel through a filter. Is grease pressure built up with clean grease within specification?	Yes	The problem is solved, perform the final check in step 11.	
			No	Go to next step.	
The mechanical function of the grease pressure release valve Y7a is disturbed	5	Remove the mechanical part from the electrical part of the grease pressure release valve Y7a. Disassemble and check the lower part and clean as necessary. Is the mechanical part of the pressure release valve blocked?	Yes	Go to next step.	
			No	Go to step 7.	
The mechanical part of the grease pressure release valve Y7a is defective	6	Repair or replace the mechanical part of the grease pressure release valve Y7a as necessary. Clean the respective contaminated grease system. Replace the grease filter and the screen.		Go to next step.	

Message -18- Central lubrication system fault

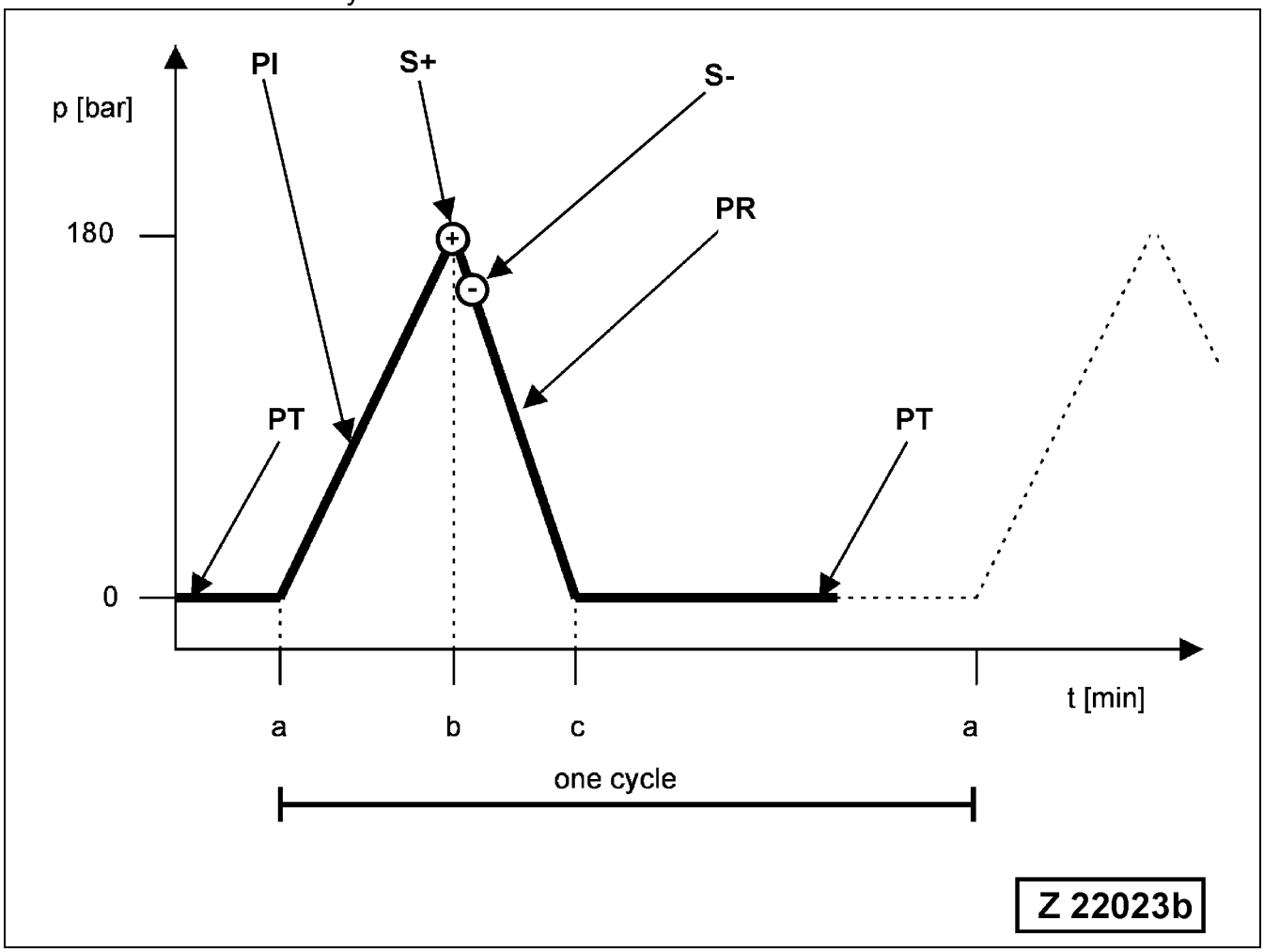
[2/7]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Grease pump: malfunction, worn or defective	7	► Keep the motor running.			
		Check the performance of the relating grease pump and arrange preparations for testing the grease pressure. For detailed test & adjustment procedures refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM. Make sure that the grease pressure release valve Y7a is closed. Does the grease pump build up grease pressure as specified?	Yes	Go to next step.	Q_{volume} 18-20 double-strokes/ min $P_{pump\ drive\ hydr.pressure}$ 45 ± 2 bar Grease max. secondary pressure 295 bar
			No	Grease pump defective, replace the grease pump.	
End-line pressure switch B43 (grease main line): wrong adjustment or defective	8	Refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM, and follow the adjustment procedure. Did you need to correct any wrong adjustment?	Yes	Go to next step.	
			No	Replace the End-line pressure switch B43. Verify that trouble message 18 -Central lubrication system fault- is not set again. Go to final step 11.	
No lubrication during the last 4.2 working hours. Bucket movement disabled due to malfunction of relay K50.	9	► Keep the main key switch 1S1in the ON position.			
		Check the function of relay K50 in the X2 switch board. - K50 active (24 V): Bucket movement enabled, LED ON. - K50 inactive (0 V): Bucket movement disabled Does the function of relay K50 meet the above specification?	Yes	Go to next step.	
			No	Replace relay K50 and check for proper system function. If the problem is not yet rectified go to next step.	
Malfunction or damage in the wiring circuit of relay K50	10	Refer to the wiring diagram on page [7/7]: - Check if B+ voltage is present - Check for proper ground connection - Check the wiring for continuity Is any problem diagnosable in the wiring circuit of relay K50?	Yes	Repair the wiring or replace any defective electrical component as necessary. Go to final step 11.	
			No	Check if any other fault message is set which may be in cross-correlation with the existing problem. If so, follow the applying trouble-shooting chart and after that recheck for proper system function.	
Final check	11	Bring the machine back into the original condition. Start the machine and verify that the fault message does not arise again. Verify that the operation cycles of the lubrication system correspond to the diagram shown on next page.			

Message -18- Central lubrication system fault

[3/7]

Function of a lubrication cycle



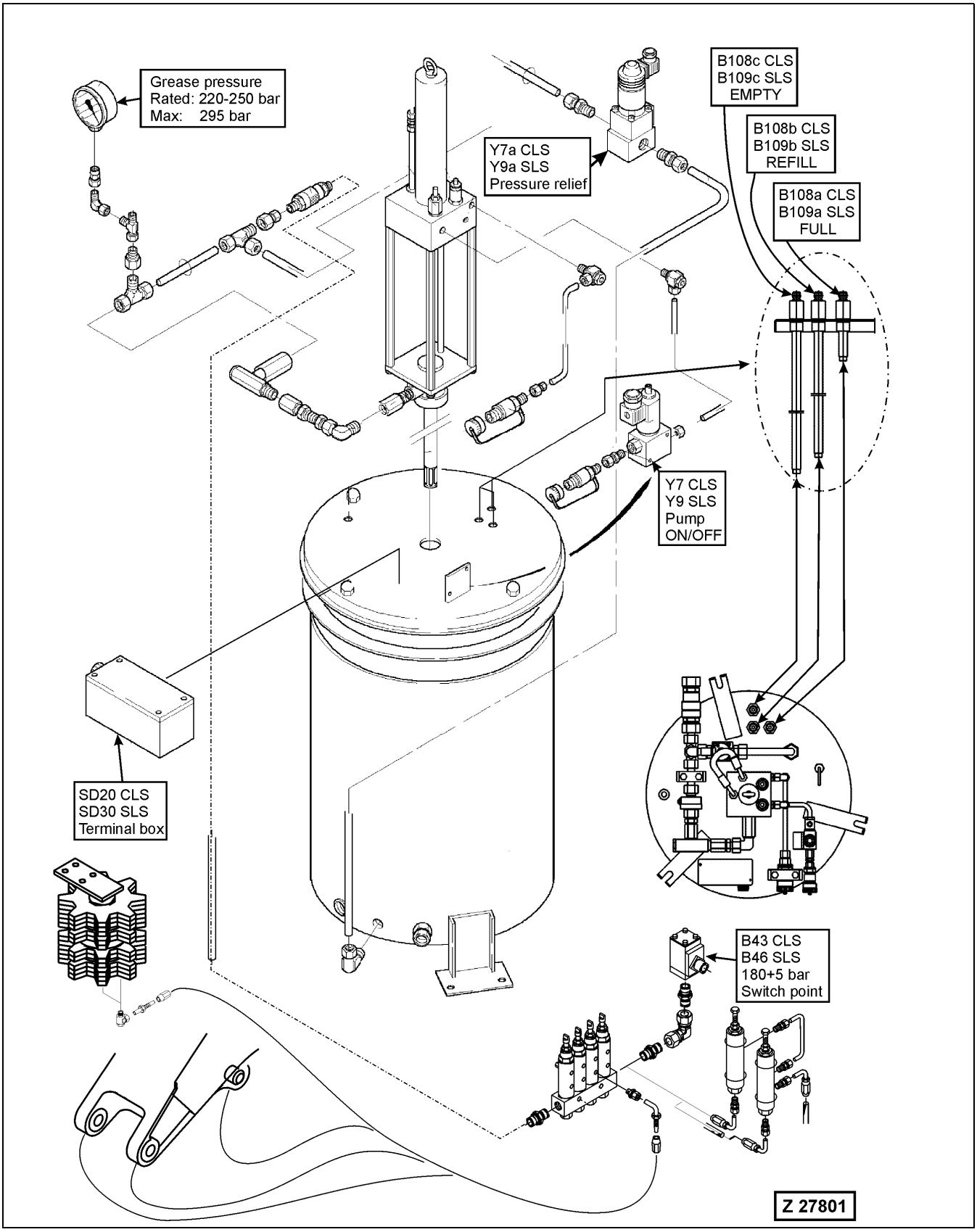
L

PT	**P**ause **T**ime
PI	**P**ressure **I**ncreasing (grease supply line)
S+	**S**witch point **ON** of the end of line pressure switch
S-	**S**witch point **OFF** of the end of line pressure switch
PR	**P**ressure **R**elieve

System circuit	Solenoid valve, hydraulic oil supply	Solenoid valve, grease pressure release	End line pressure switch
CLS	Y7	Y7a	B43
SLS	Y9	Y9a	B46

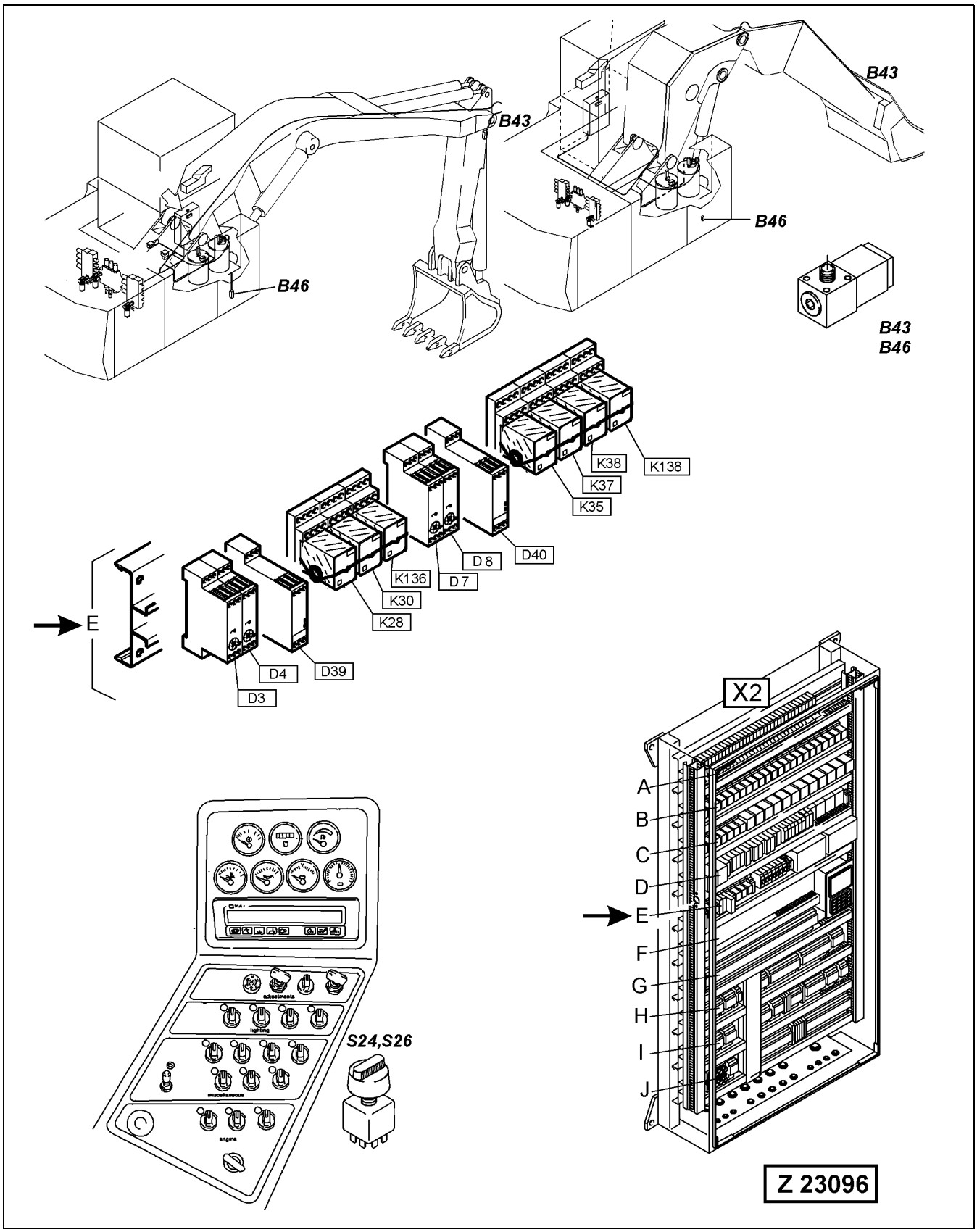
Message -18- Central lubrication system fault

View onto the grease barrel
[4/7]



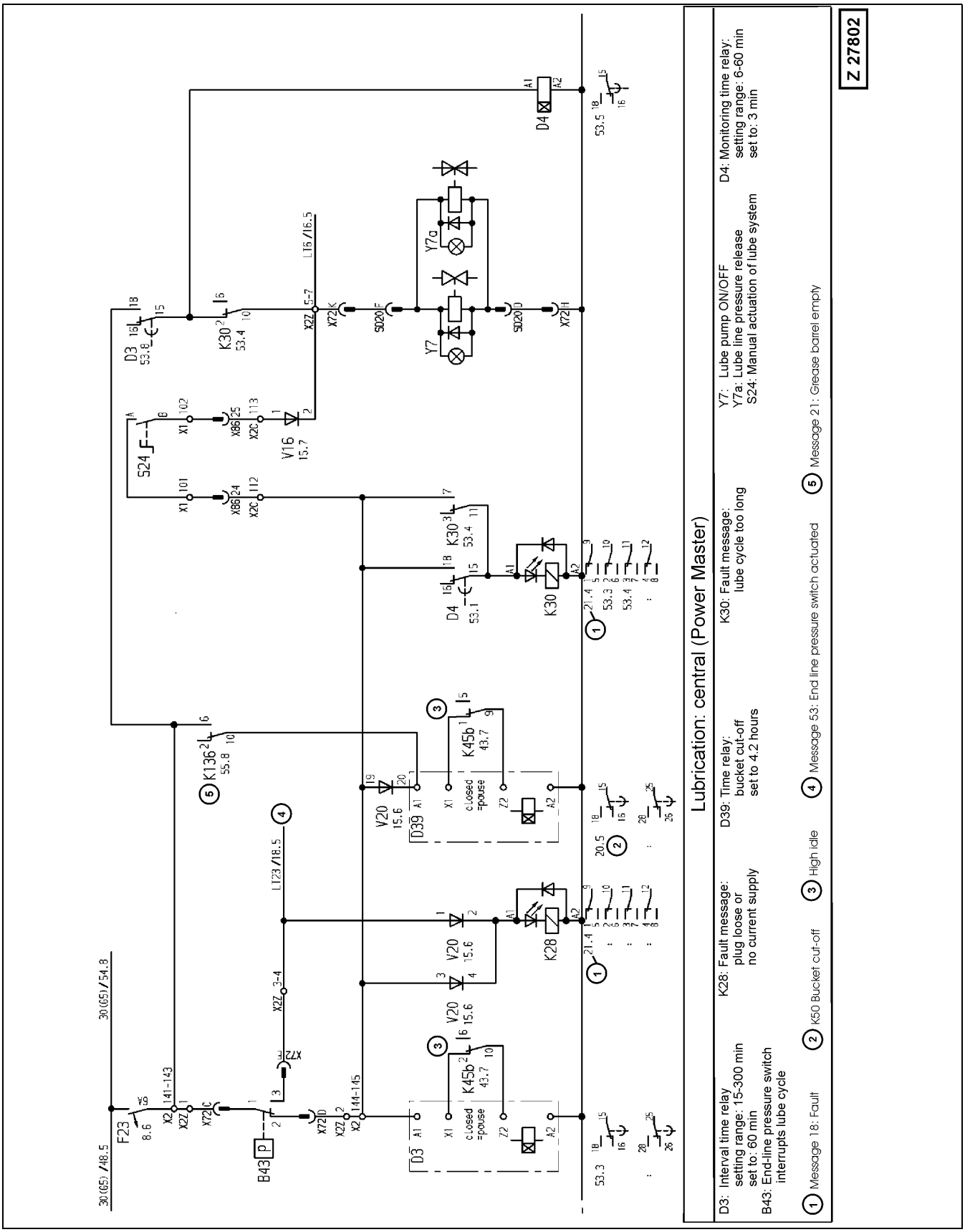
Message -18- Central lubrication system fault

View onto the X2 Box
[5/7]



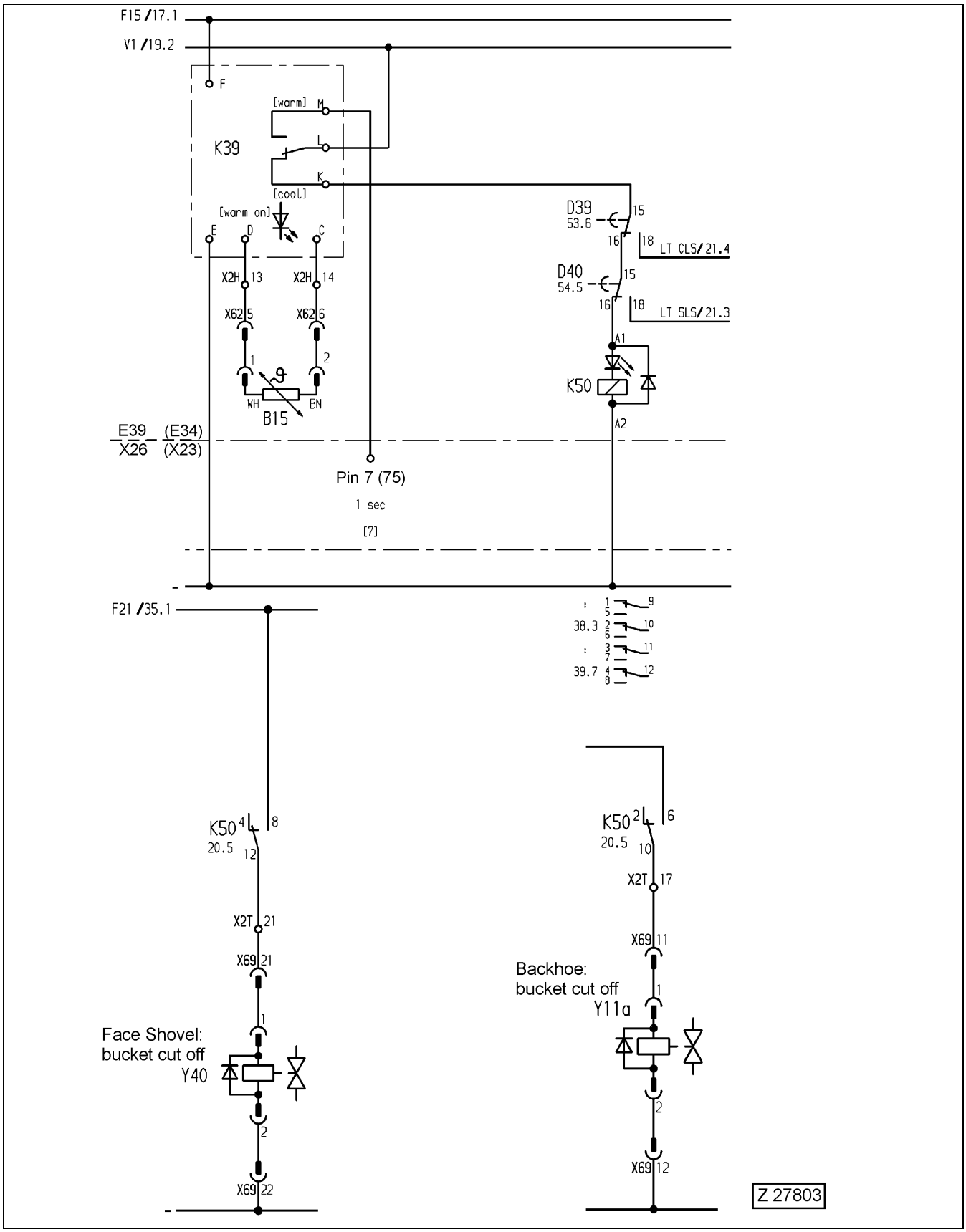
Wiring Diagram related to trouble message -18-

[6/7]



Wiring Diagram related to trouble message -18-

[7/7]



MESSAGE -19- SLEW RING GEAR LUBRICATION SYSTEM FAULT

[1/8]

Trouble code **Message page 19**	Trouble	Slew ring gear lubrication system fault
Contents of trouble	• The lubrication cycle time is too long. • No power supply. Interrupted wiring connections.	
Related information	• The lubrication cycle is not carried out in the pre-set time frame. • Check to make sure that the grease level in the barrel is within the normal range, use the dipstick provided on the grease barrel cover. It is assumed, that the message 60 -REFILL- has not been displayed. • For further details concerning the lubrication system refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM. • The illustrations on page [7/8 and 8/8] of this chart show the interacting parts of the control circuit. Use these illustrations for testing procedure of the signal wiring and connections. • The illustrations on page [5/8 and 6/8] show the location of the related parts on the machine.	
Problem that appears on machine	• Risk of damages due to insufficient lubrication. • If the machine is operated further on, after fault message 19 is set, the bucket function will be disabled after 4.2 hours.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Automatic control circuit disturbed.	1	► Keep the motor running.			
		Manually activate the swing ring gear lubrication system with switch S26 on the dashboard while swinging the superstructure. Does the system work?	Yes	Go to next step.	
			No	Go to step 8.	
Malfunction of interacting relays in the wiring loop.	2	► Motor Stopped and main key switch 1S1OFF and then in ON position.			
		Check the relevant relays in the wiring loop for proper functioning. Refer to the illustration Z27804 on page [7/8] for typical configuration of the controlling relays. Use the applying electrical diagram and the Service Manual of your machine for the checking procedure. Are the relays in proper working order?	Yes	Go to next step.	
			No	Replace defective relays. Verify that trouble message 19 -Slew ring gear lubrication system fault- is not set again. Go to final step 14.	
The electrical function of the grease pressure release valve Y9a is disturbed	3	Check the electrical function of the grease pressure release solenoid valve Y9a. Is the electrical function OK?	Yes	Go to next step.	
			No	Replace the grease pressure release valve Y9a.	
The grease may be soiled	4	Refill the grease barrel through a filter. Is grease pressure built up with clean grease within specification?	Yes	The problem is solved, perform the final check in step 14.	
			No	Go to next step.	
The mechanical function of the grease pressure release valve Y9a is disturbed	5	Remove the mechanical part from the electrical part of the grease pressure release valve Y9a. Disassemble and check the lower part and clean as necessary. Is the mechanical part of the pressure release valve blocked?	Yes	Go to next step.	
			No	Go to step 7.	

Message -19- Slew ring lubrication system fault

[2/8]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The mechanical part of the grease pressure release valve Y9a is defective	6	Repair or replace the mechanical part of the grease pressure release valve Y9a as necessary. Clean the respective contaminated grease system. Replace the grease filter and the screen.		Go to next step.	
Grease pump: malfunction, worn or defective	7	► Keep the motor running.			
		Check the performance of the relating grease pump and arrange preparations for testing the grease pressure. For detailed test & adjustment procedures refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM. Make sure that the grease pressure release valve Y9a is closed. Does the grease pump build up grease pressure as specified?	Yes	Go to next step.	Q_{volume} 18-20 double-strokes/ min $P_{pump\ drive\ hydr.pressure}$ 45 ± 2 bar Grease max. secondary pressure 295 bar
			No	Grease pump defective, replace the grease pump.	
End-line pressure switch B46 (grease main line): wrong adjustment or defective	8	Refer to the SERVICE MANUAL, section AUTOMATIC LUBRICATION SYSTEM, and follow the adjustment procedure. Did you need to correct any wrong adjustment?	Yes	Go to next step.	
			No	Replace the End-line pressure switch B46. Verify that trouble message 19 -Slew ring gear lubrication system fault- is not set again. Go to final step 14.	
Pressure switch B47 for motion detection swinging defective	9	Check Pressure switch B47 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	
Incorrect or no supply voltage at the pressure switch B47	10	► Start motor.			
		Using the T-box, measure the supply voltage for B47 at the connector terminal X69.1. Is the correct supply voltage present?	Yes	Go to step 12.	V_{min} 22V V_{max} 24V
			No	Go to next step.	

Message -19- Slew ring lubrication system fault

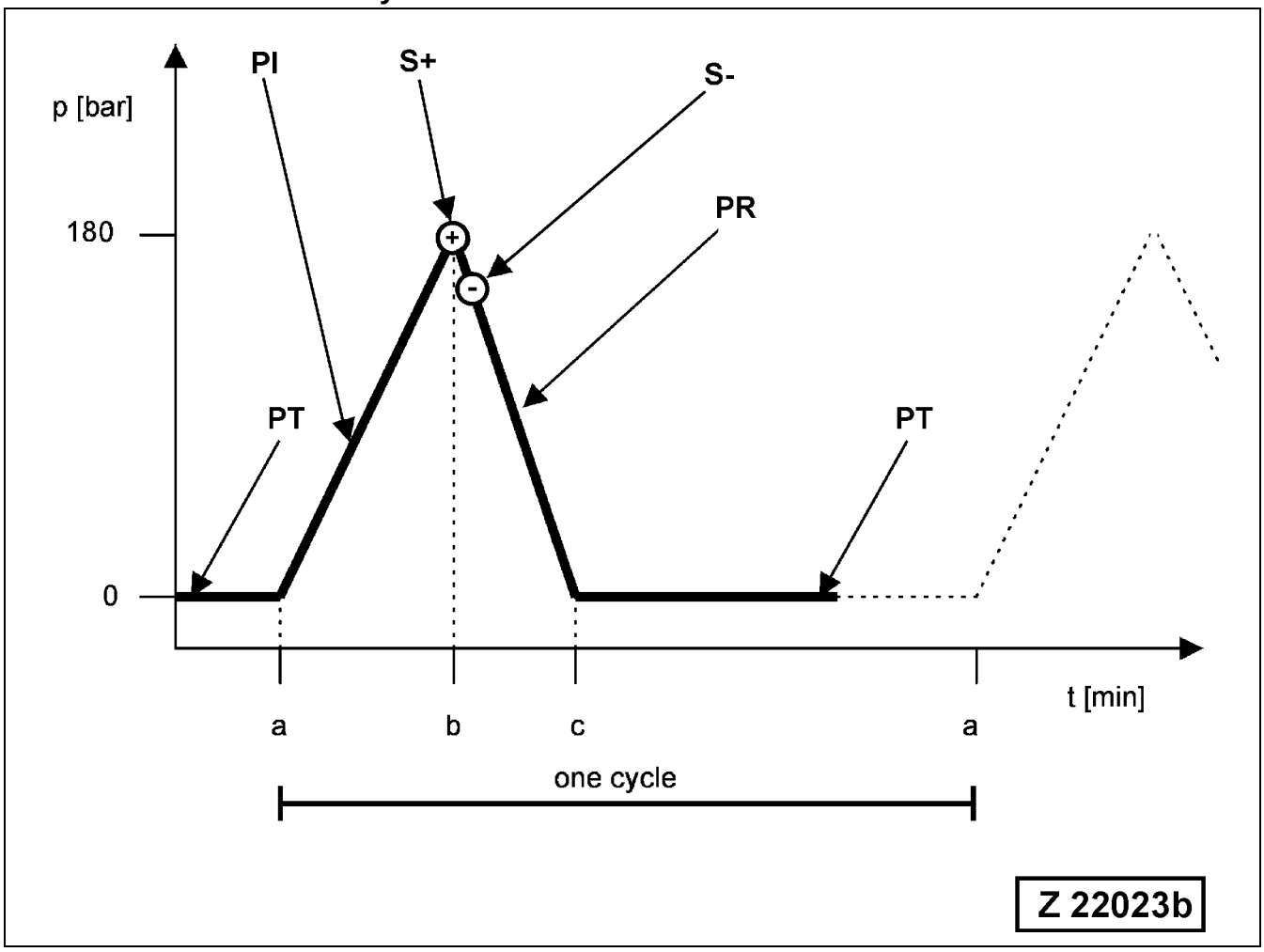
[3/8]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the voltage supply wiring between circuit breaker F21 and pressure switch B47. (disconnection or defective contact in any connector)	11	► Motor STOPPED and main key switch 1S1in OFF position.			
		Refer to the wiring diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B47 and circuit breaker F21 for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection, short circuit, or fault in the wiring between connector X69/1 and circuit breaker F21. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
No lubrication during the last 4.2 working hours. Bucket movement disabled due to malfunction of relay K50.	12	► Keep the main key switch 1S1in the ON position.			
		Check the function of relay K50 in the X2 switch box. - K50 active (24 V): Bucket movement enabled, LED ON. - K50 inactive (0 V): Bucket movement disabled Does the function of relay K50 meet the above specification?	Yes	Go to next step.	
			No	Replace relay K50 and check for proper system function. If the problem is not yet rectified go to next step.	
Malfunction or damage in the wiring circuit of relay K50	13	Refer to the wiring diagram on page [8/8]: - Check if B+ voltage is present - Check for proper ground connection - Check the wiring for continuity Is any problem diagnosable in the wiring circuit of relay K50?	Yes	Repair the wiring or replace any defective electrical component as necessary. Go to final step 14.	
			No	Check if any other fault message is set which may be in cross-correlation with the existing problem. If so, follow the applying trouble-shooting chart and after that recheck for proper system function.	
Final check	14	Bring the machine back into the original condition. Start the machine and verify that the fault message does not arise again. Verify that the operation cycles of the lubrication system correspond to the diagram shown on next page.			

Message -19- Slew ring lubrication system fault

[4/8]

Function of a lubrication cycle



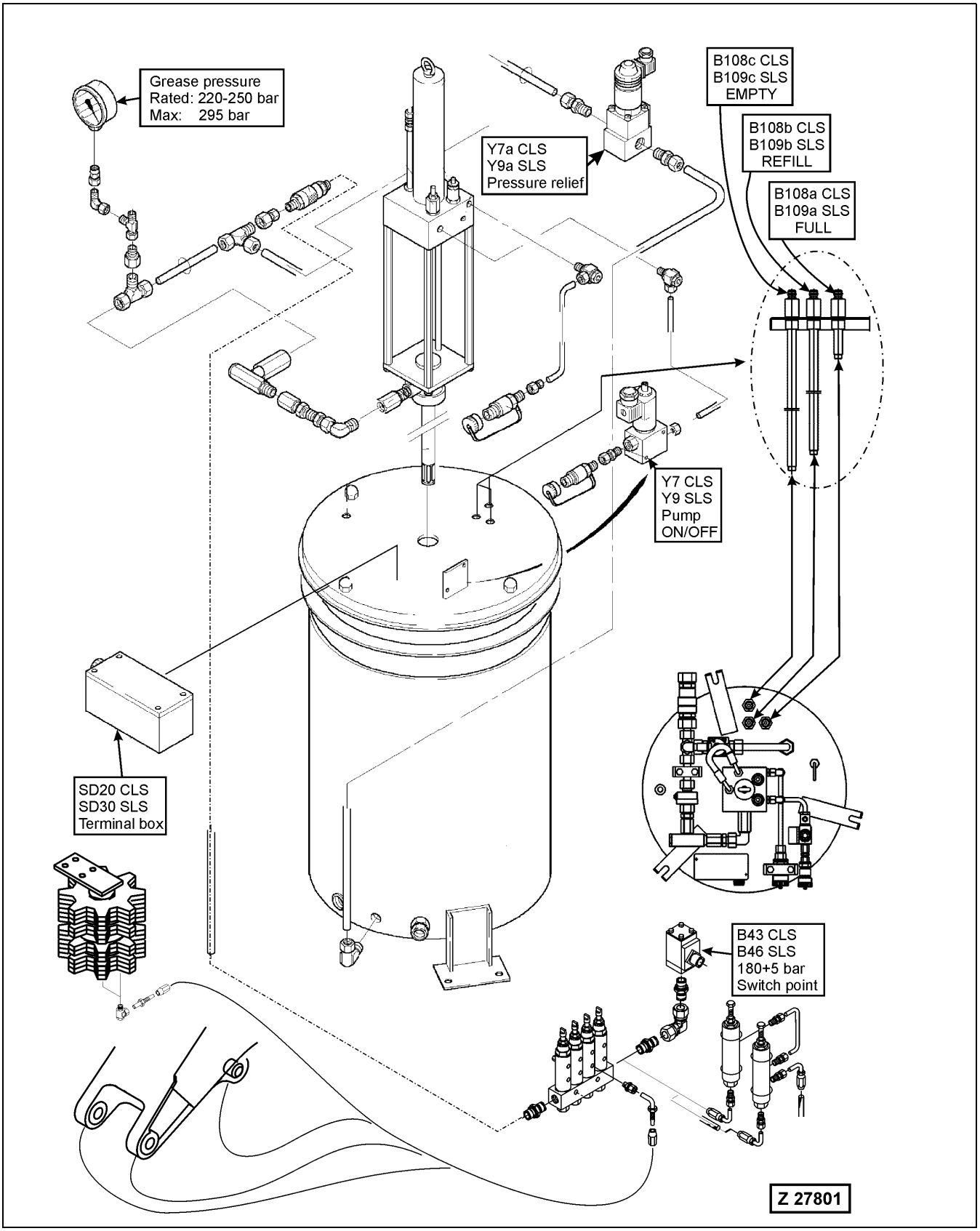
L

PT	**P**ause **T**ime
PI	**P**ressure **I**ncreasing (grease supply line)
S+	**S**witch point **ON** of the end of line pressure switch
S-	**S**witch point **OFF** of the end of line pressure switch
PR	**P**ressure **R**elieve

System circuit	Solenoid valve, hydraulic oil supply	Solenoid valve, grease pressure release	End line pressure switch
CLS	Y7	Y7a	B43
SLS	Y9	Y9a	B46

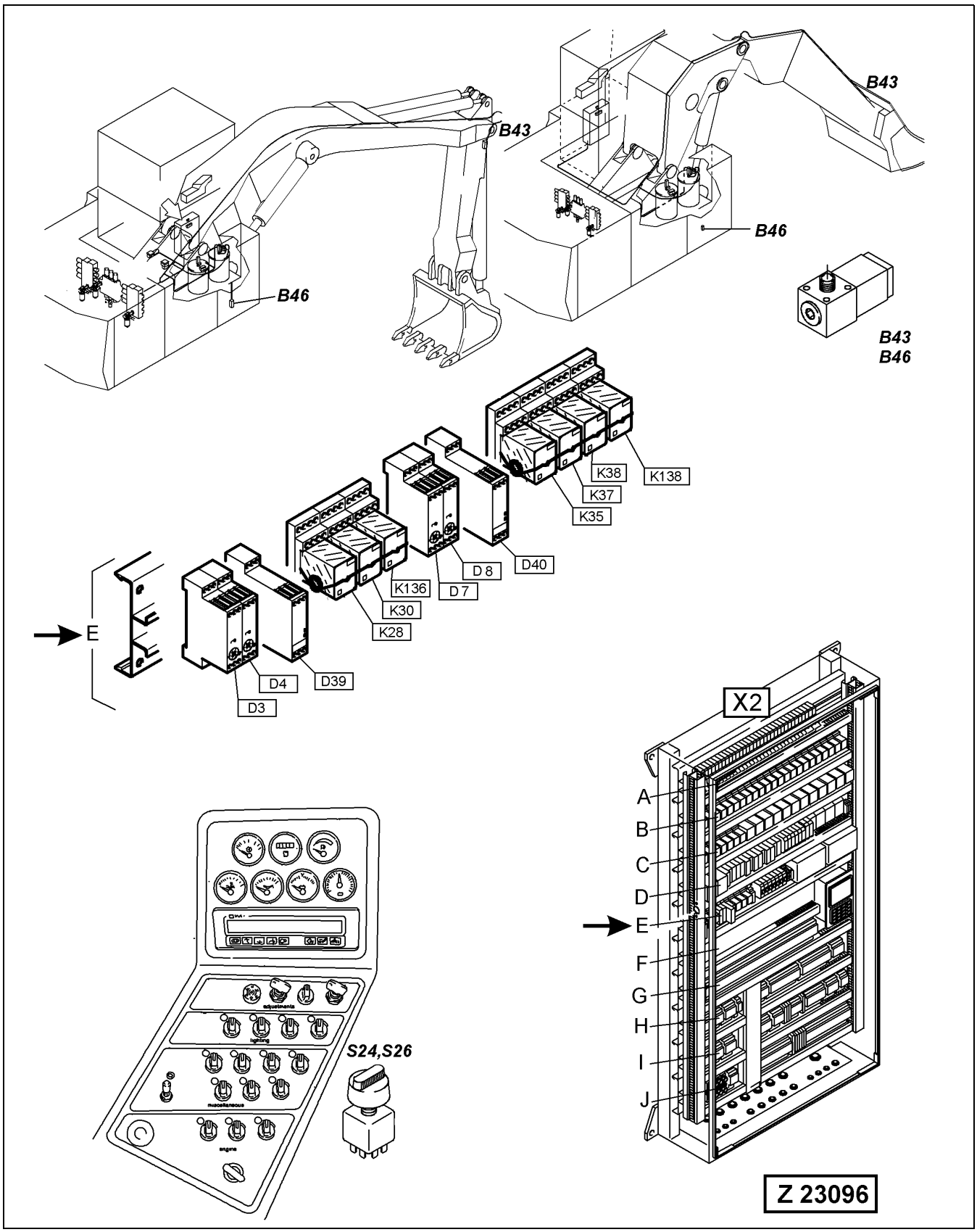
Message -19- Slew ring lubrication system fault

View onto the grease barrel
[5/8]



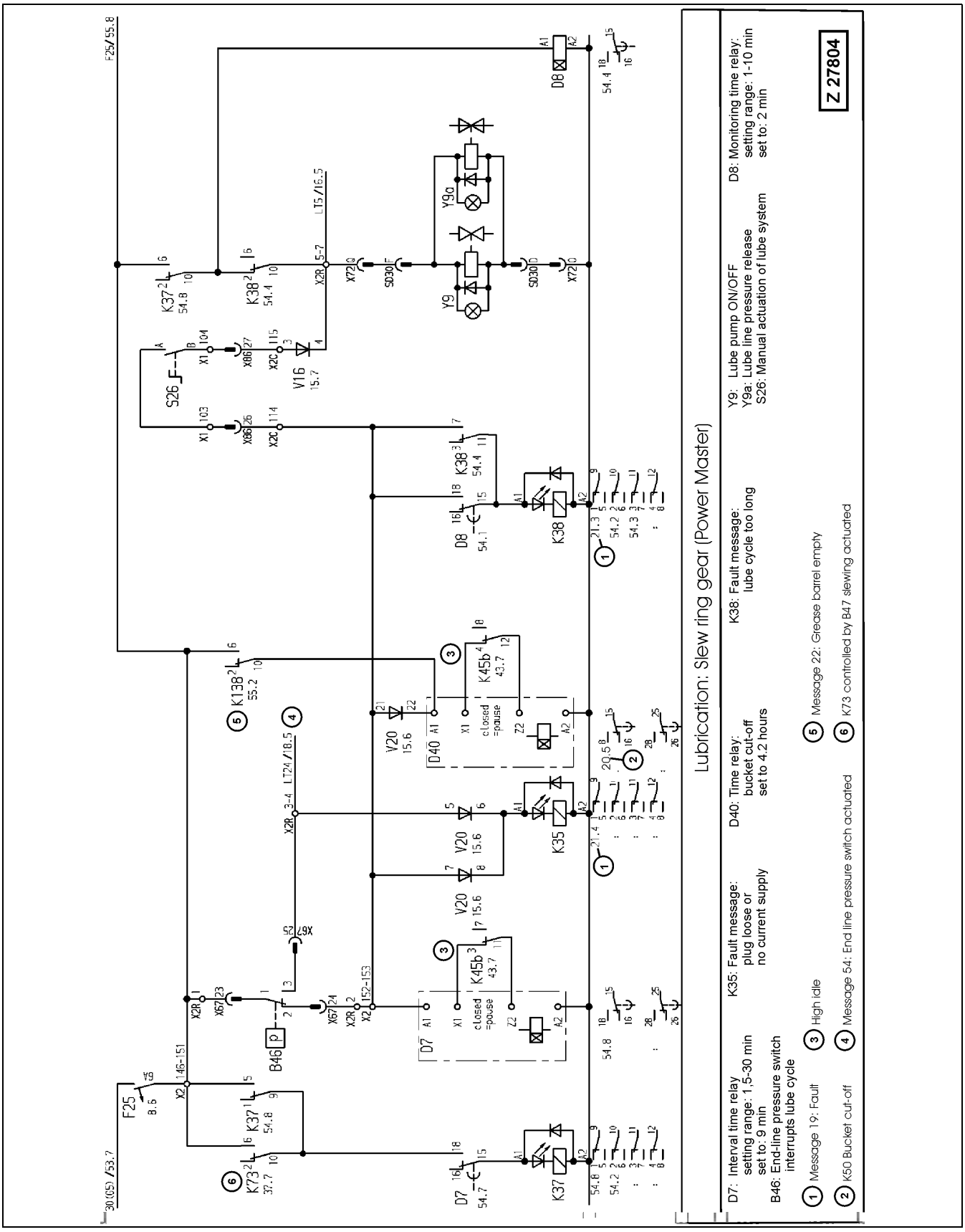
Message -19- Slew ring lubrication system fault

View onto the X2 Box
[6/8]



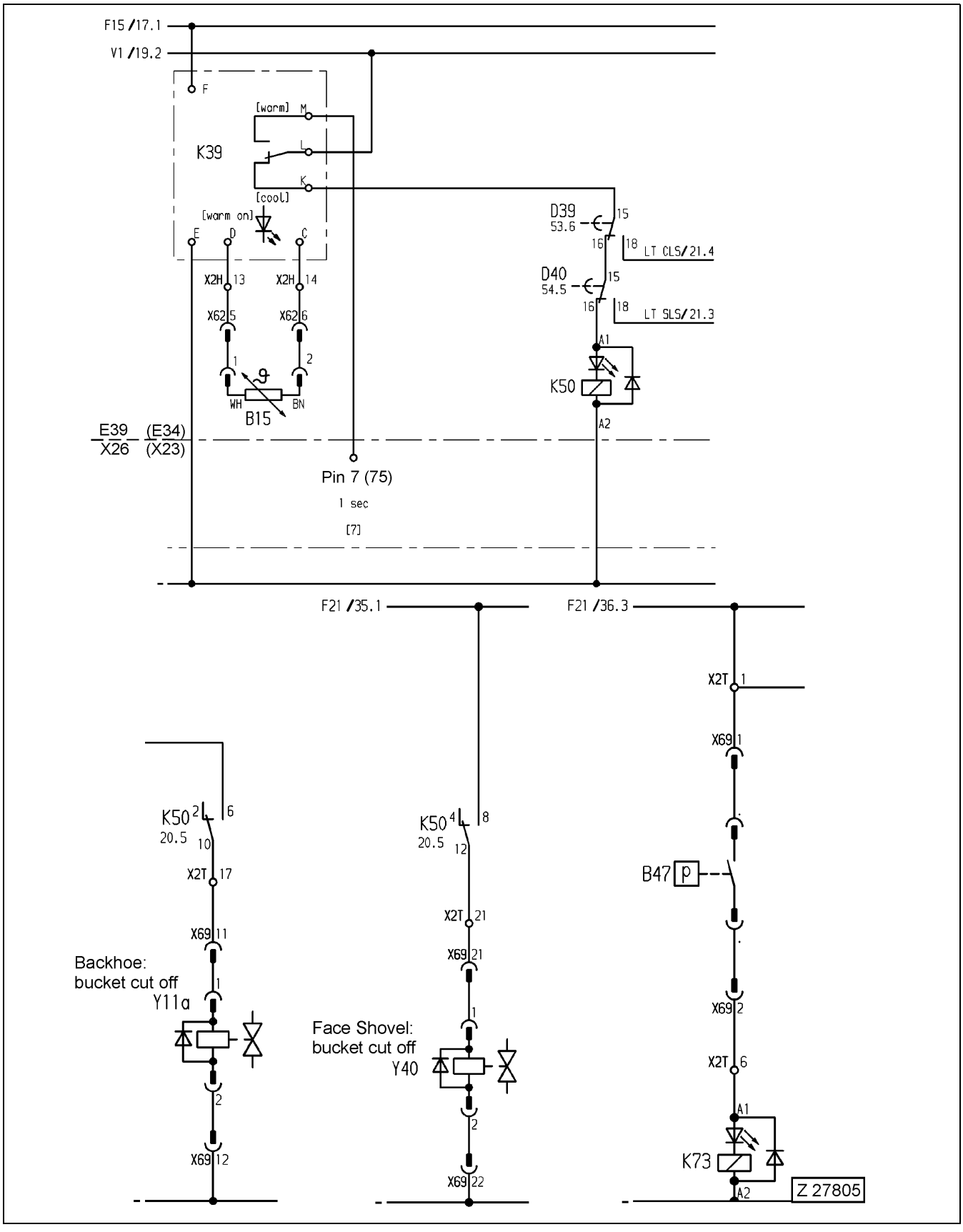
Wiring Diagram related to trouble message -19-

[7/8]



Wiring Diagram related to trouble message -19-

[8/8]



MESSAGE -20- MOTOR SHUTDOWN: MOTOR PROTECTION RELAY RELEASED (1K2)

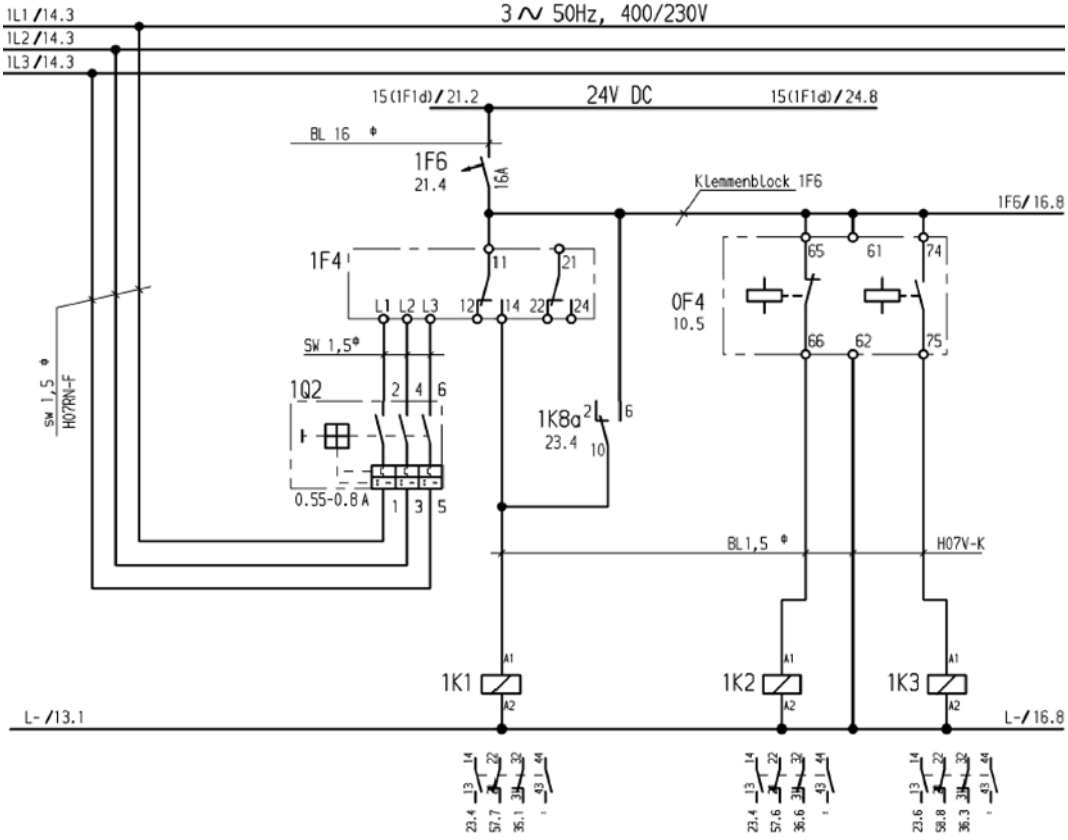
[1/4]

Trouble code **Message page 20**	Trouble	Motor shutdown: Motor protection relay released (1K2)
Contents of trouble	- The motor re-start protection relay 0F4 is triggered because overcurrent is detected, and motor is shutdown.	
Related information	- If overcurrent is detected during operation condition, contacts 65-66 in 0F4 open and motor is shutdown. - In case of overcurrent warning light 3 at the LED monitor module 6V1 is ON.	
Problem that appears on machine	Motor shutdown. Motor restart is possible if the current is as specified again.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor re-start protection relay 0F4 is triggered because overcurrent is detected	1	 **HAZARDOUS VOLTAGE!** **Access to the main transformer is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** Possible reasons for overcurrent are: 1) Netline variation (overvoltage) Inform the HV department. 2) 0F4 settings are wrong For the setting procedure refer to the SERVICE MANUAL. For the adjustment data refer to your Wiring Diagram 3) E-motor faulty Contact the Siemens service.		For troubleshooting on relay 1K2 go to next step.	
Relay 1K2 or its wiring is faulty	2	► Main key switch 1S1 ON.			
		Measure the voltage at the relay between terminals A1 and A2. Are 24V present?	Yes	Replace the defective relay.	22 - 24 V
			No	Check and repair the wiring as necessary.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

Wiring Diagram 1 (0F4 & settings and relay 1K2) related to trouble message -20-

[2/4]



I_θ = 0,95 I	I_s = 2,39 I_n	ΔI = 26 $\%I_L$
t_{6x} =10 s	t_s = 52,9 s	t_Δ = 70 s
p = 50 %	$I_{>>}$ = 6 I_n	$I_<$ = off $\%I_n$
Θ_a = 90 %	$t_{>>}$ = 0.05 s	$t_<$ = 600 s
Θ_1 = 70 %	$I_{0>}$ = 100 $\%I_n$	Σt_{si} = 13 s
k_c = 7	t_0 = 30 s	$\Delta\Sigma t_{si}$ = 108 s/h

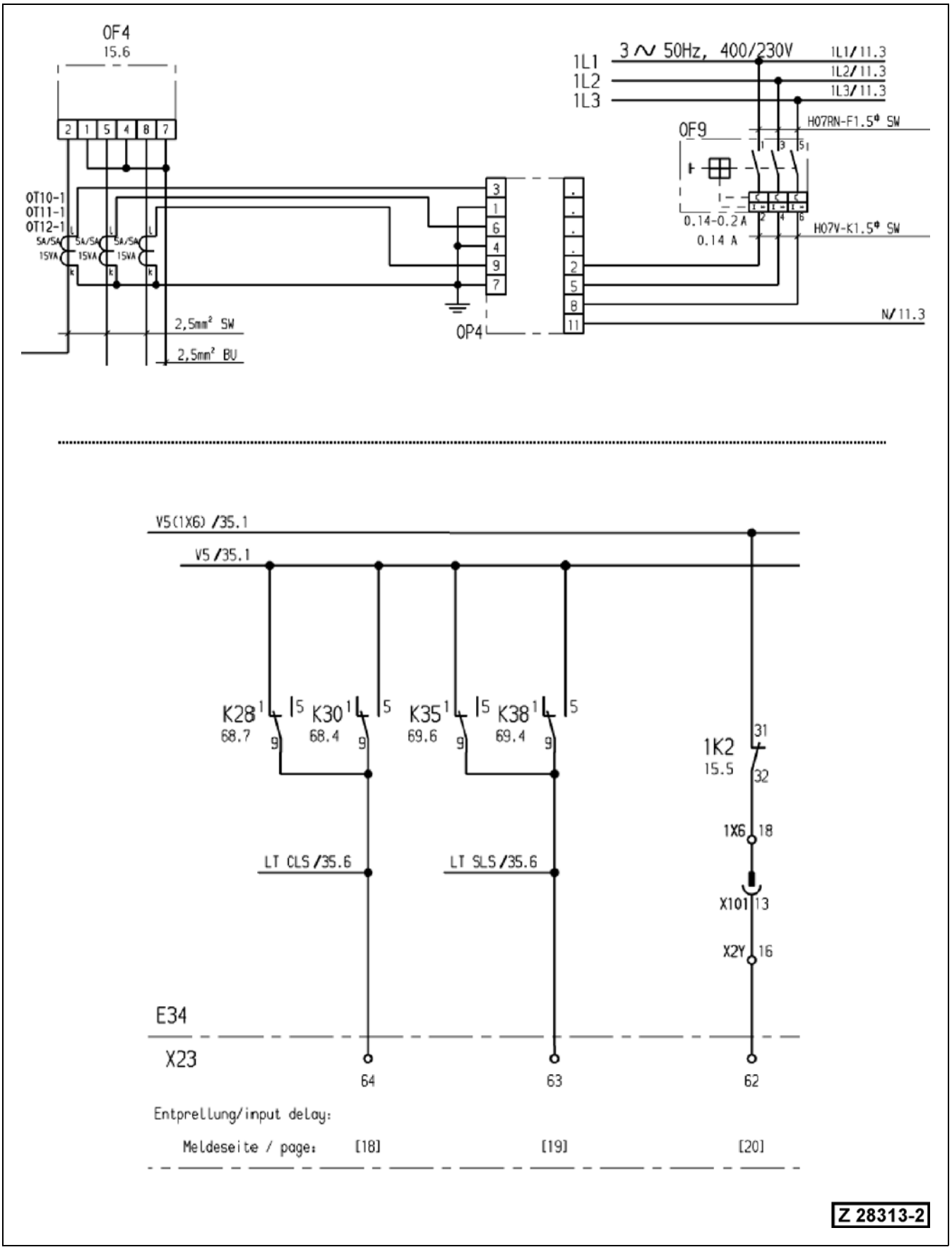
SGF		SGB		SGR1		SGR2	
1 [0] x 1	=	1 [0] x 1	=	1 [1] x 1	= 1	1 [0] x 1	=
2 [0] x 2	=	2 [0] x 2	=	2 [0] x 2	=	2 [0] x 2	=
3 [0] x 4	=	3 [0] x 4	=	3 [0] x 4	=	3 [0] x 4	=
4 [0] x 8	=	4 [0] x 8	=	4 [0] x 8	=	4 [0] x 8	=
5 [1] x 16	= 16	5 [0] x 16	=	5 [0] x 16	=	5 [0] x 16	=
6 [1] x 32	= 32	6 [0] x 32	=	6 [0] x 32	=	6 [0] x 32	=
7 [1] x 64	= 64	7 [0] x 64	=	7 [0] x 64	=	7 [0] x 64	=
8 [0] x128	=	8 [0] x128	=	8 [0] x128	=	8 [0] x128	=
Σ	=112	Σ	= 0	Σ	= 1	Σ	= 0

SG4	
1 [1] x 1	= 1
2 [0] x 2	=
3 [0] x 4	=
Σ	= 1

Z 28313-1

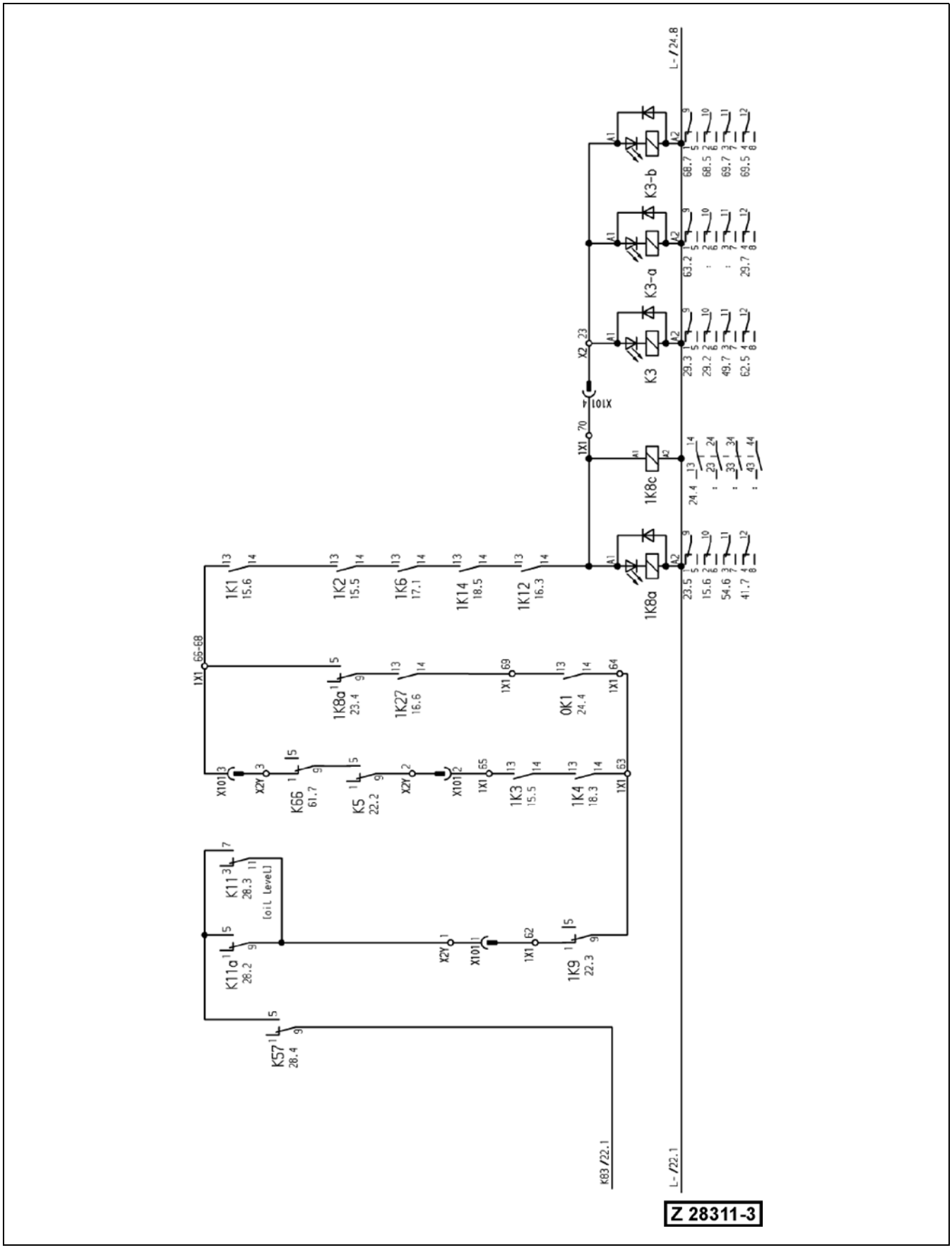
Wiring Diagram 2 (0F4 and signal input of 1K2) related to trouble message -20-

[3/4]



Wiring Diagram 3 (1K2 in the motor start/stop and interlocking circuit) related to trouble message -20-

[4/4]



MESSAGE -21- CENTRAL LUBRICATION SYSTEM, EMPTY GREASE BARREL

MESSAGE -22- SLEW RING GEAR LUBRICATION SYSTEM, EMPTY GREASE BARREL

[1/6]

Trouble code **Message page 21** **Message page 22**	Trouble	Central lubrication system, empty grease barrel Swing ring gear lubrication system, empty grease barrel
Contents of trouble	• The message is displayed if the corresponding grease barrel is empty.	
Related information	• A problem with the CLS grease level sensor (B108c) respectively the SLS grease level sensor (B109c) or connected wiring may also cause the trouble, see illustration Z27806 on page 5 and illustration Z27807 on page 6 for testing procedure of the signal wiring and connections. • The illustrations Z27801 and Z23096 on page [3 and 4] show the location of the related parts on the machine. • This trouble condition should only occur: a) if the Operator has ignored the warning message 59/60 -Refill grease barrel-, or b) if the relevant -Refill- level sensor B108b for CLS respectively the level sensor B109b for SLS failed to function. In this case check also the concerned sensors and wiring circuits of these sensors, refer to trouble message 59/60 for testing procedure.	
Problem that appears on machine	• Risk of damages due to insufficient lubrication. • If the machine is operated further on, after fault message 21 or 22 is set, the bucket function will be disabled after 4.2 hours.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The concerned grease barrel (CLS or SLS) needs refilling	1	Using the dipstick, verify if the relating grease barrel is empty. Is the grease barrel empty?	Yes	Refill with the specified grease. Verify that trouble message 21 or 22 is not displayed again.	
			No	Go to next step.	
Incorrect or no supply voltage in the circuit at the concerned grease level sensor (B108c or B109c)	2	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		Remove the cover of the terminal box SD20 for CLS or SD30 for SLS on top of the concerned grease barrel. Measure the supply voltage at terminal X1.2 inside the terminal box, see illustration Z27807 on page 6. Is supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
	3	Check if supply voltage is present at the power supply line F25. Is supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Disconnection, short circuit, or fault in the wiring between connector XB.A at the terminal box and circuit breaker F25. Check the segments of the voltage supply wiring separately, locate the problem, and repair if required.	

Message -21- Central lubrication system, empty grease barrel

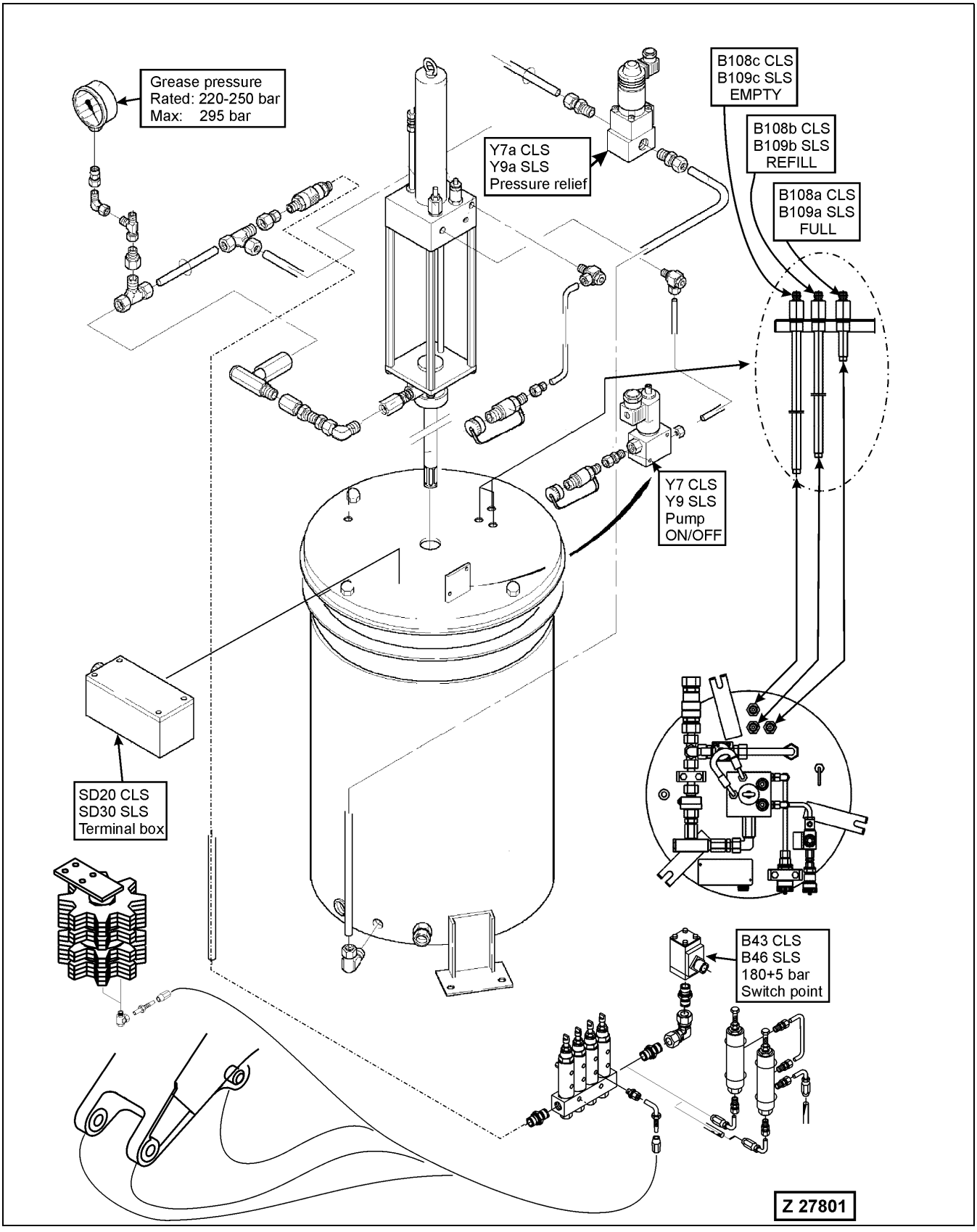
Message -22- Slew ring gear lubrication system empty grease barrel

[2/6]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the signal wiring between the concerned grease level sensor (B108c or B109c) and Input signal module E39 (slave) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-. Disconnection or defective contact in any connector or in the terminal box on top of the concerned grease barrel.	4	► Keep the main key switch 1S1in the OFF position.			
		Disconnect the signal line of the concerned grease level sensor from the input signal module. ETM version -A- Input signal module E39 (slave) Pin 21 for B108c and Pin 22 for B109c. ETM version -B- Input signal module E34 (CAN-node) Pin 61 for B108c and Pin 60 for B109c Refer to the wiring diagrams and check the signal wiring segments between the concerned grease level sensor (B108c or B109c) and the relating input signal module pin for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment of the signal line (including connectors and junction box) between the concerned grease level sensor (B108c or B109c) and the relating pin on the input signal module. Locate the problem and repair as necessary.	
Defective grease level sensor (B108c or B109c)	5	► Prepare with the main key switch 1S1in OFF position, and then carry out troubleshooting with the main key switch 1S1in ON position.			
		For checking the level sensor (B108c or B109c), it is necessary to remove the cover of the concerned grease barrel. Disconnect the current supply cable at the terminal box. Disconnect the grease and hydraulic oil lines at the quick disconnect couplings. Loosen the grease return line on the barrel bottom. Carefully lift the grease barrel cover until the level sensor (B108c or B109c) is accessible. Clean the proximity switch of the concerned sensor. Reconnect the current supply cable at the terminal box. Check the proximity switch by dipping it into a receptacle filled with grease. If the proximity switch is in proper working order, the following conditions should occur: - Proximity switch closed - 24V applied on K136 of CLS or K138 of SLS - Message -Empty grease barrel- gone Do these conditions occur?	Yes	A sporadic malfunction of the input signal module or the text display may exist. Further inspection is required.	
			No	Replace the concerned grease level sensor (B108c or B109c). Go to step 6.	
Final check	6	Bring the machine back into the original condition. Set main key switch 1S1in OFF position, and then in ON position. Start the motor. Verify that the fault message does not arise again. Recheck the function of the lubrication system by manual actuation of a lube cycle.			

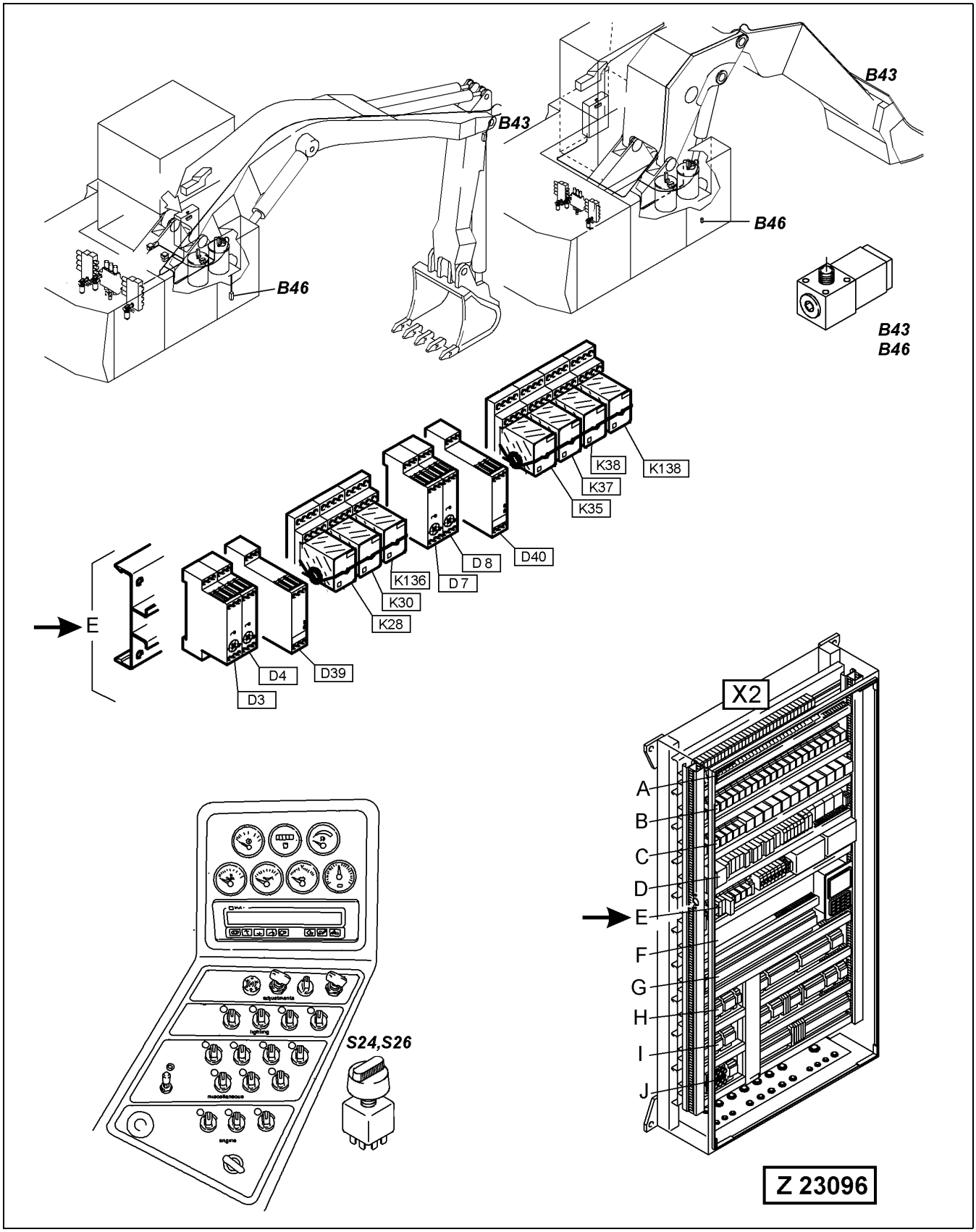
Ilustration related to trouble message -21- and -22-

View onto the grease barrel
[3/6]



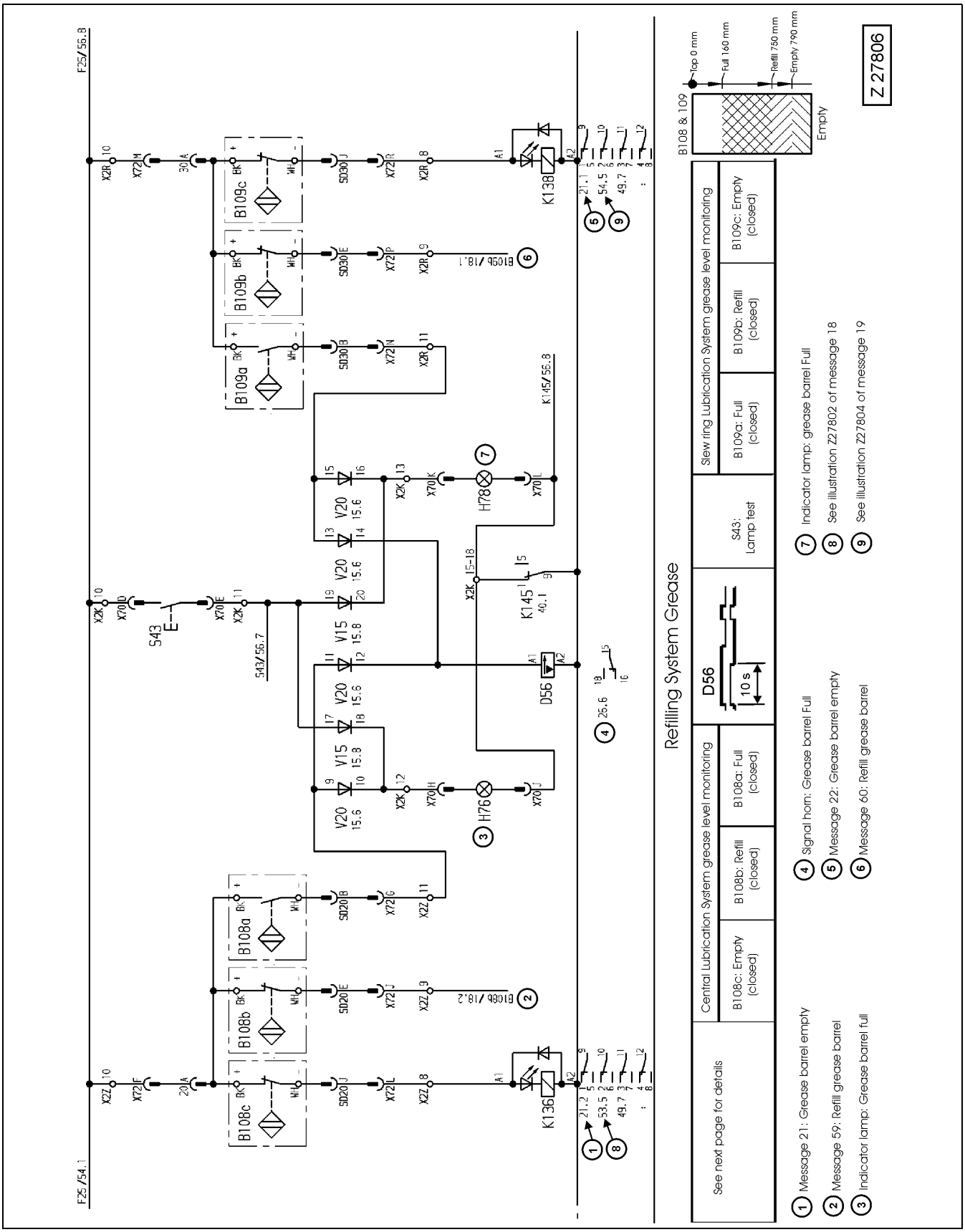
Ilustration related to trouble message -21- and -22-

View onto the X2 Box
[4/6]



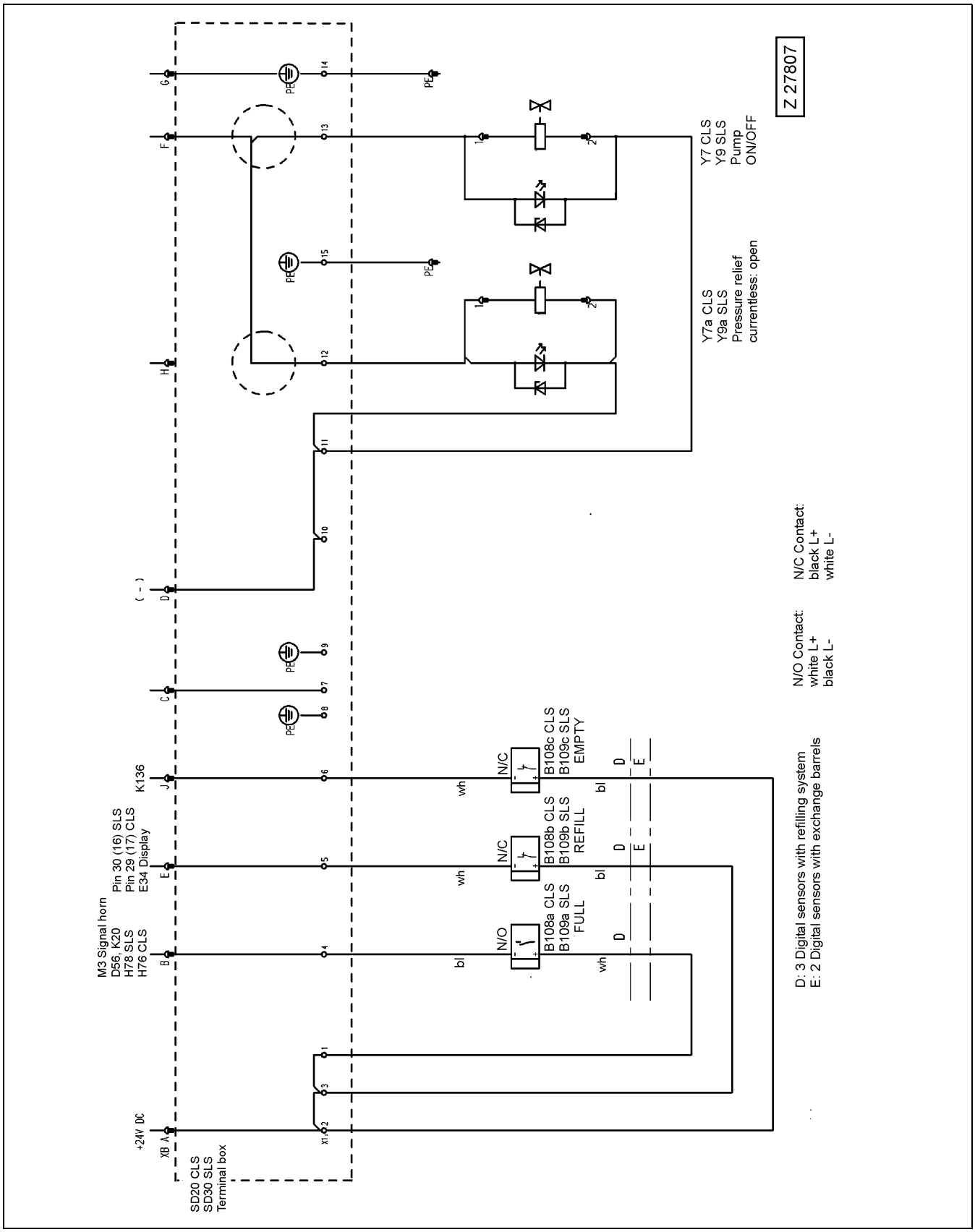
Wiring Diagram related to trouble message -21- and -22-

[5/6]



Wiring Diagram related to trouble message -21- and -22-

[6/6]



MESSAGE -23- MOTOR SHUTDOWN: MOTOR BEARING TEMPERATURE TOO HIGH (1K14)

[1/5]

Trouble code **Message page 23**	Trouble	Motor shutdown: Motor bearing temperature too high (1K14)
Contents of trouble	- Motor shutdown due to overtemperature at any motor bearing. - At one of the motor bearings (DE / NDE) a temperature above the switch point of 120°C is detected.	
Related information	- At standard operating conditions the bearing temperature is approx. 45 - 50°C. (with specified grease and maintenance in regular intervals). - Both PT100 sensors (0B3-1 & 0B3-2) send their signals to the temperature control module 1A4. - The message does not clarify if the DE, or the NDE, or both bearings are overheated. - In case of overheating warning light #5 at the LED monitor module 6V2 is ON.	
Problem that appears on machine	- Motor shutdown. - Motor restart is only possible if the bearing temperature drops below 115°C.	

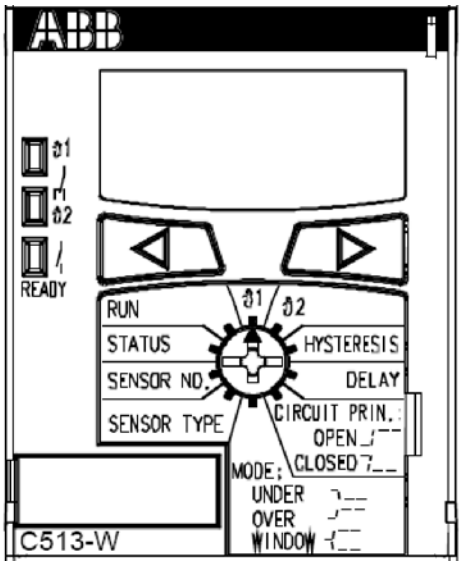
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor bearing temperature is too high	1	Possible reasons for bearing overtemperature are: 1) Wrong maintenance - Shortage of grease at the bearings - Excessive grease at the bearings - Wrong grease viscosity **NOTE:** Refer to the SERVICE MANUAL for correct bearing maintenance. 2) Bearing(s) are worn out - Over lifetime - Alignment problem **NOTE:** Refer to the SERVICE MANUAL for: - SPM measurement in order to determine the bearing wearout - If bearing(s) are defective → replace the E-motor			
Message -23- is displayed and the motor is shutdown, but the bearing temperature is in normal range	2	► Main key switch 1S1 OFF.			
		Confirm the bearing temperature using the conversion table (Ω / °C): - Using an IR temperature gun, measure the temperature in the range of the sensors. - Disconnect the sensor at connector 3X4-1, terminals 13-14 or terminals 17-18. - Measure the sensor resistance. - Determine the temperature from the conversion table. Does the measured temperature match the temperature from the conversion table?	Yes	Go to next step.	
			No	Replace the sensor	

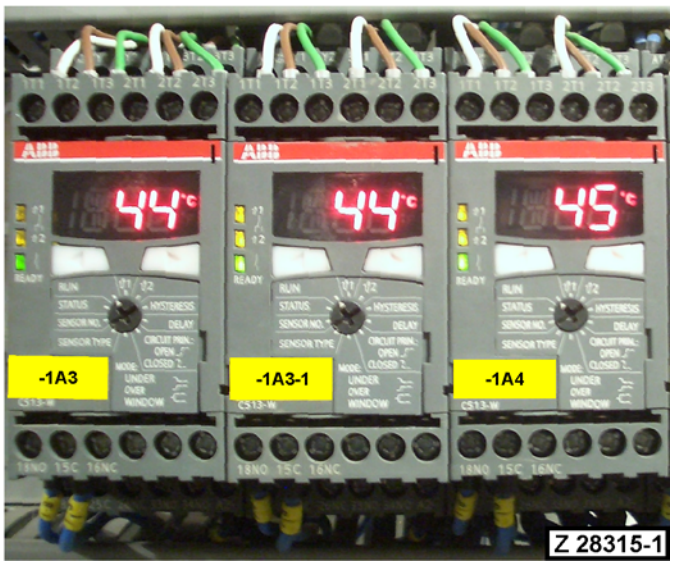
Message -23- Motor shutdown: Motor bearing temperature too high (1K14)

[2/5]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The wiring between sensors 0B3-1/2 and the input at module 1A4 is faulty	3	► Main key switch 1S1 OFF.			
		Measure the wires for continuity. Do all wires have continuity?	Yes	Go to next step.	R_{max} 1Ω
			No	Refer to the Wiring Diagram and repair the faulty wiring as necessary.	
Module 1A4 or its wiring is faulty	4	► Main key switch 1S1 ON.			
		Check if the LED at the module is ON. If the LED (READY) is ON, supply voltage (24V) and GND are present at terminals A1 and A2. If not, repair ground connection and/or power connection to 1F6 as necessary. Check the second power input at terminal 18. Is supply voltage present?	Yes	Go to next step.	22 - 24V
			No	Refer to the Wiring Diagram and repair the faulty wiring as necessary.	
Module 1A4 defective or basic setting wrong	5	► Main key switch 1S1 ON.			
		During normal operation condition 24V should be present at module 1A4 terminal 15 (closed-circuit principle). Are 24V present at terminal 15?	Yes	Replace relay 1K14 (contacts are stuck). Also check the wiring and repair if necessary.	
			No	Check the basic setting of module 1A4, and re-set if necessary. For adjustment hints refer to P&S NEWS AH09519b.	
Final check	6	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

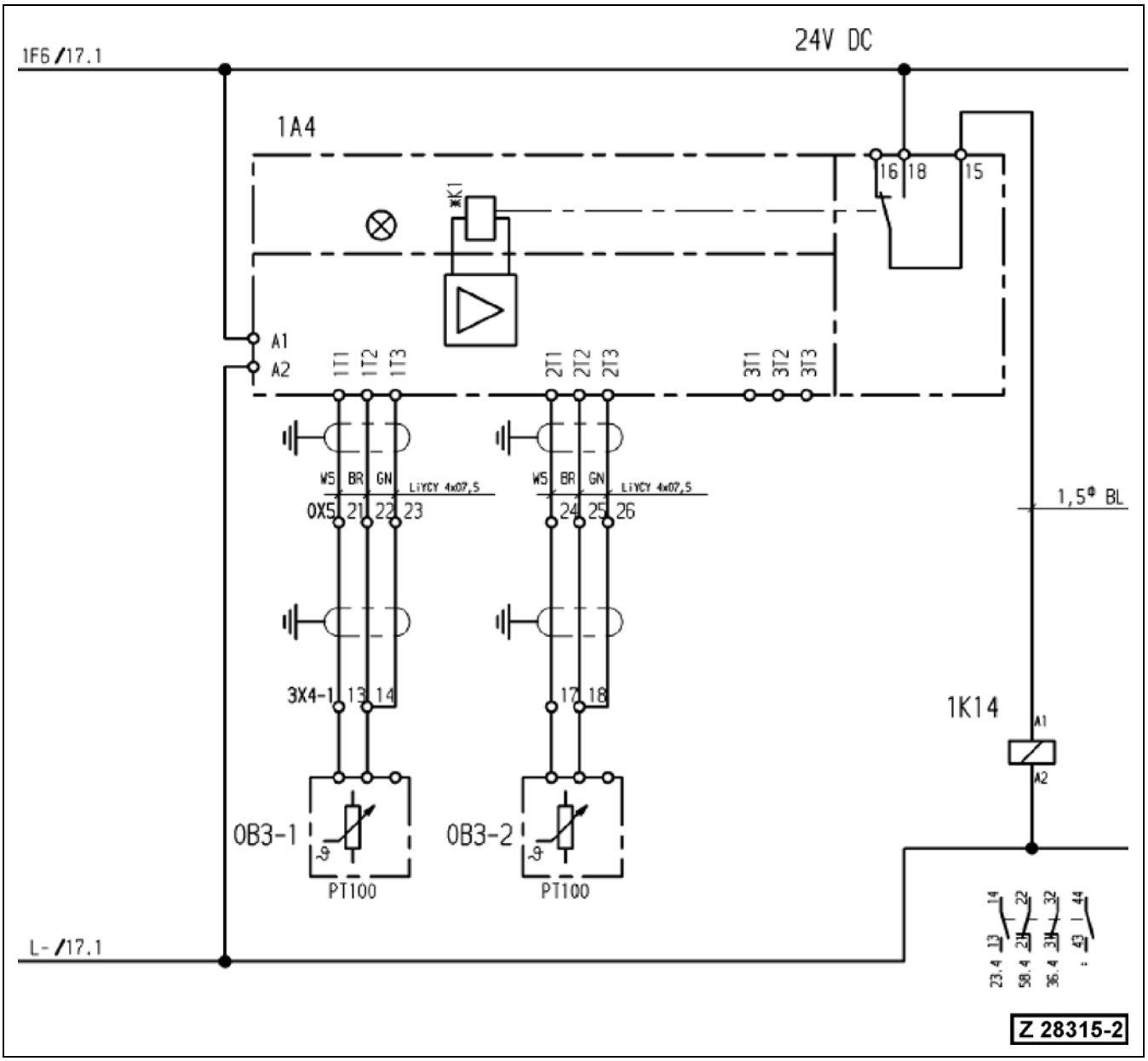
View onto the temperature control modules (1A4 for bearing temperature)





Wiring Diagram 1 (bearing temperature sensors/control module) related to trouble message -23-

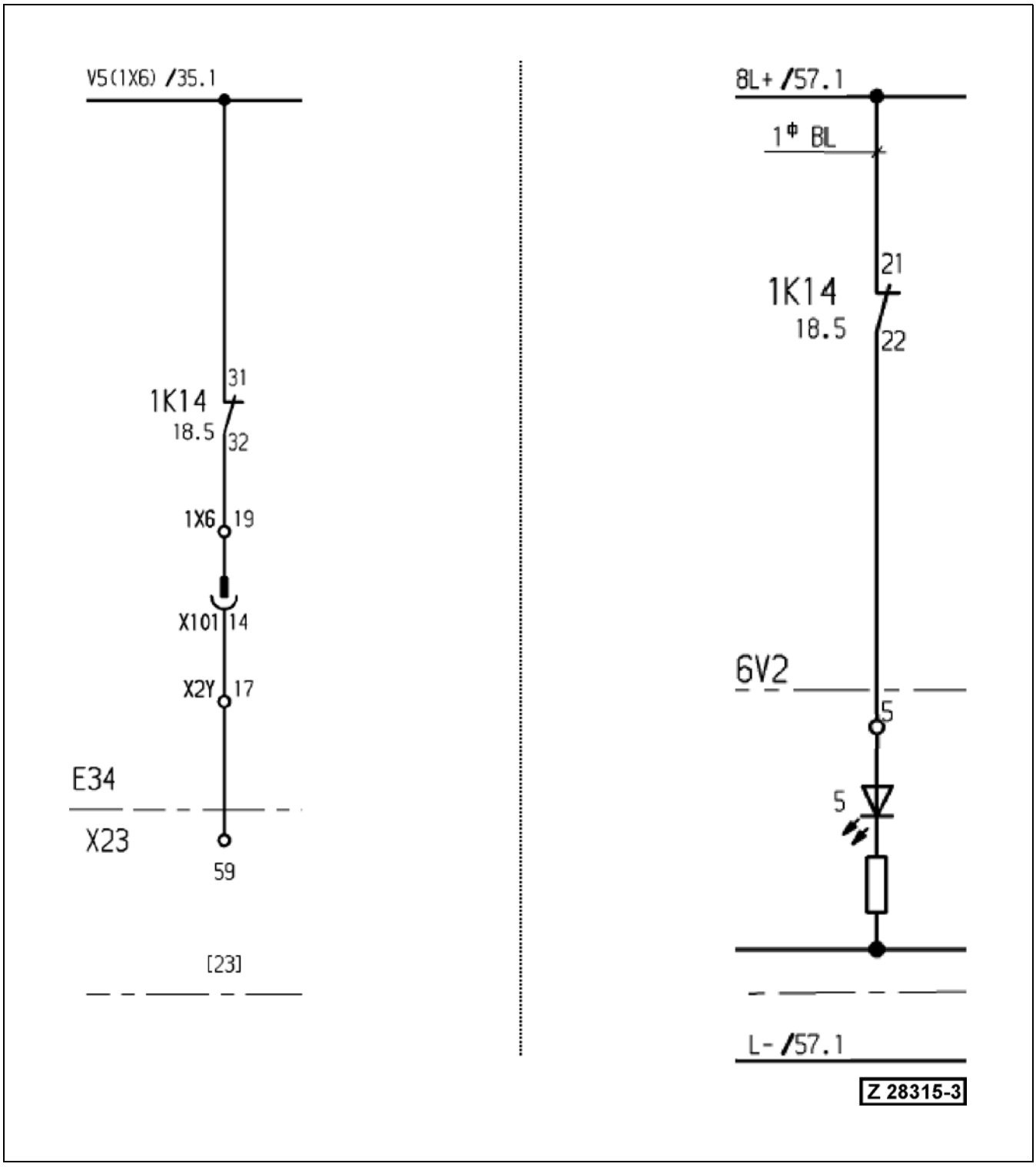
[3/5]



NOTE: Module adjustment in overtemperature mode and closed-circuit principle.

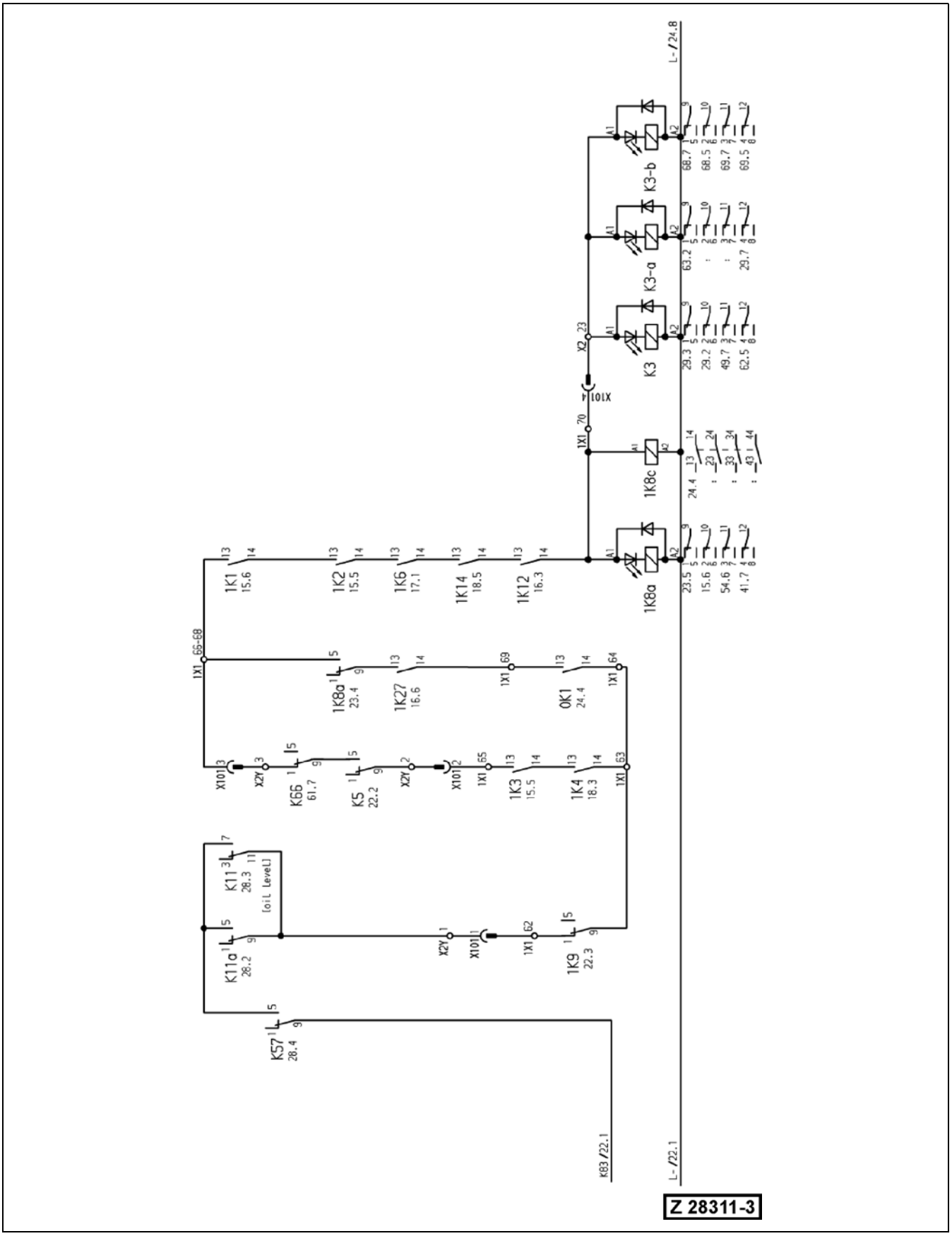
Wiring Diagram 2 (signal input at E34 and 6V2) related to trouble message -23-

[4/5]



Wiring Diagram 3 (1K14 in the motor start/stop and interlocking circuit) related to trouble message -20-

[5/5]



MESSAGE -24- CAUTION, INSULATION MONITORING ACTUATED (1K10)

[1/3]

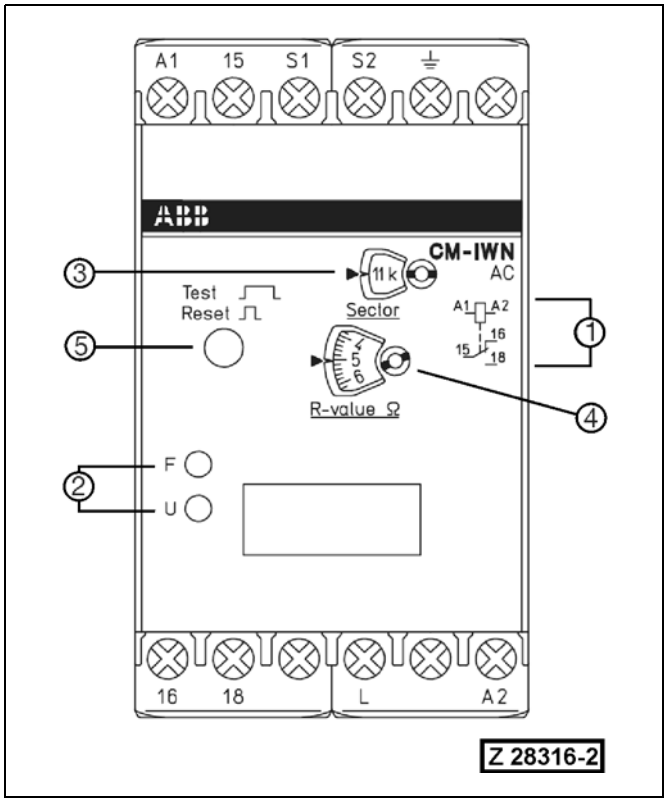
Trouble code **Message page 24**	Trouble	Caution, Insulation monitoring actuated (1K10)
Contents of trouble	- Module 1F2 (insulation meter) detects any isolation problem in the main power system.	
Related information	- Module 1F2 monitors the isolation against ground. - The switch point is at 50 kΩ (basic setting). - This message doesn't cause a shutdown. - If any isolation problem is detected, warning light #8 at the LED monitor module 6V1 is ON.	
Problem that appears on machine	 **Contact the HV department!** An isolation fault in the main power system is detected.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
An isolation fault in the main power system is detected	1	- At terminals A1 and A2 supply voltage (24V) is present. - Relay 1K10 is energized.	Yes	 Call for the HV department!	
		Check the basic setting at 1F2 (switch point = 50 kΩ). Is the setting correct?	No	Perform the setting. Alternatively check the wiring of 1F2 and repair if necessary.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

View onto 1F2

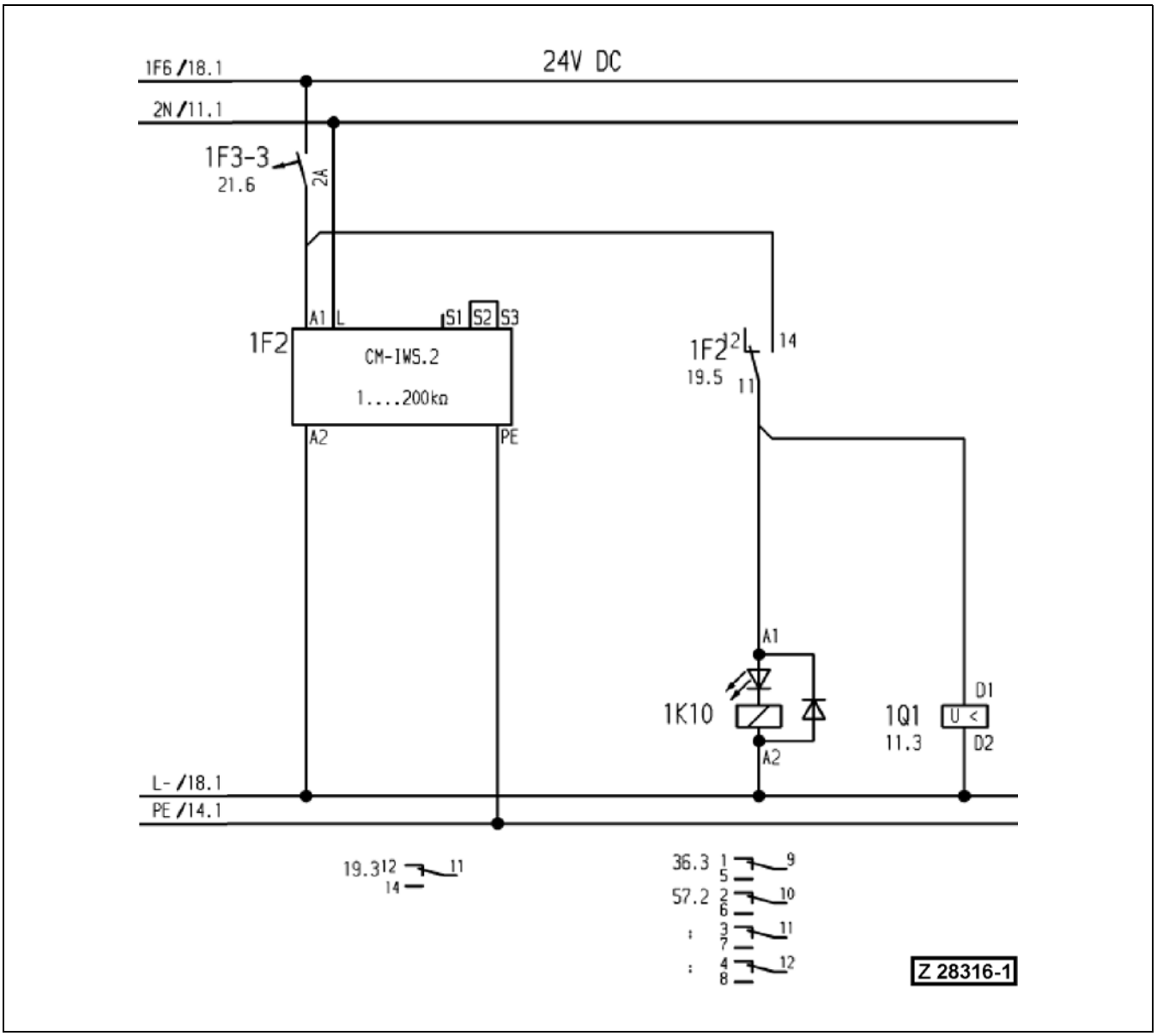
1. Circuit diagram
2. Red LED (F): Error display
   (ground leakage resistance response factor)

   Green LED (U): Power supply voltage
3. Adjustment of the resistance range 11 kΩ or 110 kΩ
4. Fine adjustment of the response factor
5. Reset/Test button



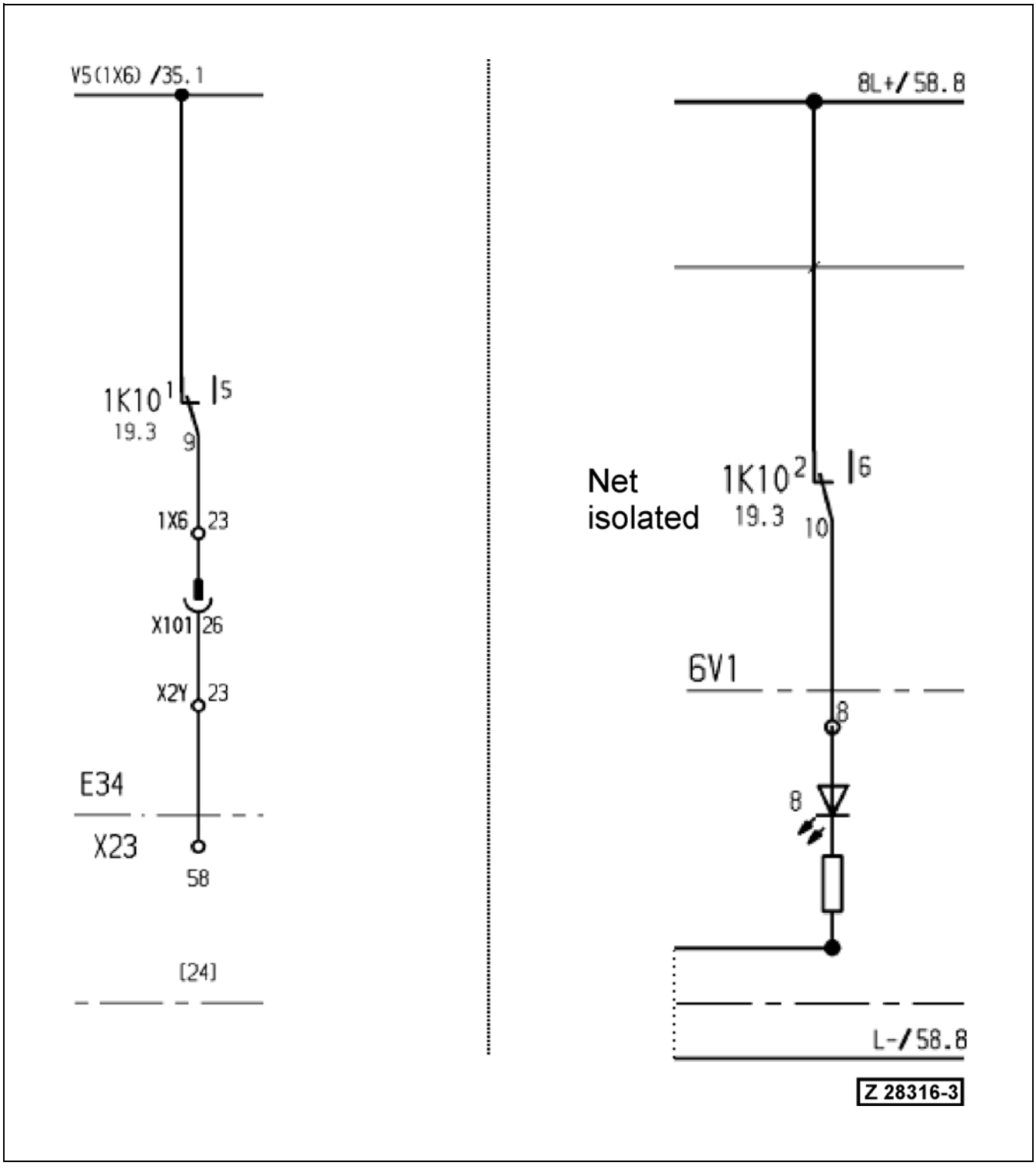
Wiring Diagram 1 (module 1F2 and relay 1K10) related to trouble message -24-

[2/3]



Wiring Diagram 2 (signal input at E34 and 6V1) related to trouble message -24-

[3/3]



MESSAGE -25- NO CLEARANCE FOR STARTING BY MOTOR PROTECTION RELAY (1K3)

[1/3]

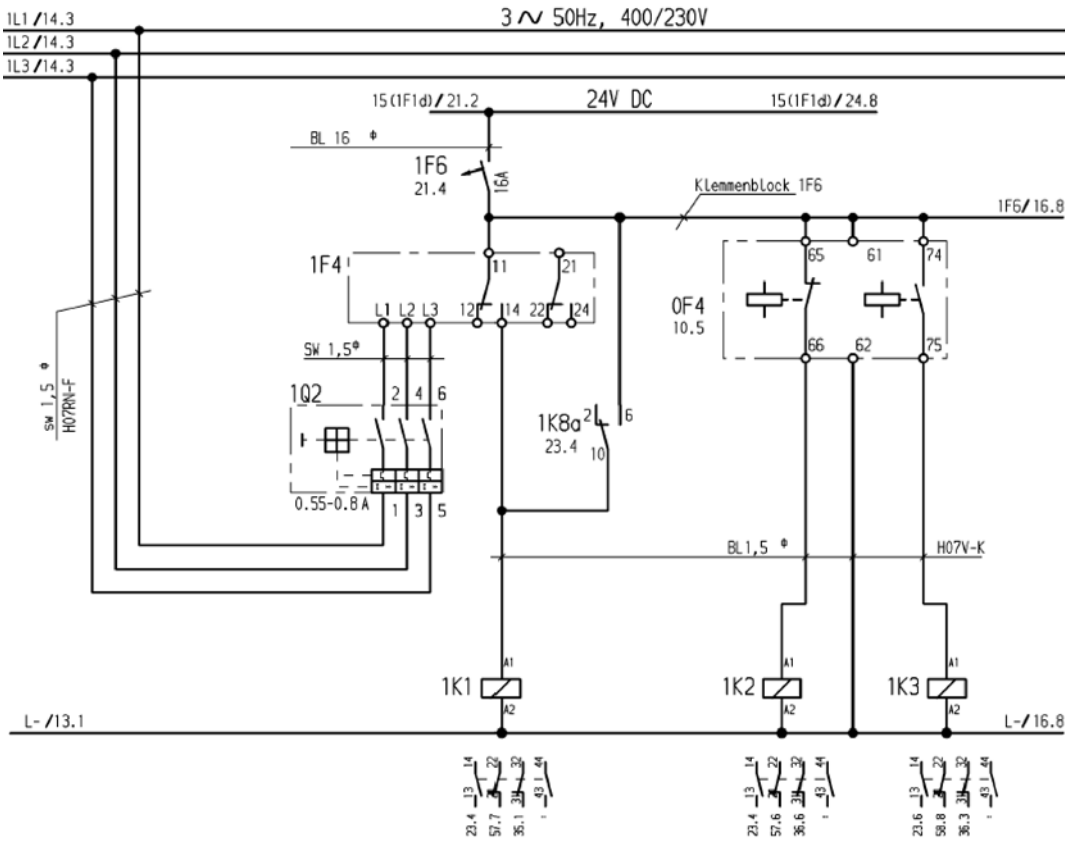
Trouble code **Message page 25**	Trouble	No clearance for starting by motor protection relay (1K3)
Contents of trouble	- After any voltage failure the motor is shutdown an restart is blocked by the motor protection relay 0F4. - 0F4 energizes the signal relay 1K3.	
Related information	- If the motor protection relay detects any voltage failure (overvoltage/undervoltage), contacts 74-75 are closed and relay 1K3 is energized. - If any voltage problem is detected, warning light #1 at the LED monitor module 6V2 is ON. - The motor protection relay is type SPAM 150 C.	
Problem that appears on machine	- As long as relay 1K3 is energized, the motor is shutdown and restart is blocked.	

<table>
<tr><th>Possible cause</th><th>Step</th><th>Test condition & Inspection</th><th>Procedure</th><th>Nominal value</th></tr>
<tr><td>Supply voltage exceeds the maximum or minimum treshold value (setting at 0F4)</td><td>1</td><td>Check the 0F4 display according to the following chart:
<table>
<tr><th>Indi-cation</th><th>Explanation</th></tr>
<tr><td>1</td><td>$\theta > \theta_a$ = The thermal level has exceeded the set prior alarm level</td></tr>
<tr><td>2</td><td>$\theta > \theta_t$ = The thermal unit has tripped</td></tr>
<tr><td>3</td><td>$\theta > \theta_i$, Σt_{si}, EINH = The thermal restart inhibit level is exceeded, the startup time counter is full or the external inhibit signal is active</td></tr>
<tr><td>4</td><td>I>> = The high-set stage of the overcurrent unit has tripped</td></tr>
<tr><td>5</td><td>ΔI = The unbalance/incorrect phase sequence protection unit has tripped</td></tr>
<tr><td>6</td><td>$I^2 \times t$ = The start-up stall protection unit has tripped</td></tr>
<tr><td>7</td><td>I_0 = The earth-fault unit has tripped</td></tr>
<tr><td>8</td><td>I< = The undercurrent unit has tripped</td></tr>
<tr><td>9</td><td>EXT.TRIP = An external tripping has been carried out</td></tr>
</table>
 Call the HV department for troubleshooting!</td><td></td><td></td></tr>
<tr><td>The basic setting of 0F4 (SPAM 150 C) is wrong</td><td>2</td><td>- Check the setting of the motor protection relay.
- For the setting values refer to the Wiring Diagram of your machine.
- Also refer to the separate Instruction Manual “Motor Protection Relay SPAM 150 C“ in volume no 2 binder for operation and maintenance instructions.
- Adjust the setting if necessary.

If the trouble message is still displayed after adjustment:
  Call the HV department for troubleshooting!</td><td></td><td></td></tr>
<tr><td>Final check</td><td>3</td><td>Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.</td><td></td><td></td></tr>
</table>

Wiring Diagram 1 (0F4 & settings and relay 1K3) related to trouble message -25-

[2/3]



I_θ = 0,95 I	I_s = 2,39 I_n	ΔI = 26 $\%I_L$
t_{6x} =10 s	t_s = 52,9 s	t_Δ = 70 s
p = 50 %	$I_{>>}$ = 6 I_n	$I_<$ = off $\%I_n$
Θ_a = 90 %	$t_{>>}$ = 0.05 s	$t_<$ = 600 s
Θ_1 = 70 %	$I_{0>}$ = 100 $\%I_n$	Σt_{si} = 13 s
k_c = 7	t_0 = 30 s	$\Delta\Sigma t_{si}$ = 108 s/h

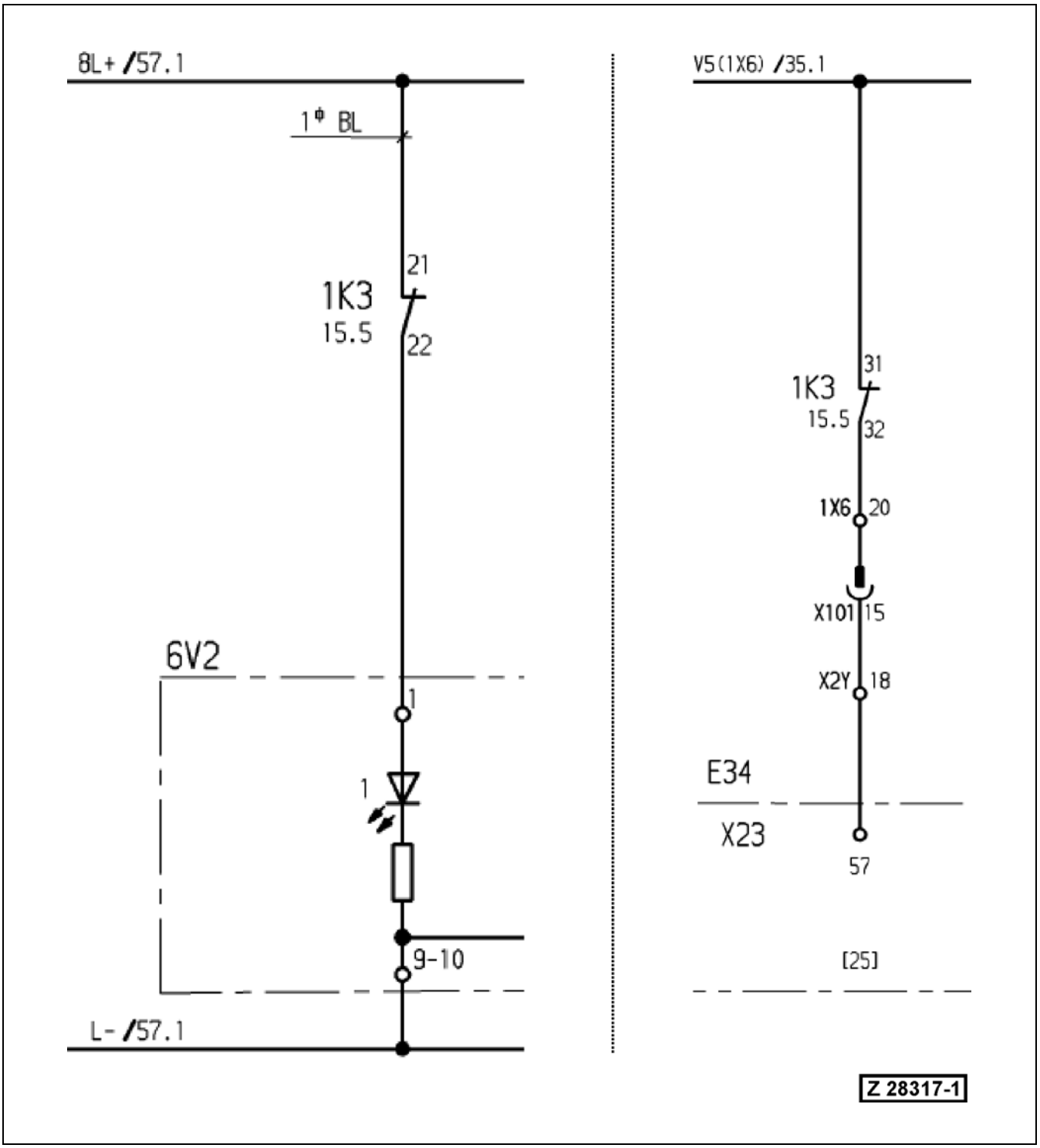
SGF		SGB		SGR1		SGR2	
1 [0] x 1	=	1 [0] x 1	=	1 [1] x 1	= 1	1 [0] x 1	=
2 [0] x 2	=	2 [0] x 2	=	2 [0] x 2	=	2 [0] x 2	=
3 [0] x 4	=	3 [0] x 4	=	3 [0] x 4	=	3 [0] x 4	=
4 [0] x 8	=	4 [0] x 8	=	4 [0] x 8	=	4 [0] x 8	=
5 [1] x 16	= 16	5 [0] x 16	=	5 [0] x 16	=	5 [0] x 16	=
6 [1] x 32	= 32	6 [0] x 32	=	6 [0] x 32	=	6 [0] x 32	=
7 [1] x 64	= 64	7 [0] x 64	=	7 [0] x 64	=	7 [0] x 64	=
8 [0] x128	=	8 [0] x128	=	8 [0] x128	=	8 [0] x128	=
Σ	=112	Σ	= 0	Σ	= 1	Σ	= 0

SG4	
1 [1] x 1	= 1
2 [0] x 2	=
3 [0] x 4	=
Σ	= 1

Z 28313-1

Wiring Diagram 2 (1K3 signal input at 6V2 and E34) related to trouble message -25-

[3/3]



MESSAGE -26- ATTENTION! OVERHEATING PROTECTION OF CAB HEATING HAS TRIGGERED

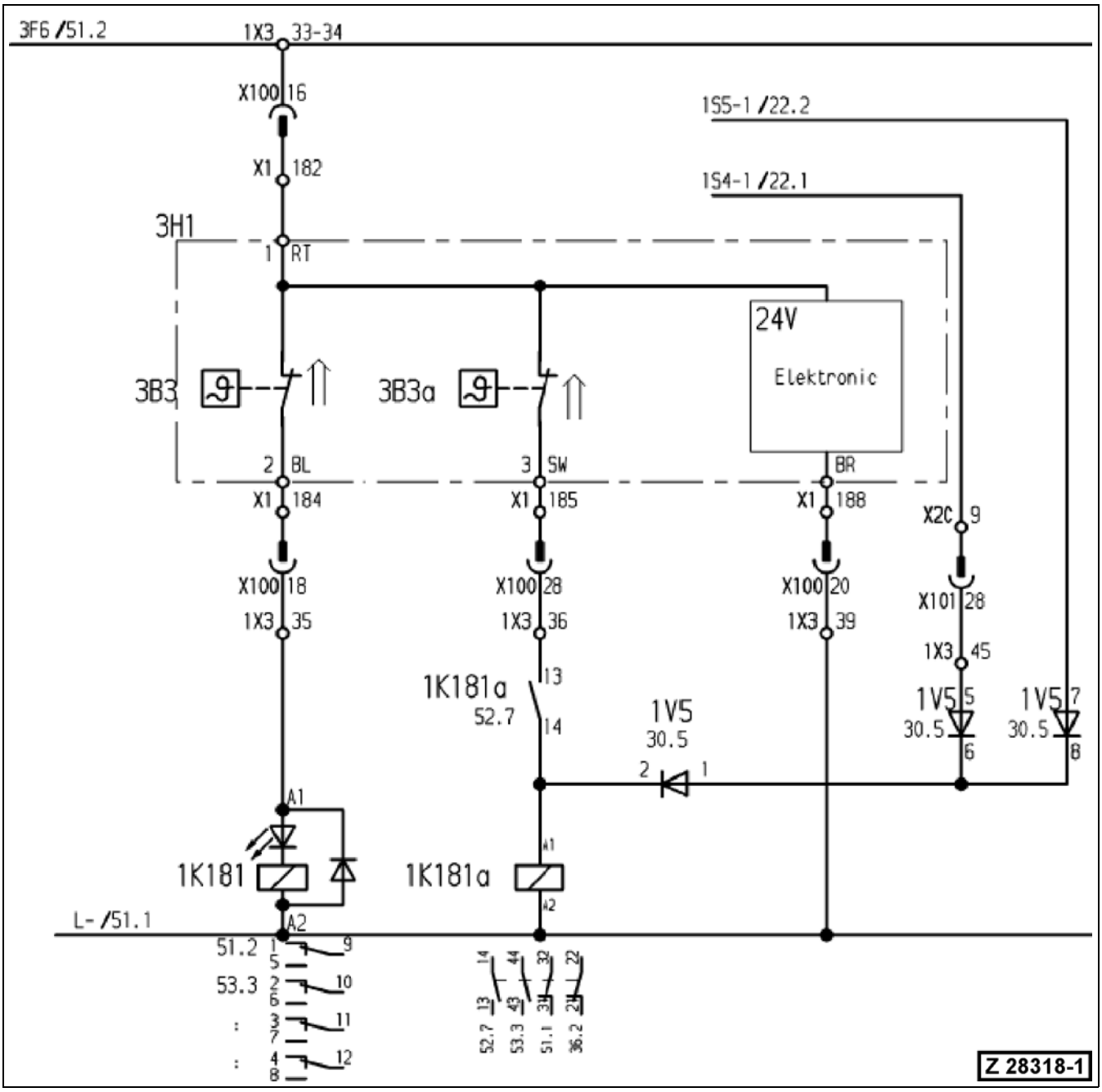
[1/4]

Trouble code **Message page 26**	Trouble	Attention! Overheating protection of cab heating has triggered
Contents of trouble	- The temperature sensors 3B3 and 3B3a inside the heater control module 3H1 are monitoring the heating cartridges below the operator's seat for overheating. - If the message is triggered (due to overtemperature), the heating cartridges are switched OFF. - If the temperature drops below the critical value, the heater is switched ON again.	
Related information	- The temperature sensors are of PTC type. 3B3 switch-off temperature = 60°C 3B3a emergency switch-off temperature = 100°C	
Problem that appears on machine	Heater is switched OFF.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Relay 1K181 / 1K181a or the wiring faulty	1	- Check the wiring and the relays. - If a trouble is found, repair/replace the wiring as necessary or replace a defective relay. Could you locate and clear any trouble?	Yes	Go to final step.	
			No	Go to next step.	
Heater control module 3H1 defective	2	If relays 1K181 and 1K181a and the wiring are OK, replace the heater control module.			
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

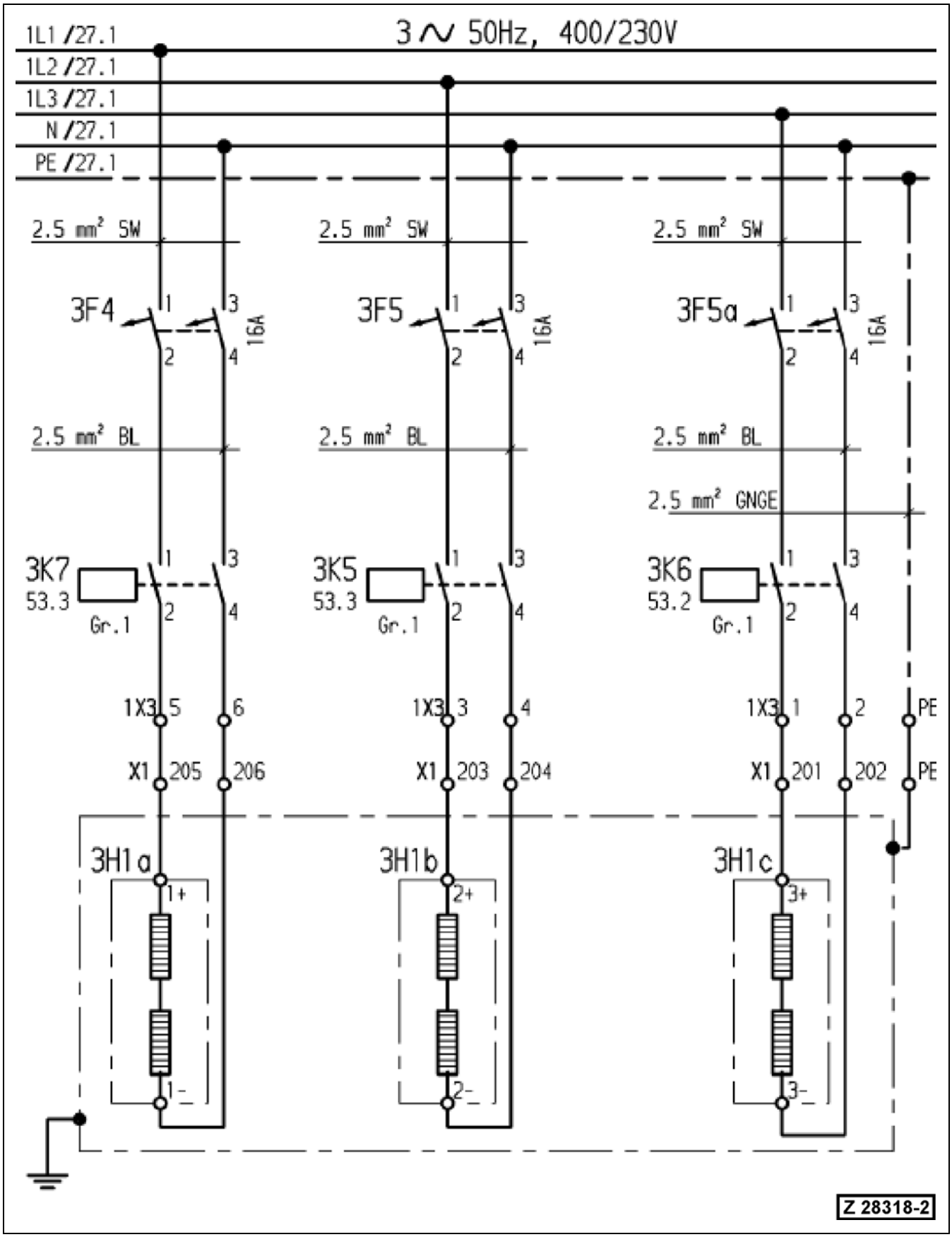
Wiring Diagram 1 (heater control module 3H1) related to trouble message -26-

[2/4]



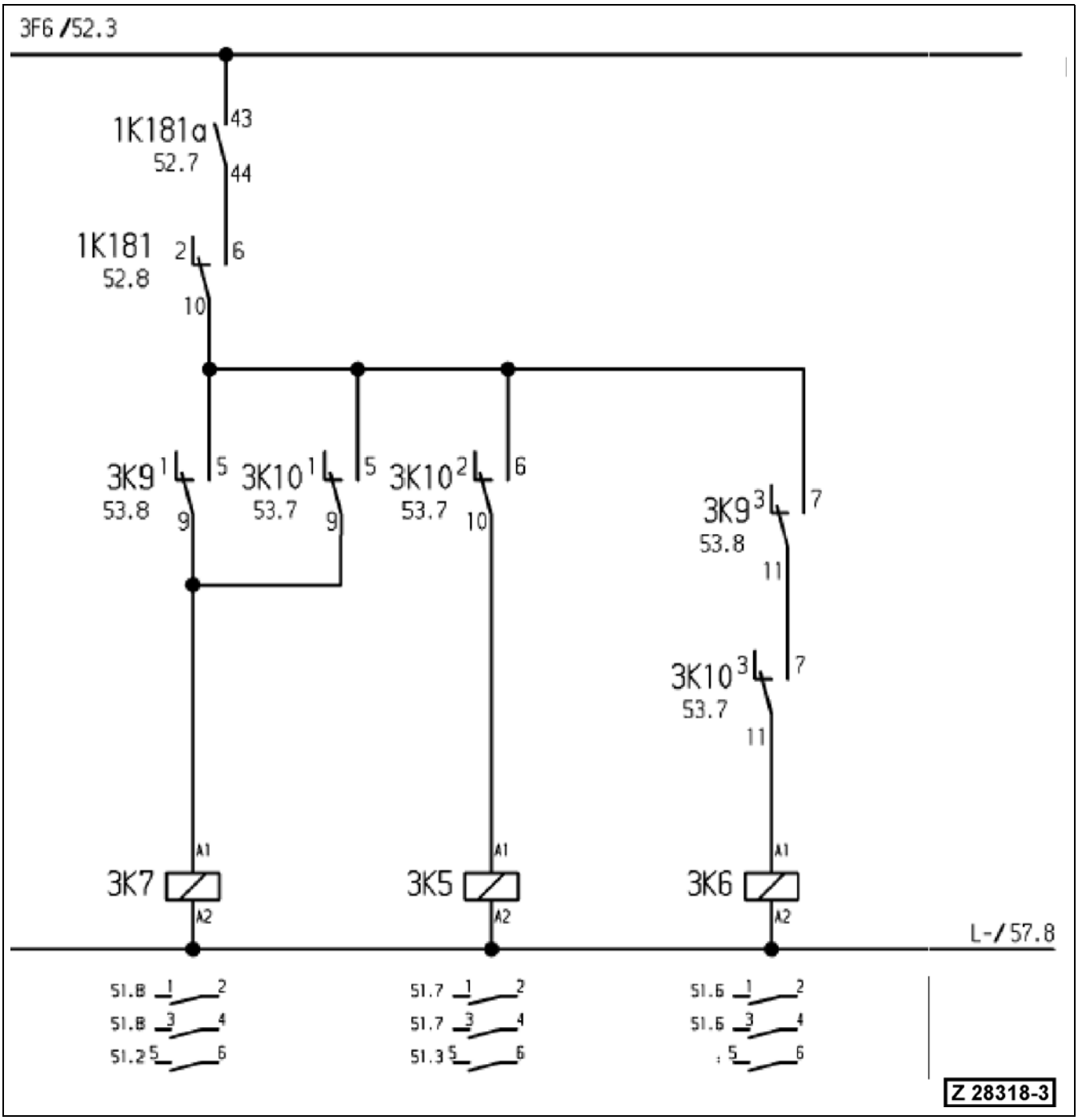
Wiring Diagram 2 (heater cartridges 2 kW each) related to trouble message -26-

[3/4]



Wiring Diagram 3 (relays 1K181 and 1K181a) related to trouble message -26-

[4/4]



MESSAGE -27- (NOT USED)

MESSAGE -28- (NOT USED)

MESSAGE -29- (NOT USED)

MESSAGE -30- (NOT USED)

MESSAGE -31- (NOT USED)

MESSAGE -32- PTO-GEAR LUBRICATION FILTER RESTRICTED (B27)

[1/3]

Trouble code **Message page 32**	Trouble	PTO-gear lubrication filter restricted (B27)
Contents of trouble	• Pressure switch B27 (NO*) signals, that the oil filter for PTO-gear lubrication is clogged. *NO = Normally Open	
Related information	• A problem with the pressure switch B27 or its wiring may also cause the trouble. • Switch point of the differential pressure switch B27 = 5 bar. • If the pressure gets higher than 5 bar, switch B27 closes and 24V are present at terminal 50 of the ETM module E34.	
Problem that appears on machine	- Unfiltered PTO gear lubrication oil flows through the bypass of the filter.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The oil filter for PTO gear lubrication is clogged	1	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked			
		Screw off the filter case and check for clogging. Is the filter element clogged?	Yes	Replace the filter element. If the filter element was clogged check the PTO gear lubrication pump for wear.	
			No	Go to next step.	
The pressure switch B27 is defective	2	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B27 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved?	Yes	Go to final step.	
			No	Go to next step.	
Defective pressure switch B27	3	Check Pressure switch B27 according to the Standard Value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace the pressure switch. After that recheck that the trouble condition does no longer exist.	

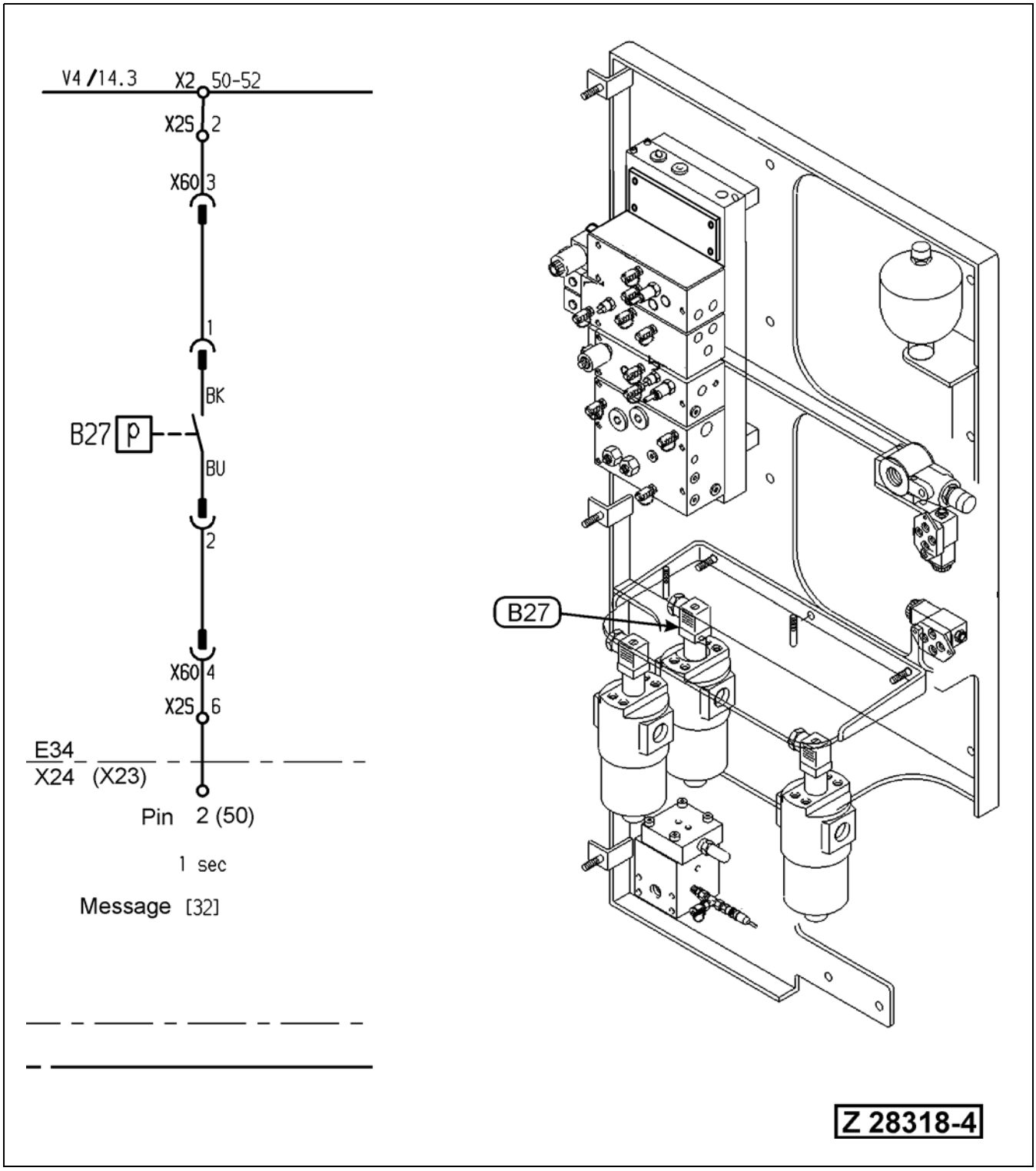
Message -32- PTO-gear lubrication filter restricted (B27)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Incorrect or no supply voltage at the pressure switch B27	4	► Start the motor.			
		Using the T-box, measure the supply voltage for B27 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 6.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B27. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1 in OFF position.			
		Refer to the Wiring Diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B27 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 820Ω
			No	Disconnection, short circuit, or fault in the wiring between connector X60/3 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B27 and Input signal module E34 (disconnection or defective contact in any connector)	6	► Main key switch 1S1 in ON position.			
		ETM version -A- Check the input signal at the Input signal module E34 (Master) pin 2. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 50. Is the signal present?	Yes	The signal wiring between pressure switch B27 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B27 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustration related to trouble message -32-

[3/3]



MESSAGE -33- COOLER FAN DRIVE FILTER RESTRICTED (B28)

[1/3]

Trouble code **Message page 33**	Trouble	Cooler fan drive filter restricted (B28)
Contents of trouble	• Pressure switch B28 (NO*) signals, that the oil filter for cooler fan drive is clogged. *NO = Normally Open	
Related information	• A problem with the pressure switch B28 or its wiring may also cause the trouble. • Switch point of the differential pressure switch B28 = 5 bar. • If the pressure gets higher than 5 bar, switch B28 closes and 24V are present at terminal 49 of the ETM module E34.	
Problem that appears on machine	- Unfiltered cooler fan drive oil flows through the bypass of the filter.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The oil filter for cooler fan drive is clogged	1	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked			
		Screw off the filter case and check for clogging. Is the filter element clogged?	Yes	Clean or replace the filter element. If the filter element was clogged check the cooler fan drive pump for wear.	
			No	Go to next step.	
The pressure switch B28 is defective	2	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked			
		The pressure switch B28 has an internal stuck in closed position. Remove the switch and replace it by a new one. Is the problem solved?	Yes	Go to final step.	
			No	Go to next step.	
Defective pressure switch B28	3	Check Pressure switch B28 according to the Standard value table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace pressure switch. After that recheck that the trouble condition does no longer exist.	
Incorrect or no supply voltage at the pressure switch B28	4	► Start the motor.			
		Using the T-box, measure the supply voltage for B28 at the connector terminal 1. Is the correct supply voltage present?	Yes	Go to step 7.	V_{min} 22V V_{max} 24V
			No	Go to next step.	

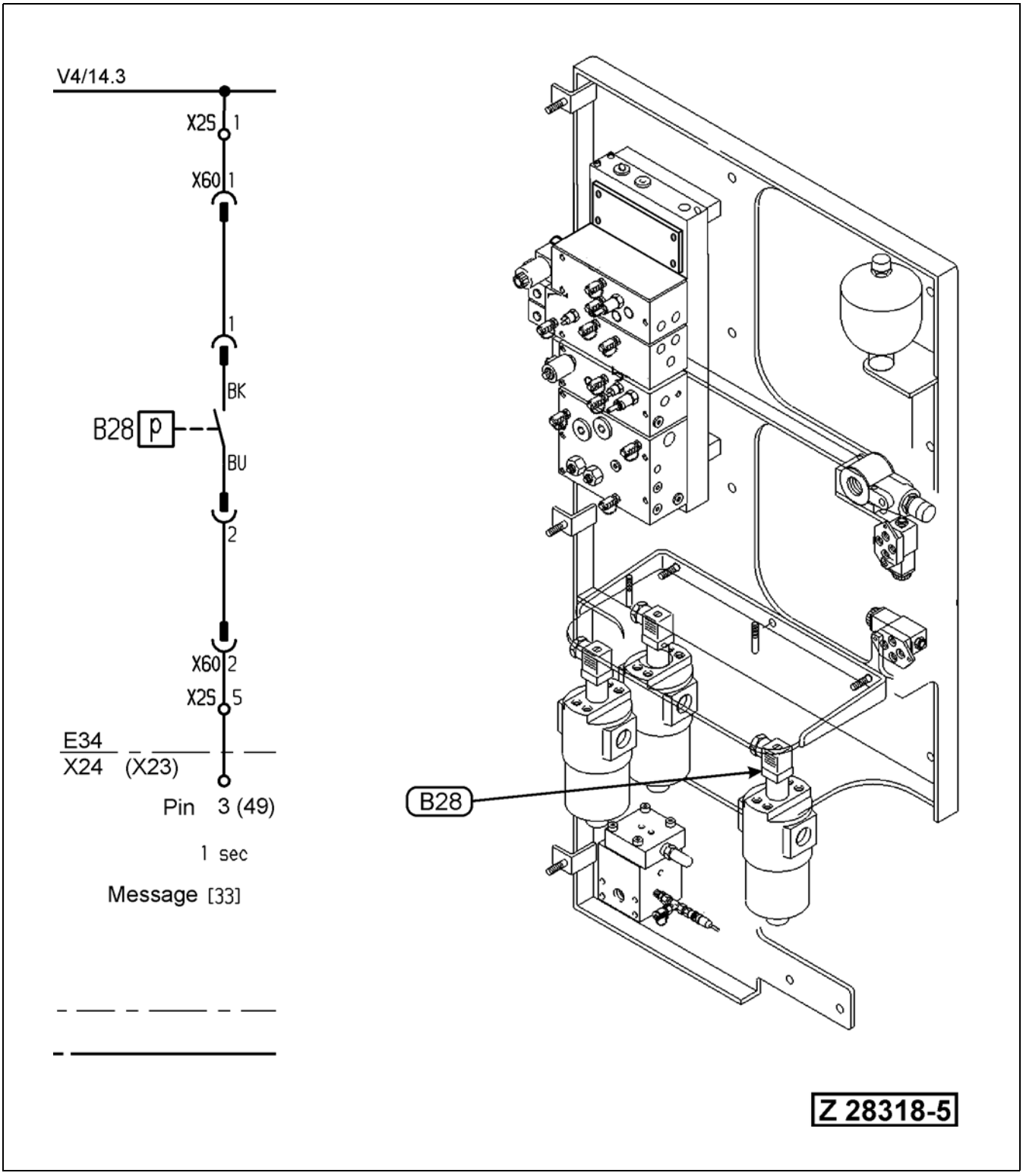
Message -33- Cooler fan drive filter restricted (B28)

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the voltage supply wiring between circuit breaker F15 and pressure switch B28. (disconnection or defective contact in any connector)	5	► Motor stopped and main key switch 1S1 in OFF position.			
		Refer to the Wiring Diagram. Using the T-box, check the voltage supply wiring segments between the pressure switch B28 and circuit breaker F15 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 820Ω
			No	Disconnection, short circuit, or fault in the wiring between connector X60/1 and circuit breaker F15. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Disconnection in the signal wiring between pressure switch B28 and Input signal module E34 (disconnection or defective contact in any connector)	6	► Main key switch 1S1 in the ON position.			
		ETM version -A- Check the input signal at the Input signal module E34 (Master) pin 3. ETM version -B- Check the input signal at the Input signal module E34 (CAN-node) pin 49. Is the signal present?	Yes	The signal wiring between pressure switch B28 and Input signal module is OK.	
			No	Disconnection, short circuit, or fault in the wiring (including connectors and pins) between pressure switch B28 and Input signal module. If necessary, check the segments of the signal wiring separately, locate the problem, and repair as necessary.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustration related to trouble message -33-

[3/3]



MESSAGE -34- (NOT USED)

MESSAGE -35- (NOT USED)

MESSAGE -36- (NOT USED)

MESSAGE -37- (NOT USED)

MESSAGE -38- (NOT USED)

MESSAGE -39- MOTOR-SHUTDOWN: SHUT-OFF (GATE) VALVE S31 CLOSED

[1/4]

Trouble code **Message page 39**	Trouble	Motor-shutdown: Shut-Off (gate) valve S31 closed
Contents of trouble	• After main key switch 1S1 is switched ON (before motor start) the proximity switch S31 monitors the gate valve between hydraulic tank and suction tank. • Motor start is disabled if the valve is not completely open. • If the proximity switch S31 senses a not completely open gate valve with the motor running, shutdown will occur.	
Related information	• A problem with the proximity switch S31 or its wiring may also cause the trouble. • If the voltage at the proximity switch S31 is 0 V, the main gate valve should be in closed position.	
Problem that appears on machine	• The motor start is blocked or if motor running, shutdown will occur.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor start is impossible, because the gate valve at the hydraulic oil tank is not fully open	1	Verify if the gate valve is fully open or not. Is the gate valve fully open?	Yes	Go to next step.	
			No	Open the gate valve completely and verify that trouble message 39 is not set again.	
The motor is stopped, because the main gate valve at the hydraulic oil tank is not fully open	2	► Keep the main key switch 1S1 in the ON position.			
		Verify if the main gate valve is fully open or not. Is the main gate valve fully open?	Yes	Go to next step.	
			No	Open the gate valve completely and verify that trouble message 39 is not set again.	
The gap between the proximity switch S31 and the open main gate valve is too large	3	► Keep the main key switch 1S1 in the ON position.			
		Check for the proper gap between the proximity switch S31 and the open main gate valve, see illustration on page [3/4]. Does the gap meet the specification?	Yes	Go to next step.	Gap: approx. 3 - 5 mm
			No	Adjust the gap. Make sure that the proximity switch S31 does not touch the gate valve when the gate valve is completely open.	
Incorrect or no supply voltage at the proximity switch S31	4	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		Disconnect the cable connector from the proximity switch S31 and insert the T-box. Measure the supply voltage at the connector terminal A. Is supply voltage present?	Yes	Go to step 7.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
	5	Check if supply voltage is present at the power supply line F12. Is supply voltage present?	Yes	Go to next step.	V_{min} 22V V_{max} 24V
			No	Check circuit between F1 and F12. Locate the problem and repair as necessary.	

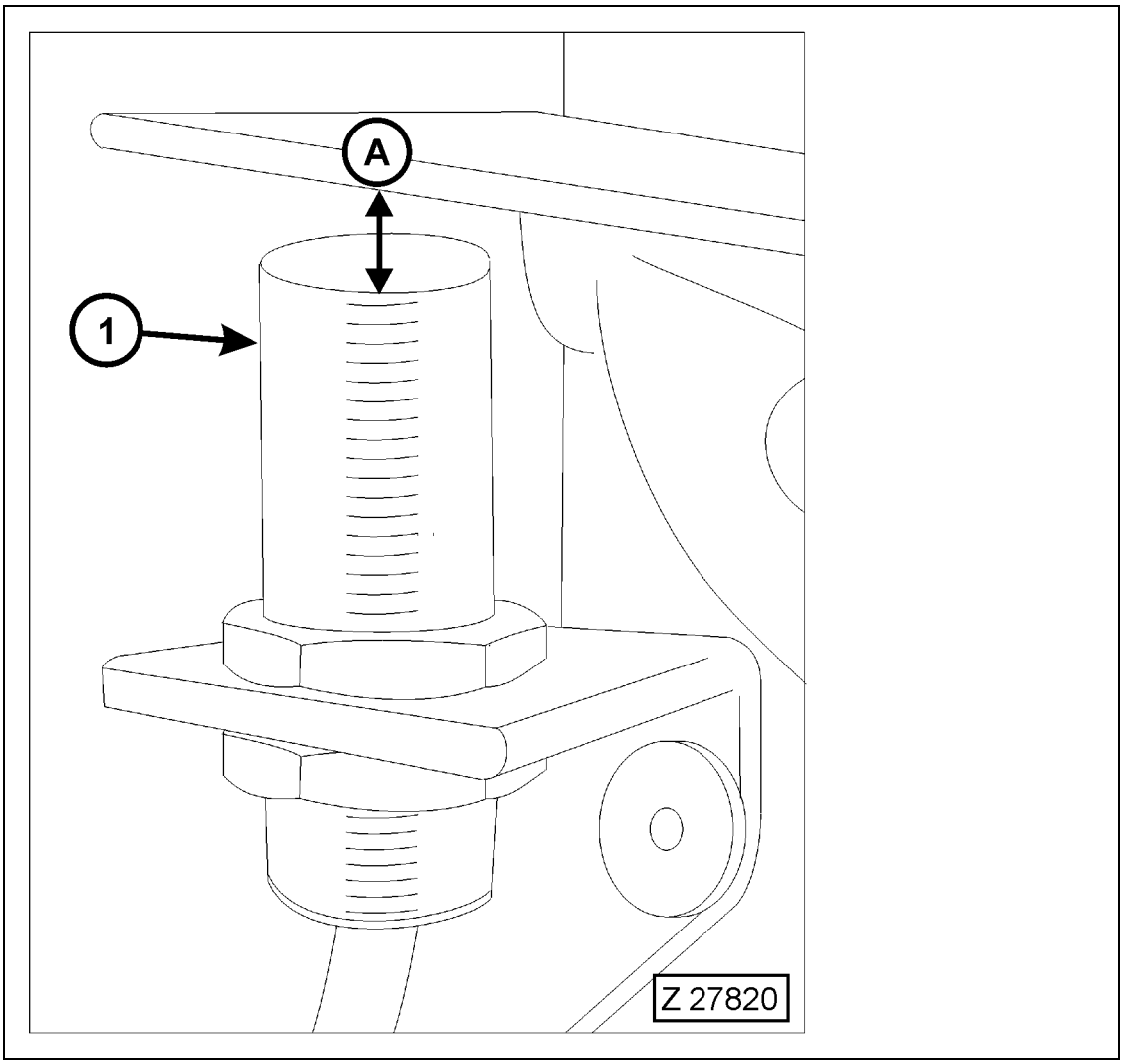
Message -39- Motor-shutdown: Shut-Off (gate) valve S31 closed

[2/4]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Malfunction in the power supply wiring between proximity switch S31 and circuit breaker F12 (disconnection or defective contact in any connector)	6	► Keep the main key switch 1S1 in OFF position.			
		Refer to the Wiring Diagram. Disconnect the connector from the proximity switch S31. Check the wire segments between proximity switch S31 and power supply line F12 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wire segment (including connectors) between proximity switch S31 and circuit breaker F12. Locate the problem and repair as necessary.	
Disconnection in the signal wiring between proximity switch S31, K57 and input signal module E34 (disconnection or defective contact in any connector).	7	► Keep the main key switch 1S1 in the OFF position.			
		Disconnect the signal line from the input signal module. ETM version -A- Input signal module E34 (Master) Pin 9. ETM version -B- Input signal module E34 (CAN-node) Pin 40. Refer to the Wiring Diagram and check the wiring segments between proximity switch S31, K57 and input signal module for continuity. Is continuity present?	Yes	Go to next step.	
			No	Disconnection or short circuit in any wiring segment (including connectors) between proximity switch S31, K57 and input signal module. Locate the problem and repair as necessary.	
Defective ground connection at the proximity switch S31	8	► Keep the main key switch 1S1 in the OFF position.			
		Disconnect the cable connector from the proximity switch S31 and check for proper ground connection at the connector terminal B. Is the ground connection OK?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment (including connectors) between proximity switch S31 and the ground line (-). Locate the problem and repair as necessary.	
Defective proximity switch S31	9	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		Disconnect the sensor connector and insert the T-box. Measure the signal voltage at terminal C with closed and open gate valve.		If the switch function is not as specified replace the proximity switch S31 and verify that the repair was successful. Go to final step 10.	V_{Signal} S31 closed: 0V V_{Signal} S31 open: 22-24V
Final check	10	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustration related to trouble message -39-

[3/4]

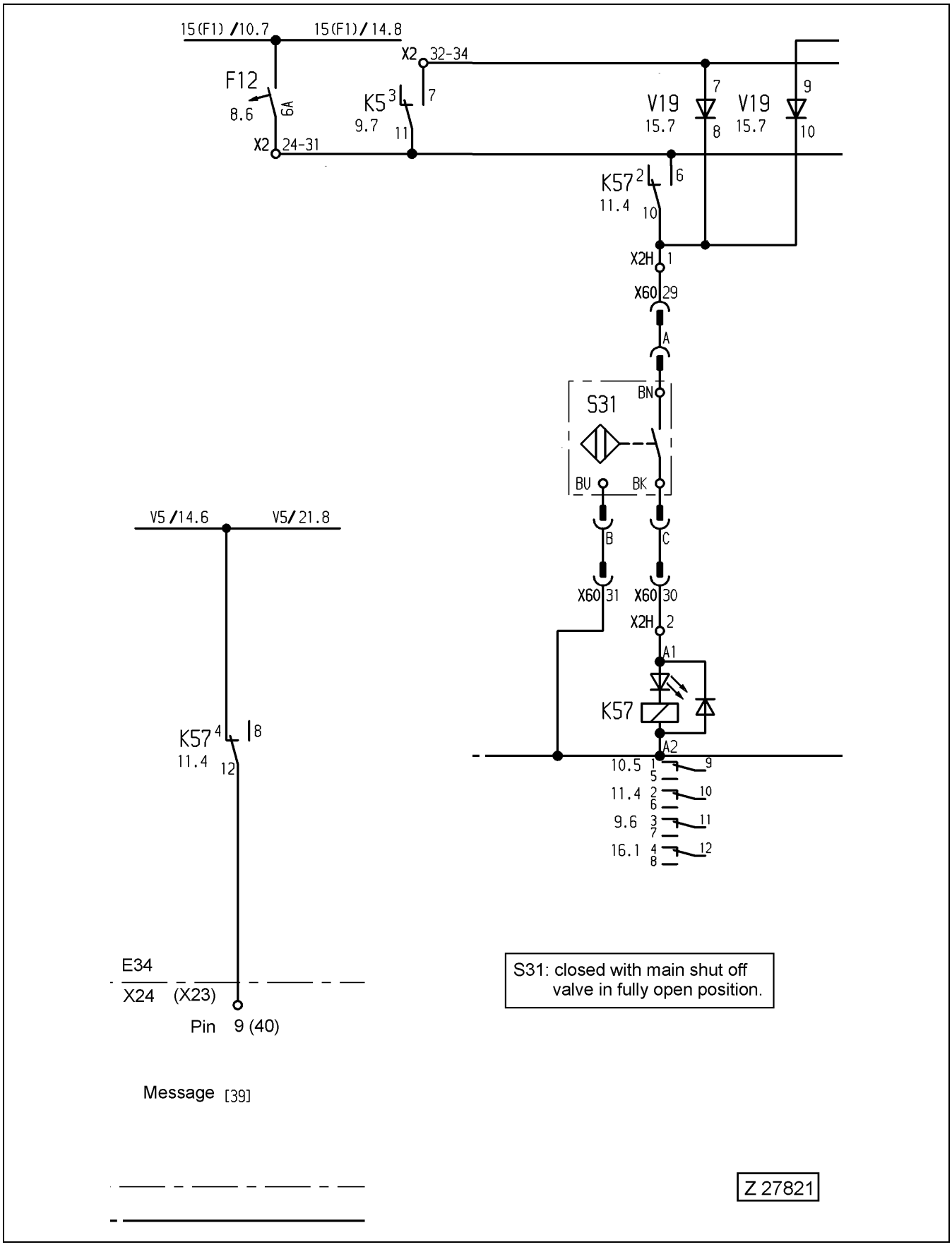


(1) Proximity switch S31

(A) Gap: 3 - 5 mm

Wiring Diagram related to trouble message -39-

[4/4]



MESSAGE -40- CAUTION, SWITCH-CABINET TEMPERATURE TOO HIGH (6B1, 6B2)

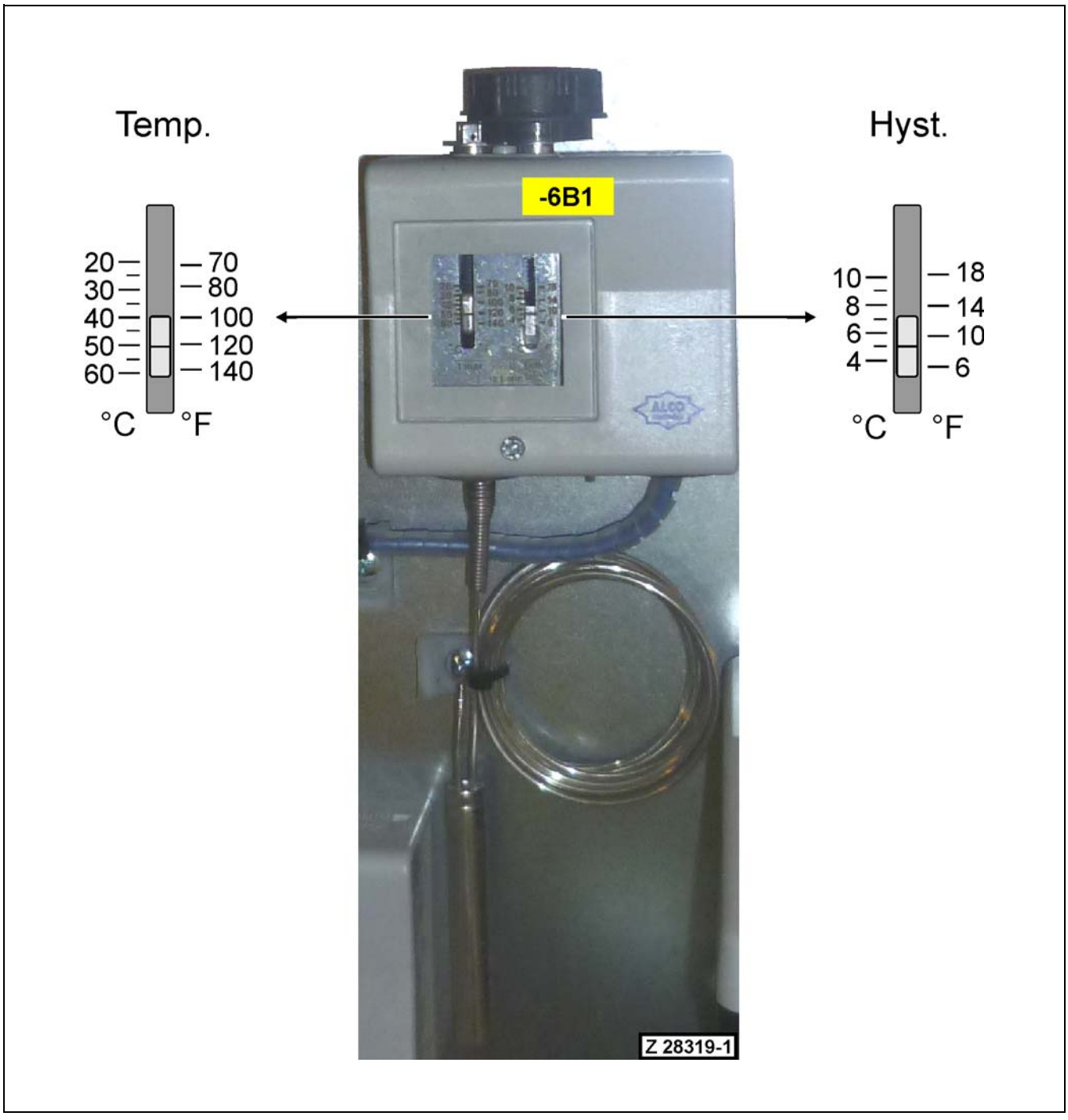
[1/4]

Trouble code **Message page 40**	Trouble	Caution, switch-cabinet temperature too high (6B1, 6B2)
Contents of trouble	- If the heater unit 3R31 or 3R25 is defective (temperature control inoperative), sensor 6B1 or 6B2 detects too high room temperature and triggers the trouble message. - If too high room temperature is detected: LV switch cabinet → Warning light #3 at the LED monitor module 6V2 is ON HV switch cabinet → Warning light #4 at the LED monitor module 6V2 is ON	
Related information	- Heater units (2 kW each) including controllers are located in both switch cabinets: 3R21 → HV switch cabinet, 3R25 → LV switch cabinet - They automatically control the temperature in the switch cabinets. - The room temperature in the switch cabinets is monitored by temp. sensors: 6B1 → LV switch cabinet, 6B2 → HV switch cabinet - Adjustment of 6B1/6B2: switch point = 50°C, hysteresis = 5°C  **Heater unit 3R21 and sensor 6B2 are located in the HV switch cabinet.** **HAZARDOUS VOLTAGE!** **High voltage is applied in the High Voltage switch cabinet.** **There is danger of serious injury or death.** **Access to the High Voltage switch cabinet is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.**	
Problem that appears on machine	- The heater in the respective switch cabinet operates without temperature control (continuously heating).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Temperature sensor 6B1 or 6B2 defective (wrong temperature signal)	1	► Main key switch 1S1 ON.			
		Check at the LED monitor module 6V2 which heater control is disturbed: Warning light #3 → LV switch cabinet Warning light #4 → HV switch cabinet. Using an IR temperature gun, measure the temperature in the range of the respective temperature sensor. Does the measured temperature confirm the overheat signal of the sensor?	Yes	Got to next step.	
			No	- Check the adjustment of 6B1 or 6B2. - Correct the adjustment if necessary. - If signal still wrong go to next step.	Switch point 50°C Hysteresis 5°C
Controller inside the heater module defective	2	► Main key switch 1S1 OFF.			
		Replace the defective heater module (3R21 or 3R25).			
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

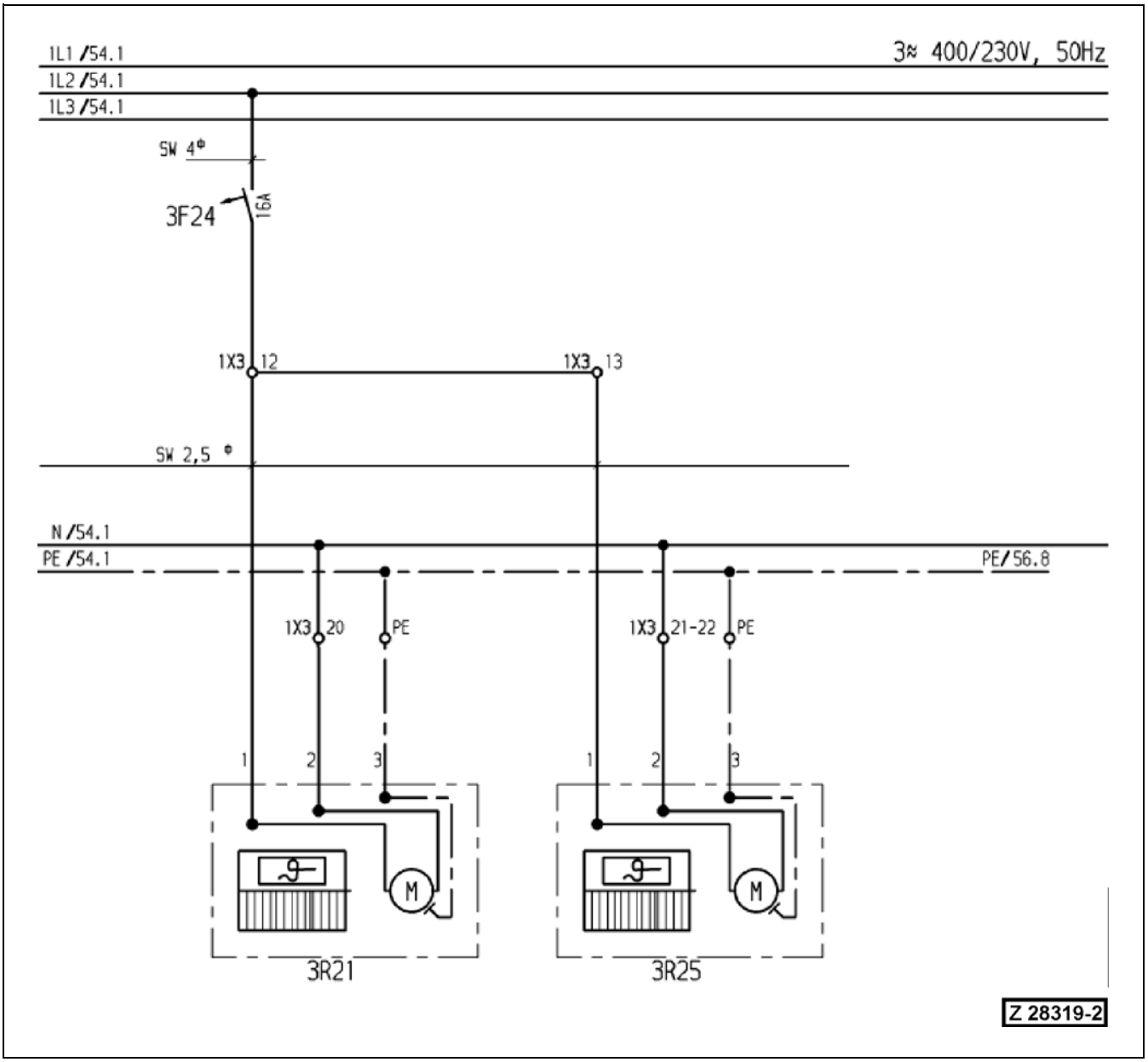
Message -40- View onto the temperature sensor 6B1 (6B2)

[2/4]

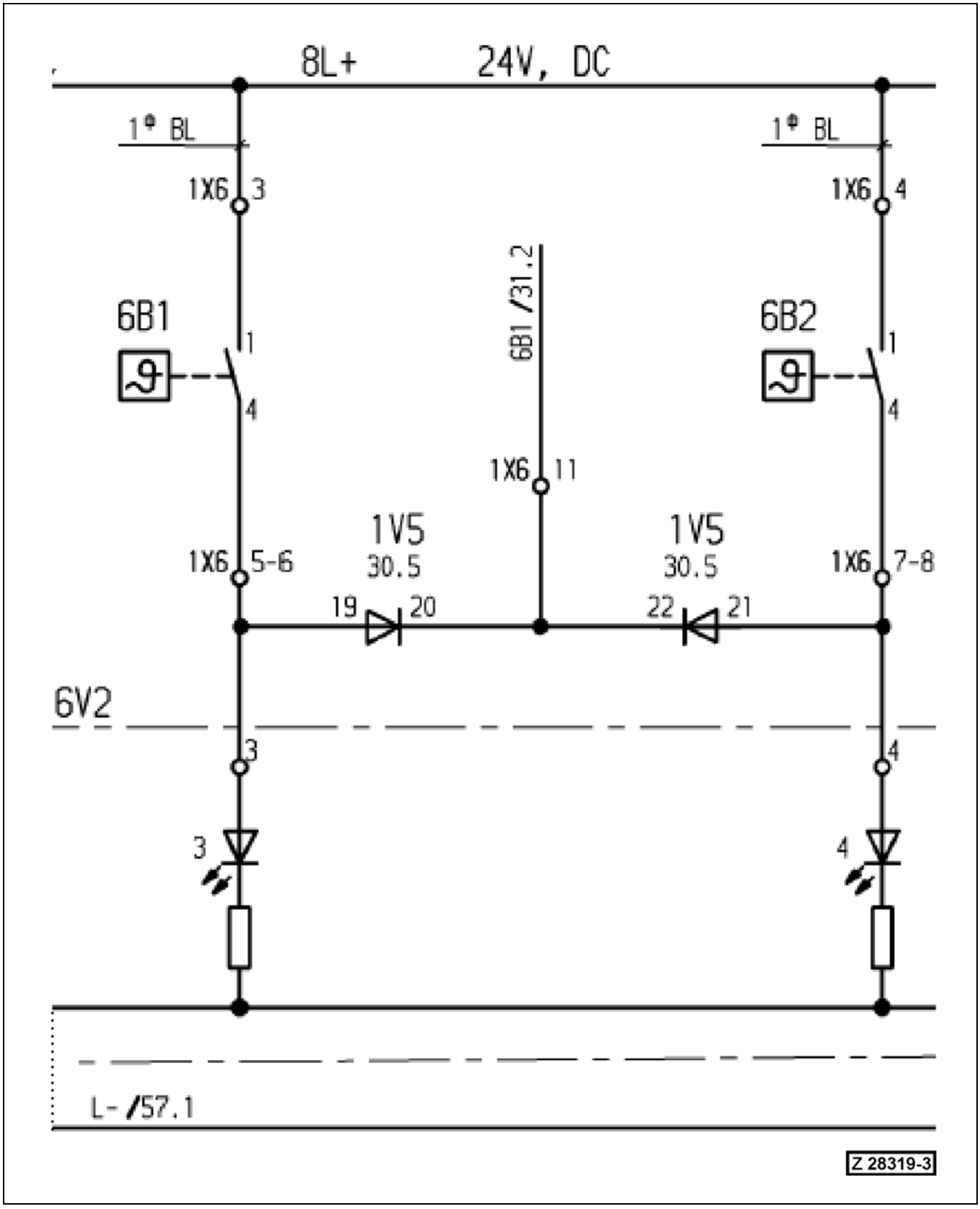


Wiring Diagram 1 (heater units 3R1 and 3R25) related to trouble message -40-

[3/4]



Wiring Diagram 2 (temperature sensors 6B1 and 6B2) related to trouble message -40- [4/4]



MESSAGE -41- MOTOR SHUTDOWN: ENGINE TEMPERATURE TOO HIGH (1K6)

[1/5]

Trouble code **Message page 41**	Trouble	Motor shutdown: Engine temperature too high (1K6)
Contents of trouble	- Motor shutdown due to overtemperature at the motor winding. - At least one of the temperature sensors at the motor winding detects overtemperature.	
Related information	- Temperature sensors 0B1-1, 0B1-2, 0B1-3 (PT100) are connected to control module 1A3. - Temperature sensors 0B2-1, 0B2-2, 0B2-3 (PT100) are connected to control module 1A3-1. - In case of overtemperature the control modules 1A3 and 1A3-1 de-energize relay 1K6 (closed-circuit principle). - Switch point = 140°C, hysteresis = 5°C, reaction time = 0 seconds - At 140°C the PT100 resistance is 153.582 Ω - In case of overtemperature warning light #6 at the LED monitor module 6V1 is ON.	
Problem that appears on machine	- Motor shutdown. - Motor restart is only possible if the winding temperature drops below 135°C.	

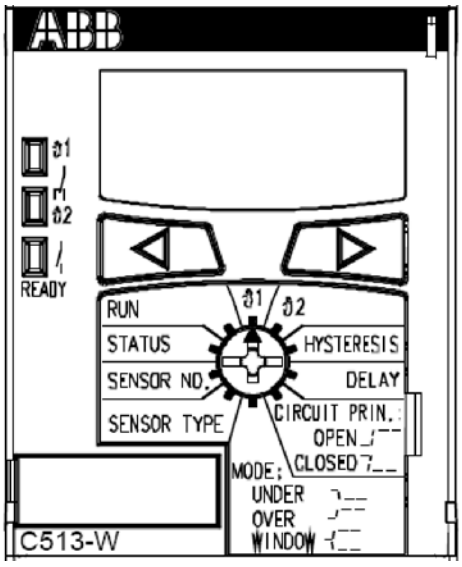
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor winding temperature is too high	1	 **HAZARDOUS VOLTAGE!** **High voltage is applied at the machine.** **There is danger of serious injury or death.** **Access to the high voltage system is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** If for example the net voltage is afflicted with highs peaks, which the SPAM can not absorb, components of the E-motor may be damaged. As a result the winding temperature may exceed the standard temperature range.  **Call the ABB service for troubleshooting!** - Components of the E-drive need to be checked in order to locate and to eliminate the reason for the increased winding temperature.			
The wiring between sensors 0B1-1/2 /3, 0B2-1/2/3 and the input at modules 1A3/1A3-1 is faulty	2	► Main key switch 1S1 OFF.			
		Measure the wires for continuity. Do all wires have continuity?	Yes	Go to next step.	R_{max} 1Ω
			No	Refer to the Wiring Diagram and repair or replace the faulty wiring as necessary.	

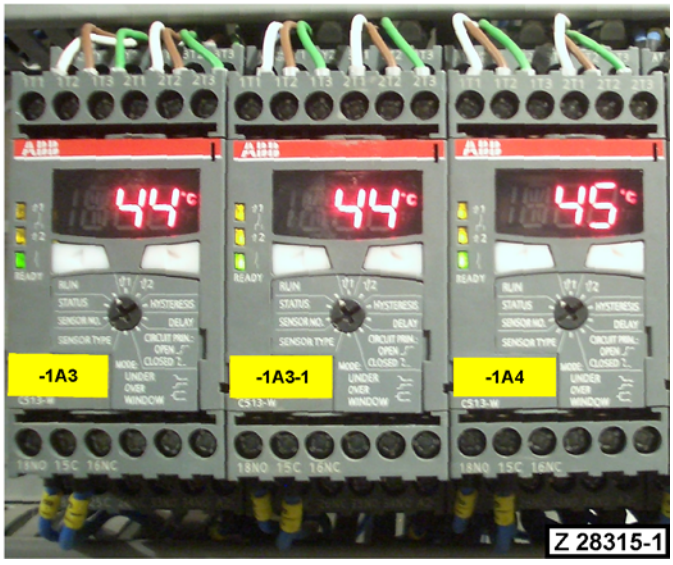
Message -41- Motor shutdown: Engine temperature too high (1K6)

[2/5]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Module 1A3 or 1A3-1 or its wiring is faulty	3	► Main key switch 1S1 ON.			
		Check if the LED at the module is ON.	Yes	Go to next step.	22 - 24V
		If the LED (READY) is ON, supply voltage (24V) and GND are present at terminals A1 and A2. If not, repair ground connection and/or power connection to 1F6 as necessary. Check the second power input at terminal 18. Is supply voltage present?	No	Refer to the Wiring Diagram and repair or replace the faulty wiring as necessary.	
a) Basic setting of modules 1A3 and/or 1A4-1 is wrong. b) Relay 1K6 or its wiring is faulty.	4	► Main key switch 1S1 ON.			
		During normal operation condition 24V should be present at the module terminal 15 (closed-circuit principle). Are 24V present at terminal 15?	Yes	Replace relay 1K6 (contacts are stuck). Also check the wiring and repair if necessary.	
			No	Check the basic setting of modules 1A3/1A3-1, and re-set if necessary. For adjustment hints refer to P&S NEWS AH09519b.	
Final check	5	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.			

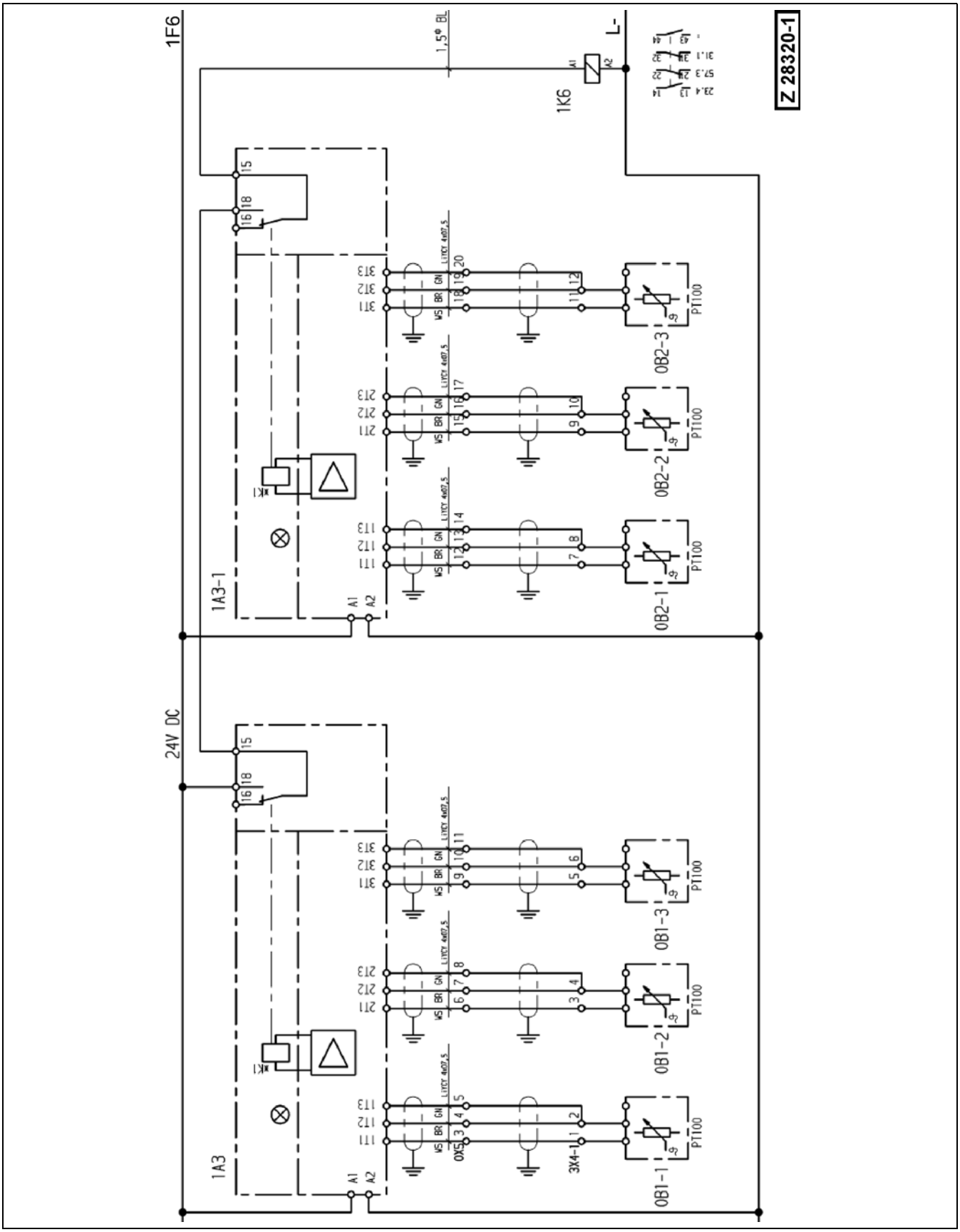
View onto the temperature control modules (1A3 and 1A3-1 for winding temperature)





Wiring Diagram 1 (winding temperature sensors and control modules) related to trouble message -41-

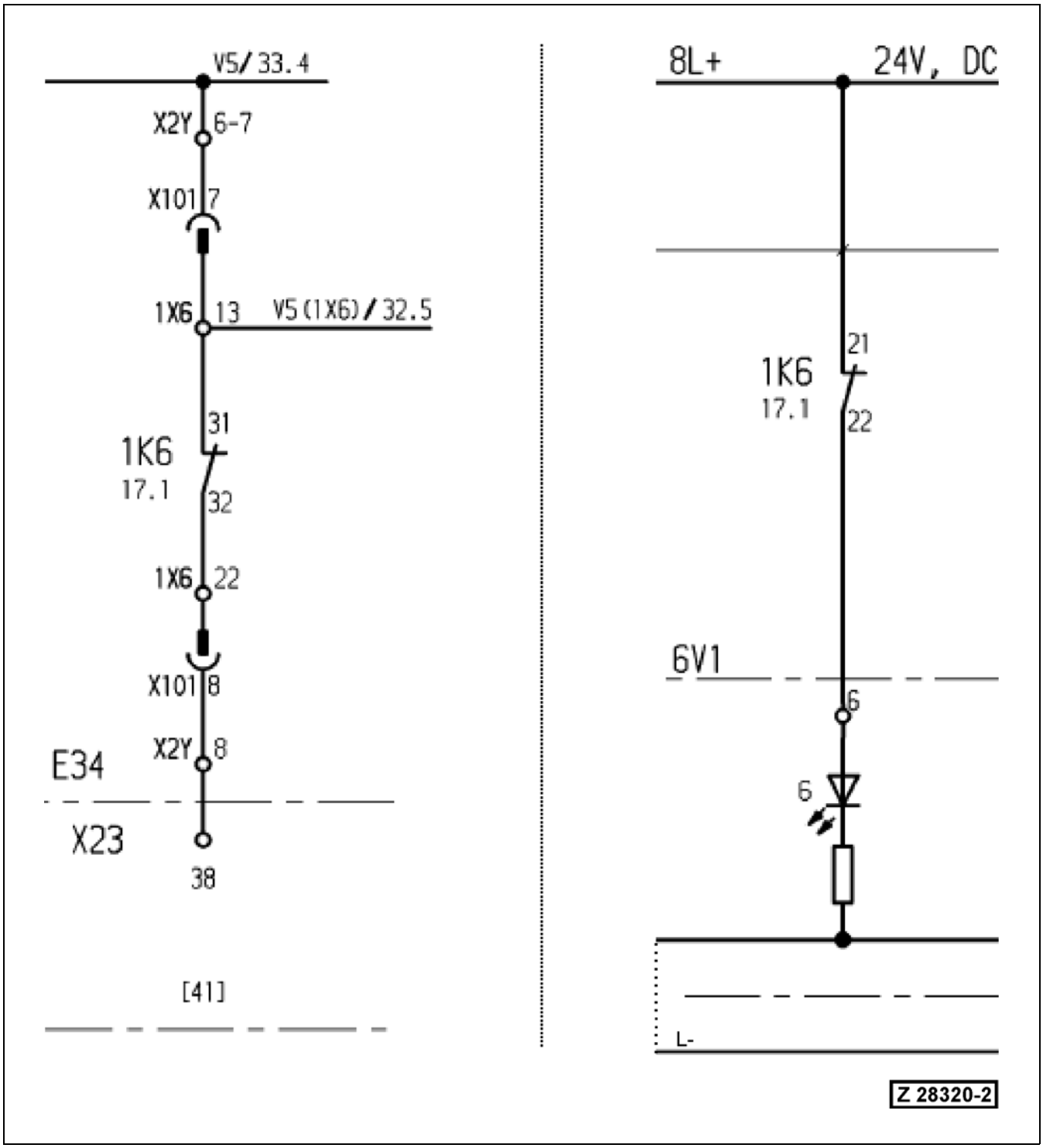
[3/5]



NOTE: Module adjustment in overtemperature mode and closed-circuit principle.

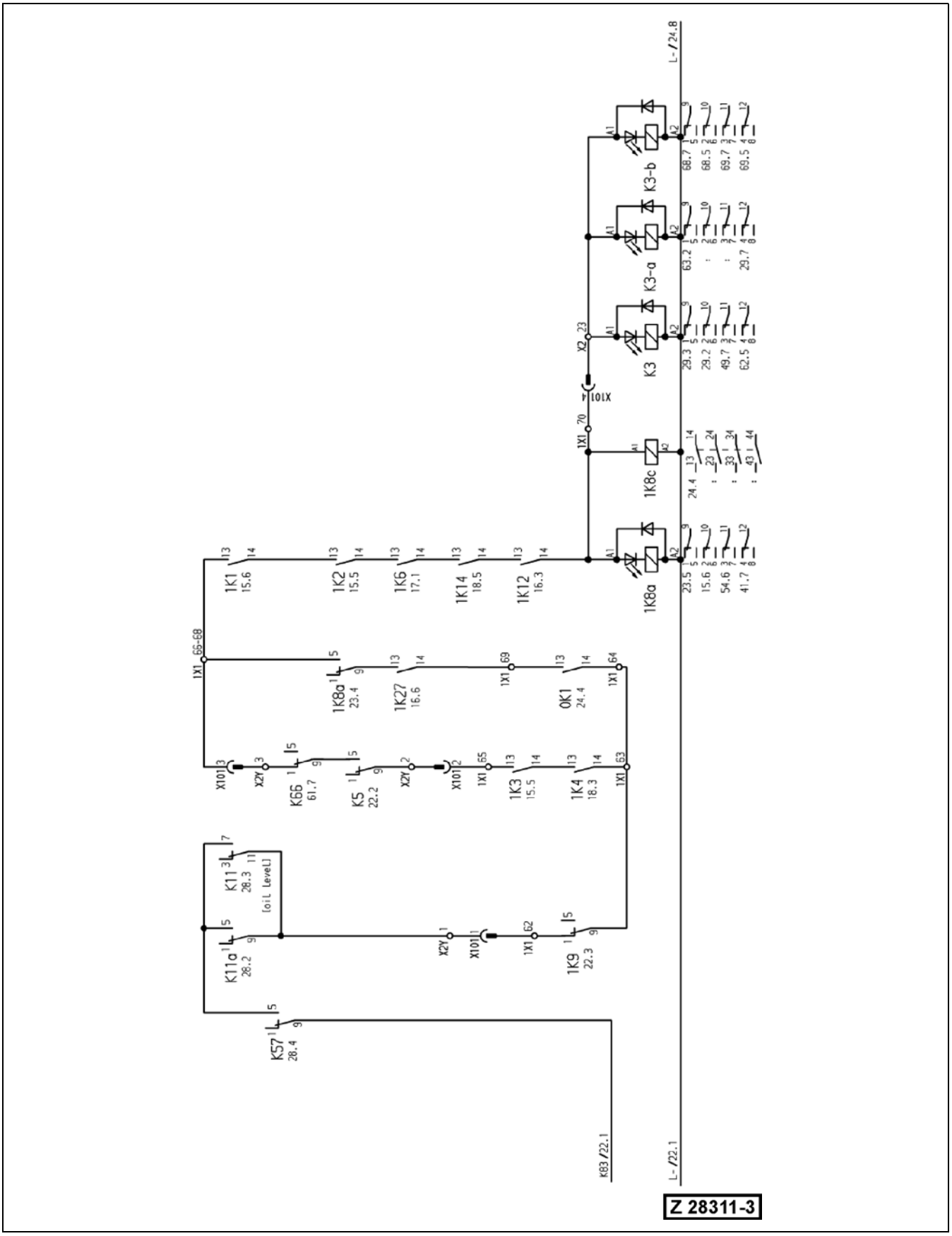
Wiring Diagram 2 (signal input at E34 and 6V1) related to trouble message -41-

[4/5]



Wiring Diagram 3 (1K6 in the motor start/stop and interlocking circuit) related to trouble message -41-

[5/5]



MESSAGE -42- (NOT USED)

MESSAGE -43- MOTOR-SHUTDOWN: EMERGENCY STOP SWITCH ACTUATED

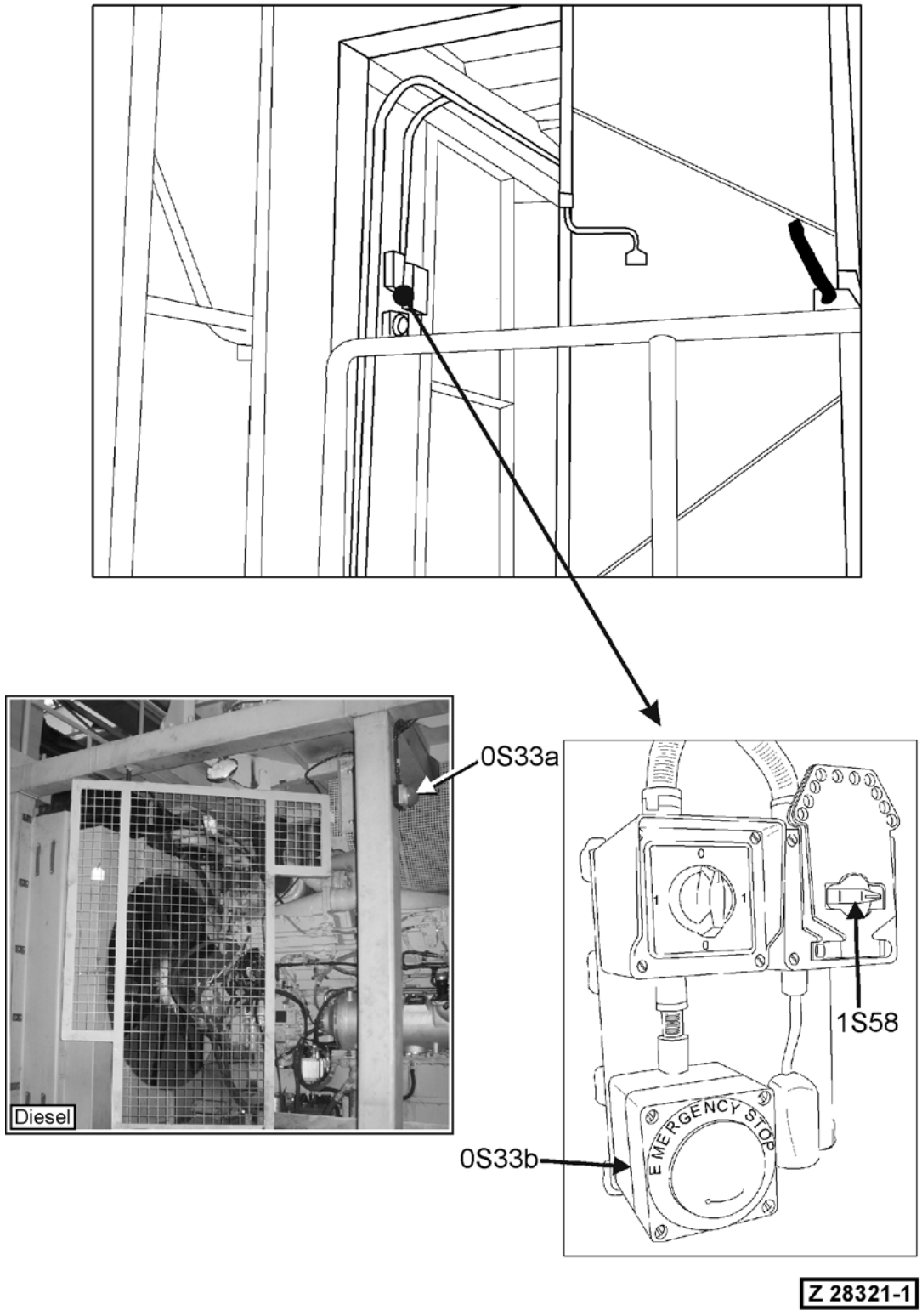
[1/4]

<table>
<tr><td>Trouble code
Message page 43</td><td>Trouble</td><td>Motor-shutdown: Emergency stop switch actuated</td></tr>
<tr><td>Contents of trouble</td><td colspan="2">• Motor shutdown or start blocked due to actuated emergency shutdown switch.
<table>
<tr><th>Emergency shutdown switch</th><th>Location</th></tr>
<tr><td>0S33a</td><td>Machinery house</td></tr>
<tr><td>0S33b</td><td>Machinery house entrance</td></tr>
<tr><td>0S36</td><td>Operator's cab</td></tr>
</table></td></tr>
<tr><td>Related information</td><td colspan="2">• Motor shutdown
• Motor start blocked</td></tr>
<tr><td>Problem that appears on machine</td><td colspan="2">• It is necessary to find the reason why the relating emergency shutdown switch was actuated.</td></tr>
</table>

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Any emergency shutdown button was activated	1	Check the cause for the emergency activation. Is any cause present?	Yes	Rectify the cause or the problem. After that release the concerned emergency shutdown switch(es).	
			No	Go to next step.	
The concerned emergency shutdown switch or its wiring may be defective	2	Visually inspect the concerned switch, the switch contact, and the respective wiring.		Refer to the Wiring Diagram. Repair or replace the concerned switch or its wiring as necessary. After repair go to final step 3.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

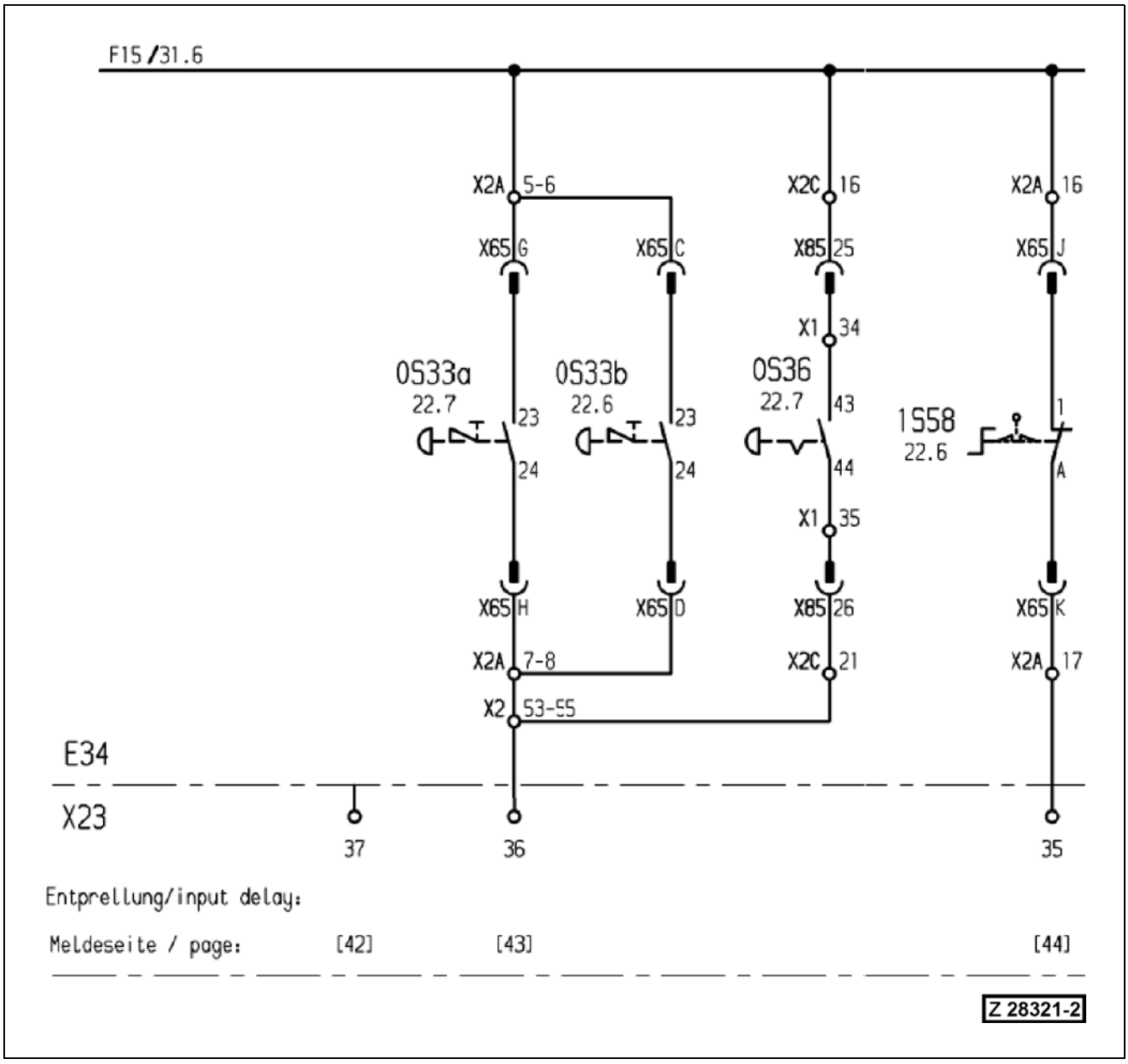
Message -43- Location of emergency stop switches in the machinery house

[2/4]



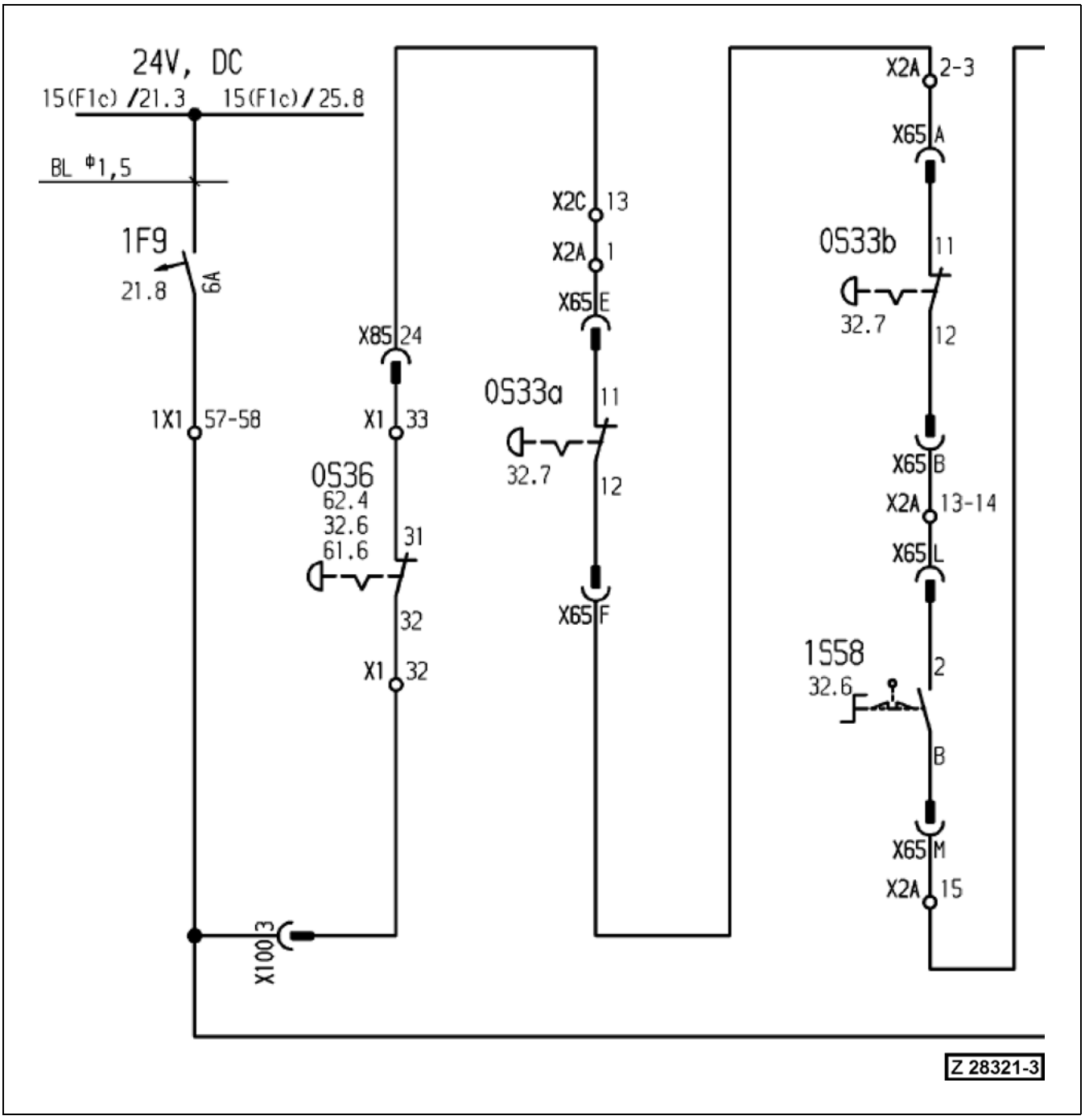
Wiring Diagram 1 (signal input at E34) related to trouble message -43-

[3/4]



Wiring Diagram 2 related to trouble message -43-

[4/4]



MESSAGE -44- MOTOR-SHUTDOWN: SAFETY SWITCH ACTUATED (1K58)

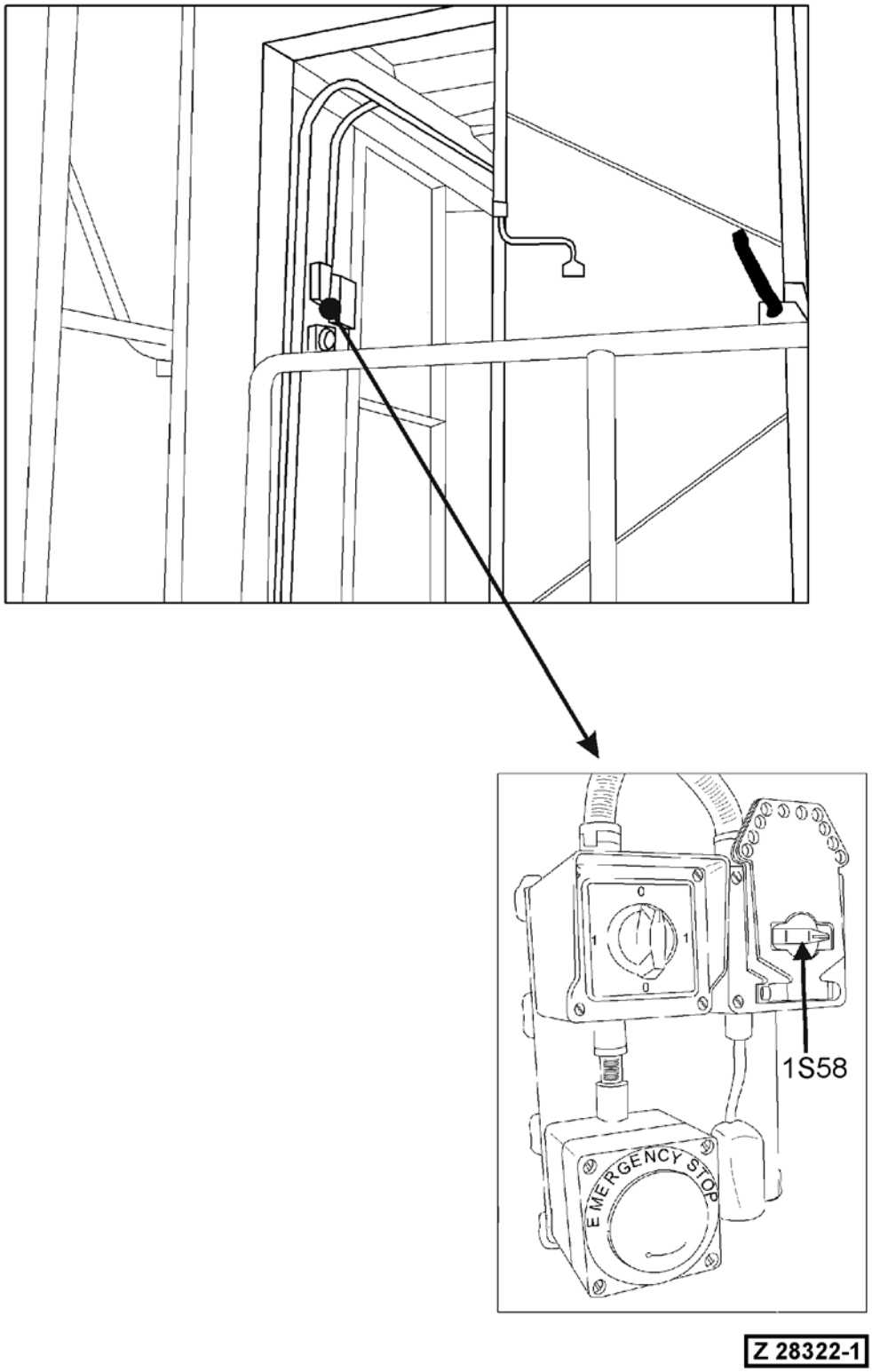
[1/4]

Trouble code **Message page 44**	Trouble	Motor-shutdown: Safety switch actuated (1K58)
Contents of trouble	• Motor shutdown or start blocked due to actuated maintenance safety switch 1S58. Maintenance safety switch: 1S58 — Location: Machinery house entrance	
Related information	• The motor is stopped because of emergency shut off via the maintenance safety switch 1S58. • Switch 1S58 is the maintenance safety switch which can be secured with padlocks.	
Problem that appears on machine	• Motor shutdown. • Motor start blocked. • It is necessary to find the reason why the maintenance safety switch was actuated.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The maintenance safety switch 1S58 has been switched off intentionally	1	► Keep the main key switch 1S1 in the OFF position			
		Check the condition of the maintenance safety switch 1S58. Is the maintenance safety switch 1S58 switched OFF?	Yes	Contact your supervisor and clarify the reason why 1S58 is switched off.	
			No	Go to next step.	
Malfunction of the maintenance safety switch 1S58 or its wiring	2	► Main key switch 1S1 in the OFF position			
		Check the function of the maintenance safety switch 1S58 and its wiring. Could you identify any trouble?	Yes	Switch 1S58 Repair or replace as necessary. Wiring of 1S58 Refer to the Wiring Diagram. Locate the problem and repair as necessary.	
			No	Check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

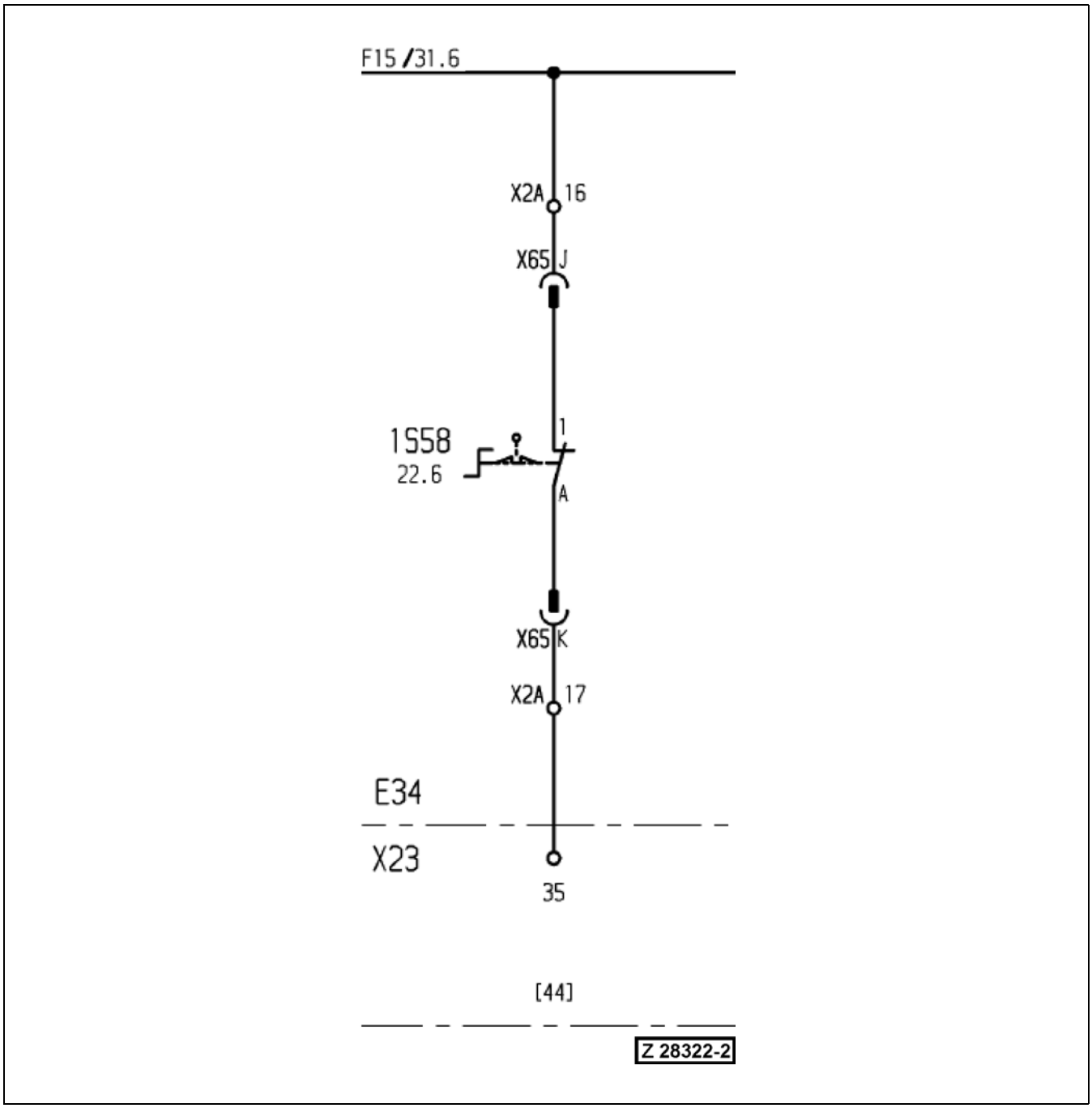
Message -44- Location of safety switch 1S58 in the machinery house

[2/4]



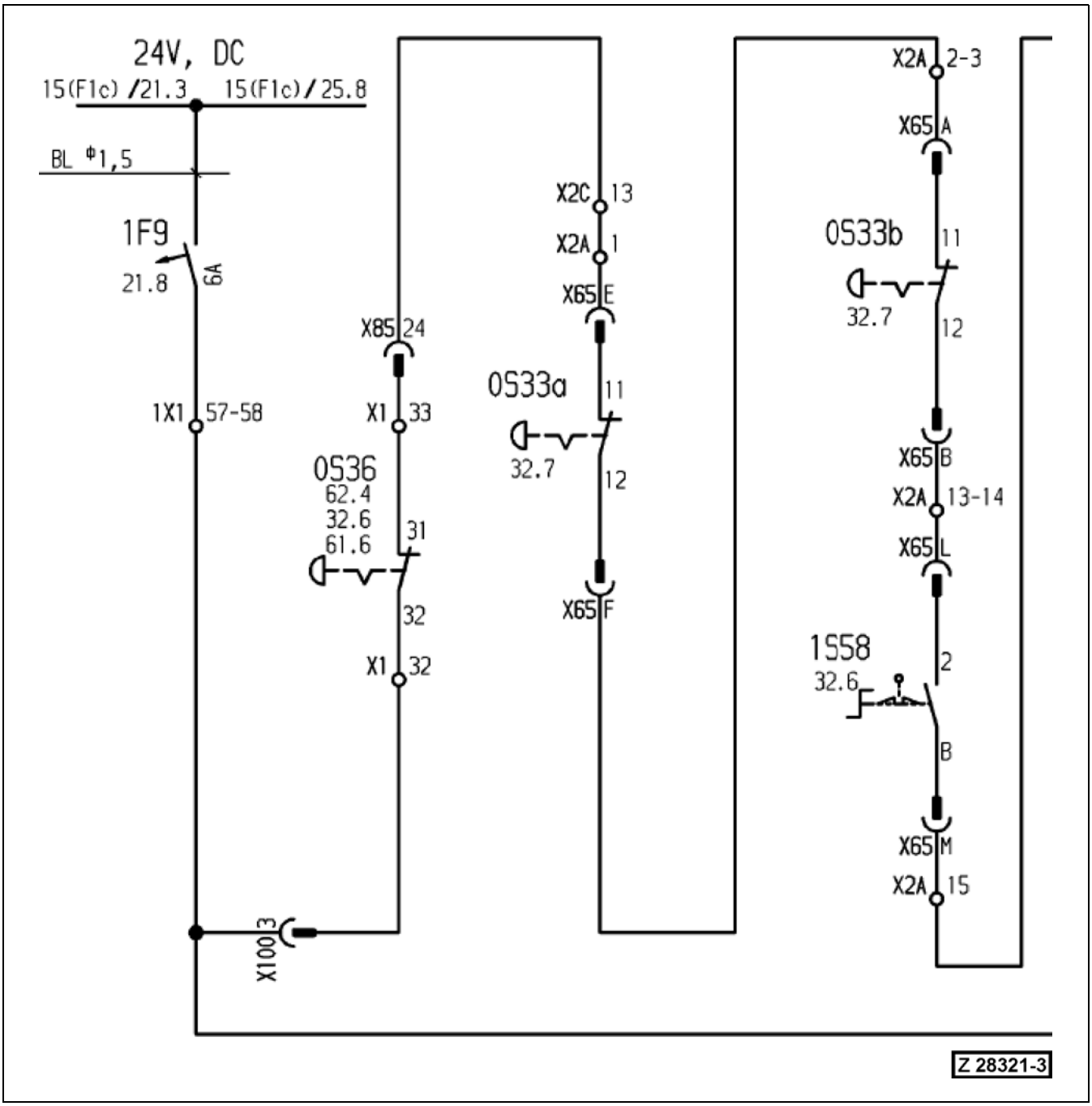
Wiring Diagram 1 (signal input at E34) related to trouble message -44-

[3/4]



Wiring Diagram 2 (safety switch 1S58) related to trouble message -44-

[4/4]



MESSAGE -45- MOTOR-SHUTDOWN: CAPACITOR FAILED (1K12)

[1/4]

Trouble code **Message page 45**	Trouble	Motor-shutdown: Capacitor failed (1K12)
Contents of trouble	- Sensor 0S1 or 0S2 at the capacitor detected a capacitor failure. - 0S1 and 0S2 are NC switches (normally closed). - In case of a capacitor failure the switch is opened by high pressure. - Motor shutdown is triggered by relay 1K12.	
Related information	The capacitor assembly operates as an *operating capacitor* to generate the phase displacement for the E-motor during motor start sequence, in order to compensate the apparent power.  **High voltage is applied to the motor capacitors!** **HAZARDOUS VOLTAGE!** **There is danger of serious injury or death.** **Access to the capacitors is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.**	
Problem that appears on machine	- Motor shutdown. - Motor re-start blocked.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Capacitor failed	1	► Main key switch 1S1 OFF.			
		 **Call the HV department for troubleshooting!** **HAZARDOUS VOLTAGE!** **There is danger of serious injury or death.** **Access to the capacitors is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** **The grounding procedure is necessary to discharge the capacitor.** **Disconnection from the supply must only be performed by authorized personnel.** Does the capacitor inspection confirm the displayed trouble?	Yes	Capacitor needs replacement by the HV department.	
			No	Go to next step.	
The sensor wiring may be defective (disconnection or short circuit)	2	► Main key switch 1S1 OFF.			
		- Disconnect the sensor connectors. - Check the wiring for continuity. Is continuity present?	Yes	Reconnect the sensor connectors and go to next step.	R_{max} 1Ω
			No	- Refer to the Wiring Diagram and locate the trouble. - Repair or replace the wiring as necessary. - Recheck the wiring for proper condition.	

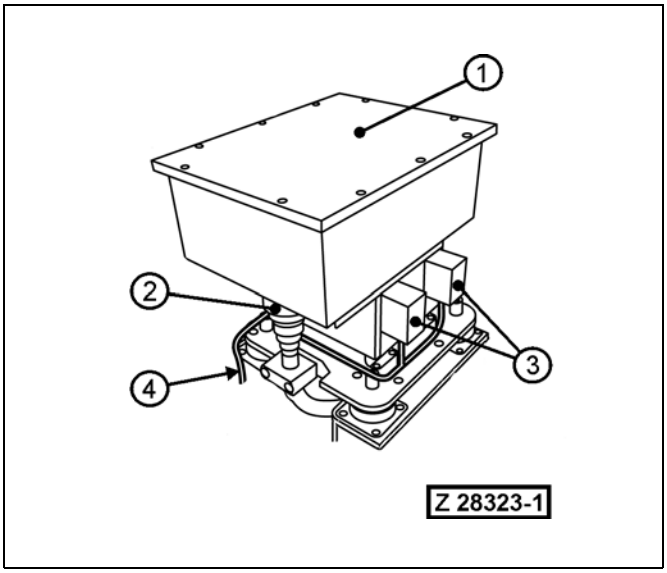
Message -45- Motor-shutdown: Capacitor failed (1K12)

[2/4]

Possible cause	Step	Test condition & Inspection	Procedure	Nominal value
Relay 1K12 or its wiring defective. The relay contacts may stuck.	3	► Main key switch 1S1 ON.		
		Measure the voltage between relay terminals A1 and A2. Provided that 1F6 delivers 24V and A2 is connected to GND, standard operating voltage (24V) should be present. Otherwise the wiring or the relay is defective. - Check and repair the wiring if necessary (refer to the Wiring Diagram). - If the wiring is OK, replace the defective relay 1K12.		22 - 24 V
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the trouble code does not arise again.		

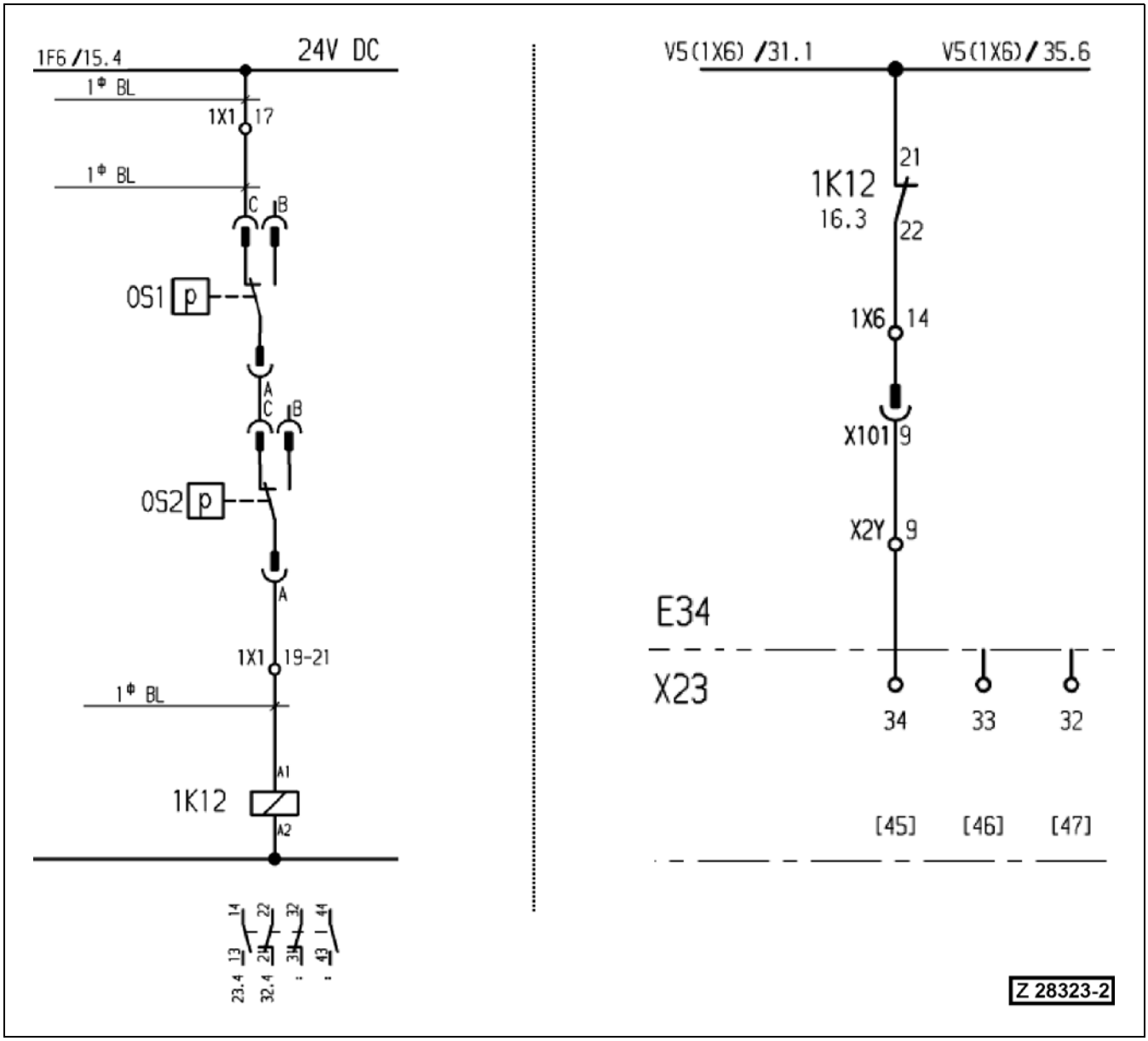
Exemplary view onto a capacitor

1. Capacitor (2 plates)
2. Main power cable gland
3. Pressure sensor (capacitor monitoring)
4. Cable for pressure sensors



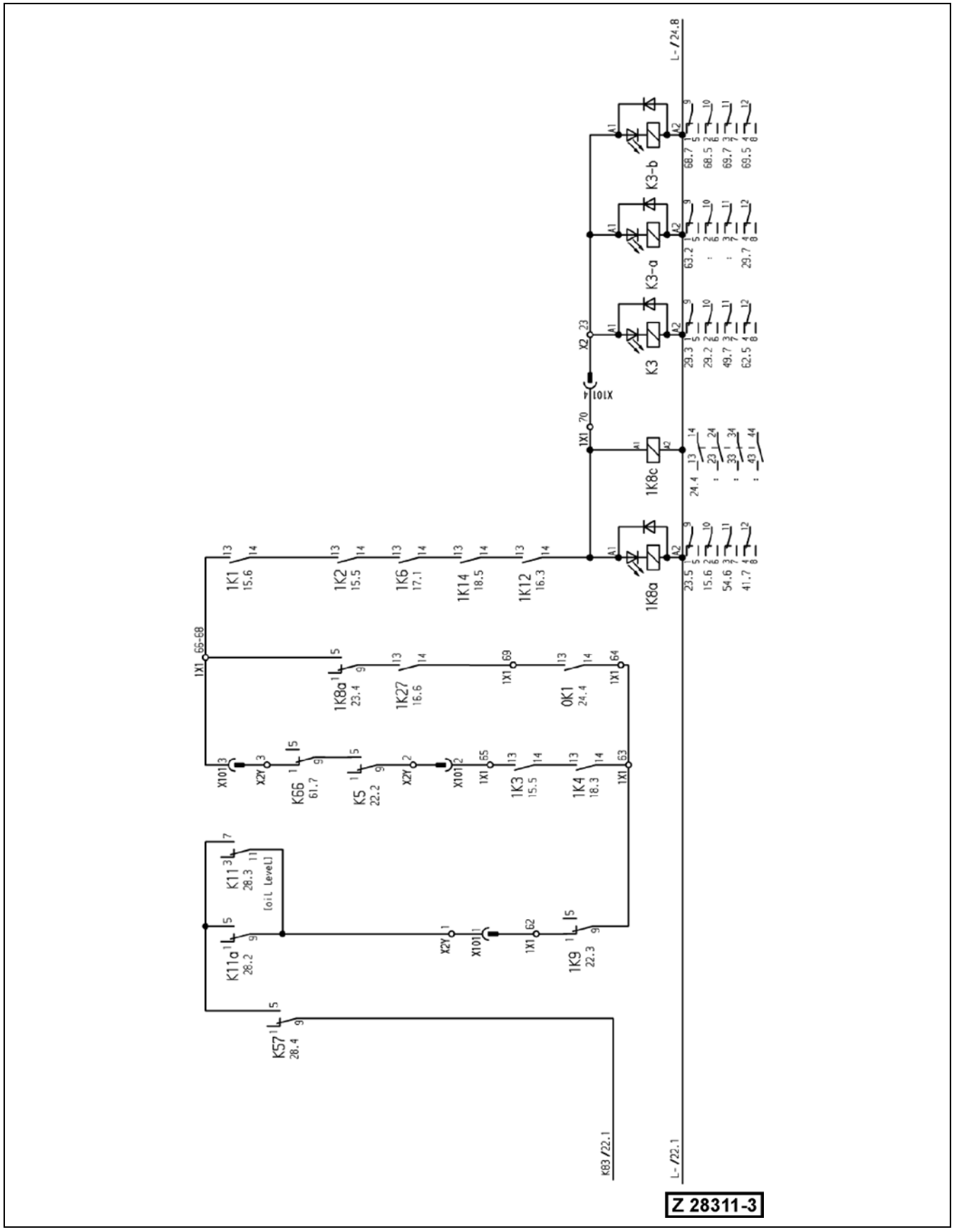
Wiring Diagram 1 (sensors 0S1 & 0S2 and signal input of 1K12 at E34) related to trouble message -45-

[3/4]



Wiring Diagram 2 (1K12 in the motor start/stop and interlocking circuit) related to trouble message -45-

[4/4]



MESSAGE -46- (NOT USED)

MESSAGE -47- (NOT USED)

MESSAGE -48- TOO HIGH PTO-GEAR OIL TEMPERATURE (B49)

[1/3]

Trouble code **Message page 48**	Trouble	Too high PTO-gear oil temperature (B49)
Contents of trouble	• Overheated gear oil is detected.	
Related information	• The PTO gear oil temperature sensor B49 detects an oil temperature above the maximum permissible operating temperature as shown in the temperature chart below. • A problem with the gear oil temperature sensor B49 (PT100 type) or its wiring may also cause the trouble, see table [2/3] for testing procedure of the signal wiring and connections. For more information see also the illustration on page [3/3].	
Problem that appears on machine	• The gear oil is overheated. • Lubrication capability of the gear oil is reduced. Risk of PTO gear damage. • No safety or cut-off function.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The PTO gear oil is overheated	1	► Keep the motor running, with max. fan speed.			
		Recheck the gear oil temperature at the temperature sensor B49, see illust. Z27883 using an infrared temperature gauge. For the maximum permissible operating temperature refer to the temperature chart below. Does the measured gear oil temperature indicate overheating as well?	Yes	Go to next step.	
			No	Check for other related faults starting with step 3.	
PTO gear oil level too high	2	Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked.			
		Check the PTO gear oil level with the dipstick as described in the OPERATION & MAINTENANCE MANUAL. Is the PTO gear oil level within the specified range?	Yes	Go to next step.	OPERATION & MAINTE-NANCE MANUAL
			No	Drain PTO gear oil until the target level is reached. Then monitor the PTO gear oil temperature again.	
Wrong setting of temperature switch point at switch unit K43 in relation to the gear oil viscosity	3	► Motor OFF and main key switch 1S1 in OFF position.			
		Check the temperature setting in relation to the gear oil viscosity used in the machine. Values are given in the temperature chart below. Are the oil viscosity settings correct?	Yes	Continue with step 4.	
			No	Adjust K43 to the correct switching point. Refer to the SERVICE MANUAL and to the PM-CLINIC sheet for setting instructions. Verify that trouble message -48- is not set again.	

Temperature Chart: adjust switch unit K43 to the respective oil viscosity grade as follows:

Oil viscosity grade according to ISO	T32	CLP 150	75W-140
-K43- Adjust temperature limit value to switching point °C	+60°	+95°	+95°

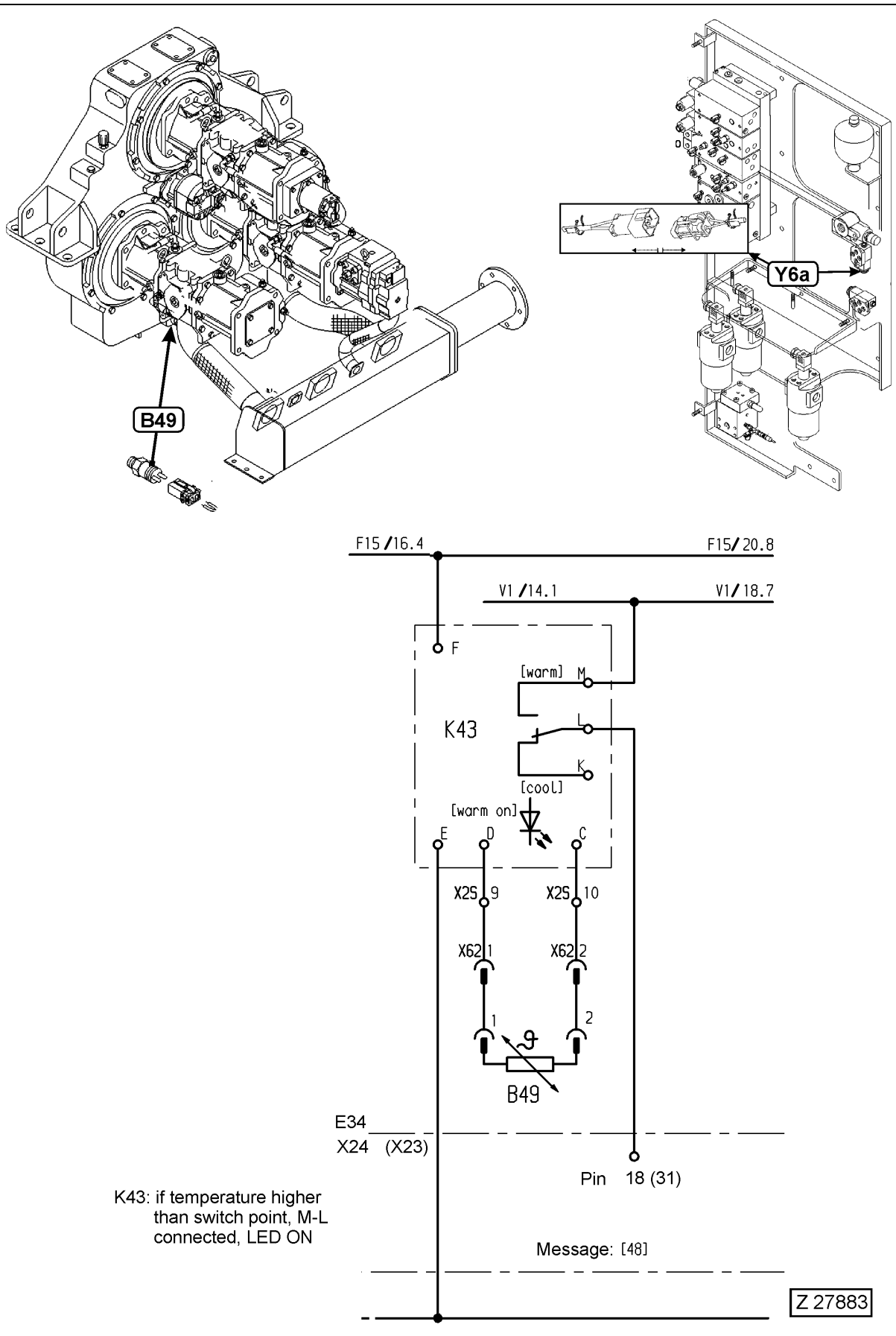
Message -48- Too high PTO-gear oil temperature

[2/3]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Defective temperature sensor B49 (PT100 type)	4	Check temperature sensor B49 according to the Standard Value Table in this section. Is the test result OK?	Yes	Go to next step.	
			No	Replace temperature sensor. After that recheck that the trouble condition does no longer exist.	
The hydraulic oil cooler fans are not driven with maximum speed (gear oil coolers are part of the hydraulic oil coolers)	5	► Keep the motor running.			
		Maximum fan speed is achieved if no voltage (0 V) is applied at solenoid valve Y6a, see illust. Z27883. Therefore disconnect the cable connector from solenoid valve Y6a. If the fan speed does not increase, check the hydraulic settings for the fan speed (refer to the SERVICE MANUAL). Connect cable connector to the solenoid valve Y6a. Is the setting for max. fan speed correct?	Yes	Go to next step.	
			No	Refer to the SERVICE MANUAL, section Hydraulic Oil Cooling, and perform the setting of the max. fan speed. If the setting is not as specified follow the adjustment and/or inspection procedures in the SERVICE MANUAL as necessary.	RPM max. 1250 ± 50
Any soiled gear oil cooler segment of the hydraulic oil coolers caused the overheating of the gear oil	6	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked.			
		⚠ This test step must only be performed with the motor at standstill and maintenance safety switch in OFF position and locked. Perform a visual inspection of the oil coolers. Also open the swing out doors of the cooler frame and check for deposits between these door wings. Is intense pollution present?	Yes	Clean the oil coolers with a pressure cleaner (max. 60 bar) as necessary.	
			No	Go to next step.	
Overheating due to any malfunction in the PTO lubrication system	7	► Keep the motor running.			
		Refer to the SERVICE MANUAL, chapter 2, section *PTO Lubrication and Cooling* and follow the given adjustment procedures. Are the adjustments as specified?	Yes	Go to next step.	
			No	Carry out adjustments as necessary. After that verify that the PTO gear oil temperature is in normal range.l	
Final check	8	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Message -48- Sensor B49 and signal input of K43 at E34

[3/3]

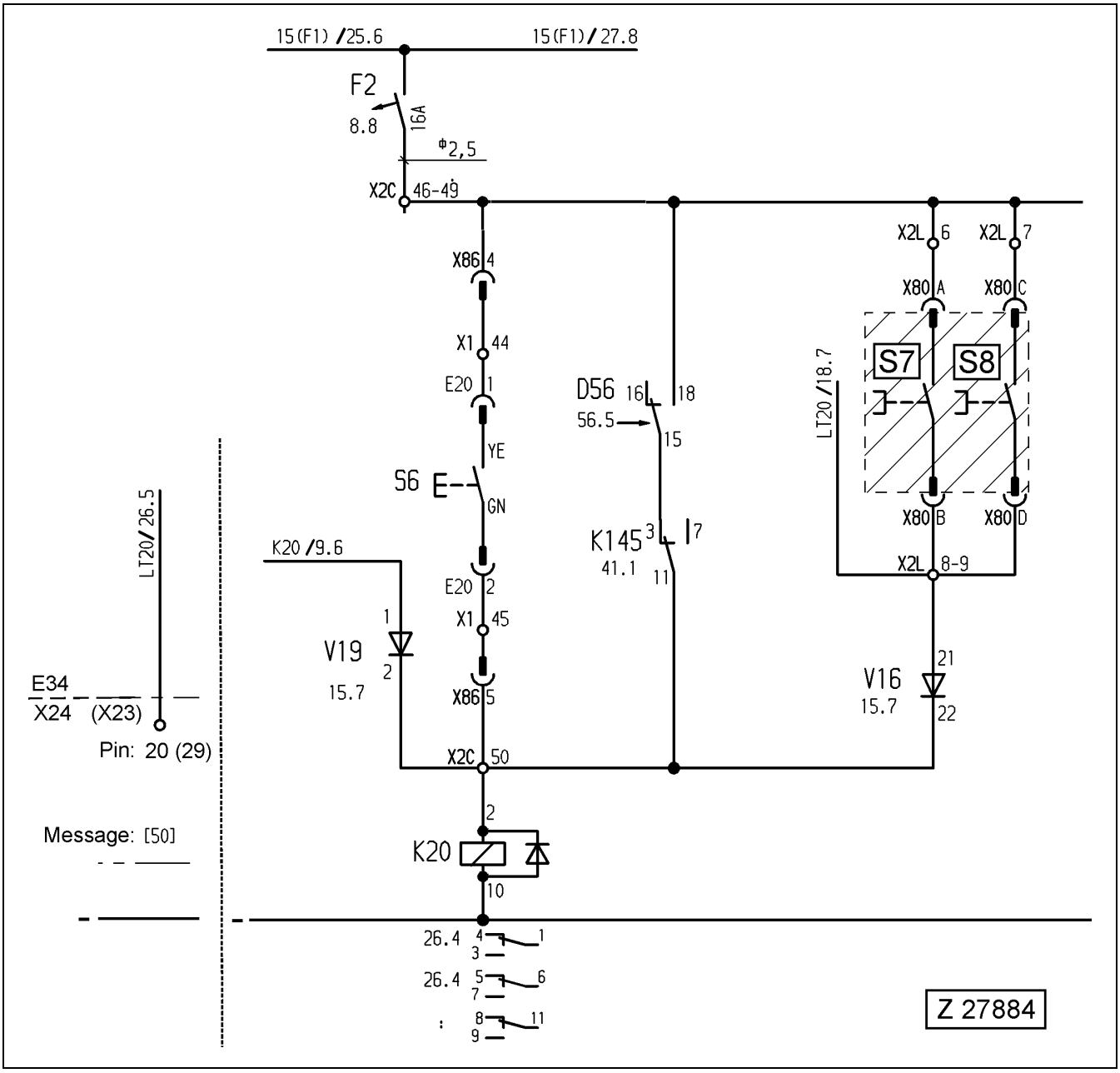


MESSAGE -50- CAUTION, PULL SWITCH FROM GROUND MAN ACTUATED

[1/1]

Trouble code **Message page 50**	Trouble	Caution, pull switch from ground man actuated
Contents of trouble	• The operator warning system has been actuated from the ground man via switch S7 and/or S8 (if equipped).	
Related information	• Via switches S7/S8 (optional equipment) this signal can be set from ground manually in order to activate an information message for the operator. • For resetting pull switch S7/S8 again.	
Problem that appears on machine	• The operator is asked to contact the ground man.	

Wiring Diagram related to trouble message -50-



MESSAGE -51- MOTOR SHUTDOWN: PULL SWITCH FROM GROUND MAN ACTUATED

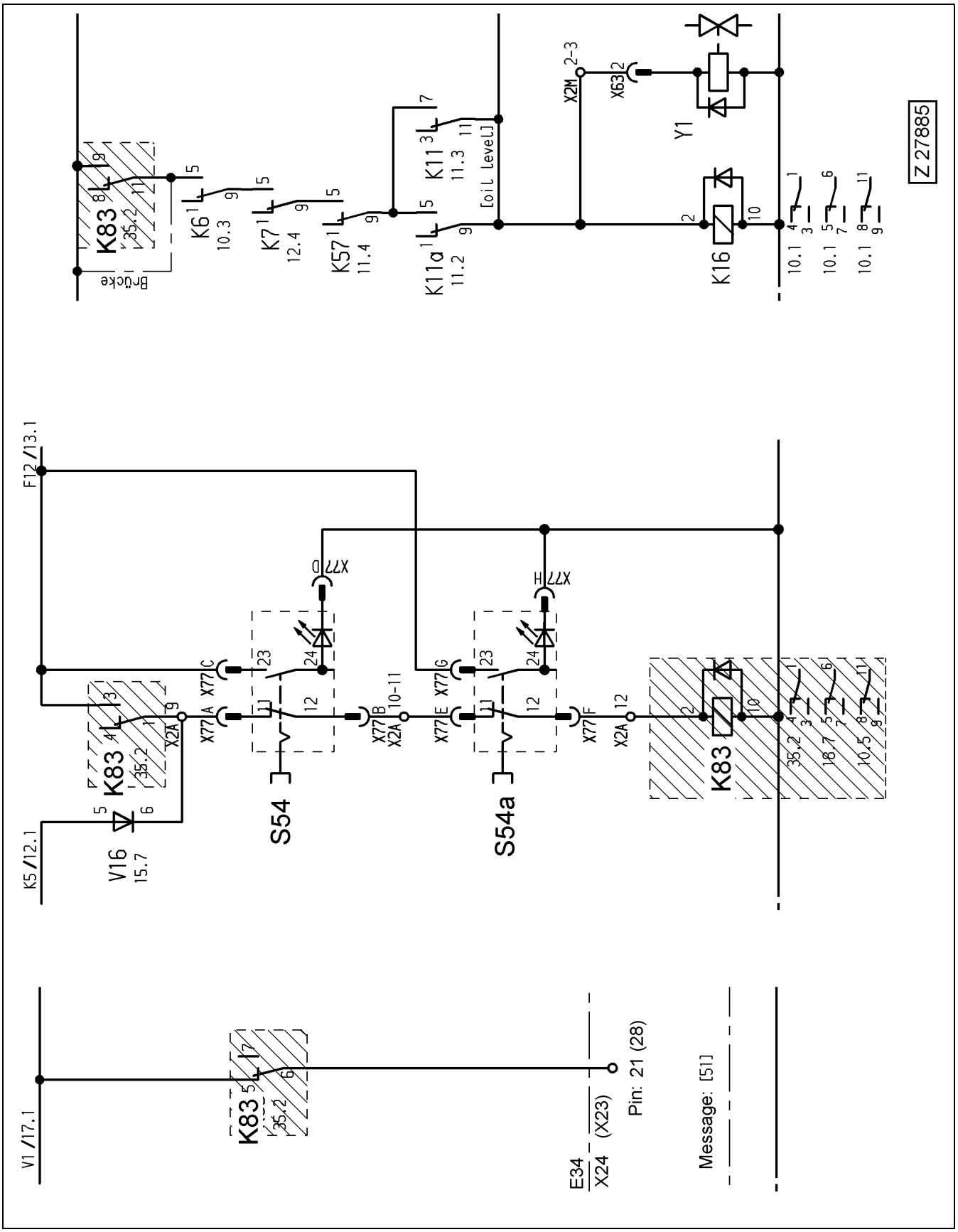
[1/2]

Trouble code **Message page 51**	Trouble	Motor shutdown: pull switch from ground man actuated
Contents of trouble	• The motor has been stopped manually from the ground man via switch S54 and/or S54a (if equipped).	
Related information	• If switch S54 or S54a is activated from ground while the motor is running, the relay K83 shuts down the motor. • After switch S54 or S54a has been activated from ground under the condition motor stopped & MAIN KEY SWITCH 1S1 ON, motor start is not possible.	
Problem that appears on machine	• The motor is shut off and restart is not possible until switch S54 and/or S54a is manually switched back into neutral position.	

<table>
<tr><th>Possible cause</th><th>Step</th><th>Test condition & Inspection</th><th></th><th>Procedure</th><th>Nominal value</th></tr>
<tr><td rowspan="3">The motor shutdown was manually activated from ground</td><td rowspan="3">1</td><td colspan="3">► Keep the main key switch 1S1 in ON position.</td><td></td></tr>
<tr><td>Check the cause for the manually activated motor shutdown.
Was switch S54 and/or S54a manually activated from ground?</td><td>Yes</td><td>Contact your supervisor and clarify the situation. After the cause for the motor shutdown is clarified, actuate switch S54 and/or S54a again and restart the motor.</td><td></td></tr>
<tr><td>No</td><td>Go to next step.</td><td></td></tr>
<tr><td rowspan="3">No input signal at the input signal module E34.</td><td rowspan="3">2</td><td colspan="3">► Keep the main key switch 1S1 in ON position.</td><td></td></tr>
<tr><td>Actuate switch S54/S54a and verify if its signal is present at the pin 21 of the input signal module E34 (Master) of ETM version -A- or at the pin 28 of the Input signal module E34 (CAN-node) of ETM version -B-.
<table><tr><th>Relay K83</th><th>Voltage</th></tr><tr><td>Contacts 8 - 11 → closed</td><td>24V</td></tr><tr><td>Contacts 8 - 11 → open</td><td>0V</td></tr></table>Does the signal of switch S54/S54a correspond to the above table?</td><td>Yes</td><td>Switch S54/S54a may have been activated unintentionally.
Go to final step 3.</td><td></td></tr>
<tr><td>No</td><td>Refer to the wiring diagram. Check switch S54/S54a and its wiring.
Locate the problem and repair as necessary. After repair recheck switches S54 and S54a for proper function.</td><td></td></tr>
<tr><td>Final check</td><td>3</td><td colspan="3">Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.</td><td></td></tr>
</table>

Wiring Diagrams related to trouble message -51-

[2/2]



MESSAGE -52- LOAD LIMITING CLAM DEFECTIVE

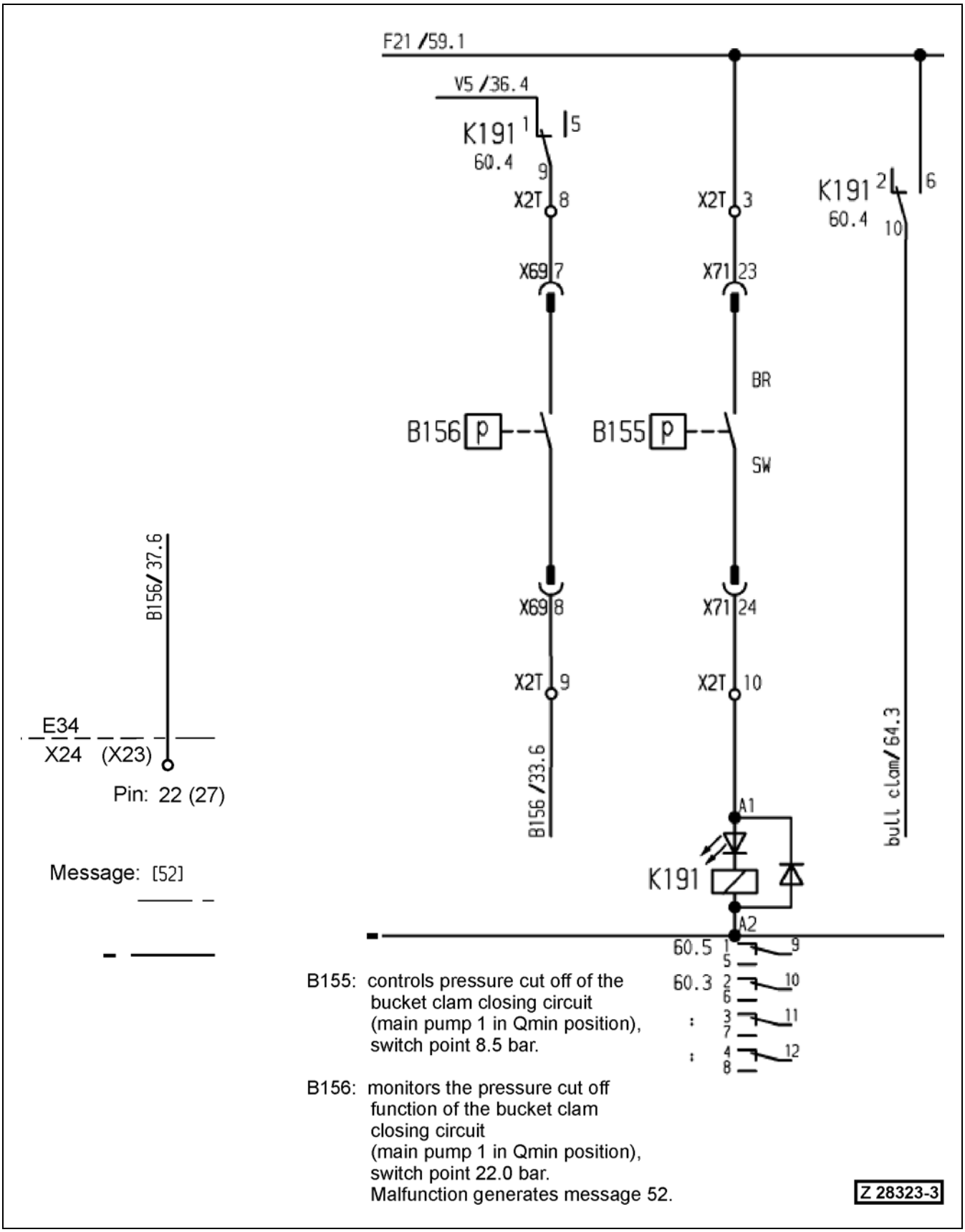
[1/3]

Trouble code **Message page 52**	Trouble	Load limiting clam defective
Contents of trouble	• Pressure switch B156 detects a malfunction in the clam closing circuit. (Face shovel machines only)	
Related information	• Pressure switch B156 monitors the pressure cut-off function of the bucket clam closing circuit (main pump 1 in Qmin position), switch point 22 bar. Malfunction generates message no. 52. • Pressure switch B155 controls the pressure cut-off of the bucket clam closing circuit (main pump 1 in Qmin position), switch point 8.5 bar.	
Problem that appears on machine	• Bucket clam may close to fast, creating strong shock loads when hitting the bucket back wall. • Irregular clam closing movement.	

Possible cause	Step	Test condition & Inspection	Procedure	Nominal value
Pressure switch B155 and/or B156 malfunction	1	► Motor OFF. Main key switch 1S1 OFF and then ON.		
		Check the clam closing circuit.	Refer to the SERVICE MANUAL and follow the adjustment and/or repair procedures as necessary. If OK go to final step 3.	
Pressure switch B155 and/or B156 and/or relay K191 or the belonging wiring defective	2	► Motor OFF. Main key switch 1S1 OFF or ON for testing.		
		<u>Sensors B155 & B156</u> - B155 & B156 are NO switches (normally open). - If the wiring is interrupted, the switch will close. - Also B155 or B156 may be stuck in *open* position. In these cases there is no cut off function.	Refer to the Wiring Diagram and check the wiring. Locate the wiring trouble and repair or replace as necessary.	
		<u>Relay K191</u> - The wiring of K191 may be interrupted. - K191 may be stuck in *open* position. In these cases there is no cut off function.	If the wiring is OK replace the defective pressure switch or the relay.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again and that the bucket clam closing function is OK.		

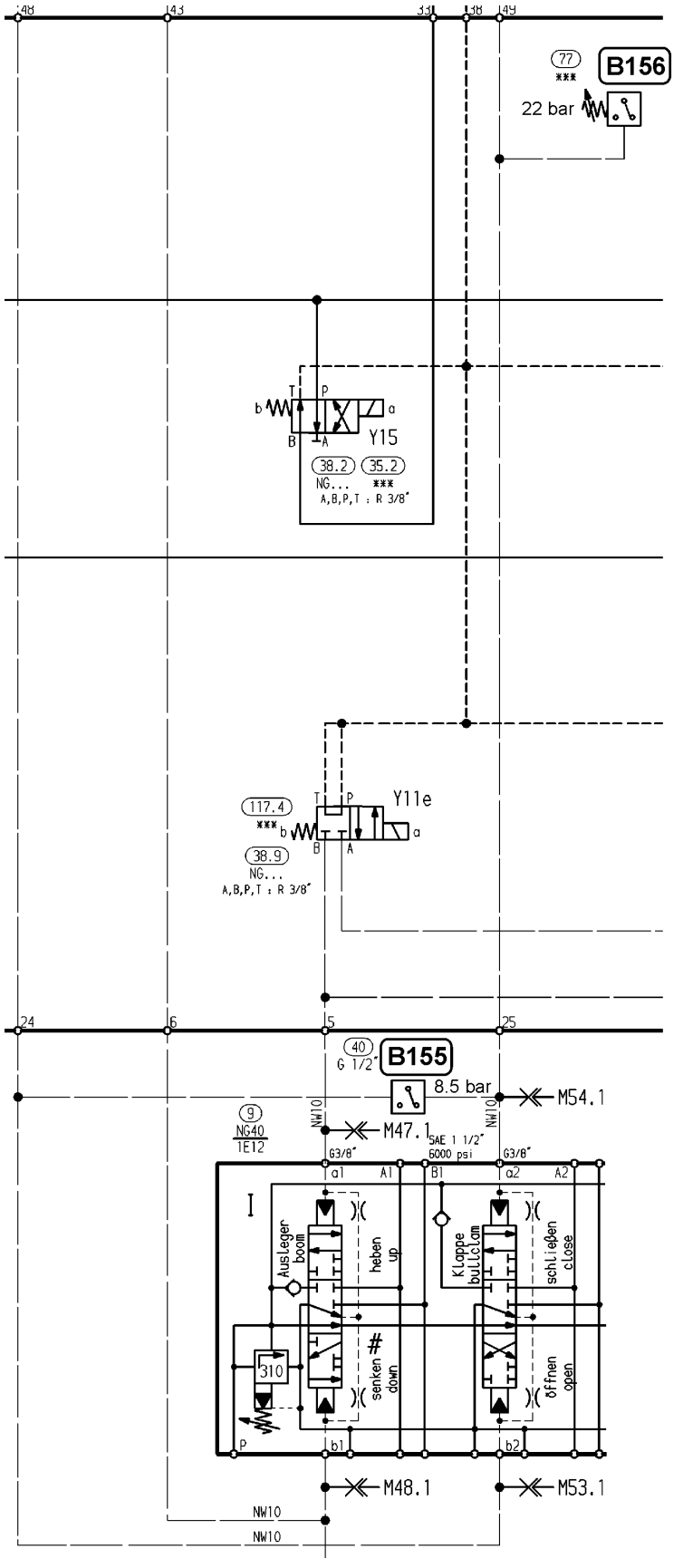
Message -52- Sensors B155 & B156, relay K191, and signal input at E34

[2/3]



Message -52- Hydraulic Diagram (B155 & B156)

[3/3]



MESSAGE -53- PRESSURE SWITCH OF THE CENTRAL LUBRICATION SYSTEM ACTUATED (B43)

MESSAGE -54- PRESSURE SWITCH OF THE SLEW RING GEAR LUBRICATION SYSTEM ACTUATED (B46)

[1/1]

Trouble code	Trouble	
Message page 53 **Message page 54**		Pressure switch of the central lubrication system actuated (B43) Pressure switch of the slew ring gear lubrication system actuated (B46)
Contents of trouble	• The grease pressure line is not released within the determined time frame.	
Related information	• Message 53: Refer to the Trouble Shooting Chart of Message -18- CENTRAL LUBRICATION SYSTEM FAULT for inspection and repair procedures. • Message 54: Refer to the Trouble Shooting Chart of Message -19- SLEW RING GEAR LUBRICATION SYSTEM FAULT for inspection and repair procedures.	
Problem that appears on machine	• Risk of damages due to insufficient lubrication.	

MESSAGE -55- REFILL HYDRAULIC OIL

[1/4]

Trouble code **Message page 55**	Trouble	Refill hydraulic oil
Contents of trouble	• Sensor B50 indicates that the hydraulic oil level is below the normal operating range.	
Related information	• If the sensor B50 does not sense oil, its signal voltage turns to 0 V. • A problem with the sensor or connected wiring may also cause the trouble, see table [2/4] for testing procedure of the signal wiring and connections. • For more information see also the illustrations on page [3/4 and 4/4].	
Problem that appears on machine	• Risk to run out of hydraulic oil	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The hydraulic oil level is below the normal operating level	1	► Check with stopped motor and the attachment in its defined position.			
		Verify if the hydraulic oil level in the main oil reservoir is above or below the oil level sensor B50, see illustration Z27888 on page 3 of this chart. Refer also to illustration Z28000 on page 4 of this troubleshooting chart, which shows the hydraulic oil tank label with the defined positions at the bottom of the label and the relevant oil levels on the left and right side of the label. Is the oil level below the sensor B50?	Yes	Refill specified hydraulic oil and verify that trouble message 55 is not displayed again.	
			No	Go to next step.	
Incorrect or no supply voltage in the circuit of the oil level sensor B50	2	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		Disconnect the cable connector from the oil level sensor B50, insert the T-box, and measure the supply voltage at the cable connector terminal A. Is supply voltage present?	Yes	Go to step 5.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
	3	Check if supply voltage is present at the power supply line F12. Is supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Disconnection or short circuit in any wire segment (including connectors) between sensor B50 and the main power supply from F1. Locate the problem and repair as necessary	

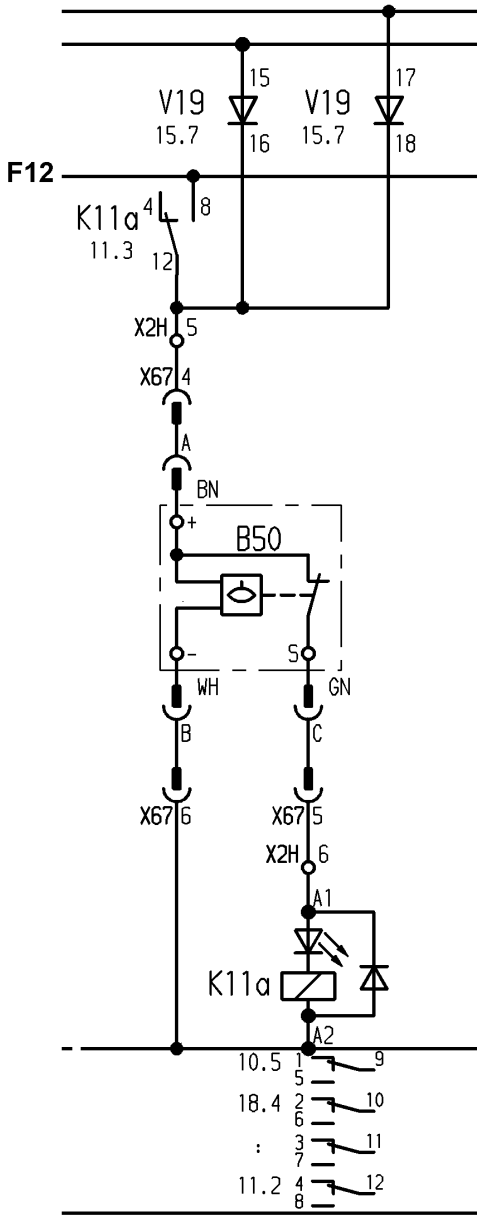
Message -55- Refill hydraulic oil

[2/4]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the wiring segments between the oil level sensor B50 and the power supply line of circuit breaker F12 (disconnection or defective contact in any connector)	4	► Keep the main key switch 1S1 in the OFF position.			
		Refer to the wiring diagram and check the wiring segments between oil level sensor B50 and the power supply line F12 for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment (including connectors) between oil level sensor B50 and the line of the circuit breaker F12. Locate the problem and repair as necessary.	
Defective ground connection at the oil level sensor B50	5	► Keep the main key switch 1S1 in the OFF position.			
		Disconnect the cable connector from the oil level sensor B50 and check for proper ground connection at the connector terminal B. Is the ground connection OK?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment (including connectors) between oil level sensor B50 and the ground line (-). Locate the problem and repair as necessary.	
Defective oil level sensor B50	6	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		Using the T-box measure the signal voltage at the connector terminal C of the oil level sensor B50 when the sensor is dipped into oil (sensor closed), and when it is not touching any oil (sensor open). Do the measurements meet the specifications?	Yes	A sporadic malfunction may exist. Further inspection is required.	V_{Signal} open: 0V V_{Signal} closed: 22 - 24V
			No	Replace the oil level sensor B50 and verify that the repair was successful. Go to next step.	
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Illustrations related to trouble message -55-

[3/4]



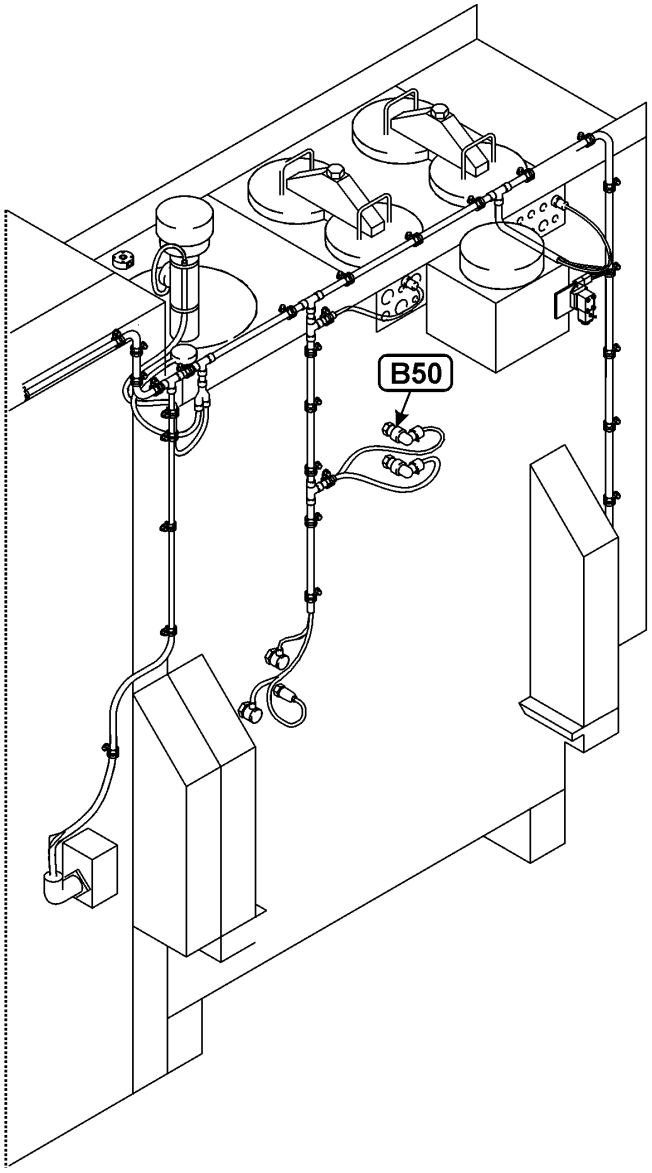
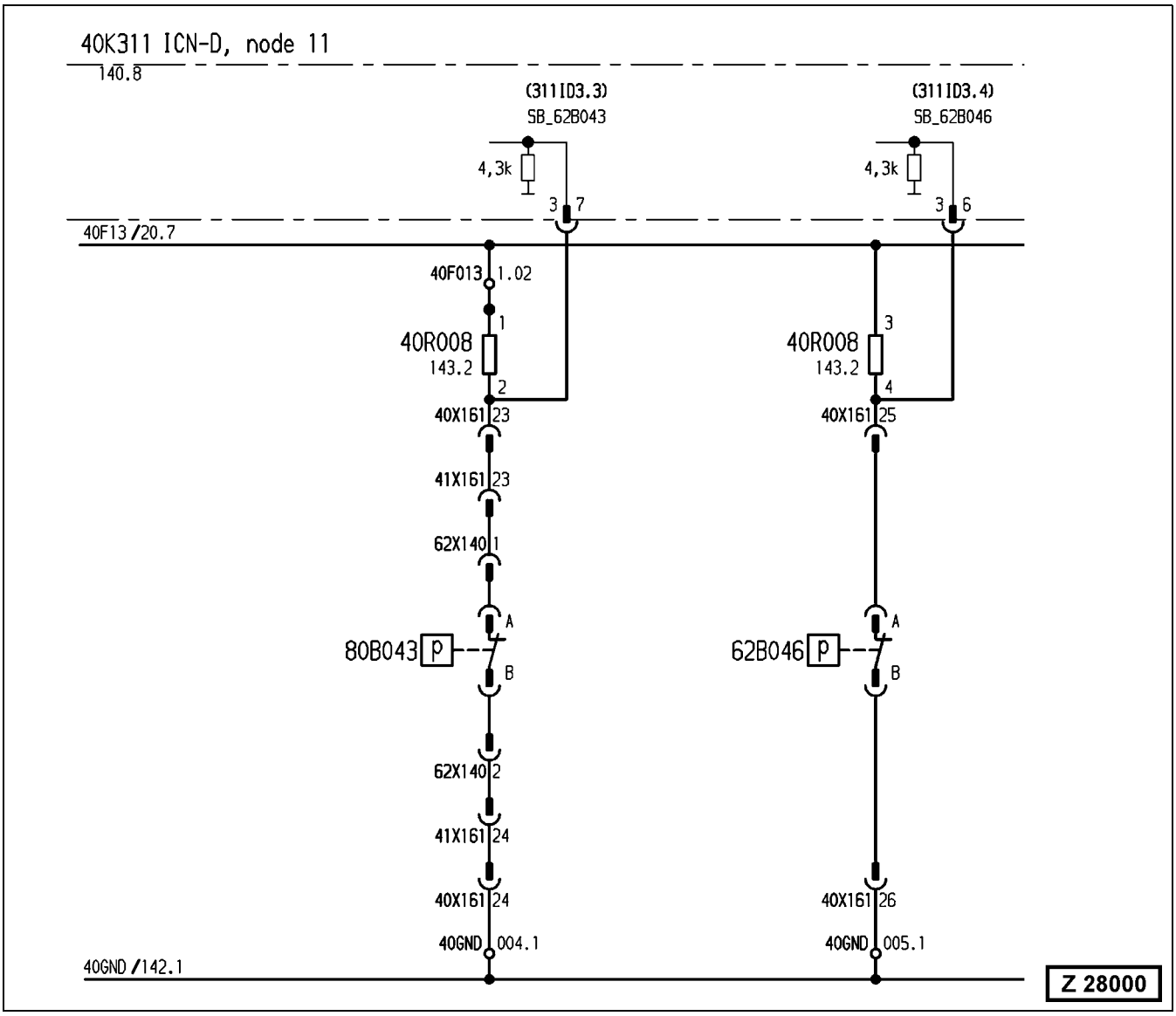


Illustration related to trouble message -55-

[4/4]



MESSAGE -56- PILOT CONTROL CUT OUT

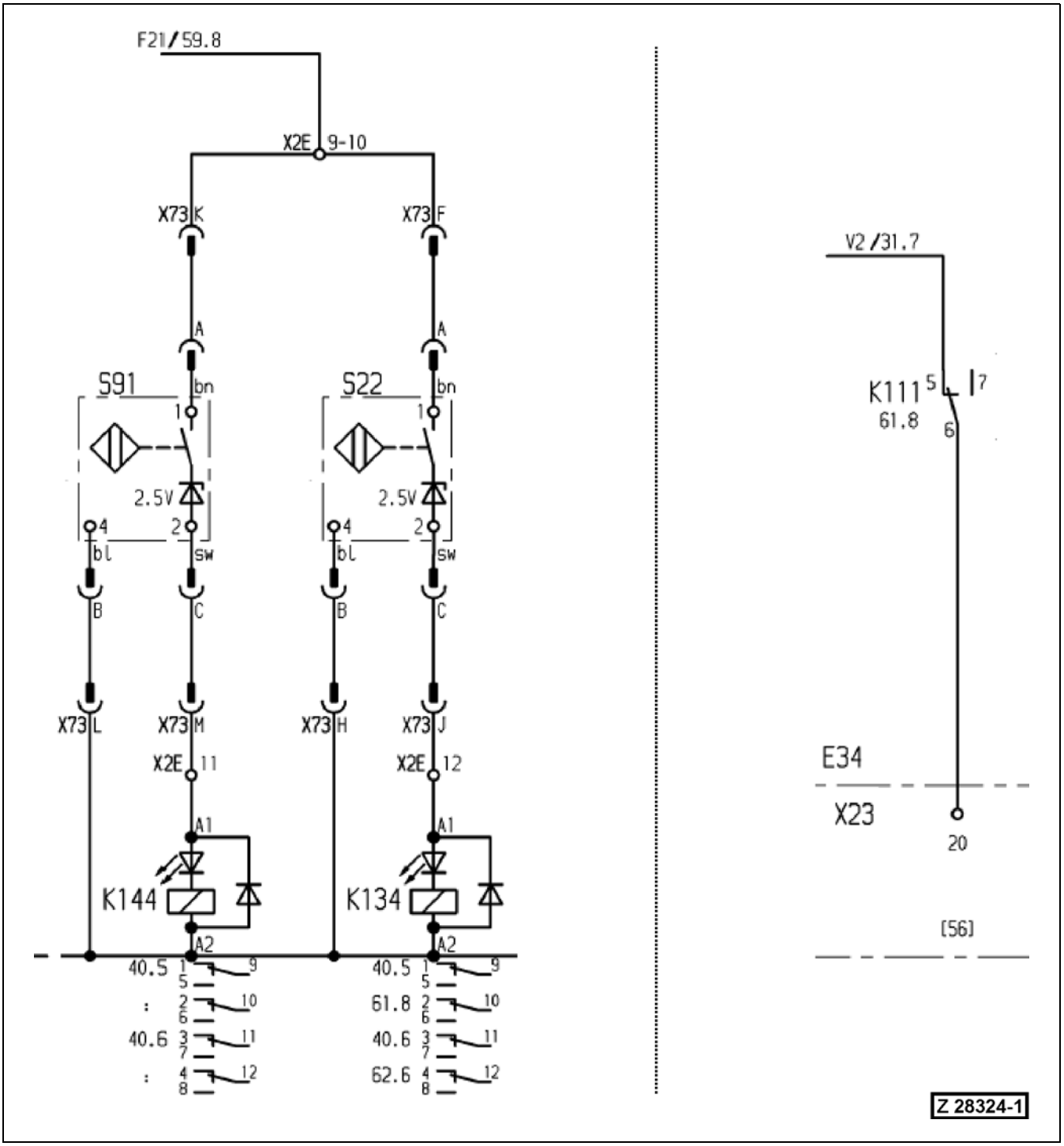
[1/4]

Trouble code **Message page 56**	Trouble	Pilot control cut out
Contents of trouble	Motor running - Pilot control cut out because lock lever (S105) in upper position. - Ladder in down position. Condition for motor start: - The lock lever must be in upper position (disabled), in order to allow K66 to close the safety circuit.	
Related information	- With the lock lever the pilot control is switch ON/OFF via relay K111. - The ladder circuit is controlled by K134 (ladder up signal). - If the ladder is down, pilot control is not available.	
Problem that appears on machine	- Pilot control function is not available. - Motor start not possible.	

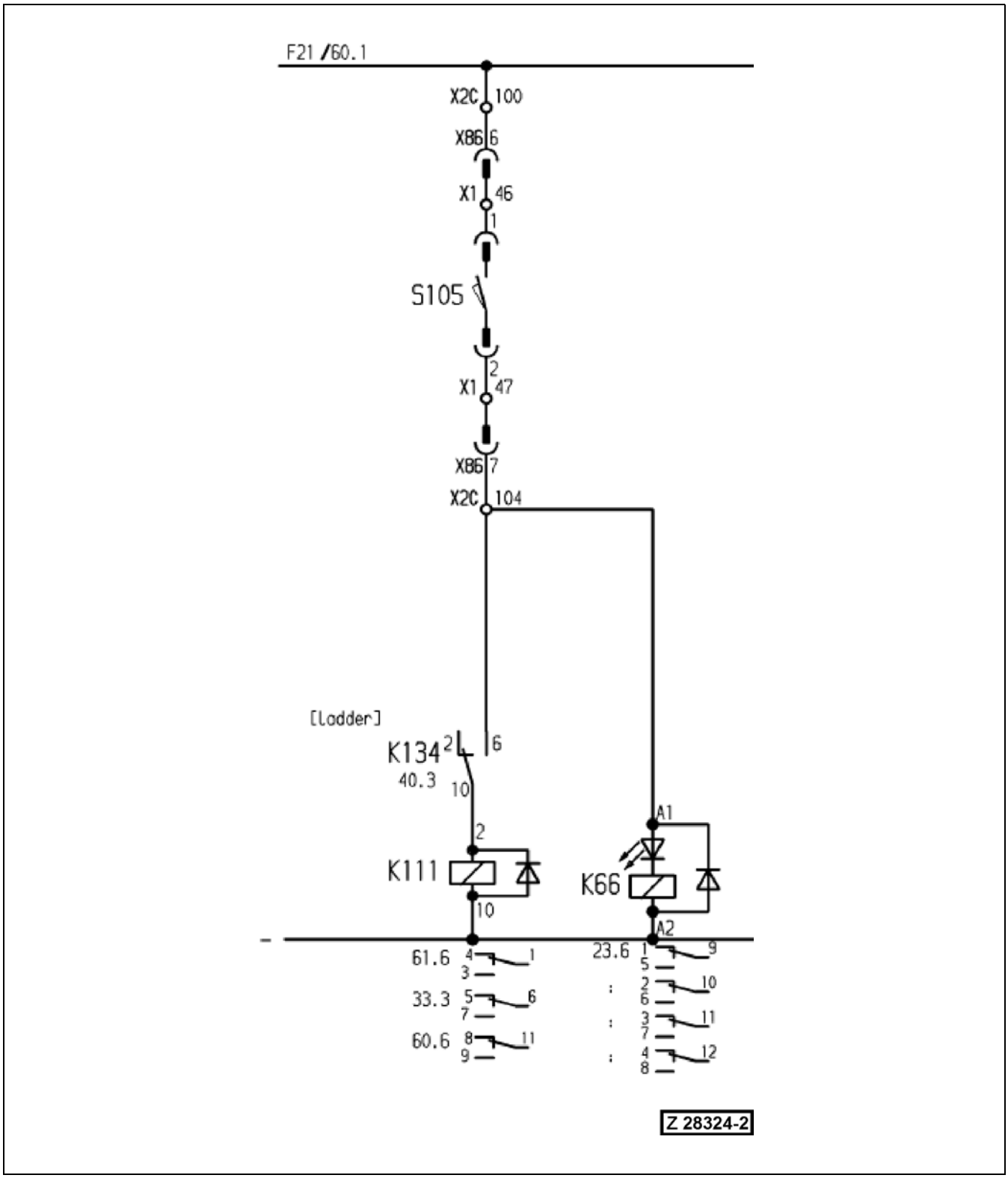
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
- Lock lever not down - Lock lever switch (S105) not closed - Wiring of S105 faulty	1	► Keep the motor running.			
		Move the lock lever down. Is pilot control available?	Yes	Faulty machine operation.	
			No	Refer to the Wiring Diagram: - Check lock lever switch S105 for proper function. - Check its wiring for proper condition. - Repair or replace the wiring as necessary or replace S105 if it is faulty.	
Ladder is down	2	► Keep the motor running.			
		Check the ladder position Is the ladder down?	Yes	Faulty machine operation.	Ladder up = S22 closed
			No	Check the circuit of the upper ladder end-switch (S22) leading to relay K134. If a problem is found, repair or replace faulty components as necessary.	
Relay K111 or its wiring defective (possibly the relay contacts are stuck)	3	► Keep the motor running.			
		No pilot control even the ladder is up and the lock lever is down. Refer to the Wiring Diagram: - Check the relay for proper function. - Check its wiring for proper condition. - Repair or replace the wiring as necessary or replace K111 if it is faulty.			

Wiring Diagram 1 (sensor S22, relay K134 and signal input of K111 at E34) related to trouble message -56-

[2/4]

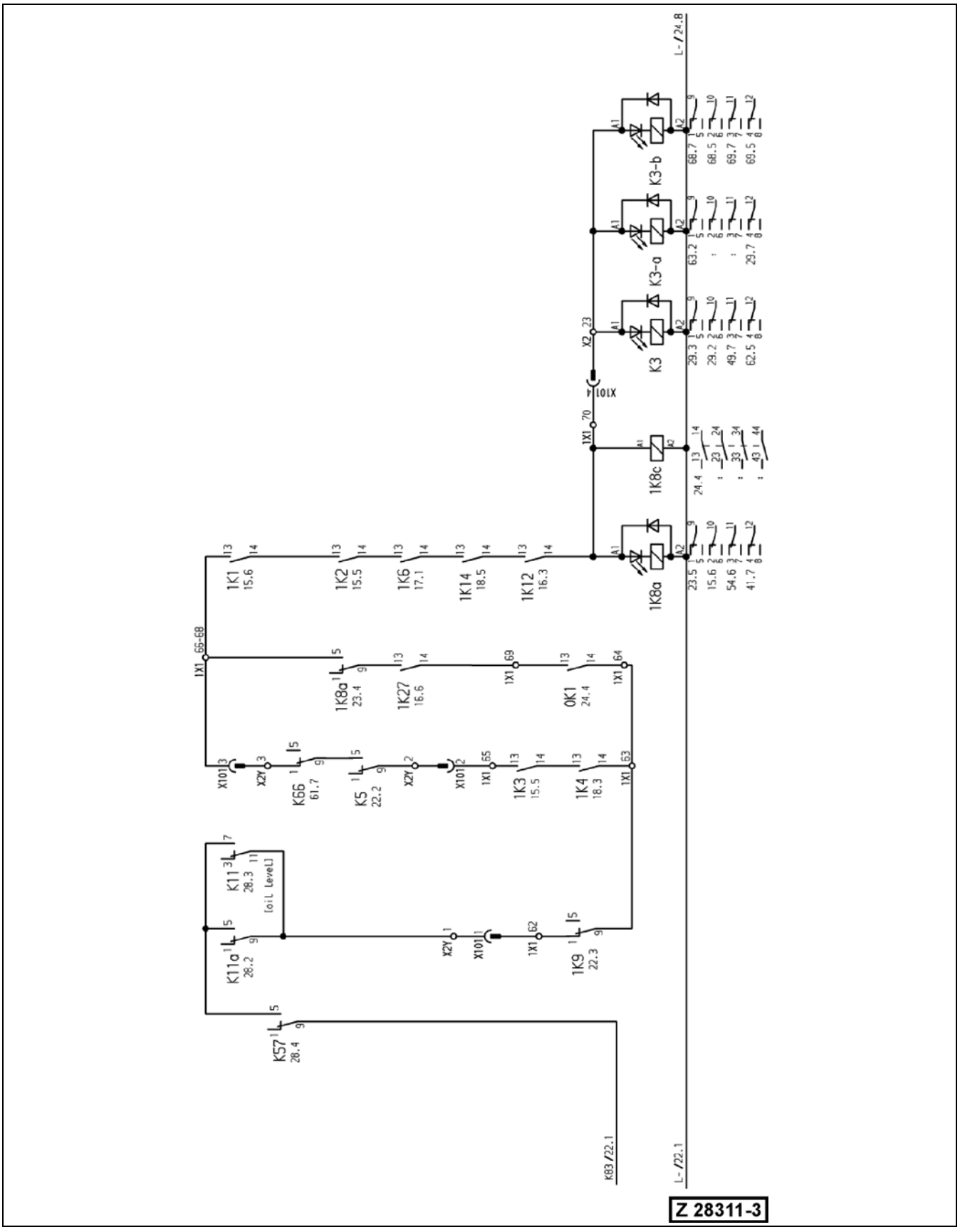


Wiring Diagram 2 (lock lever S105, relays K134 and K111) related to trouble message -56- [3/4]



Wiring Diagram 3 (motor start/stop circuit) related to trouble message -56-

[4/4]



MESSAGE -57- PUMP REGULATION FAULTY

[1/1]

Trouble code **Message page 57**	Trouble	Pump regulation faulty
Contents of trouble	• Malfunction in the electronic pump regulation system	
Related information	• Refer to the Service Manual, section -7.4 ELECTRONIC PUMP REGULATION- and follow the adjustment and/or repair procedures as necessary	
Problem that appears on machine	• Irregular operation of the hydraulic system.	

MESSAGE -58- STRAINER OIL COOLER RESTRICTED (B165)

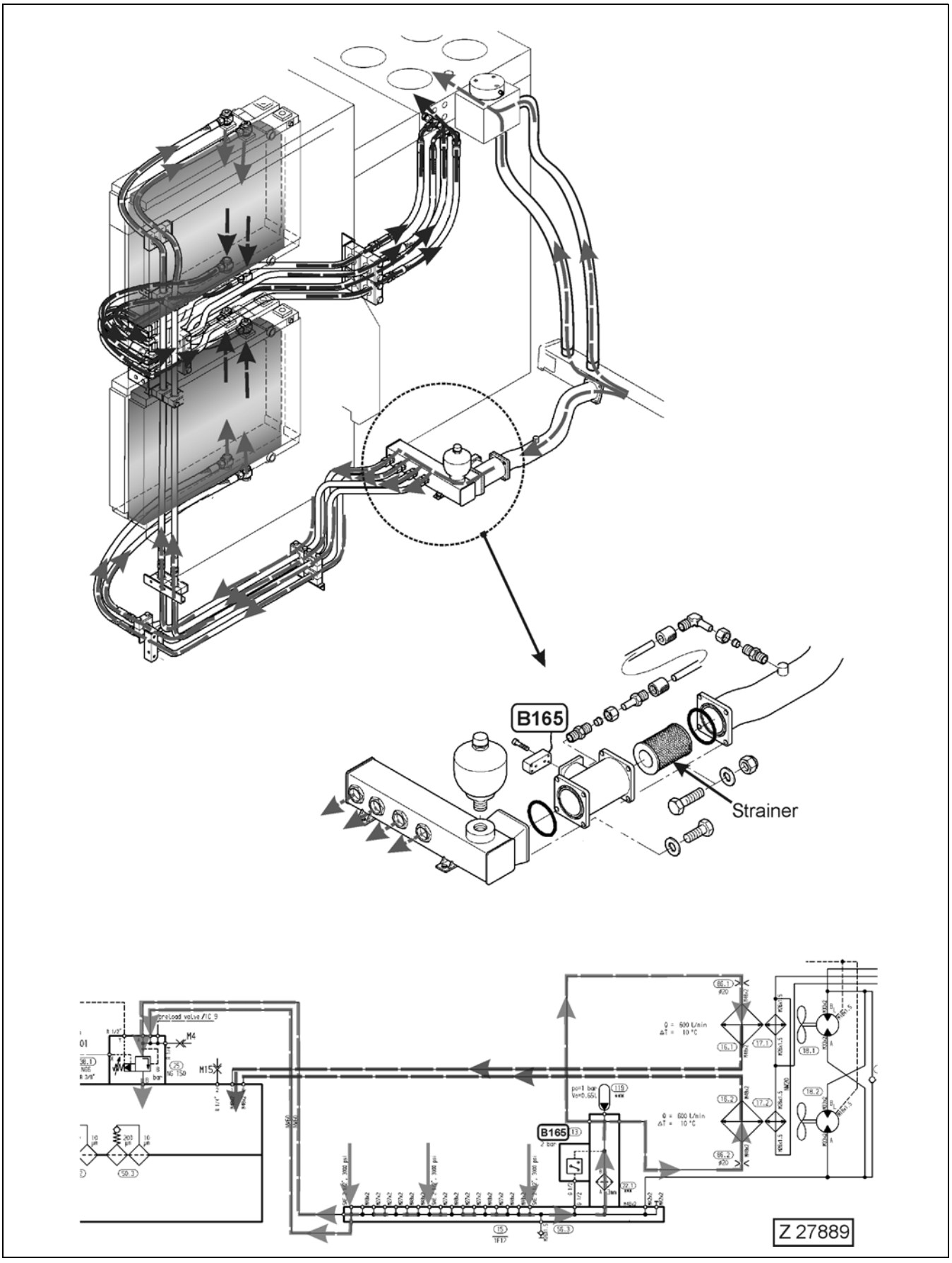
[1/2]

Trouble code **Message page 58**	Trouble	Strainer oil cooler restricted (B165)
Contents of trouble	• Pressure switch B165 signals a pressure difference of more than 2 bar (before ↔ behind the strainer in the intermediate pipe leading to the hydraulic oil cooler).	
Related information	• Before checking the strainer for clogging, first check pressure switch B165 for proper function according to PARTS & SERVICE NEWS AH07524 (latest version).	
Problem that appears on machine	• Risk to run the machine without sufficient cooling of the hydraulic oil.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Pressure switch B165 may have a malfunction	1	Check pressure switch B165 for proper function according to PARTS & SERVICE NEWS AH07524 (latest version). Is pressure switch B165 faulty?	Yes	Replace pressure switch B165 and recheck for proper function.	
			No	Go to next step.	
The strainer in the intermediate pipe (oil flow → to the oil cooler) is clogged	2	► Motor OFF - Main key switch 1S1 OFF - Maintenance safety switch 1S58 OFF & locked.			
		Remove the oil strainer in the intermediate pipe. Clean or replace the strainer as necessary. Remark: If the strainer is clogged the cause for the clogging needs to be found. Locate the problem and take adequate measures.			
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the trouble message does not arise again.			

Message -58- Illustration for B165

[2/2]



MESSAGE -59- CENTRAL LUBRICATION SYSTEM, REFILL GREASE BARREL

MESSAGE -60- SLEW RING GEAR LUBRICATION SYSTEM, REFILL GREASE BARREL

[1/6]

Trouble code **Message page 59** **Message page 60**	Trouble	Central lubrication system, refill grease barrel Slew ring gear lubrication system, refill grease barrel
Contents of trouble	• The message is displayed if the grease level in the corresponding grease barrel is at minimum.	
Related information	• A problem with the CLS grease level sensor (B108b) respectively the SLS grease level sensor (B109b) or connected wiring may also cause the trouble, see illustration Z27806 on page 5/6 and illustration Z27807 on page 6/6 for the testing procedure of the signal wiring and connections. • The illustrations Z27801 and Z23096 on page [3/6 and 4/6] show the location of the related parts on the machine.	
Problem that appears on machine	• Risk of damages due to insufficient lubrication.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The concerned grease barrel (CLS or SLS) needs refilling	1	Using the dipstick, verify if the grease level of the relating grease barrel is at minimum. Is the grease at minimum?	Yes	Refill with the specified grease. Verify that trouble message 59 or 60 is not displayed again.	
			No	Go to next step.	
Incorrect or no supply voltage in the circuit at the concerned grease level sensor (B108b or B109c)	2	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		Remove the cover of the terminal box SD20 for CLS or SD30 for SLS on top of the concerned grease barrel. Measure the supply voltage at terminal X1.3 inside the terminal box, see illustration Z27807 on page 6. Is supply voltage present?	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
			No	Go to next step.	
	3	► Main key switch 1S1 ON.	Yes	Go to step 4.	V_{min} 22V V_{max} 24V
		Check if supply voltage is present at the power supply line F25. Is supply voltage present?	No	Disconnection, short circuit, or fault in the wiring between connector XB.A at the terminal box and circuit breaker F25. If necessary, check the segments of the voltage supply wiring separately, locate the problem, and repair as necessary.	

Message -59- Central lubrication system, refill grease barrel

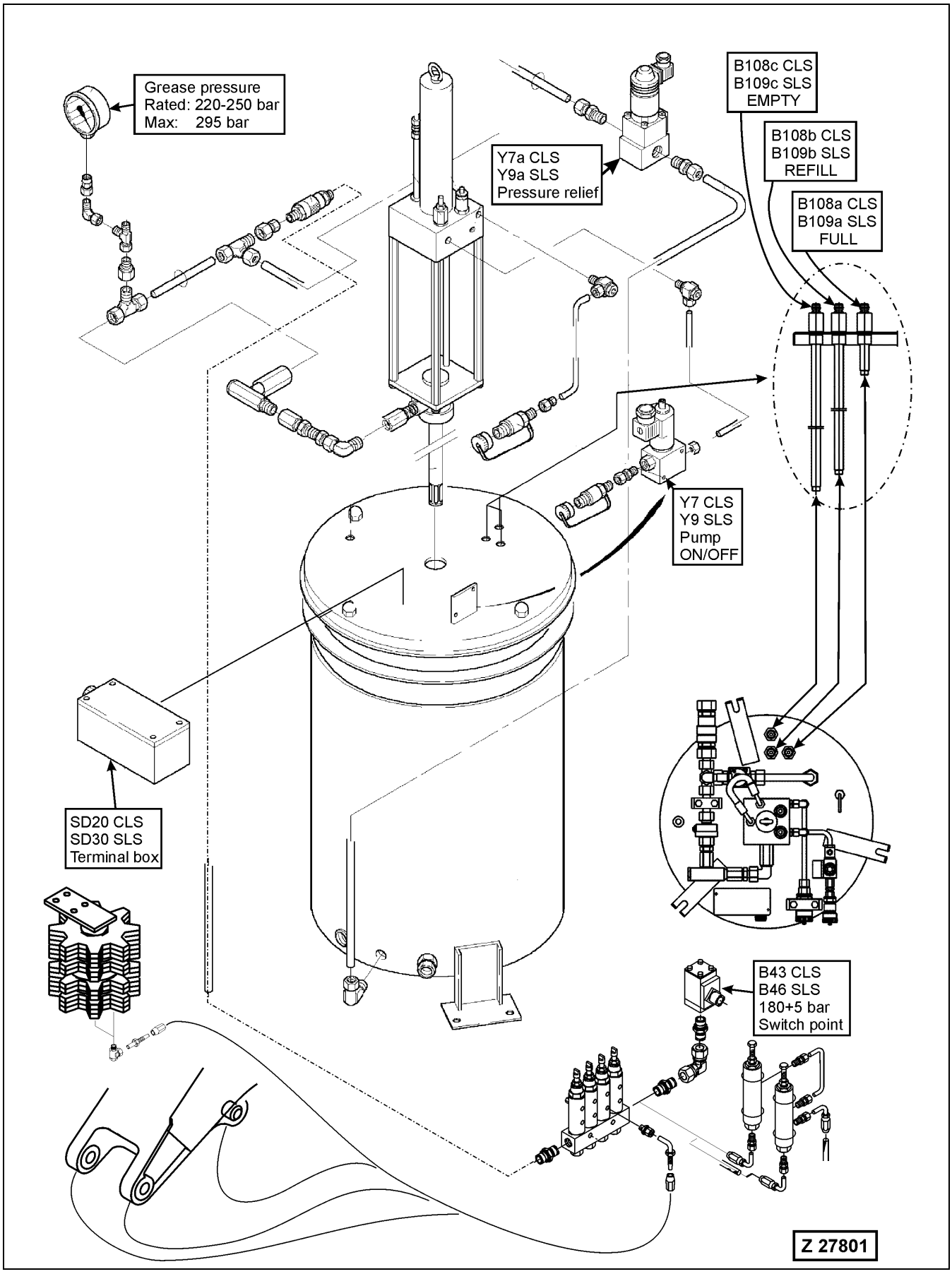
Message -60- Slew ring gear lubrication system refill grease barrel

[2/6]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Disconnection in the signal wiring between the concerned grease level sensor (B108b or B109b) and Input signal module E34 (Master) of ETM version -A- or Input signal module E34 (CAN-node) of ETM version -B-. Disconnection or defective contact in any connector or in the terminal box on top of the concerned grease barrel.	4	► Keep the main key switch 1S1 in the OFF position.			
		Disconnect the signal line of the concerned grease level sensor from the input signal module. ETM version -A- Input signal module E34 (Master) Pin 29 for B108b and Pin 30 for B109b. ETM version -B- Input signal module E34 (CAN-node) Pin 17 for B108b and Pin 16 for B109b. Refer to the Wiring Diagram and check the signal wiring segments between the concerned grease level sensor (B108c or B109c) and the relating input signal module pin for continuity. Is continuity present?	Yes	Go to next step.	R_{max} 1Ω
			No	Disconnection or short circuit in any wiring segment of the signal line (including connectors and junction box) between the concerned grease level sensor (B108b or B109b) and the relating pin on the input signal module. Locate the problem and repair as necessary.	
Defective grease level sensor (B108b or B109b)	5	► Prepare with the main key switch 1S1 in OFF position, and then carry out troubleshooting with the main key switch 1S1 in ON position.			
		For checking the level sensor (B108b or B109b), it is necessary to remove the cover of the concerned grease barrel. Disconnect the current supply cable at the terminal box. Disconnect the grease and hydraulic oil lines at the quick disconnect couplings. Loosen the grease return line on the barrel bottom. Carefully lift the grease barrel cover until the level sensor (B108b or B109b) is accessible. Clean the proximity switch of the concerned sensor. Reconnect the current supply cable at the terminal box. Check the proximity switch by dipping it into a receptacle filled with grease. If the proximity switch is in proper working order, the following conditions should occur: - Proximity switch closed - 24V applied on K136 of CLS or K138 of SLS - Message -Refill grease barrel- gone Do these conditions occur?	Yes	A sporadic malfunction of the input signal module or the text display may exist. Further inspection is required.	
			No	Replace the concerned grease level sensor (B108b or B109b). Go to step 6.	
Final check	6	Bring the machine back into the original condition. Set main key switch 1S1 in OFF position, and then in ON position. Start the motor. Verify that the fault message does not arise again. Recheck the function of the lubrication system by manual actuation of a lube cycle.			

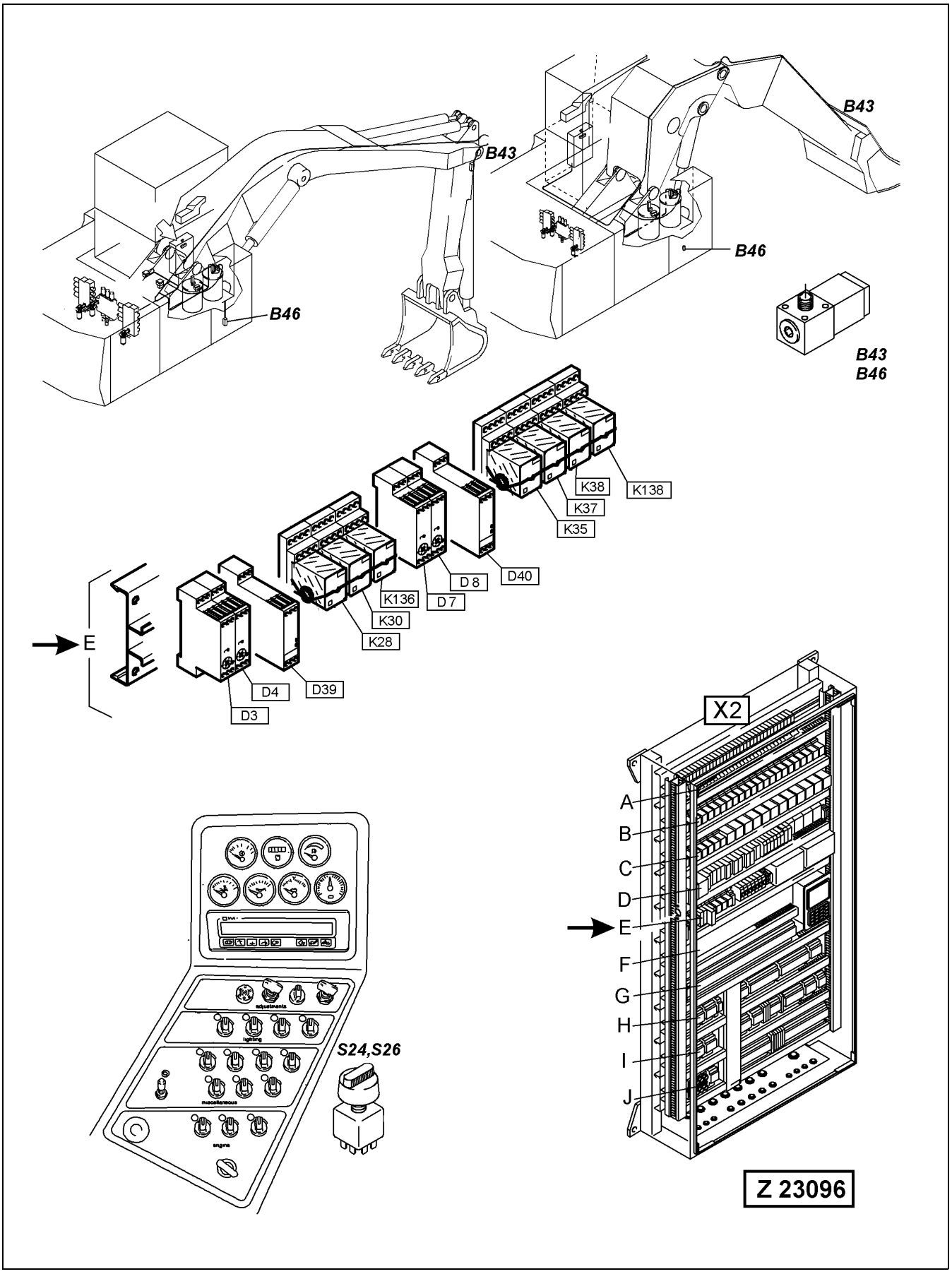
Message -59- and -60- View onto the grease barrel

[3/6]



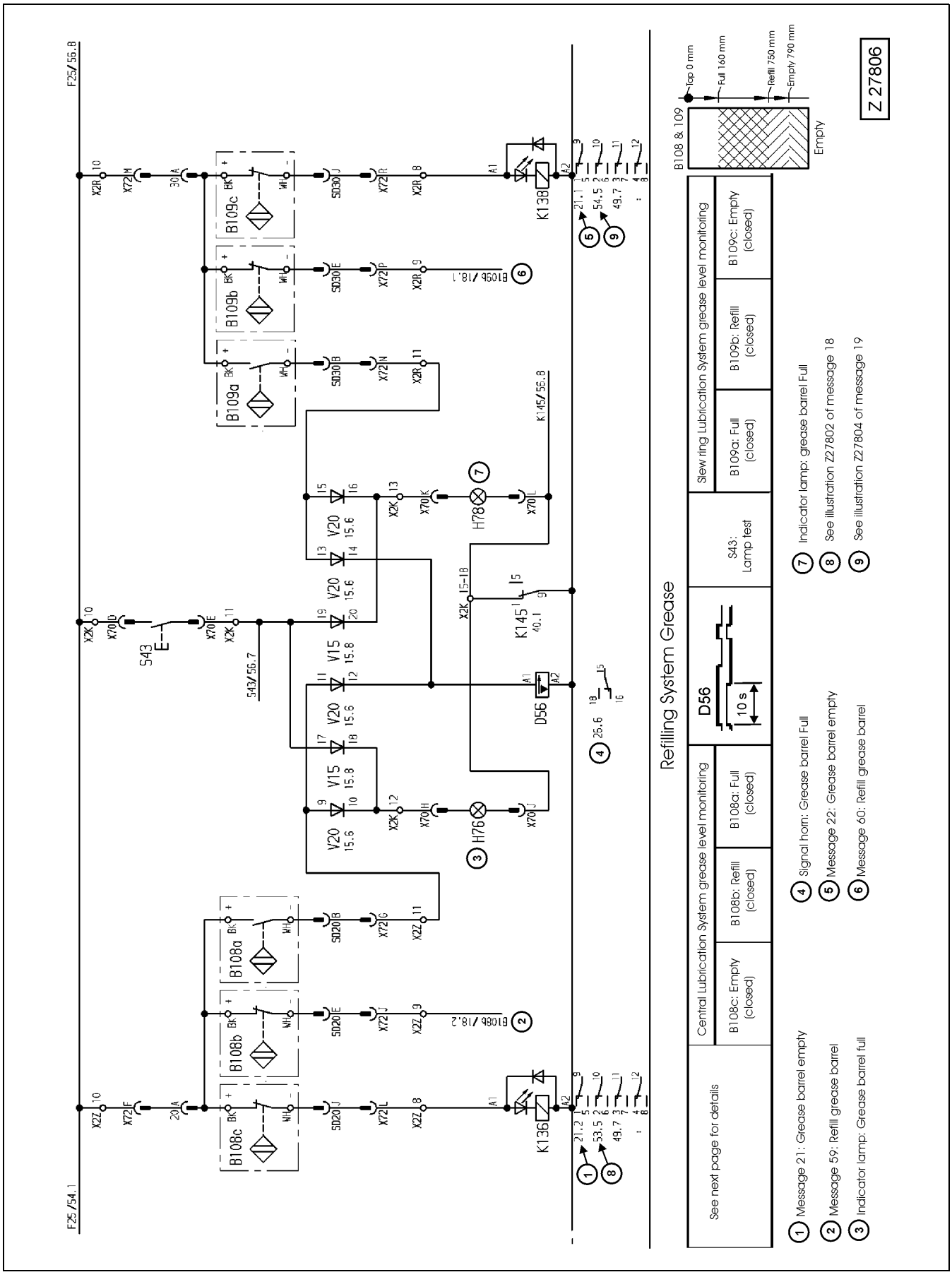
Message -59- and -60- View onto the X2 Box

[4/6]



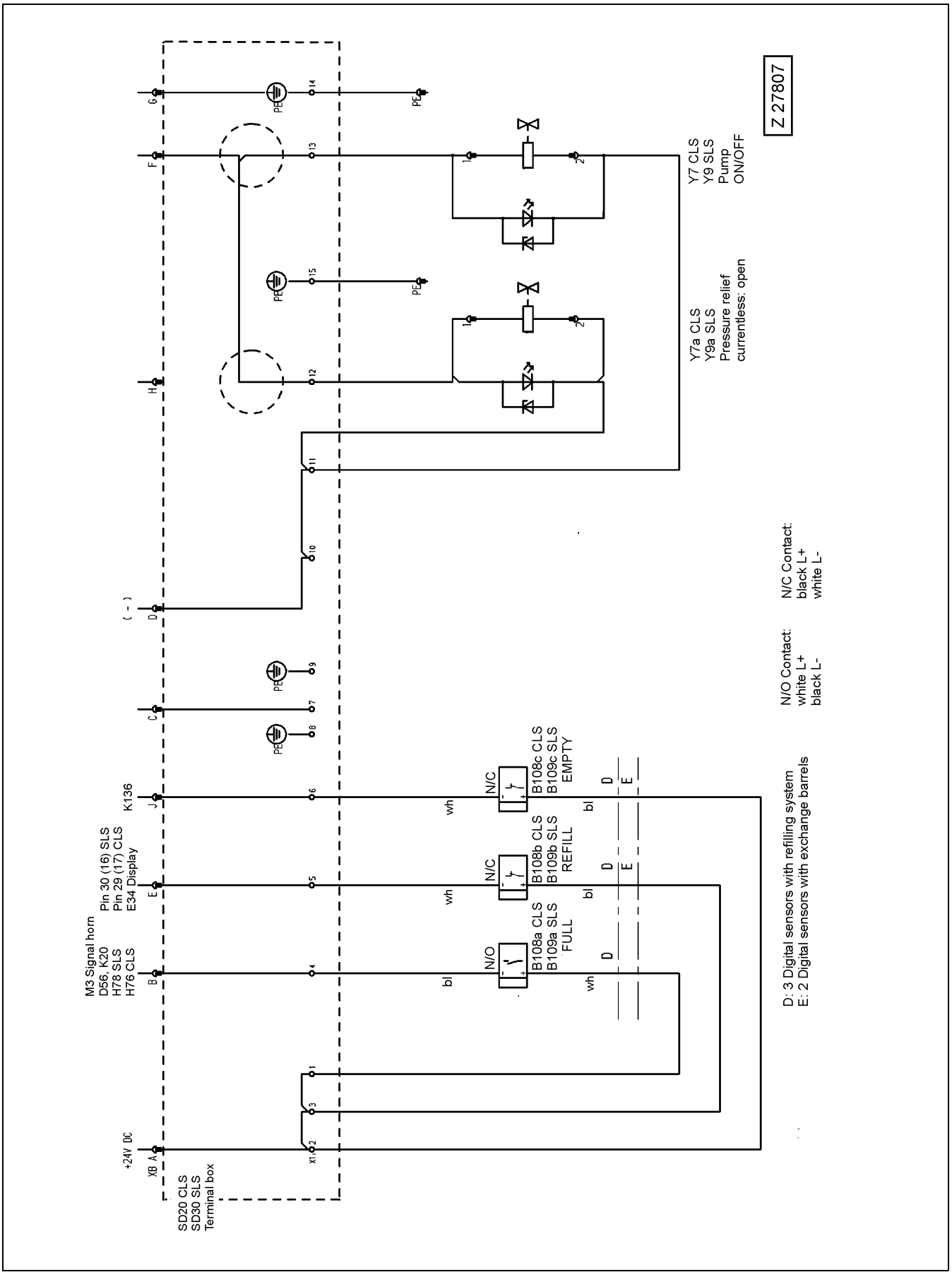
Wiring Diagram 1 related to trouble message -59- and -60-

[5/6]



Wiring Diagram 2 related to trouble message -59- and -60-

[6/6]



TROUBLESHOOTING BY SYMPTOMS

SYMPTOMS OF THE ELECTRIC MOTOR AND RELATED ELECTRICAL SYSTEM

SEM001 - MOTOR DOES NOT START

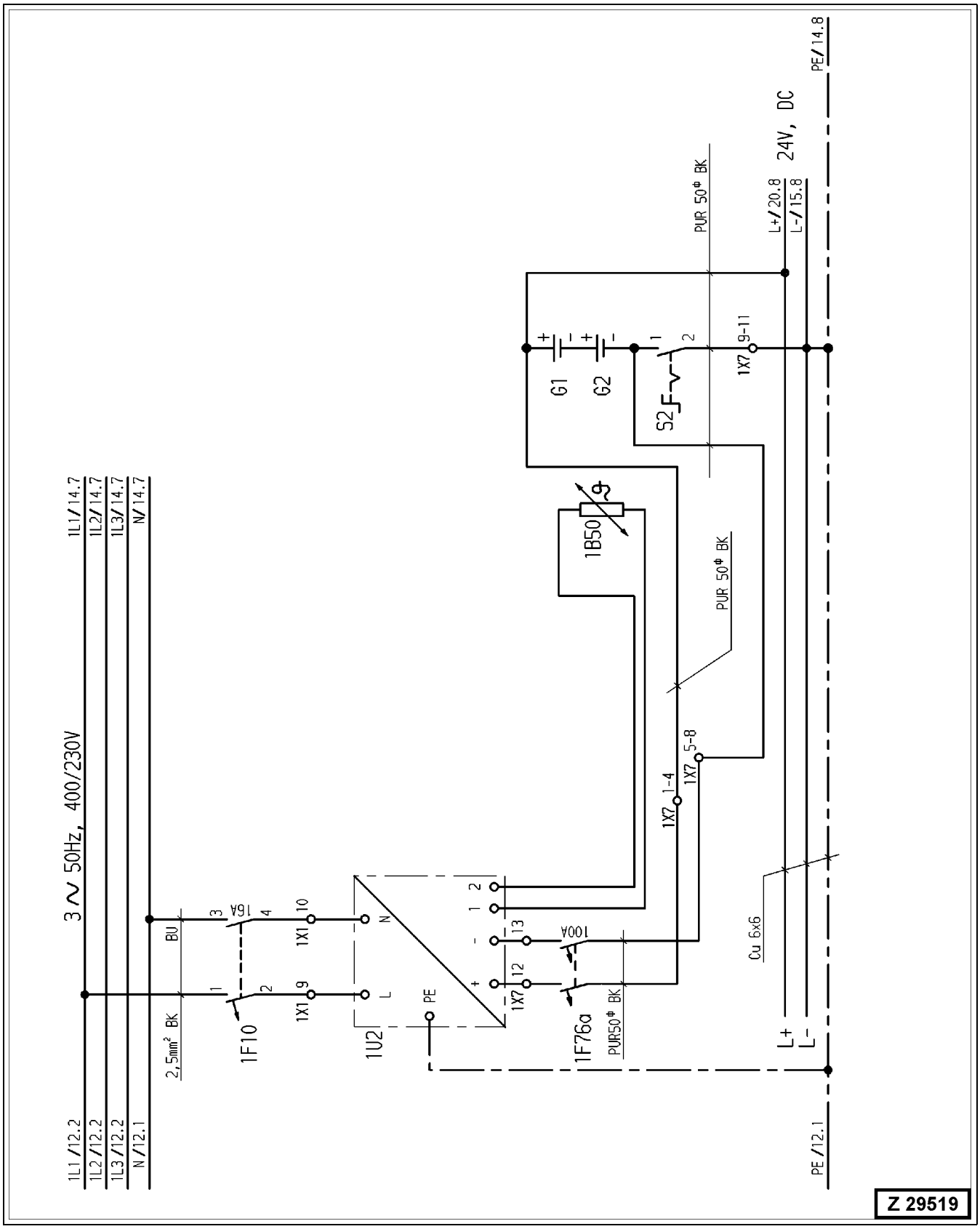
[1/4]

Trouble code	Trouble	Motor does not start
SEM001		
Contents of trouble	• The electric motor does not start. • A corresponding trouble message is not displayed.	
Related information	• The following blocking mechanisms in the motor start circuit prevent the motor start: - No or not enough battery power is present (below 17 Volt) at the ELV board. 1K8a (Self holding) not actuated. - The safety lock lever (S105) is in down position during motor start. - The safety lock lever switch (S105) is faulty.	
Problem that appears on machine	• The electric motor does not start.	
Possible causes	• Missing battery maintenance, internal short circuit in any battery or low battery voltage (below 17 V) • Handling error by the operator), safety lock lever in the wrong (down) position • Safety lock switch (S105) or its wiring faulty	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Missing battery maintenance Internal short circuit in any battery or low battery voltage (below 17 V)	1	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check the capacity and condition of batteries G1-G2. Is any battery defective?	Yes	Replace the concerned battery/batteries.	22 - 24 V Electrolyte specific gravity >1.26
			No	Go to next step.	
Handling error by the operator Safety lock lever in the wrong (down) position	2	► Main key switch 1S1ON			
		Check for the correct lock lever position (up) during motor start. Is lock lever S105 in the correct position (up) for motor start?	Yes	Go to next step.	
			No	Move the lock lever in the up position and repeat the motor start.	
Safety lock switch (S105) or its wiring faulty	3	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Refer to the wiring diagram, check the switch function (S105) and the belonging wiring. Are safety lock switch S105 and its wiring ok?	Yes	Go to next step.	
			No	Repair or replace as necessary.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

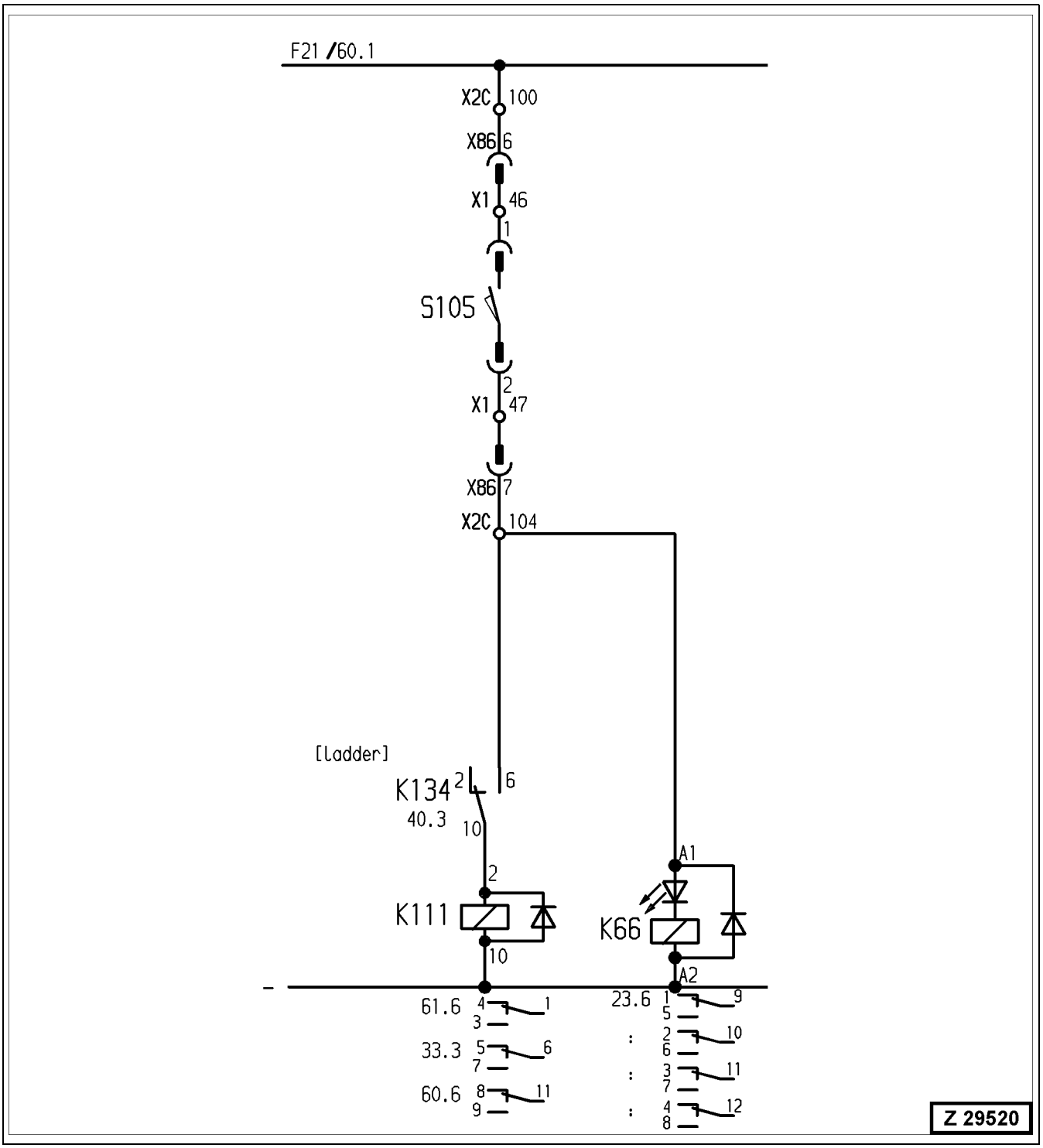
Wiring Diagram 1 for trouble code SEM001

[2/4]



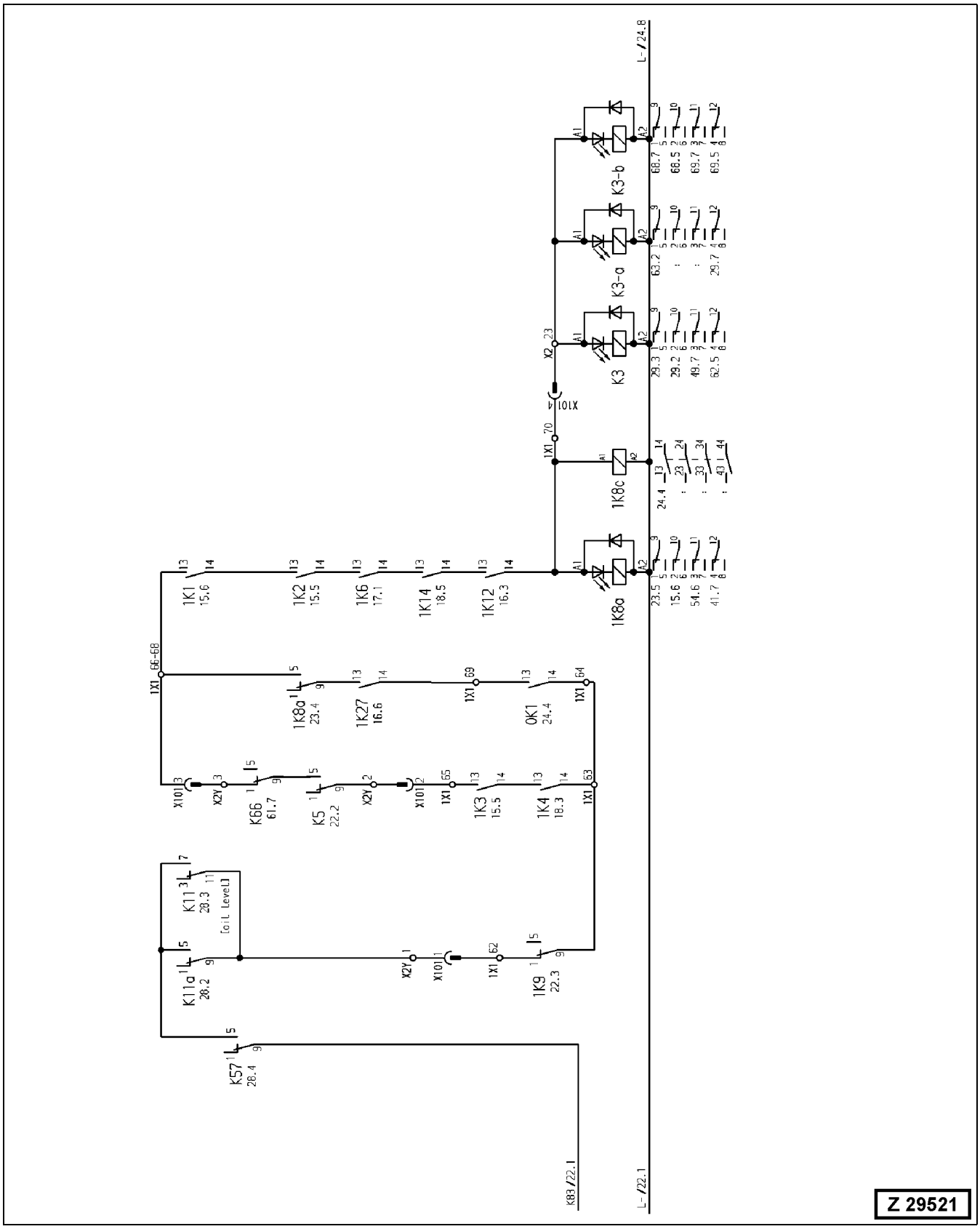
Wiring Diagram 2 for trouble code SEM001

[3/4]



Wiring Diagram 3 for trouble code SEM001

[4/4]



SEM002 - MOTOR DOES NOT STOP

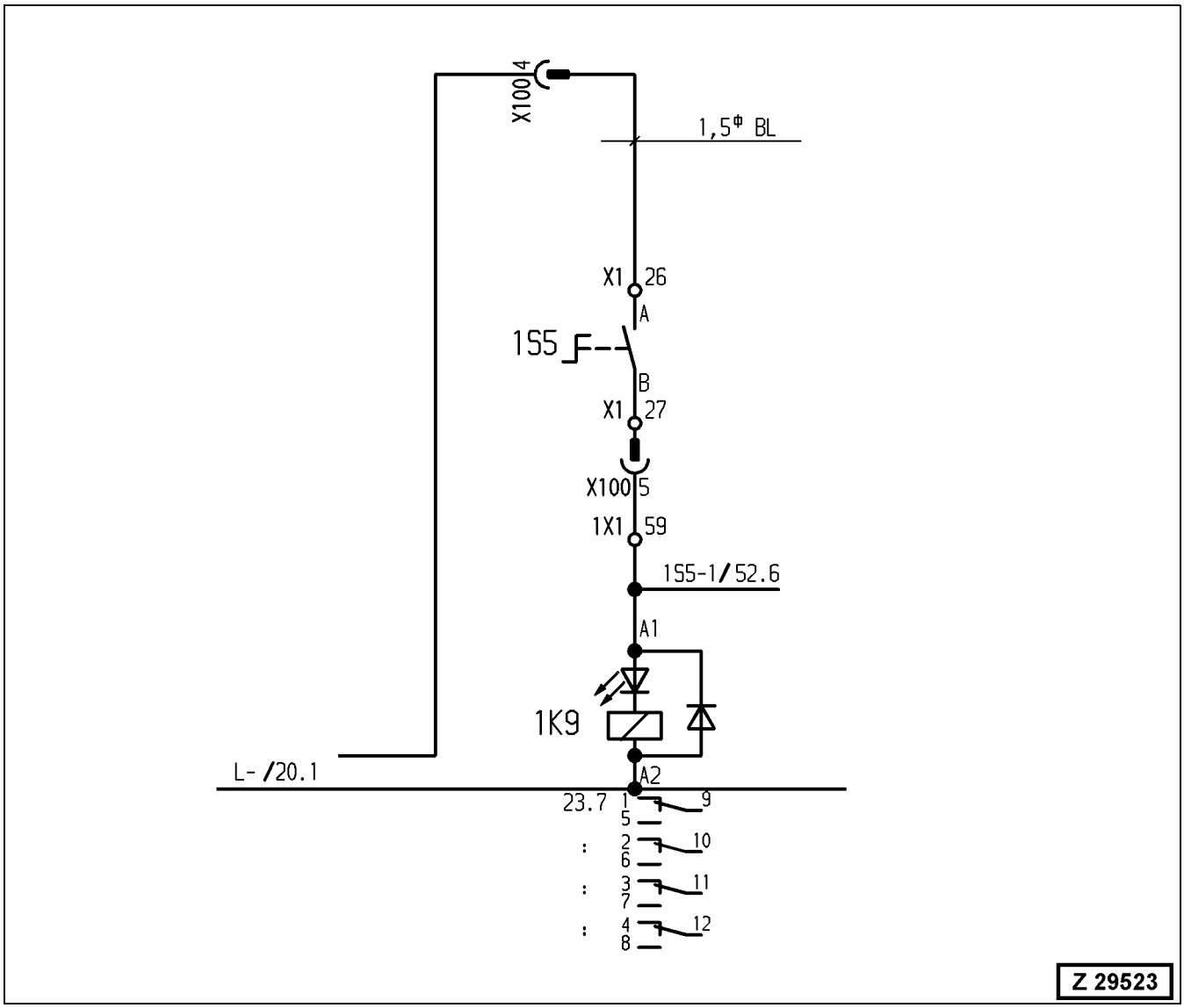
[1/2]

Trouble code	Trouble	Motor does not stop
SEM002		
Contents of trouble	• The electric motor does not stop during the motor stop switch 1S5 is activated.	
Related information	• A corresponding trouble message is not displayed.	
Problem that appears on machine	• Motor shut-down is not possible.	
Possible causes	• Motor stop switch switch (1S5) or its wiring faulty	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor stop switch switch (1S5) or its wiring faulty.	1	► Main key switch 1S1ON			
		- Activate switch 1S5.	Yes	Go to next step.	22 – 24 V
		- Refer to the wiring diagram, check the circuit of 1S5 and the function of 1K9 relay. Does the circuit function properly?	No	Repair or replace as necessary	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

Wiring Diagram for trouble code SEM002

[2/2]



SEM003 - EXCESSIVE MOTOR VIBRATION

[1/1]

Trouble code	Trouble	Excessive motor vibration
SEM003		
Contents of trouble	• Abnormal or excessive vibration coming from Motor or related parts.	
Related information	• First check all related parts of function and tightening. • Check the alignment between motor and coupling (PTO) according to the Parts & Service News No. AH11517 & AH11510 latest versions.	
Problem that appears on machine	• Abnormal vibration causes bearing & coupling failure of the electric motor. • Excessive wear of the components - risk of machine break down.	
Possible causes	• Electric motor mountings are loose. • Inproper alignment of Motor to coupling & PTO. • Coupling defective.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Electric motor mountings are loose	1	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check all mounting bolts at the electric motor. Also refer to the Parts & Service News No. AH11510. Are the motor mountings ok?	Yes	Go to next step.	Refer to the torque list in the Parts & Service News AH11510.
			No	Check and re-torque the motor mountings and carry out the alignment check and inspect the coupling.	
Inproper alignment of Motor to coupling & PTO	2	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Carry out the proper procedure of the alignment and inspection of electric motors. - Check the coupling of wear. - Check the PTO input shaft of wear. - Check the rotation of the Rotor on the DE site of the motor. If all this above items ok carry out an alignment according the Parts & Service News No. AH11517. Is the alignment ok?	Yes	Go to next step.	Refer to the Parts & Service News AH11517.
			No	Carry out an alignment according the Parts & Service News No. AH11517.	
Coupling defective	3	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check the coupling of wear / dust or improper alignment. Is the coupling condition ok?	Yes	Go to step 2.	Coupling lifetime = motor lifetime (25000hrs)
			No	Replace coupling. Go to step 2.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SEM004 - ABNORMAL NOISE FROM MOTOR

[1/1]

Trouble code	Trouble	Abnormal noise from motor
SEM004		
Contents of trouble	• Motors makes abnormal noise (bearing rotating noise).	
Related information	• First check all related parts of function and tightening. • Check the grease (quality & quantity) at the NDE & DE bearing of the motor. • Check the grease line (must be free and clean).	
Problem that appears on machine	• Excessive bearing wear – risk of motor break down.	
Possible causes	• Motor bearing at the NDE or DE ran out of grease. • Inproper alignment of Motor and PTO (see SEM003).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Motor bearing at the NDE or DE ran out of grease	1	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check the quality and the quantity of the used grease. Is the grease quality & quantity ok?	Yes	Go to next step.	For more information see type plate for the motor bearings on the motor.
			No	Measure the bearing condition (Test tool Vibrotip).  If the bearing index is out of the specification replace the electric motor.	Bearing Index: up to 1.3 mm/s is ok.
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SEM005 - MOTOR LACKS OUTPUT (LOW POWER)

[1/1]

Trouble code	Trouble	Motor lacks output (low power)
SEM005		
Contents of trouble	• The rated motor output power (Ampere on the dashboard) does not fit the specification.	
Related information	• A corresponding trouble message is not displayed. • The shovel is working with proper electronic pump regulation – Please confirm! • The net line power supply is not more than 5% below or above the rated voltage.	
Problem that appears on machine	• Sufficient loading or moving is not possible.	
Possible causes	• Electric motor defective.	

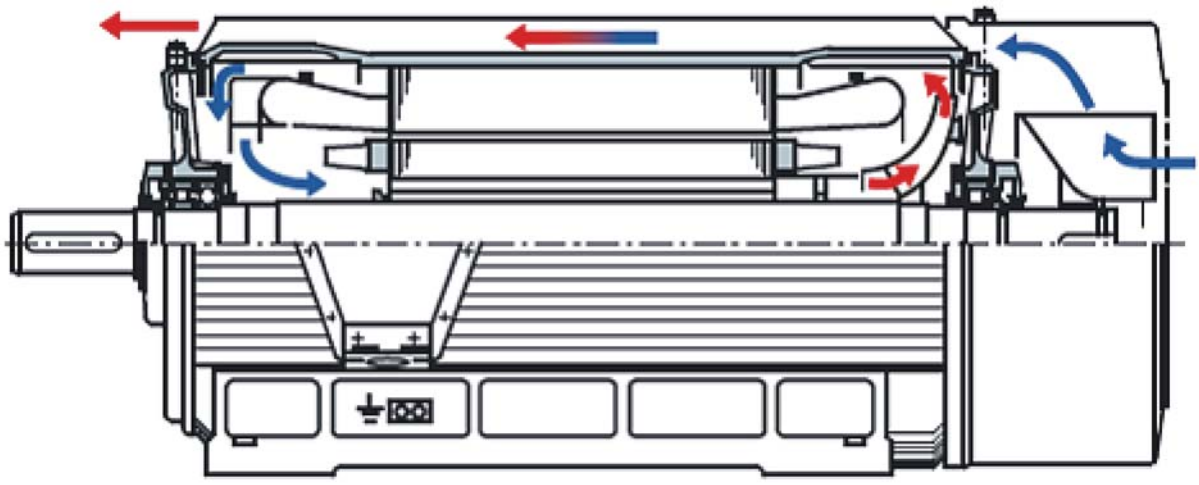
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Electric motor defective	1	► Main key switch 1S1ON – Power check.			
		Carry out a motor power check according to the PM-CLINIC procedure. Is the test successfully?	Yes	Re check the adjustments of the pump regulation system (Q_{min} & Q_{max} solenoids) using the multi monitor. Refer to the SERVICE MANUAL.	See PM-CLINIC sheet for electric motors
			No	Carry out a motor diagnosis with the Siemens service. Replace the electric motor if necessary.	Siemens service
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SEM006 - AIR TO AIR COOLING SYSTEM NOT SUFFICIENT

[1/1]

Trouble code	Trouble	Air to Air cooling system not sufficient
SEM006		
Contents of trouble	• The cooling system called “air to air system” is not sufficient. • The max. temperature of rotor or winding is reached immediately.	
Related information	• The trouble messages no. 23 and 41 are displayed • Motor is shut down.	
Problem that appears on machine	• Motor is shut down. • Excessive bearing wear.	
Possible causes	• The motor’s “air to air” cooling system is clogged.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The motor’s “air to air” cooling system is clogged.	1	► Main key switch 1S1ON – motor running without load.			
		Check the air flow through the motor using smoke cartridges and verify volume and leaks (Smoke cartridges for tests can ordered by all Siemens dealers). Is the cooling system ok?	Yes	Carry out a motor diagnosis with the Siemens service. Replace the electric motor if necessary.	See drawing
			No	Check the cooling system and clean up. Re check the cooling function.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			



Z 29524

SEM007 - MOTOR SHUTS DOWN DURING START

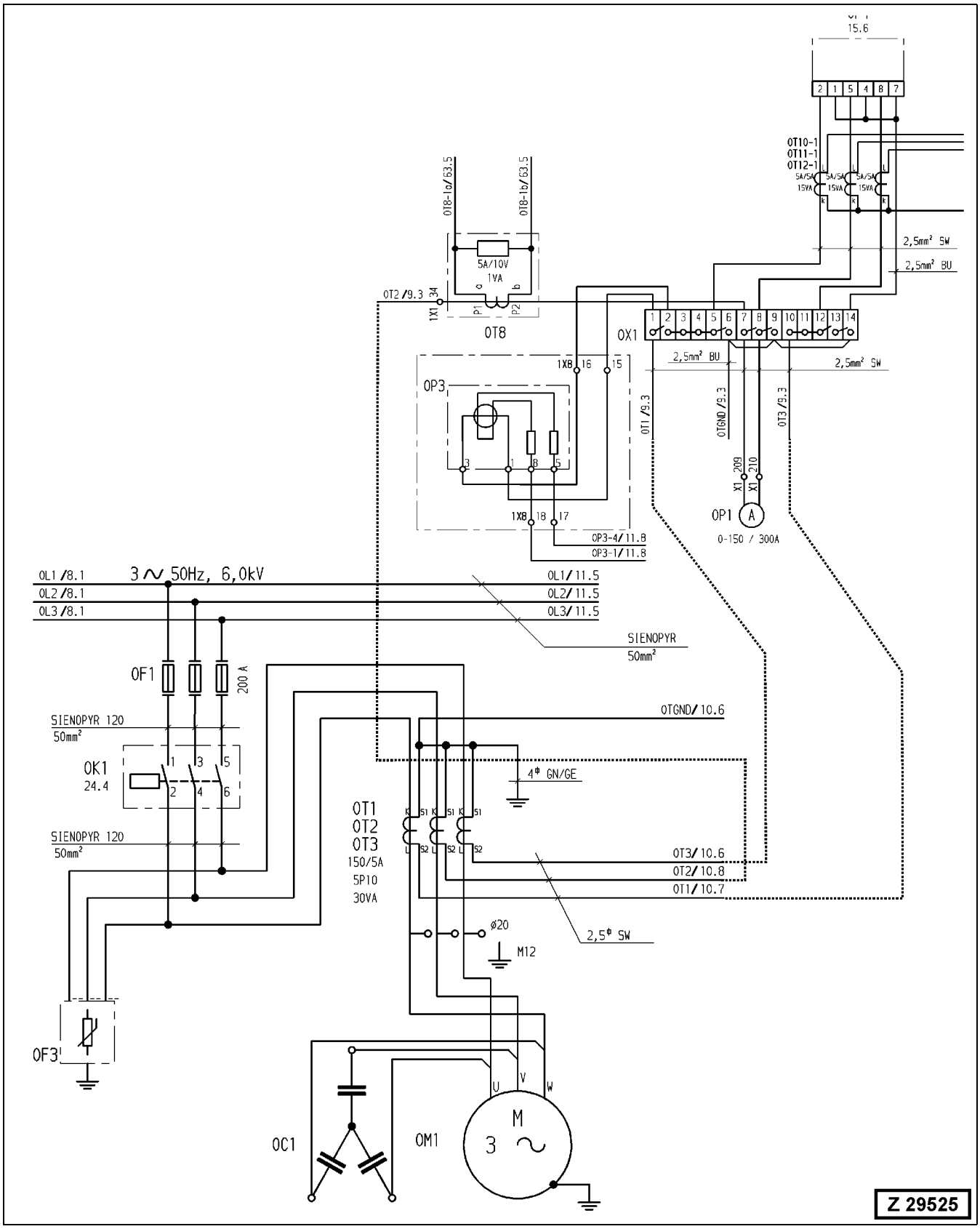
[1/2]

Trouble code	Trouble	Motor shuts down during start
SEM007		
Contents of trouble	• The electric motor is shut down during start. • The starting switch (circuit) is enabled.	
Related information	• A corresponding trouble message is not displayed. • Motor starts up but the motor running is blocked.  **HAZARDOUS VOLTAGE!** **High voltage is applied at the transformers.** **There is danger of serious injury or death.** **Access to the main transformers is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.**	
Problem that appears on machine	• The electric motor does not start (running).	
Possible causes	• No phase signal from the motor to the motor protection relay.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No phase signal from the motor to the motor protection relay.	1	► Main key switch 1S1ON – motor start sequence.			
		Check the phase signals from the motor at the motor protection relay (SPAM) during the start sequence. Are all three signals present?	Yes	Check standard values at the motor protection relay (SPAM) (Basic setting according wiring diagram).	All three phase signals present at the SPAM.
			No	Go to next step.	
	2	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		 **HAZARDOUS VOLTAGE!** **Access to the transformers is only permitted for authorized service staff with permission to work on medium and high voltage systems, after carrying out the grounding procedure.** The HV department needs to check the signal wiring between main transformers 0T1-0T3 and 0F4 (SPAM) the for continuity. Do all wires have continuity?	Yes	Go to next step.	R_{max} 1Ω
			No	Repair or replacement needs to be done by the HV department if necessary.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

Wiring Diagram for trouble code SEM007

[2/2]



SEM008 - SUB STATION MAIN CIRCUIT BREAKER TRIPS

[1/2]

Trouble code	Trouble	Sub Station main circuit breaker trips
SEM008		
Contents of trouble	• The substation circuit breaker trips under load of the shovel.	
Related information	• Current peak trips the main circuit breakers in the substation (mine responsibility). • The HV department (mine) needs to check the basic adjustment on the main circuit breakers.  **HAZARDOUS VOLTAGE!** **High voltage is applied in the sub station.** **There is danger of serious injury or death.** **Access to the sub station is only permitted for authorized mine service staff with permission to work on medium and high voltage systems.**	
Problem that appears on machine	• Power shut down	
Possible causes	• Faulty adjustment of the main circuit breakers in the substation	

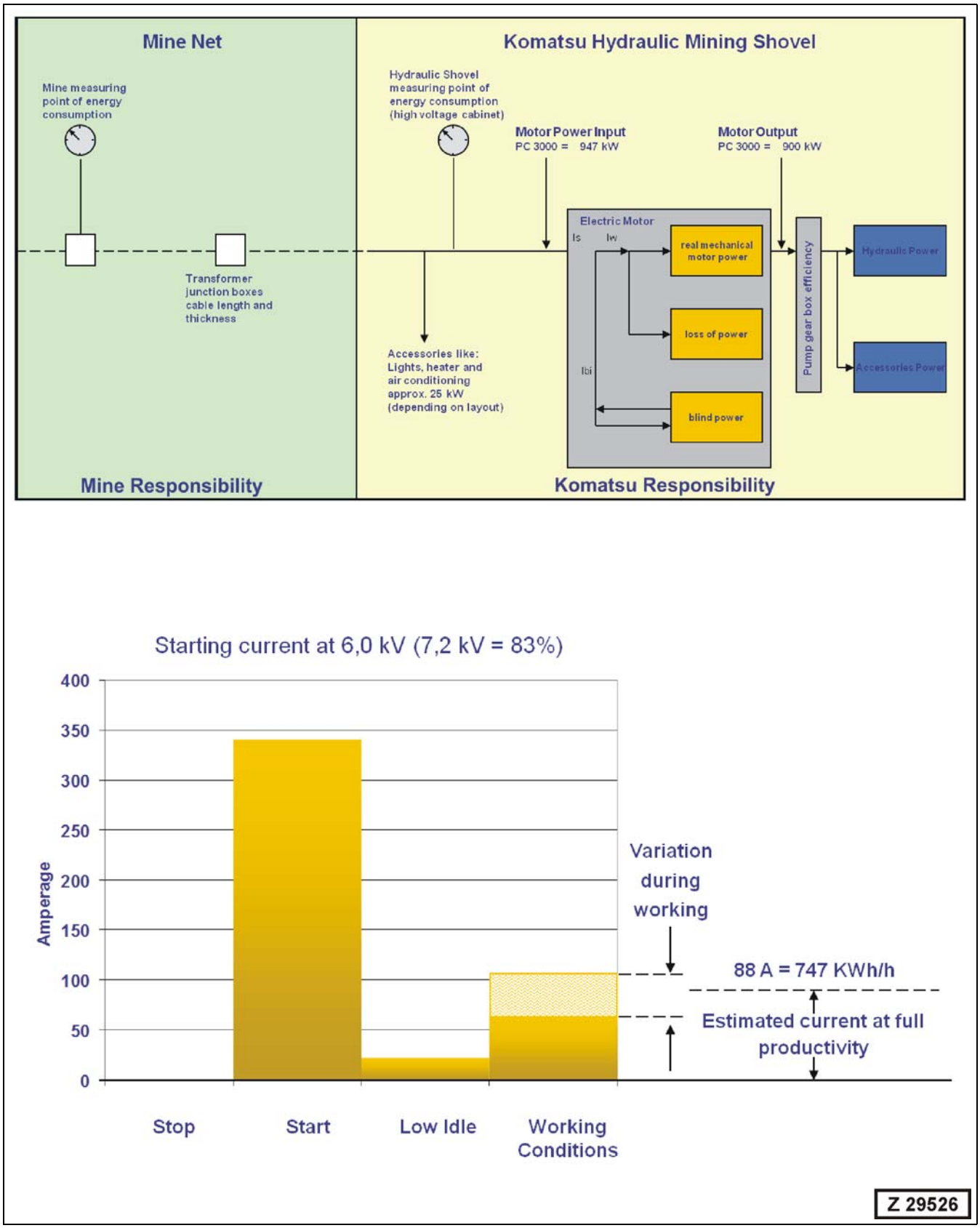
Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Faulty adjustment of the main circuit breakers in the substation	1	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		 **HAZARDOUS VOLTAGE!** **High voltage is applied in the sub station.** **Access to the sub station is only permitted for authorized mine service staff with permission to work on medium and high voltage systems.** The HV department (mine) needs to verify the basic adjustment on the main circuit breakers. Are the basic calibration ok?	Yes	Go to final step 2.	See related information
			No	The HV department (mine) needs to re-calibrate the main circuit breakers.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

Related information for trouble code SEM008

Electric Energy Consumption in KW/h						
Motor manufacturer	**Motor type**	**Rated motor output**	**Mining Conditions**			**Max. consumption**
			easy	**average**	**heavy**	
			70%	75%	80%	100%
Siemens	1 LA 452	900	663	711	758	947

Related information for trouble code SEM008

[2/2]



SYMPTOMS OF THE HYDRAULIC SYSTEM

SHY001 - ALL THE WORK EQUIPMENT SHOWS LACK OF POWER OR SLOW MOVEMENT (ESPECIALLY THE BOOM UP FUNCTION)

[1/2]

Trouble code	Trouble	All the work equipment shows lack of power or slow movement (especially the BOOM UP function)
SHY001		
Contents of trouble	• The work equipment shows lack of power or its movement (especially the BOOM UP function) is too slow. • A corresponding trouble message is not displayed.	
Related information	Before starting any troubleshooting work in detail: • Ask the operator for conspicuous failure symptoms and for the history. • Bring the shovel to the operating temperature → hydraulic oil temperature min. 60C°. • Recheck yourself all oil levels and functions and inspect for any leakage. • Before starting any adjustment work, operate the shovel yourself and test all affected functions. Check for noise or abnormal reactions. • Using a stop watch, test the cycle times of the work equipment (beginning at the movement start, measure the time between start stroke and end stroke of the cylinder). • Always use the PM-CLINIC sheet for comparing the taken measurements with the default values. • When carrying out tests use the recommended adjustment values only.	
Problem that appears on machine	• The work performance of the shovel is poor and/or too slow.	
Possible causes	• Wrong hydraulic adjustments (SRV, MRV) or defective valves • Any hydraulic cylinder defective (internal leak) • Low motor power output	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wrong adjustment of the pump regulation system	1	► Follow the respective requirements of the SERVICE MANUAL.			
		Check the adjustments of the pump regulation system (Q_{min} & Q_{max} solenoids) using the multi monitor. Refer to the SERVICE MANUAL. Are the offset values correct?	Yes	Go to next step.	SERVICE MANUAL
			No	Adjust of the pump regulation system (Q_{min} & Q_{max} solenoids) using the multi monitor. Refer to the SERVICE MANUAL.	
Wrong hydraulic adjustments (SRV, MRV) or defective valves	2	► Main key switch 1S1ON – motor running without load.			
		Check the adjustments of the SRV (Safety Relief Valves) and MRV (Main Relief Valves) according to the SERVICE MANUAL and the PM-CLINIC sheet. Is any valve adjustment below the default value?	Yes	Adjust valves as necessary according to the PM-CLINIC sheet and as described in the SERVICE MANUAL.	
			No	Go to next step.	
Any hydraulic cylinder defective (internal leak)	3	► Motor running.			
		Using an infrared temperature gauge, check the boom cylinder temperature (refer to - Cylinder bypass test- in this manual) in order to locate an internal leakage. Could you determine any abnormality?	Yes	If any cylinder leakage is found replace the defective hydraulic cylinder.	
			No	Go to next step.	

SHY001 - All the work equipment shows lack of power or slow movement (especially the BOOM UP function)

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Low motor power output	4	► Main key switch 1S1ON – motor running.			
		Check the rated motor output according to the PM-CLINIC sheet. Despite all correct adjustment about the recommended values. Check voltage supply from the main net line. Is the motor rated output power ok?	Yes	Recheck the hydraulic System of malfunctions.	PM-CLINIC
			No	Call the Siemens service for a complete motor check up Replace motor ic nessecary.	
Final check	5	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY002 - NO WORK EQUIPMENT, TRAVEL OR SWING MOVEMENT

[1/1]

Trouble code	Trouble	No work equipment, travel or swing movement
SHY002		
Contents of trouble	• Work equipment, travel or swing movement is impossible. • A corresponding trouble message is not displayed.	
Related information	• Even no trouble code is displayed, any movement of the work equipment is impossible. • Ladder and service arm are in their correct upper position. • The motor is running and all safety functions are unlocked.	
Problem that appears on machine	• No movement is possible.	
Possible causes	• Relay K111 is inoperative or burnt. • Pilot pressure is not available (PPC pump defective).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Relay K111 is inoperative or burnt	1	► Main key switch 1S1in ON position.			
		Check the circuits of K111 for proper function. Check the function of relay K111 manually. Could you identify any problem in the electrical system?	Yes	Repair or replace relay K111 and/or its socket if damaged.	Output 24V
			No	Go to next step.	
Pilot pressure is not available	2	► Motor running.			
		Check the pilot pressure at the valve and cartridge block as described in the SERVICE MANUAL. Make sure that the pilot pressure is in the specified working range. Is specified pilot pressure available?	Yes	Check for proper movement of the work equipment.	X4 = 60bar X2 = 35bar
			No	Check the drive input of pilot pump. Check the pilot pump for damage (internal fault). Repair or replace as necessary.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY003 - WORK EQUIPMENT HAS TOO MUCH HYDRAULIC DRIFT

[1/2]

Trouble code	Trouble	Work equipment has too much hydraulic drift
SHY003		
Contents of trouble	• In the neutral position of the control levers the work equipment has too much visible drift.	
Related information	• According to section -Cylinder bypass test- check and measure the existing drift at the cylinder circuit. Locate the affected hydraulic circuit. • Install pressure gauges at the main blocks of the affected circuit and monitor the pilot pressure for the respective main spool.	
Problem that appears on machine	• Risk, to loose control of the shovel operation. • Damage or breakdown is possible due to contacting dump trucks.	
Possible causes	• No or incorrect main spool movement (neutral spring damaged). • FSA - Floating valve internal leak or damage • Hydraulic cylinder damaged (internal leak).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No or incorrect main spool movement (neutral spring damaged)	1	► Motor running & all PPC sections on the main blocks equipped with pressure gauges.			
		On the main blocks check all functions of the work equipment for increasing and decreasing times of the pilot pressure control, simultaneously to the lever movements. Is the pilot pressure increasing and decreasing simultaneously?	Yes	Go to next step.	30 - 35 bar at the main block spools (when the lever is moved 'full stroke').
			No	Check the components of the pilot pressure circuit for proper function	
FSA - Floating valve internal leak or damage	2	► Motor running & all PPC sections on the main blocks equipped with pressure gauges.			
		In the floating valves Y132b (stick) or Y132c (boom) any leakage or damage of the inner spool is possible. Inspect the floating valves as described in the SERVICE MANUAL. Can you identify any leakage or damage?	Yes	Replace the affected valve and adjust as described in the SERVICE MANUAL and PM-CLINIC procedures.	SERVICE MANUAL PM-CLINIC
			No	Go to next step.	

SHY003 - Work equipment has too much hydraulic drift

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Hydraulic cylinder damaged (internal leak)	3	► Motor running.			
		The function of the main blocks is ok. Refer to section -Cylinder bypass test- and check the suspicious cylinders for defective internal seals. Can you find out significant temperature differences during the cylinder bypass test?	Yes	Replace any defective cylinder by a new one. Bleed the new hydraulic cylinder(s). Perform the bypass test again in order to make sure hat no other cylinder is affected.	Min. test period 20 minutes
			No	Check the SRV and suction valves for proper function. Repair or replace if necessary and after that verify that the hydraulic drift of the work equipment meets the specified range.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY004 - WORK EQUIPMENT HAS BIG TIME LAG

[1/1]

Trouble code	Trouble	Work equipment has big time lag
SHY004		
Contents of trouble	• The work equipment has a big time lag between lever actuation and movement of the attachment.	
Related information	• The reaction time of the attachment after lever movement is delayed.	
Problem that appears on machine	• Due to a big time lag the machine can not be operated properly.	
Possible causes	• Wrong adjustment of the relief valve 252.1 and/or 252.2.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wrong adjustment of the relief valves 252.1 and/or 252.2.	1	► Motor running.			
		Perform the testing and adjustment of the relief valves 252.1 and 252.2 according to the procedures described in the SERVICE MANUAL. Is the setting of the relief valves correct?	Yes	Perform the complete checking of the pilot circuit. Refer to the SERVICE MANUAL and the applying hydraulic diagram.	
			No	Adjust both relief valves to the correct values.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY005 - MACHINE DEVIATES DURING TRAVEL MOVEMENT

[1/1]

Trouble code	Trouble	Machine deviates during travel movement
SHY005		
Contents of trouble	• During machine traveling the shovel deviates to one side. • The shovel is traveling only (no other movement is done). • Precondition: flat ground and same grip on both sides.	
Related information	• The PM-CLINIC procedures were performed successfully. • The SRV and MRV adjustments meet the nominal values of the hydraulic diagram. • The left and right track wear measurement is on the same level.	
Problem that appears on machine	• During machine traveling the shovel deviates to one side. • The operator has to stop the shovel after a couple of meters in order to correct the direction of travel to straight forward.	
Possible causes	• Defective hydraulic travel motor. • Internal leak in the rotary joint.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Defective hydraulic travel motor Internal leak in the rotary joint	1	► Motor running.			
		Connect high pressure gauges (400 bar) to the main blocks. Disconnect the parking brake solenoid Y16 in order to close the brake. Actuate the travel pedals and read the current pressure at the gauges. Is there any pressure difference between left and right, or loss of pressure?	Yes	Disconnect the ports A and B at the travel motors (in order to exclude the travel motors). Plug the hoses and the inlet ports of the travel motors. Now check the pressure again. If the pressure values are still the same as before, replace the travel motor(s).	SRV: 310 bar MRV: 315 bar
			No	If the pressure values are not the same as before, repair or replace the rotary joint due to an internal leakage.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY006 - MACHINE DOES NOT SWING

[1/1]

Trouble code	Trouble	Machine does not swing
SHY006		
Contents of trouble	• The machine does not swing when the lever is moved. • The swing parking brake is released (otherwise a corresponding info message is displayed on the ETM display). • All other operating functions are going well.	
Related information	• No swing movement (left or right) is possible.	
Problem that appears on machine	• No swing movement is possible.	
Possible causes	• No pressure increase at the 2-stage valve.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No pressure increase at the two-stage valve	1	► Motor running - swing lock switch ON - attachment in ground position			
		Follow the instructions for testing and adjusting the swing circuit as described in the SERVICE MANUAL. For better adjustment of the 2-stage valve disconnect and plug the pilot lines of the 2-stage valve. 1. Adjustment of the 330 bar stage 2. Adjustment of the 150 bar stage Could you successfully adjust the 2-stage valve?	Yes	Reconnect the pilot pressure hoses to the 2-stage valve and put the shovel back to operation.	For the target adjustment values refer to PM-CLINIC sheets
			No	Replace the 2-stage valve and perform the basic adjustments.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY007 - SWING ACCELERATION IS POOR

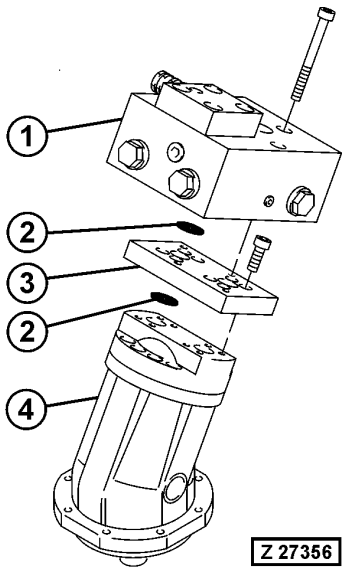
[1/2]

Trouble code	Trouble	Swing acceleration is poor
SHY007		
Contents of trouble	• The machine swing acceleration is poor (too long acceleration time or too short acceleration time). • The swing parking brake is released (otherwise a corresponding info message is displayed on the ETM display). • All other operating functions are going well.	
Related information	• For the basic adjustment procedures refer to the SERVICE MANUAL, section -Checks and Adjustments for the Swing Circuit-. • For the target adjustment values refer to PM-CLINIC sheets.	
Problem that appears on machine	• The swing acceleration is poor and any fast swing & load movement is not possible.	
Possible causes	• The 2-stage valve is defective or its adjustment is wrong. • The swing brake valve is defective (internal leak in the anti-cavitation/suction valves). • The O-ring at the spacer plate is leaking (oil flows from A to B side). • The swing motor is defective (internal leakage).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The 2-stage valve is defective or its adjustment is wrong	1	► Motor running & swing lock switched ON.			
		Perform the basic adjustments of the swing circuit according the SERVICE MANUAL and the PM-CLINIC sheets. Could you adjust the correct swing acceleration (swing cycle time)?	Yes	Go to next step.	PM-CLINIC sheets
			No	Replace the 2-stage valve and perform the basic adjustments. If - after replacement of the 2-stage valve - the swing acceleration is still poor, replace the complete swing brake valve blocks on top of the swing motors. Refer to next step.	
The swing brake valve is defective (internal leak in the anti-cavitation/suction valves)	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Perform the following steps according to the Safety Notes in the SERVICE MANUAL. Disconnect the high pressure hoses from the swing brake block. Remark: Before mounting the new block in the next work step, also see step 3 in this troubleshooting chart. Replace the complete block with new 2-stage valve and O-rings. Then perform the basic adjustments according to the SERVICE MANUAL and the PM-CLINIC sheets. Could you now adjust the correct swing acceleration (swing cycle time)?	Yes	Go to final step 5.	SERVICE MANUAL, PM-CLINIC sheets
			No	Go to next step	

SHY007 - Swing acceleration is poor

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The O-ring at the spacer plate is leaking (oil flows from A to B side)	3	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Check the O-rings on the spacer plate and on the lower side of the swing brake block. Are these O-rings ok?	Yes	Go to next step.	SERVICE MANUAL, PM-CLINIC sheets
			No	Check the spacer plate for indications of corrosion or pitting. If necessary replace the spacer plate. Replace the O-rings and mount the swing brake valve. Then perform the basic adjustments for the swing system according the SERVICE MANUAL and the PM-CLINIC sheets.	
				**Legend for figure Z 27356** (1) Swing brake valve (2) O-ring (3) Spacer plate (4) Swing motor	
The swing motor is defective (internal leakage)	4	► For testing the leak oil keep the motor running.			
		Check the leak oil flow at the swing motor. - Disconnect the leak oil hose from swing motor and plug the hose opening. - Mount an external hose to the swing motor leak oil port and drain the leak oil into a suitable container (min. volume: 20 ℓ). - Operate the swing movement for 1 minute and measure the leak oil amount (approx. 5 ℓ/min). Does the amount of leak oil exceed the above recommendation?	Yes	Replace the swing motor.	
			No	Check the swing motor drive shaft input at the swing gearbox. If any malfunction is found, repair as necessary.	
Final check	5	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY008 - EXCESSIVE OVERRUN WHEN STOPPING SWING

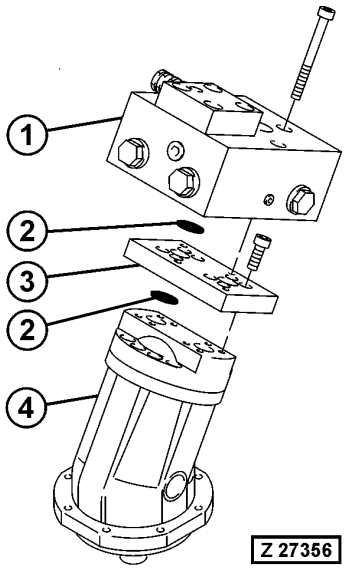
[1/2]

Trouble code	Trouble	Excessive overrun when stopping swing
SHY008		
Contents of trouble	• There is an excessive overrun when the swing movement of the superstructure is stopped.	
Related information	• If during swing movement the control lever is moved into neutral position, the superstructure swing movement should be stopped within an angle of 70°. • If the superstructure does not stop within this angle, the superstructure overrun can be considered as to be excessive.	
Problem that appears on machine	• The swing deceleration is poor (too long) and a correct positioning of the attachment is not possible.	
Possible causes	• The 2-stage valve is defective or its adjustment is wrong. • The swing brake valve is defective (internal leak in the anti-cavitation/suction valves). • The O-ring at the spacer plate is leaking (oil flows from A to B side). • The swing motor is defective (internal leakage).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The 2-stage valve is defective or its adjustment is wrong	1	► Motor running & swing lock switched ON.			
		Perform the basic adjustments of the swing circuit according the SERVICE MANUAL and the PM-CLINIC sheets. Could you adjust the correct swing deceleration?	Yes	Go to next step.	SERVICE MANUAL, PM-CLINIC sheets
			No	Replace the 2-stage valve and perform the basic adjustments. If - after replacement of the 2-stage valve - the swing deceleration is still poor, replace the complete swing brake valve block on top of the swing motor. Refer to next step.	
The swing brake valve is defective (internal leak in the anti-cavitation/suction valves)	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Perform the following steps according to the Safety Notes in the SERVICE MANUAL. Disconnect the high pressure hoses from the swing brake block. Remark: Before mounting the new block in the next work step, also see step 3 in this troubleshooting chart. Replace the complete block with new 2-stage valve and O-rings. Then perform the basic adjustments according to the SERVICE MANUAL and the PM-CLINIC sheets. Could you now adjust the correct swing deceleration?	Yes	Go to final step 5.	SERVICE MANUAL, PM-CLINIC sheets
			No	Go to next step	

SHY008 - Excessive overrun when stopping swing

[2/2]

<table>
<tr><th>Possible cause</th><th>Step</th><th colspan="2">Test condition & Inspection</th><th>Procedure</th><th>Nominal value</th></tr>
<tr><td rowspan="4">The O-ring at the spacer plate is leaking (oil flows from A to B side)</td><td rowspan="4">3</td><td colspan="2">► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.</td><td></td><td></td></tr>
<tr><td rowspan="2">Check the O-rings on the spacer plate and on the lower side of the swing brake block.
Are these O-rings ok?</td><td>Yes</td><td>Go to next step.</td><td rowspan="2">SERVICE MANUAL, PM-CLINIC sheets</td></tr>
<tr><td>No</td><td>Check the spacer plate for indications of corrosion or pitting.
If necessary replace the spacer plate.
Replace the O-rings and mount the swing brake valve.
Then perform the basic adjustments for the swing system according the SERVICE MANUAL and the PM-CLINIC sheets.</td></tr>
<tr><td colspan="3"> 

Legend for figure Z 27356
(1) Swing brake valve
(2) O-ring
(3) Spacer plate
(4) Swing motor</td></tr>
<tr><td rowspan="3">The swing motor is defective (internal leakage)</td><td rowspan="3">4</td><td colspan="2">► For testing the leak oil, keep the motor running.</td><td></td><td></td></tr>
<tr><td rowspan="2">Check the leak oil flow at the swing motor.
- Disconnect the leak oil hose from the swing motor and plug the hose opening.
- Mount an external hose to the swing motor leak oil port and drain the leak oil into a suitable container (min. volume: 20 ℓ).
- Operate the swing movement for 1 minute and measure the leak oil amount (approx. 5 ℓ/min).
Does the amount of leak oil exceed the above recommendation?</td><td>Yes</td><td>Replace the swing motor.</td><td rowspan="2"></td></tr>
<tr><td>No</td><td>Check the swing motor drive shaft input at the swing gearbox.
If any malfunction is found, repair as necessary.</td></tr>
<tr><td>Final check</td><td>5</td><td colspan="3">Bring the machine back into the original condition. Start the machine and verify that the repair was successful.</td><td></td></tr>
</table>

SHY009 - FLOATING SYSTEM WITHOUT FUNCTION

[1/1]

Trouble code	Trouble	Floating system without function
SHY009		
Contents of trouble	• The floating system for the functions -Stick In- and/or -Boom Down- is inoperative and/or its deactivation is impossible.	
Related information	• The floating system is divided in the two functions -Stick In- or -Boom Down-. • Floating valve Y132c is installed in the boom circuit. • Floating valves Y132b is installed in the stick circuit. • The electric circuit is faultless, but the floating system is inoperative and/or its deactivation is impossible (the spools in the hydraulic valves Y132c and Y132b do not move).	
Problem that appears on machine	• No floating function for -Stick In- and/or -Boom Down-.	
Possible causes	• The spools in the floating valves are mechanically stuck.	

Possible cause	Step	Test condition & Inspection	Procedure	Nominal value
The spools in the floating valves are mechanically stuck	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked - the attachment must be in down and full reach position		
		- Find out in which circuit the problem occurs (-Stick In- and/or -Boom Down-). - Check the electrical signals at the solenoid valves. - Disassemble the defective floating valve and check the state of the inner spool and spring.	Replace or repair the defective valve. Then perform the basic adjustment. Refer to the SERVICE MANUAL and to the PM-CLINIC sheets.	SERVICE MANUAL, PM-CLINIC
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.		

SHY010 - ABNORMAL NOISE COMING FROM SRV OR MRV

[1/1]

Trouble code	Trouble	Abnormal noise coming from SRV or MRV
SHY010		
Contents of trouble	• Abnormal noise is coming from the SRV or MRV sections.	
Related information	• Due to a possible defect of the mechanical inner parts (spring or piston) in the SRV or MRV sections, an abnormal noise is possible when the valve is activated. • Also a wrong adjustment of SRV or MRV may cause abnormal noise in the system.	
Problem that appears on machine	• Abnormal noise in the hydraulic system occurs during operation. • Any malfunction of the hydraulic circuit is possible. • Overheating of the hydraulic oil is possible.	
Possible causes	• Wrong adjustment or malfunction of SRV or MRV. • Malfunction of any throttle valve or suction valve at the main blocks or at the manifold.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wrong adjustment or malfunction of SRV or MRV.	1	► Motor running.			
		Perform a sectional troubleshooting and check the adjustment of SRV and MRV. Follow the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. Could you find out any functional problem?	Yes	Adjust or replace as necessary.	SERVICE MANUAL, PM-CLINIC sheets
			No	Go to next step.	
Malfunction of any throttle valve or suction valve at the main blocks or at the manifold	2	► Motor running.			
		Check the suspicious hydraulic circuit for noise and locate the problem (temperature difference) with the infrared temperature gauge. Is there a difference in the temperature?	Yes	Replace or repair as necessary.	
			No	Perform the complete PM-CLINIC procedures.	PM-CLINIC sheets
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY011 - TRACK TENSIONING SYSTEM WITHOUT FUNCTION

[1/1]

Trouble code	Trouble	Track tensioning system without function
SHY011		
Contents of trouble	• The automatic track tensioning system does not work.	
Related information	• The automatic track tensioning system works under the following condition: the motor is running and the ladder is in up position. • During a travel distance of approx. 20 m the track tension is built up automatically. • When the engine is stopped a system pressure of 35 bar is kept up, and in combination with an accumulator safety system (shock absorber) the correct track tension is always present.	
Problem that appears on machine	• If the track tensioning is too low, the wear-out on track pads and sprockets increases. • Track-off may occur during traveling or any movements due to loose tracks.	
Possible causes	• No or too low track tensioning pressure. • Accumulators insufficiently charged or defective. • Defective track tensioning cylinder(s).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No or too low track tensioning pressure	1	► Motor running - ladder in up position			
		Test and adjust the track tensioning pressure according to the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. Is adjustment possible and does the track tensioning system work orderly?	Yes	Bring the machine back to operation.	SERVICE MANUAL, PM-CLINIC sheets
			No	Go to next step.	
Accumulators insufficiently charged or defective	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked			
		Test and adjust the charging of the accumulators according to the instructions and specifications in the SERVICE MANUAL. Also refer to the latest version of PARTS & SERVICE NEWS AH03533. Is the function -charging the accumulators- possible and does the track tensioning system work orderly?	Yes	Bring the machine back to operation.	SERVICE MANUAL, PARTS & SERVICE NEWS AH03533 (latest version)
			No	Go to next step.	
Defective track tensioning cylinder(s)	3	► Motor running - ladder in up position			
		The tests and adjustments in steps 1 and 2 have been completed successfully (that means a track tensioning pressure of 35 bar is present and the accumulators are charged orderly). But the track tensioning cylinders can not hold the supplied pressure. Disassemble the track tensioning cylinders and check the seals at the pistons. Are the seals in the track tensioning cylinders defective?	Yes	Replace the seals. Flush the pipes and hoses of the track tensioning lines.	
			No	Check the track tensioning cylinders for swollen partitions. If necessary repair or replace the barrels - alternatively replace complete track tensioning cylinders.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY012 - LADDER DOES NOT MOVE, OR SLOW MOVEMENT

[1/2]

Trouble code	Trouble	Ladder does not move, or slow movement
SHY012		
Contents of trouble	• The ladder does not move when the function -ladder up- or -ladder down- is requested, or the ladder movement is too slow.	
Related information	• The electrical function of this circuit is already tested and ok. • The ladder pressure supply is charged by X4 pressure. • A 70 bar SRV is mounted in the ladder circuit.	
Problem that appears on machine	• The ladder does not move when the function -ladder up- or -ladder down- is requested, or the ladder movement is too slow.	
Possible causes	• No or too low X4 pressure. • Too low SRV pressure or the valve is defective. • Valve Y123a, Y123b and/or Y125 is mechanically stuck. • The ladder cylinder has an internal leakage (piston seal defective). • The ladder bearing is defective or it is tightened too strong.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No or too low X4 pressure	1	► Motor running.			
		Check the X4 pressure according to the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. Is the X4 pressure adjustment correct?	Yes	Go to next step.	60 bar
			No	Re-adjust and verify that the ladder is operated orderly.	
Too low SRV pressure or the valve is defective	2	► Motor running.			
		Check the SRV pressure in the ladder circuit according to the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. Is the SRV pressure adjustment correct?	Yes	Go to next step.	70 bar
			No	Re-adjust and verify that the ladder is operated orderly.	
Valve Y123a, Y123b and/or Y125 is mechanically stuck	3	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Disassemble the solenoid valve Y123a/b and Y125 and check the inner part of the spool for proper condition. Is the inner part of the spool in good condition?	Yes	Go to next step.	
			No	Replace valve Y123a/b and Y125.	
The ladder cylinder has an internal leakage (piston seal defective)	4	► Motor running.			
		Activate the function -ladder up- or -ladder down- until the cylinder end position is reached, and measure the temperature difference between the cylinder rod and the cylinder bottom side with an infrared temperature gauge. Do you measure a temperature difference in the piston area which is higher than 10°C?	Yes	Repair or replace the ladder cylinder.	
			No	Go to next step.	

SHY012 - Ladder does not move, or slow movement

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The ladder bearing is defective or it is tightened too strong	5	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Check the ladder bearing for proper condition and for the hand-tight tightening torque. Also refer to the ASSEMBLY MANUAL. Are the bearings in good condition and are they tightened correctly?	Yes	Inspect: - Is the standard ladder frame used (standard weight & size)? - Is the ladder or the ladder support defective or damaged by an accident? Replace or repair as necessary.	ASSEMBLY MANUAL
			No	Repair or replace the bearings as necessary and/or correct the tightening torque.	
Final check	6	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY013 - THE HYDRAULIC DRIVEN GREASE PUMP DOES NOT MOVE

[1/1]

Trouble code	Trouble	The hydraulic driven grease pump does not move
SHY013		
Contents of trouble	• The hydraulic driven grease pump does not start or move correctly (CLS or SLS).	
Related information	• The electrical function of this circuit is already tested and ok. • The hydraulic driven grease pump pressure supply is charged by X4 pressure. • The hydraulic oil flow and pressure at the hydraulic driven grease pump is reduced. • Also refer to the information in the SERVICE MANUAL section -Lubrication Systems-.	
Problem that appears on machine	• The grease system is inoperative because the pump does not work. • Any relevant trouble message may be displayed on ETM monitor.	
Possible causes	• No or too low X4 pressure. • No or insufficient performance of the relief valve on the grease pump. • No or insufficient performance of the volume reducing valve on the grease pump. • Internal leakage or damage at the grease pump. • Wrong grease pressure adjustment (grease pressure too high). • The vent valve (Y7) causes a mechanical jam (grease pump overload).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No or too low X4 pressure	1	► Motor running.			
		Check the X4 pressure according to the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. Is the X4 pressure adjustment correct?	Yes	Go to next step.	60 bar
			No	Readjust and verify that the grease pump operation is orderly.	
No or insufficient performance of the relief valve on the grease pump	2	► Motor running.			
		Check the pressure of the grease pump relief valve according to the instructions and specifications in the SERVICE MANUAL. Is the pressure adjustment of the grease pump relief valve correct?	Yes	Go to next step.	45 bar
			No	Readjust and verify that the grease pump operation is orderly.	
No or insufficient performance of the volume reducing valve on the grease pump	3	► Motor running.			
		Check the volume adjustment of the grease pump volume reducing valve according to the instructions and specifications in the SERVICE MANUAL. Is the volume adjustment correct?	Yes	Go to next step.	16 - 19 double strokes / min.
			No	Readjust and verify that the grease pump operation is orderly.	
Internal leakage or damage at the grease pump	4	► Motor running.			
		Check the adjustment and operation of the grease pump according to the instructions and specifications in the SERVICE MANUAL. Is the grease pump adjustment and operation correct?	Yes	Go to next step.	
			No	Repair or replace the grease pump as necessary.	
Wrong grease pressure adjustment (grease pressure too high)	5	► Motor running.			
		Check the grease pressure adjustment according to the instructions and specifications in the SERVICE MANUAL.. Is the grease pressure correct?	Yes	Go to next step.	Max. 180 bar
			No	Readjust or replace the end of line switch.	
The vent valve (Y7) causes a mechanical jam (grease pump overload)	6	► Motor running.			
		Test the vent valve operation according to the instructions and specifications in the SERVICE MANUAL.	Repair or replace the vent valve and test and adjust the grease system if necessary.		
Final check	7	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY014 - ABNORMAL TRACK CHAIN MOVEMENT DURING DIGGING

[1/1]

Trouble code	Trouble	Abnormal track chain movement during digging
SHY014		
Contents of trouble	• Abnormal track chain movement in reverse direction during bucket filling - FSA. • Abnormal track chain movement in forward direction during bucket filling - BHA.	
Related information	• During bucket filling movement (FSA) the track chain(s) make a self-movement in reverse direction. That means, the forward loading force will be eliminated by a self-reversing movement of the track system. • During bucket filling movement (BHA) the track chain(s) make a self-movement in forward direction. That means the reverse loading force will be eliminated by a self-forwarding movement of the track system.	
Problem that appears on machine	• Insufficient digging or loading performance.	
Possible causes	• Internal leakage at the rotary joint.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Internal leakage at the rotary joint	1	► Motor running.			
		Test and adjust the SRV pressure according to the instructions and specifications in the SERVICE MANUAL and PM-CLINIC sheets. → Measure the SRV pressure at the travel system with connected travel main lines (310 ± 5 bar). Is a pressure adjustment necessary and possible?	Yes	Test and adjust the SRV pressure at the travel system with closed travel parking brake (Y16 disconnected). Adjust or change the SRV.	Travel SRV pressure: 310 ± 5 bar
			No	Repair or replace the rotary joint.	
Final check	2	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY015 - SWING PARKING BRAKE WITHOUT FUNCTION

[1/1]

Trouble code	Trouble	Swing parking brake without function
SHY015		
Contents of trouble	• The swing parking brake is inoperative (the superstructure moves) although the swing parking brake switch S29 is switched to position 1 (brake closed). • The X2 pressure circuit is already tested and ok (X2 pressure 35 bar at valve Y5). • The electrical circuit is already tested and ok.	
Related information	• The swing parking brake is a disc brake, mounted on the top of the swing gearbox. • The -Siebenhaar- swing gearbox has an external swing brake housing with a separate oil filling. • The -L&S- swing gearbox has an integrated swing brake system which is lubricated by the gearbox oil. • Both swing parking brakes are working as a -Spring Loaded System-. That means the pilot pressure opens the swing parking brakes. Without pilot pressure the springs close the swing parking brakes.	
Problem that appears on machine	• The swing parking brake is inoperative.	
Possible causes	• The pilot pressure in the swing parking brake circuit is not released. • The brake discs are burnt and/or the piston seals are defective.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The pilot pressure in the swing parking brake circuit is not released	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		The solenoid valve Y5 is mechanically stuck. The pilot pressure in the swing parking brake line is not released to the hydraulic oil tank. Is the piston of the solenoid valve stuck?	Yes	Replace the solenoid valve Y5.	0 bar
			No	Go to next step.	
The brake discs are burnt and the piston seals are defective	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Perform the following steps: - Disconnect the pilot pressure supply hoses for the swing parking brakes from the swing gearbox. - Install a manual hydraulic pump with 60 bar manometer in order to test the swing parking brake piston seals. - Use a manual pump and increase the pressure up to 60 bar. - The built-up pressure must be held for 15 minutes. Can you build-up and hold the pressure for 15 minutes?	Yes	Check for leaks or mixed-up connections in the pilot supply lines for the swing parking brake system.	60 bar for 15 minutes
			No	Repair or replace the swing parking brake and check the swing parking brake for proper operation. **NOTICE** **Do not t forget to bleed the pilot line to the swing parking brake housing correctly.**	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SHY016 - TRAVEL PARKING BRAKE WITHOUT FUNCTION

[1/1]

Trouble code	Trouble	Travel parking brake without function
SHY016		
Contents of trouble	• The travel parking brake is inoperative. • The X2 pressure circuit is already tested and ok (X2 pressure 35 bar at valve Y16). • The electrical circuit is already tested and ok.	
Related information	• The travel parking brake is a mounted disk brake between both travel motors and the L&S gearboxes. (wet brakes). • The travel parking brake is a mounted disk brake between both travel motors and the Zollern gearboxes. (dry brakes). • Both travel parking brakes are working as a -Spring Loaded System-. That means the pilot pressure opens the travel parking brakes. Without pilot pressure the springs close the travel parking brakes.	
Problem that appears on machine	• There is no automatic travel parking brake function when the ladder is down or the engine is stopped.	
Possible causes	• The pilot pressure in the travel parking brake circuit is not released. • Low X2 pressure. • The brake discs are burnt and/or the piston seals are defective.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The pilot pressure in the travel parking brake circuit is not released	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		The solenoid valve Y16 is mechanically stuck. The pilot pressure in the travel parking brake line is not released to the hydraulic oil tank. Is the piston of the solenoid valve stuck?	Yes	Replace the solenoid valve Y16.	0 bar
			No	Go to next step.	
Low X2 pressure	2	► Motor running.			
		Check and adjust the X2 pressure according the Shop Manual and PM Clinic sheet. Is a pressure adjustment possible?	Yes	Go to next step.	35 bar
			No	Replace the X2 pressure valve.	
The brake discs are burnt and/or the piston seals are defective	3	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Perform the following steps: - Disconnect the pilot pressure supply hoses from the travel parking brake housings. - Install a manual hydraulic pump with 60 bar manometer in order to test the travel parking brake piston seals (perform the test for both travel brake housings). - Using the manual pump increase the pressure up to 60 bar. - The built-up pressure must be held for 15 minutes. Can you build-up and hold the pressure for 15 minutes?	Yes	Check for leaks or mixed-up connections in the pilot supply lines for the travel parking brake system.	60 bar for 15 minutes
			No	Repair or replace the travel parking brakes and check the travel parking brakes for proper operation. **NOTICE** **Do not t forget to bleed the pilot line to the travel parking brake housing correctly.**	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SYMPTOMS OF THE MECHANICAL SYSTEM

SME001 - ABNORMAL NOISE AROUND THE PTO / HYDRAULIC PUMPS

[1/2]

Trouble code	Trouble	Abnormal noise around the PTO / hydraulic pumps
SME001		
Contents of trouble	• Abnormal or excessive noise coming from the PTO or from the hydraulic pumps.	
Related information	• At first check oil level and oil quality of the PTO. • Refer to the OPERATION & MAINTENANCE MANUAL for detailed information concerning the correct oil specification and oil quality.	
Problem that appears on machine	• Abnormal noise. • The noisy component may cause an unexpected breakdown.	
Possible causes	• No oil or the oil level is too low in the PTO • No oil or low oil level at the Pump bearing housing • Main pump defective (internal mechanical damage) • Air entering the hydraulic system or air bubbles inside the hydraulic suction lines	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
No oil or the oil level is too low in the PTO	1	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check the oil level in the PTO housing using the dipstick. For details refer to the OPERATION & MAINTENANCE MANUAL. Is the oil level correct between the MIN and MAX marks?	Yes	Go to next step.	Between MIN & MAX
			No	Refill specified oil up to the specified level and recheck if the abnormal noise disappeared.	
No oil or low oil level at the Pump bearing housing	2	► Main key switch 1S1OFF and maintenance safety switch 1S58 locked.			
		Check and verify the correct oil level in the pump bearing housing on the PTO gear box. Refer also the OMM. The levels and oil quality ok? (also refer P&S News AH07515 latest edition)	Yes	Go to next step.	See OMM and Parts & Service News for refill level and spec. oil quality.
			No	Refill specified oil up to the specified level and recheck if the abnormal noise disappeared.	
Main pump defective (internal mechanical damage)	3	► Motor running & move the attachment in working condition.			
		Check for abnormal noise coming from around or directly from the main pumps. Use a vibration analyzer, take measurements at all main pumps and compare the measurement results. Does any main pump show excessive noise generation?	Yes	Double-check the suspicious main pump for excessive noise generation. If it is confirmed, replace the noisy main pump.	
			No	All main pumps generate too much noise simultaneously. Got to next step.	

SME001 - Abnormal noise around the PTO / hydraulic pumps

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Air entering the hydraulic system or air bubbles inside the hydraulic suction lines	4	► Motor running & move the attachment in working condition.			
		Check the sight glass at the hydraulic tank for air bubbles. Also check the suction lines and clamps of sealed connections for air leakage. Remark: Suction lines always must be mounted with silicon applied to the pipes and they must be fixed with double-clamps. Could you locate air in the hydraulic tank or air bubbles in the suction lines?	Yes	If you located air bubbles in the hydraulic system refer to the latest version of PARTS & SERVICE NEWS AH01513 & AH00504 for bleeding the hydraulic cylinders.	There is no air in the hydraulic oil tank and in the suction lines.
			No	Check the lifetime of the PTO and the main pumps. The abnormal noise may be caused by normal wear due to component aging.	
Final check	5	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME002 - THERE IS A BIG SHOCK WHEN STOPPING SWING

[1/1]

Trouble code	Trouble	There is a big shock when stopping swing
SME002		
Contents of trouble	• A big shock occurs when stopping the swing movement with the swing foot brake or by the counter swing function.	
Related information	• A big shock occurs when the superstructure changes its swing direction.	
Problem that appears on machine	• Excessive wear and in consequence any damage may occur.	
Possible causes	• Wear in vertical & horizontal direction at the boom foot bushings and pins. • The swing circle causes excessive wear at the bearing or backlash.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wear in vertical & horizontal direction at the boom foot bushings and pins	1	► Main key switch 1S1ON – motor running without load.			
		Move the attachment smoothly and check for excessive wear at the boom foot bushings. Is the movement too large?	Yes	Replace pins and bushings at the connection between boom and superstructure. After replacement measure the axial clearance at the connection between boom and superstructure. If the axial play is still out of range repair with the welding and lathe tool.	Boom foot bushings & pins, max. clearance 2.5 mm Axial play between boom and super-structure, max. 2.5 mm
			No	Go to next step.	
The swing circle causes excessive wear at the bearing or backlash	2	► Main key switch 1S1ON – motor running without load.			
		Move the attachment smoothly and check for excessive wear and movement of the swing circle bearing and backlash. Also refer to the latest version of PARTS & SERVICE NEWS AH02513. Is the wear out of range?	Yes	Replace the swing circle.	P&S NEWS AH02513 (latest version)
			No	- Check the teeth around the swing circle. - Also check the lubrication crosses for the swing circle teeth (tightening). - Go to SME006 and refer to its troubleshooting procedures.	
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME003 - THERE IS A BIG ABNORMAL NOISE WHEN MOVING/STOPPING SWING

[1/2]

Trouble code	Trouble	There is a big abnormal noise when moving/stopping swing
SME003		
Contents of trouble	• A big abnormal noise occurs when moving/stopping swing. • The abnormal noise may also exist permanently during the swing movement.	
Related information	• First check all oil levels of the swing gearbox. • Verify the oil level and the viscosity of the oil used in the swing gearbox (CLP 220 only). • On -Siebenhaar- swing gear verify the oil level and the viscosity of the oil used in the swing parking brake housing (HLP 32 only). • Verify the oil level and the viscosity of the oil used in the motor adapter housing (HLP 32 only). • Also refer to the OPERATION & MAINTENANCE MANUAL and to the SERVICE MANUAL.	
Problem that appears on machine	• Abnormal noise and in consequence any damage may occur.	
Possible causes	• Swing gearbox - internal damage. • Faulty bottom bearing of the swing gearbox pinion (external grease lubrication). • Swing parking brake - internal damage (discs are welded together). • Damaged mechanical connection between the swing hydraulic motor and the swing gearbox (motor adapter housing, excessive wear at the connection).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Swing gearbox - internal damage	1	► Motor running & smooth swing movement of the superstructure.			
		Check for proper tightening of the swing gearbox and retighten if necessary. During swing movement check and listen for noise at the gearbox. Can you identify any abnormal noise coming from the swing gearbox?	Yes	Replace the swing gearbox.	SERVICE MANUAL → Tightening chart
			No	Go to next step.	
Faulty bottom bearing of the swing gearbox pinion (external grease lubrication)	2	► Motor running & smooth swing movement of the superstructure.			
		During swing movement check the movement of the swing gearbox pinion. During swing movement also check and listen for noise at the lower side of the swing gearbox. Check the grease supply and grease amount (injector adjustment) at the pinion bearing. Can you identify any abnormal noise or abnormal movement of the swing gearbox pinion?	Yes	Replace the swing gearbox.	SERVICE MANUAL → Component structure and grease system.
			No	Go to next step.	

SME003 - There is a big abnormal noise when moving/stopping swing

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Swing parking brake - internal damage (discs are welded together)	3	► Motor running & smooth swing movement of the superstructure.			
		Check the swing parking brake for proper condition (parking brake switch closed). Carefully move the control lever to start swing movement. Can you identify any abnormal noise or abnormal swing movement?	Yes	Although the swing parking brake is closed the superstructure moves and causes any noise or shock during stopping or starting the swing movement. - Inspect and replace the defective swing parking brake as necessary. - Also check for contamination of the gearbox oil. Drain contaminated oil and refill fresh specified gearbox oil.	SERVICE MANUAL
			No	Go to next step.	
Damaged mechanical connection between the swing hydraulic motor and the swing gearbox (motor adapter housing, excessive wear at the connection)	4	► Motor running & smooth swing movement of the superstructure.			
		Check the swing movement for proper condition. (starting swing - stopping swing - counter swing - foot brake actuation - fast swing speed - low swing speed) Can you identify any abnormal noise, abnormal swing movement, or abnormal vibration?	Yes	Disconnect the hydraulic swing motor from the swing gear box and check the connection between both components. If the connection is damaged, replace the swing gearbox and the hydraulic motor.	SERVICE MANUAL
			No	Check the hydraulic swing adjustment (two stage valve). Refer to the SERVICE MANUAL and the PM-CLINIC sheets. Also see the troubleshooting charts of SHY008-010.	SERVICE MANUAL, PM-CLINIC
Final check	5	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME004 - ABNORMAL NOISE COMING FROM A TRAVEL GEARBOX

[1/2]

Trouble code	Trouble	Abnormal noise coming from a travel gearbox
SME004		
Contents of trouble	• Abnormal noise occurs when the shovel starts or stops travelling. • The abnormal noise may also exist permanently when the shovel is travelling continuously.	
Related information	• First check the oil levels at both travel gearboxes. • Verify the oil level and the viscosity of the oil used in the travel gearboxes (CLP 220 only). • Make sure that the vent valves at the end of the breather hoses are clean (inside car body). • Drain the oil of the travel gearboxes and check for particles, chippings, or pieces due to wear. • Also refer to the OPERATION & MAINTENANCE MANUAL and to the SERVICE MANUAL.	
Problem that appears on machine	• Abnormal noise and in consequence any capital damage may occur.	
Possible causes	• Gearbox broken (housing or internal parts defective). Short term failure. • A wrong oil viscosity is used (any planetary gear teeth and/or bearing defective). Long term failure. • Hydraulic travel motor connection defective (no oil in the motor adapter housing).	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Gearbox broken (housing or internal parts defective). Short term failure.	1	► Motor running.			
		Take the track off the ground - during rotation check each side for abnormal noise or shocks. Can you identify any abnormal noise or shock during track rotation?	Yes	Refer to the SERVICE MANUAL & PM-CLINIC sheets and carry out the basic adjustment of the travel system. If the noise continues after the basic adjustment, replace the concerned travel gearbox.	SERVICE MANUAL, PM-CLINIC
			No	Go to next step.	
A wrong oil viscosity is used (any planetary gear teeth and/or bearing defective). Long term failure.	2	► Motor running.			
		Take the track off the ground - during rotation check each side for abnormal noise or shocks. Can you identify any abnormal noise or shock during track rotation?	Yes	Refer to the SERVICE MANUAL & PM-CLINIC sheets and carry out the basic adjustment of the travel system. If the noise continues after the basic adjustment, replace the concerned travel gearbox.	SERVICE MANUAL, PM-CLINIC
			No	Go to next step.	

SME004 - Abnormal noise coming from a travel gearbox

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Hydraulic travel motor connection defective (no oil in the motor adapter housing)	3	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Disconnect the concerning travel motor from the adapter housing and check the connection at the motor and inside the motor adapter housing for any damage. Can you identify any damage (because of missing oil)?	Yes	Replace the concerning travel motor and replace the planetary gear pinion at the gear box.	
			No	Check the travel motor(s) for internal and external leakage and replace if necessary.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME005 - ABNORMAL NOISE COMING FROM THE WORK EQUIPMENT (LACK OF GREASE)

[1/2]

Trouble code	Trouble	Abnormal noise coming from the work equipment (lack of grease)
SME005		
Contents of trouble	• Abnormal noise occurs when the attachment is moving. • The abnormal noise occurs due to lack of grease (CLS) at the pins and the bushings of the work equipment	
Related information	• First check the grease level at the CLS grease barrel. • Verify that the barrel contains grease of the specified quality and viscosity. • Verify that all injectors are in good condition and properly adjusted. • Verify that the setting of the pause time is correct. • Also refer to the OPERATION & MAINTENANCE MANUAL and to the SERVICE MANUAL.	
Problem that appears on machine	• Abnormal noise and in consequence any capital damage of the pins and bushings may occur.	
Possible causes	• A wrong grease quality is used. • Contaminated grease is used. • Daily fault messages of the CLS system occur - Download the ETM protocol memory data and verify the system status.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
A wrong grease quality is used	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Recheck the grease quality and specification under consideration of the KMG recommendations in the OPERATION & MAINTENANCE MANUAL and in the latest version of the PARTS & SERVICE NEWS AH02539. Does the used grease correspond to the KMG recommendations?	Yes	Go to next step.	SERVICE MANUAL, PM-CLINIC P&S NEWS AH02539 (latest version)
			No	Replace wrong grease by specified grease and replace the defective pins and bushings on the attachment side. Check all grease injectors for proper function. Pre-lubricate all pins and bushings through the injectors until the new grease flows out.	

SME005 - Abnormal noise coming from the work equipment (lack of grease)

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Contaminated grease is used (grease barrel refilling without refilling the filter)	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		The injectors are worn out due to contaminated grease in the barrel. No or not enough grease causes early wear and noise at the pins and bushings of the attachment. Are pins and bushings worn out due to lacking grease and defective injectors?	Yes	Replace the contaminated grease by specified grease. Clean or replace the barrel as necessary. Replace the defective pins and bushings on the attachment side. Check all grease injectors for proper function. Pre-lubricate all pins and bushings through the injectors until the new grease flows out.	
			No	Go to next step.	
Daily fault messages of the CLS system occur. Download the data from the protocol memory of the ETM and verify the system status	3	► Keep the main key switch 1S1in ON position.			
		Download the data from the protocol memory of the ETM and verify the system status. Do the memory data show a lot of -bucket cut off- messages due to missing grease?		Repair or replace any defective parts of the CLS system as necessary. Replace the defective pins and bushings on the attachment side (standardization as primary state). Check all grease injectors for proper function. Pre-lubricate all pins and bushings through the injectors until the new grease flows out.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME006 - ABNORMAL IDLER MOVEMENT

[1/1]

Trouble code	Trouble	Abnormal idler movement
SME006		
Contents of trouble	• Abnormal idler movement occurs at the supports.	
Related information	• First check the idler axle and bushings for wear. • Verify that the specified oil quality and viscosity is used inside the idler (CLP 220). • Verify that the track tensioning system is working properly. • Also refer to the OPERATION & MAINTENANCE MANUAL and to the SERVICE MANUAL.	
Problem that appears on machine	• Abnormal idler movement and in consequence any capital damage of axles, bushings, or supports may occur.	
Possible causes	• Leak at the life time seals (axle & bushing worn out). • Idler supports are below the wear limit.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Leak at the life time seals (axle & bushing worn out)	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Open the track chain and remove the idler. Disconnect the supports of the idler and check the lifetime seals and the wear of axle and bushings. Are axle and bushings worn out?	Yes	Replace the complete idler including new supports.	
			No	Go to next step.	
Idler supports are below the wear limit	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Check for excessive wear between the idler supports and the track frame.	If excessive wear is found replace the idler supports. Also check the lifetime of the idler, and if the lifetime is exceeded replace worn components as necessary.		
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SME007 - LEAK AT THE PTO SHAFT SEAL

[1/2]

Trouble code	Trouble	Leak at the PTO shaft seal
SME007		
Contents of trouble	• Leak at the PTO shaft seal (after changing the oil viscosity).	
Related information	Temperatur Getriebeöl (TG) / temperature gear oil (TG) — Z 26718 Ölsorte nach ISO / kind of oil according to ISO	

temperature in C	T32	CLP150	Hypoid Oil 75W-140
TypNr. / type no.	1	2	3
TG2	60	95	95
TG1	15	48	54

☞ **Standard oil specification: CLP 150**

Remark:
In some areas it is possible to change to Hypoid Oil 75W-140.
Before changing the oil type contact KMG.

Problem that appears on machine	• Leak at the PTO shaft seal.
Possible causes	• The bearing of the drive shaft is defective (too much movement), check the PTO lifetime. • The shaft seal is burnt due to use of a wrong oil viscosity. • Coupling mounting or coupling defective (misalignment).

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
The bearing of the drive shaft is defective (too much movement), check the PTO lifetime	1	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Remove the coupling. Check the input shaft bearing of the PTO gearbox. Is abnormal bearing movement (horizontal & vertical) detectable?	Yes	Replace or repair as necessary.	Movement of max. 1.2 mm
			No	Go to next step.	
The shaft seal is burnt due to use of a wrong oil viscosity	2	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Clean up the PTO drive shaft area and check for any leakage. In some areas the oil viscosity recommend by KMG is not available. If a wrong oil viscosity was used the seal ring on the drive shaft may be burnt, causing a leakage. Was oil type CLP 150 used on your shovel?	Yes	Check the PTO lifetime and replace the PTO if necessary. Remanufacture the old PTO according the Reman Guide Line.	Oil specification: CLP 150
			No	Replace / reseal - remanufacture the PTO, and from now on use the correct oil specification CLP 150.	

SME007 - Leak at the PTO shaft seal

[2/2]

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Coupling mounting or coupling defective (misalignment)	3	► Main key switch 1S1OFF & maintenance safety switch 1S58 locked.			
		Remove the coupling. Check the coupling lifetime and check the coupling for any defect Is any problem present?	Yes	Replace the coupling and observe the correct alignment.	Refer to the SERVICE MANUAL
			No	Due to a long term PTO overheating (permanently over 80 C°) the seal ring on the input shaft may be hardened and brittle. Check the PTO cooling system - verify that the PTO temperature does not exceed 65 C° during normal operation.	
Final check	4	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

SYMPTOMS OF ADDITIONAL SYSTEMS

SAD001 - AIR CONDITIONER DOES NOT WORK

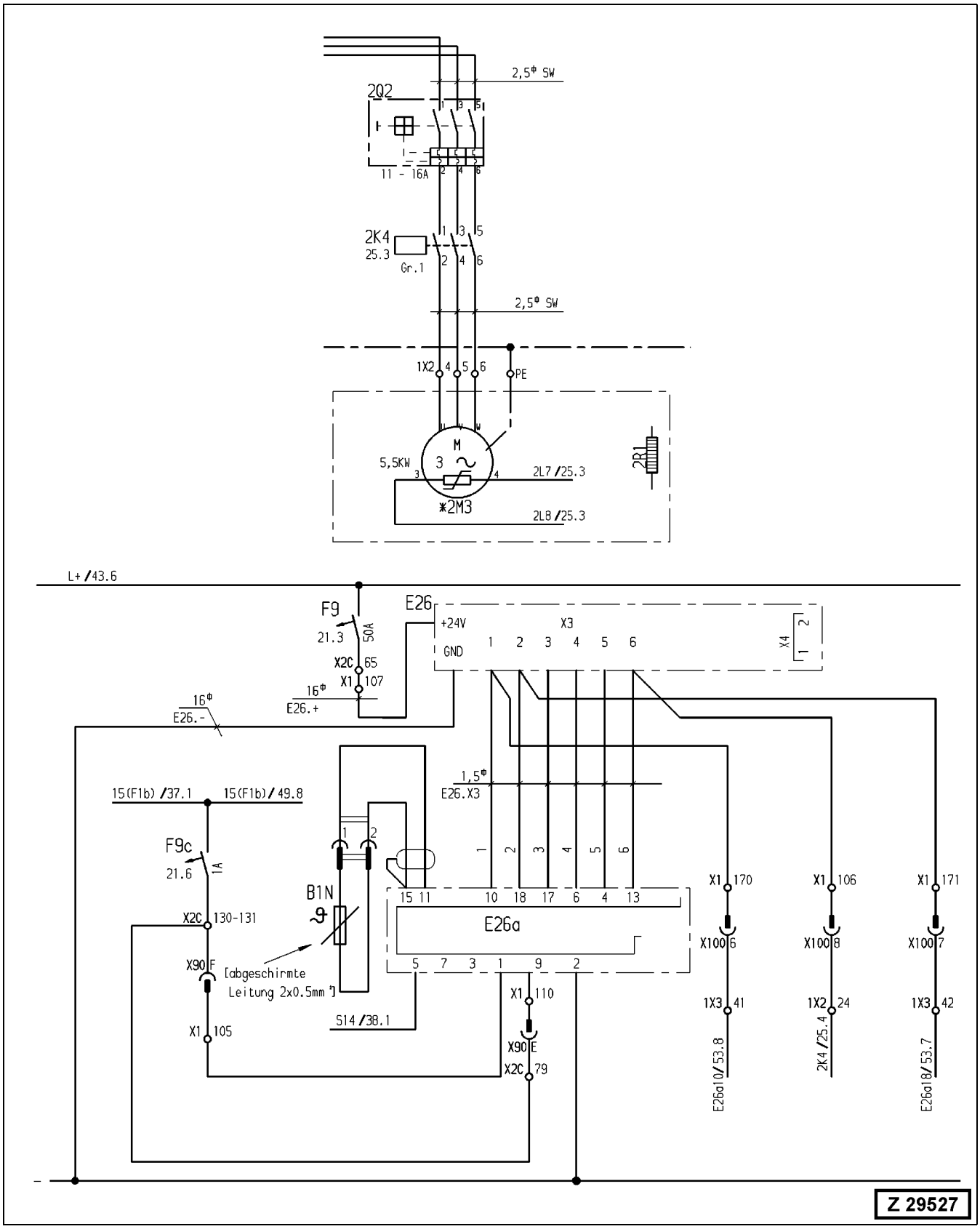
[1/2]

Trouble code	Trouble	Air conditioner does not work
SAD001		
Contents of trouble	• The AC system doesn't work properly or it shuts off.	
Related information	• For maintenance hints and the filling procedure refer also to the Operation and Maintenance Manual of the air conditioner manufacturer.	
Problem that appears on machine	• AC does not work and excessive heat in the operator's cab is possible. • Shovel breakdown may happen (due to excessive heat in the operator's cab).	
Possible causes	• Wrong or no maintenance of the AC system. • Any electrical problem causes a disconnection or short circuit in the wiring or any component of the AC system.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wrong or no maintenance of the AC system	1	► Motor running & AC switched ON.			
		Check the maintenance history. Was the maintenance done as scheduled?	Yes	Go to next step.	
			No	Using the AC test tools, perform the standard checking procedure.	
Any electrical problem causes a disconnection or short circuit in the wiring or in any component of the AC system	2	► Motor running & AC switched ON.			
		WARNING When working at he AC circuit or when handling refrigerant R134a always observe the safety hints given in the AC manual and in the data sheet of the refrigerant manufacturer. a) Using the AC test tools perform a standard check of the AC system. b) Check for trouble codes displayed in the AC console (for details refer to the AC manual). c) Check for electrical problems. Could you identify any problem?		a) Check the AC system together with an AC specialist and repair as necessary. b) Refer to the AC manual and follow the given procedures. c) Refer to the wiring diagram and check the main fuse box and the relays in the AC unit. Check the wiring for disconnection or short circuit. Locate the problem and repair as necessary.	AC manual Wiring diagrams
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

Wiring Diagram for trouble code SAD001

[2/2]



SAD002 - AIR CONDITIONER HV / LV CABINET DOES NOT WORK

[1/2]

Trouble code	Trouble	Air conditioner does not work
SAD002		
Contents of trouble	• The AC system of the HV / LV cabinet does not work properly or it shuts off.	
Related information	• For maintenance hints and the filling procedure refer also to the Operation and Maintenance Manual of the air conditioner manufacturer.	
Problem that appears on machine	• AC does not work and excessive heat in the HV / LV cabinet is possible. • Shovel breakdown may happen (due to eelectrical problems n the HV / LV cabinet).	
Possible causes	• Wrong or no maintenance of the AC system. • Any electrical problem causes a disconnection or short circuit in the wiring or any component of the AC system.	

Possible cause	Step	Test condition & Inspection		Procedure	Nominal value
Wrong or no maintenance of the AC system	1	► Motor running & AC switched ON.			
		Check the maintenance history. Was the maintenance done as scheduled?	Yes	Go to next step.	
			No	Using the AC test tools, perform the standard checking procedure.	
Any electrical problem causes a disconnection or short circuit in the wiring or in any component of the AC system	2	► Motor running & AC switched ON.			
		⚠ WARNING When working at he AC circuit or when handling refrigerant R134a always observe the safety hints given in the AC manual and in the data sheet of the refrigerant manufacturer. a) Using the AC test tools perform a standard check of the AC system. b) Check for trouble codes displayed in the AC console (for details refer to the AC manual). c) Check for electrical problems. Could you identify any problem?		a) Check the AC system together with an AC specialist and repair as necessary. b) Refer to the AC manual and follow the given procedures. c) Refer to the wiring diagram and check the main fuse box and the relays in the AC unit. Check the wiring for disconnection or short circuit. Locate the problem and repair as necessary.	AC manual Wiring diagrams
Final check	3	Bring the machine back into the original condition. Start the machine and verify that the repair was successful.			

Wiring Diagram for trouble code SAD002

[2/2]

